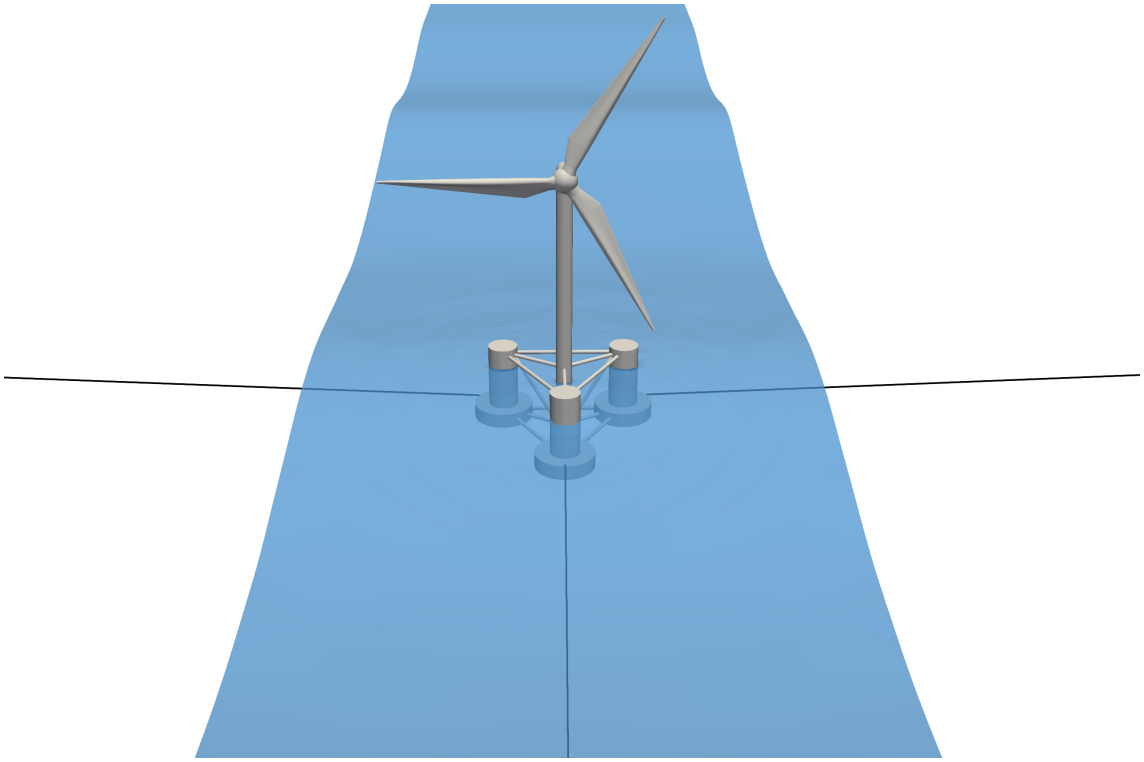
REEF3D :: User Guide



REEF3D 26.09

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AI declaration

This User Guide has been partially created with the assistance of Claude (Anthropic, model claude-opus-5-5).

Parts of the REEF3D open-source software have been created with the assistance of Claude (Anthropic, model claude-opus-5-5).

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1. Introduction

REEF3D is an open-source hydrodynamics framework for coastal, marine and hydraulic engineering. It combines several flow solvers that cover the relevant length scales, from ocean waves travelling over tens of kilometres to the detailed flow around a single structure:

- **REEF3D::CFD**: three-dimensional two-phase Navier-Stokes solver with interface capturing,
- **REEF3D::NHFLOW**: three-dimensional non-hydrostatic wave and flow model on a σ-grid,
- **REEF3D::FNPF**: fully nonlinear potential flow wave model,
- **REEF3D::SFLOW**: depth-averaged non-hydrostatic shallow water model.

All modules share the same input file, wave generation library, parallelization and output routines. The mesh and the domain decomposition are generated with the accompanying open-source pre-processor DIVEMesh, which also handles solid geometries from STL files and bathymetry from measured data points. Multiphysics solvers for floating bodies (6DOF) with mooring lines and nets, sediment transport, porous media and turbulence are coupled to the flow solvers. Typical applications are wave propagation and transformation in coastal areas, wave loads on offshore and coastal structures including offshore wind foundations and floating structures, open channel and river flow, scour and sediment transport, and aquaculture structures.

REEF3D is released under the GNU General Public License, version 3 or later (GPL v3). The source code, the release notes and this guide are available from `www.reef3d.com` and from the GitHub repository `https://github.com/REEF3D/REEF3D`; DIVEMesh is available from `https://github.com/REEF3D/DIVEMesh`. Open source means that the code can be used at no cost, that users can see exactly which equations and algorithms are solved, and that developments benefit students, researchers and engineers working on academic and commercial problems alike.

REEF3D was designed under the following premises:

1. Hydraulic, coastal, marine and environmental engineering applications
   - robust modeling of complex free-surface flows and breaking waves
   - numerical wave tank and open channel boundary conditions
   - sediment transport and fluid-structure interaction
2. Ease of use: mesh generation should not limit user productivity
   - rectilinear grids with immersed boundaries instead of body-fitted meshes
   - STL file input for solid geometries
   - straightforward handling of natural bathymetries

3. Large computations can only be exploited in parallel
    - full parallelization based on domain decomposition and MPI
4. Stable, accurate and fast numerical simulations
    - high-order WENO discretization in space and TVD Runge-Kutta schemes in time
    - staggered grid with tight velocity-pressure coupling (CFD)
    - adaptive time stepping for stability
5. Maintainability and extensibility
    - modular, object-oriented C++
    - MPI and HYPRE as industry-standard libraries for high-performance computing

How this guide is organised

- Chapter 2 (Model Overview) describes the four hydrodynamic modules and helps to choose the right one for a given problem.
- Chapter 3 (Getting Started) walks through a first simulation.
- Chapter 4 (Theory and Numerical Methods) summarizes the governing equations and numerical methods of each module.
- Chapter 5 (Multiphysics) summarizes the theory of the multiphysics models: floating bodies, moorings and nets, flexible structures, sediment transport, porous media and vegetation, heat and concentration, rheology, Lagrangian particles, sea ice and wind forcing.
- Chapter 6 (Functions) is the reference for all commands of the input file `ctrl.txt`, sorted by letter.
- Chapter 7 (Compiling the Code) explains how to install the dependencies and build REEF3D and DIVEMesh on Linux, macOS, Windows (WSL) and with Docker.
- Chapter 8 (Running the Code) explains the simulation folder, how to run DIVEMesh and REEF3D, and how to run on HPC clusters.
- Chapter 9 (Post-Processing) describes the output files and their visualization in ParaView.
- The remaining chapters contain tutorials for SFLOW, FNPF, NHFLOW and CFD, which are also provided as input files in the `Tutorials` folder of the repository.

2. Model Overview

The hydrodynamic module is selected with **A 10** in `ctrl.txt`. Table 2.1 summarises the main differences between the modules, Section 2.1 gives guidance for the choice, and Chapter 4 describes the governing equations and numerical methods of each module. As a rule of thumb, the cheapest module that still resolves the relevant physics should be chosen: SFLOW or FNPF for large areas and long durations, NHFLOW for nearshore wave processes with breaking and run-up, and CFD where the three-dimensional flow, the two-phase interface or the detailed flow around structures matters.

Table 2.1: Overview of the hydrodynamic modules.

	SFLOW	**FNPF**	**NHFLOW**	**CFD**
A 10	2	3	5	6
Governing equations	depth-averaged non-hydrostatic shallow water equations	Laplace equation for the velocity potential, fully nonlinear free-surface boundary conditions	non-hydrostatic Navier-Stokes equations, Poisson equation for the dynamic pressure	incompressible RANS/Navier-Stokes equations, two-phase
Vertical resolution	none (2D horizontal grid)	σ-coordinates	σ-coordinates	full 3D rectilinear grid
Free surface	water level η	single-valued η	single-valued η	level set (or VOF), overturning possible
Wave breaking	breaking detection	breaking model (detection and damping)	shock-capturing (HLL-type Riemann solver)	resolved by the two-phase flow
Wetting and drying	yes	coastline algorithm, optional wetting/drying (A 343)	yes	implicit through interface capturing
Floating bodies (6DOF)	one-way body motion (e.g. ship waves)	yes, resolved bodies with mooring	yes, with mooring and nets	yes, with mooring and nets
Sediment transport	bedload	no	bedload and suspended load	bedload and suspended load
Further physics	turbulence and diffusion, bed friction	coupling to CFD (HDC)	turbulence (RANS/LES), VRANS porous media, wind forcing	turbulence (RANS/LES), VRANS porous media, heat and concentration transport, non-Newtonian rheology, FSI, coupling from FNPF (HDC)
Relative cost	lowest	low	medium	highest
Typical use	open channel and river flow, long waves over large areas	wave propagation from deep to shallow water, design sea states, wave input for CFD	nearshore waves with breaking and run-up, wave-structure interaction	detailed flow around structures, breaking wave loads, scour, dam break

2.1 Choosing a Module

The module is selected with A 10: 2 = SFLOW, 3 = FNPF, 5 = NHFLOW, 6 = CFD (default). As a first orientation:

REEF3D::FNPF for wave propagation from deep to shallow water over large areas and complex coastlines, irregular and multi-directional sea states and wave kinematics for design, as long as the free surface is single-valued and viscous effects are not important. It is the most efficient module for waves; the Laplace equation is solved on a σ-grid with only a few cells in the vertical, and its results can be used as boundary conditions for CFD (hydrodynamic coupling, HDC). Theory: Section 4.4.

REEF3D::NHFLOW for waves and currents in coastal areas where wave breaking, run-up with wetting and drying, the vertical flow structure or floating bodies matter, with a single-valued free surface. The shock-capturing finite volume scheme handles breaking waves and bores without an extra breaking model. Theory: Section 4.3.

REEF3D::SFLOW for depth-averaged problems such as open channel and river flow, long waves and bedload sediment transport, where the vertical flow structure is not important. The non-hydrostatic pressure extends the range to shallow and intermediate water waves. Theory: Section 4.5.

REEF3D::CFD when the free surface can overturn or break violently, for wave impact and slamming, the detailed flow around structures and the forces on them, scour and two-phase flow in general. It is the most expensive module: restrict its domain to the region of interest and use FNPF to bring the waves there. Theory: Section 4.2.

All modules share the grid generation with DIVEMesh, the MPI parallelization, the high-order discretization schemes and the wave generation library (Section 4.1). A first simulation is described in Chapter 3, and each module has its own tutorial chapter (Chapters 10 to 13).

3. Getting Started

This chapter takes you from an installed code to a first simulation and explains the steps you will repeat for every new case. It assumes that DIVEMesh and REEF3D have been compiled (Chapter 7) and that the executables `DiveMESH` and `REEF3D` are on your PATH (Section 8.1). The example is the smallest wave case of the tutorials: regular waves in a 2D numerical wave tank computed with REEF3D::FNPF. It runs in a few minutes on a laptop.

3.1 The Workflow at a Glance

Every REEF3D simulation follows the same five steps (Fig. 3.1):

1. Create a simulation folder for the case.
2. Describe the domain, the grid, solid geometries and the domain decomposition in `control.txt` and run **DiveMESH**. It writes one grid file per subdomain into `DIVEMesh_Grid`.
3. Describe the physics, numerics and output in `ctrl.txt`: which module (A 10), boundary conditions and waves (B), water level (F 60), simulation time (N 41), output (P) and number of processes (M 10).
4. Run REEF3D in parallel: **mpirun -n** *M 10* **REEF3D**.
5. Look at the results: 3D fields in ParaView, time series from gauges and probes with Python, Matlab or similar (Chapter 9).

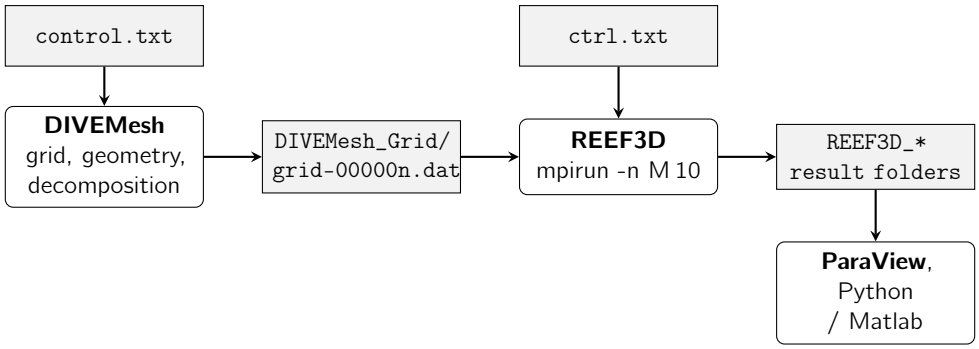

Figure 3.1: Workflow of a REEF3D simulation. The number of subdomains M 10 must be the same in `control.txt`, in `ctrl.txt` and in the mpirun command.

Both input files are plain text files with one command per line: a capital letter for the group, a number for the option and the values, e.g. `F 60 2.0`. Everything after a / is a comment. The syntax and its pitfalls are explained in Chapter 6; the most important points are summarized in Section 3.5.

3.2 Your First Simulation: Regular Waves in a 2D Wave Tank

The case is tutorial `Tutorials/REEF3D_FNPF/9_1 Regular Wave Propagation` of the code repository. A 2nd-order Stokes wave with height $H = 0.04\,\text{m}$ and wave length $L = 4.0\,\text{m}$ is generated at the left end of a 40 m long tank with a water depth of $d = 2.0\,\text{m}$ ($kd \approx 3.1$, i.e. close to deep water) and absorbed at the right end. The simulated surface elevation is compared with the wave theory at three gauges.

Step 1: Create the simulation folder

Create an empty folder, e.g. `first_case`, open a terminal in it and create the two input files as described below (or copy them from the tutorial folder).

Step 2: Grid and domain – control.txt

```
C 11 6
C 12 3
C 13 3
C 14 7
C 15 21
C 16 3

B 1 0.1
B 2 400 1 10
B 10 0.0 40.0 0.0 0.1 0.0 2.0

B 103 5
B 113 2.5
B 116 2.0

M 10 6
M 20 2
```

C 11 – C 16 Boundary conditions of the six sides of the domain: 1 = x_{min} (left), 2 and 3 = the two sides in y-direction, 4 = x_{max} (right), 5 = bottom, 6 = top. Here: wave generation (6) on the left, numerical beach (7) on the right, a wall (21) at the bottom and symmetry planes (3) elsewhere. For a 2D case, the y-sides are symmetry planes.

B 1, B 2 Cell size (0.1 m) and number of cells in x, y and z ($400 \times 1 \times 10$). If B 2 is given, it determines the grid. One cell in y-direction makes the case 2D.

B 10 Rectangular domain: x from 0 to 40 m, y from 0 to 0.1 m, z from 0 to 2 m.

B 103, B 113, B 116 Vertical grid stretching (point focus with sinh, stretching factor 2.5, focus at $z = 2.0\,\text{m}$), so that the ten vertical cells are concentrated near the free surface. For the sigma-grid models FNPF and NHFLOW, the vertical grid is stretched between the bed and the free surface; the water depth itself comes from F 60 in `ctrl.txt`.

M 10, M 20 Decompose the domain into 6 subdomains with decomposition method 2.

Run **DiveMESH** in the folder. It creates `DIVEMesh_Grid` with the files `grid-000001.dat` to `grid-000006.dat`. The folder `DIVEMesh_Paraview` can be opened in ParaView to check the grid and the decomposition.

Step 3: Physics, numerics and output – ctrl.txt

```
A 10 3
A 310 3
A 311 4
A 343 0

B 90 1
B 92 4
B 91 0.04 4.0
B 96 4.0 8.0
B 98 2
B 99 1

F 60 2.0

N 41 40.0
N 47 1.0

M 10 6

P 10 1
P 30 0.5
P 50 15.0 0.005
P 50 20.0 0.005
P 50 25.0 0.005
P 51 15.0 0.005
P 51 20.0 0.005
P 51 25.0 0.005
P 52 0.005
P 53 1
P 54 100

W 22 -9.81
```

A 10 3 Selects REEF3D::FNPF. A 310 and A 311 select 3rd-order Runge-Kutta time stepping and 5th-order WENO for the free surface boundary conditions; A 343 0 switches wetting and drying off (there is no coastline in this case).

B 90 – B 99 Wave generation: B 90 1 activates wave generation, B 92 4 selects 2nd-order Stokes waves, B 91 gives wave height and wave length in m (B 93 would give height and period instead). B 96 sets the lengths of the generation zone (4 m, one wave length) and of the numerical beach (8 m, two wave lengths). B 98 2 and B 99 1 select the relaxation methods for generation and absorption.

F 60 Still water level $z = 2.0$ m, i.e. the water depth.

N 41, N 47 Simulate 40 s with a CFL safety factor of 1.0 (the default is 0.3; FNPF is typically run with larger values than CFD).

M 10 Number of processes, the same as in `control.txt`.

P 10, P 30 Write 3D results in VTU format every 0.5 s. Without an output interval (P 20, P 30 or similar) no field output is written.

P 50, P 51 Wave gauges at $x = 15, 20, 25$ m: P 50 records the wave theory, P 51 the simulated surface elevation.

P 52 – P 54 Write the free surface along the line $y = 0.005$ m every 100 iterations, including the theoretical surface (P 53 1).

W 22 Gravitational acceleration in z-direction.

Step 4: Run REEF3D

```
mpirun -n 6 REEF3D
```

REEF3D first prints `read ctrl`, then the iteration number, the simulated time and the time step. The simulation ends after 40 s of simulated time. If your computer has fewer than six cores, set M 10 to a smaller number in both input files and run DIVEMesh again.

Step 5: Check the results

After the run, the simulation folder contains, among others:

REEF3D_FNPF_VTU the 3D fields. Open the `.pvtu` files in ParaView (they combine the per-process `.vtu` files) and colour by `eta` or the velocity.

REEF3D_FNPF_WSF the gauge time series: `REEF3D-FNPF-WSF-HG.dat` (simulated, P 51) and `REEF3D-FNPF-WSF-HG-THEORY.dat` (theory, P 50).

REEF3D_FNPF_WSFLINE the free surface along the tank at the P 54 interval.

Plot the simulated and the theoretical elevation at the three gauges (see Section 9.5 for a Python example). After the waves have reached the gauges, both curves should agree closely. This comparison of P 50 and P 51 is a quick check that wave generation, grid resolution and absorption are adequate, and it is worth repeating for every new wave case.

3.3 Structure of ctrl.txt

The commands in `ctrl.txt` are grouped by their letter. Table 3.1 shows where to look for which setting. Commands that are not given keep their default value, so a `ctrl.txt` file only needs to contain the settings that differ from the defaults. In the reference chapter (Chapter 6), four markers next to each command show for which module it is valid: S (SFLOW), F (FNPF), N (NHFLOW) and C (CFD).

Group	Content	Typical first commands
A	module selection and module-specific numerics (Section 6.1)	A 10 (module), wave breaking, wetting and drying
B	boundary conditions, inflow and outflow, wave generation and absorption (Section 6.2)	B 90 – B 99 (waves), B 60 (flow)
C	concentration / pollutant transport (Section 6.3)	C 10
D	discretization of the CFD equations (Section 6.4)	D 10, D 20, D 30
F	free surface and initial water level (Section 6.5)	F 60 (still water level), F 30 (level set)
H	heat transfer (Section 6.6)	H 10
I	initialization and restart (Section 6.7)	I 10, I 40
L	Lagrangian particle tracking in NHFLOW and CFD, e.g. marine plastics (Section 6.8)	L 10, L 21 – L 23
M	parallel computing (Section 6.9)	M 10 (number of processes)
N	solvers, time stepping and run length (Section 6.10)	N 41 (simulated time), N 47 (CFL factor)
P	output: 3D fields, gauges, probes, forces (Section 6.11)	P 10, P 30, P 51
Q	particle model for sediment (Section 6.12)	Q 10
S	sediment transport (Section 6.13)	S 10
T	turbulence modelling in CFD (Section 6.14)	T 10
W	fluid and material properties, gravity (Section 6.15)	W 1, W 22
X	floating bodies (6DOF) and moorings (Section 6.16)	X 10
Z	fluid-structure interaction (Section 6.17)	Z 10

Table 3.1: Command groups in `ctrl.txt`.

3.4 Setting Up Your Own Case

The easiest way to set up a new case is to start from the tutorial that is closest to your problem (Chapters 10 to 13) and to change it step by step. Go through the following checklist:

1. **Module**: A 10 (Section 2.1).

2. **Domain and grid** (`control.txt`): B 10 for the domain, B 1 or B 2 for the resolution, stretching with B 101 – B 126 where needed, solids from primitives (S commands of DIVEMesh) or STL files, bathymetry from data points. Check the grid in `DIVEMesh_Paraview` before running REEF3D.

3. **Boundaries** (`control.txt`): C 11 – C 16 for the six sides (inflow, outflow, wave generation, numerical beach, wall, symmetry plane).

4. **Water level and flow**: F 60 for the still water level; for currents or open channel flow, the inflow and outflow settings in the B section (B 60 and following) and the outflow water level F 62 with its ramp F 64.

5. **Waves**: B 90 1, the wave theory B 92, wave height and length (B 91) or period (B 93), or a spectrum for irregular waves, the zone lengths B 96 and the methods B 98 / B 99. The tutorials use a generation zone of about one wave length and a numerical beach of about two wave lengths.

6. **Run length and time step**: N 41 (simulated time) or N 45 (maximum iterations), whichever is reached first ends the run; N 47 is the CFL safety factor. Reduce N 47 if a simulation becomes unstable.

7. **Output**: P 10 with an interval (P 20 iterations or P 30 seconds; SFLOW uses P 182) for field data, P 51 wave gauges, probes and force output. Output files can become large; choose the intervals accordingly.

8. **Parallelization**: M 10 in both files, at most the number of physical cores.

Grid resolution

The tutorials keep the grids small so that they run on a laptop; they are not grid-converged. Table 3.2 lists their horizontal resolution as a starting point. For every new case, check the sensitivity of the results to the grid, e.g. by comparing the P 51 gauges with the P 50 theory for a regular wave or by repeating the simulation with a finer grid.

Tutorial	Module	Δx [m]	L/Δx	kd
8_1 Stokes wave propagation	SFLOW	0.02	200	0.79
9_1 Regular wave propagation	FNPF (10 cells vertically)	0.10	40	3.14
9_2 Submerged bar	FNPF (10 cells vertically)	0.04	120	0.52
2 Submerged bar	NHFLOW (3 cells vertically)	0.029	128	0.67
11_6 Rectangular wave tank	CFD	0.05	80	0.79
11_10 Submerged bar	CFD	0.01	373	0.67

Table 3.2: Horizontal resolution of selected wave tutorials: cell size Δx, cells per wave length $L/\Delta x$ and relative water depth kd at the wave generation.

3.5 Common Pitfalls and Troubleshooting

Symptom	Cause and remedy
REEF3D stops with `Inconsistent M 10 parameter`	The number of MPI processes, M 10 in `ctrl.txt` and M 10 in `control.txt` differ. Make them equal and run DIVEMesh again after changing `control.txt`.
REEF3D stops with `Could not open DIVEMesh grid file`	DIVEMesh has not been run in this folder, or REEF3D is started from a different folder.
REEF3D stops after `read ctrl` with `missing input parameter in ctrl.txt`	A command has fewer values than required. Check the number of values against Chapter 6. If the incomplete command is on the last line of the file, REEF3D does not stop and the command keeps its default value.
A setting seems to have no effect	The command was not read. Unknown command numbers and lines starting with a lower-case letter (e.g. `b 120`) are ignored without a warning. Check spelling, letter case and the number of values. Also check whether the setting is changed automatically (Section 3.6) or whether the command is valid for the selected module (S F N C markers).
A comment is interpreted as a command	Comments must start with /. Free text without a leading / is read token by token; a capital letter in it can be taken as a command. Use // for all comments.
An integer command gets a decimal value	For `int` commands a value like `2.5` is cut to its integer part.
No result files are written	By default no periodic field output is written. Set P 10 together with an interval (P 20 or P 30; for SFLOW P 181 or P 182).
Old and new results are mixed	Result folders from a previous run are not deleted: files with the same name are overwritten, others remain. Delete or move old `REEF3D_*` folders before a new run.
The simulation becomes unstable or produces NaN	Reduce the CFL factor N 47, check the grid quality and resolution near the free surface and structures, check that the wave is resolved (Table 3.2) and that it is not too steep for the chosen wave theory, and check that the generation and absorption zones fit into the domain.
Waves are reflected at the end of the tank	The numerical beach is too short. Use a beach length (B 96) of at least about two wave lengths or active absorption (B 99 with the corresponding boundary condition in `control.txt`).
The wave height decreases along the tank	The grid is too coarse for the wave length, or the vertical resolution near the free surface is insufficient. Refine the grid (Table 3.2) and compare P 51 with the P 50 theory.

3.6 Inputs that are Changed Automatically

After reading `ctrl.txt`, REEF3D adjusts some settings to keep the input consistent (file `control_parse.cpp`). These changes are not reported in the terminal output. The affected entries in Chapter 6 mention them as well.

Condition	Automatic change
B 98 1	changed to B 98 2
NHFLOW (A 10 5) and B 98 3	changed to B 98 4 (Dirichlet generation with active absorption)
NHFLOW (A 10 5) and B 98 5	changed to B 98 3

Condition	Automatic change
X 10 $>$ 0 (6DOF)	D 22 is set to 2; N 40 3 and 23 are changed to 4, N 40 13 to 14
F 80 $>$ 0 (VOF) and F 35 $>$ 0	F 35 is set to 0
I 10 1	I 11 = 1, I 12 = 2, I 13 = 1
I 10 2	I 11 = 2, I 12 = 2, I 13 = 1
I 40 $>$ 0 (restart)	I 10, I 11, I 12 and I 13 are set to 0
T 10 0	I 13 is set to 0
S 10 $\geq$ 1 or topography in the grid file	P 27 is set to 1 (bed printed to the VTU files)
A 550 $>$ 1 or A 560 $>$ 0	A 512 is set to 2 (implicit diffusion)

4. Theory and Numerical Methods

This chapter summarizes the governing equations and the main numerical methods of the four hydrodynamic modules. It is meant as a compact background for choosing and understanding the input commands, not as a complete derivation; the commands that select a method are given in parentheses, e.g. (A 311), and are described in Chapter 6. For details and validation, see the cited publications. Section 4.1 describes the methods that are shared by several modules (grids, WENO discretization, Runge-Kutta time stepping, linear solvers, wave generation and absorption); the following sections describe each module in turn.

Throughout the chapter, $x_i = (x, y, z)$ are the Cartesian coordinates with z pointing upwards, $u_i = (u, v, w)$ the velocities, t the time, ρ the density, ν and ν_t the kinematic and eddy viscosity, p the pressure and g the gravitational acceleration. d is the still water depth, η the free surface elevation and $h = \eta + d$ the total water depth. In REEF3D::CFD, ϕ denotes the level set function; in REEF3D::FNPF, Φ is the velocity potential.

4.1 Common Numerical Methods

The four hydrodynamic modules share the grid, the parallelization, the high-order discretization schemes, the linear solvers and the wave generation library, which are summarized here.

4.1.1 Grid, data layout and parallelization

All modules use a rectilinear grid generated by DIVEMesh. The cell sizes can vary in each direction (stretching with B 101–B 126 in the DIVEMesh `control.txt`), and all schemes use the local cell sizes. CFD uses a staggered grid: u, v and w are stored at the cell faces in their direction, while p, the level set function ϕ and the turbulence quantities are stored at the cell centres. NHFLOW and SFLOW are finite volume methods with cell-centred unknowns and fluxes at the faces. FNPF stores Φ at the horizontal cell centres and at the vertical grid nodes, so that the bed and the free surface are node levels. For the σ-grid models NHFLOW and FNPF, the vertical node coordinates from DIVEMesh are normalized to $\sigma \in [0, 1]$ from the bed to the free surface, keeping the vertical stretching (B 103, B 113, B 116).

All fields have three layers of ghost cells in each direction, which is enough for the WENO stencils. The ghost cells hold the boundary conditions at walls, inflow and outflow, and the values of the neighbouring subdomains. DIVEMesh splits the grid into M 10 subdomains, one per MPI process, and the ghost cells are updated by MPI exchange with the neighbours after each field update. M 10 must be identical in DIVEMesh, in `ctrl.txt` and in the MPI start command (Section 6.9).

4.1.2 Spatial discretization: WENO schemes

Convection terms and free surface gradients are discretized with the fifth-order weighted essentially non-oscillatory (WENO) scheme of Jiang and Shu [1996]. The scheme is fifth-order accurate on a smooth solution. Near steep gradients, such as the free surface, breaking fronts or bores, it falls back to the smoothest third-order sub-stencil and does not create spurious oscillations. For a positive face velocity $u_{i+1/2} > 0$, the face value of a variable φ is a weighted combination of three candidate stencils:

$$\begin{aligned}\varphi_{i+1/2} &= \omega_1\Big(\tfrac{1}{3}\varphi_{i-2} - \tfrac{7}{6}\varphi_{i-1} + \tfrac{11}{6}\varphi_i\Big) + \omega_2\Big(-\tfrac{1}{6}\varphi_{i-1} + \tfrac{5}{6}\varphi_i + \tfrac{1}{3}\varphi_{i+1}\Big) \\ &\quad + \omega_3\Big(\tfrac{1}{3}\varphi_i + \tfrac{5}{6}\varphi_{i+1} - \tfrac{1}{6}\varphi_{i+2}\Big), \qquad (4.1)\end{aligned}$$

$$\begin{aligned}IS_1 &= \tfrac{13}{12}\left(\varphi_{i-2} - 2\varphi_{i-1} + \varphi_i\right)^2 + \tfrac{1}{4}\left(\varphi_{i-2} - 4\varphi_{i-1} + 3\varphi_i\right)^2, \\ IS_2 &= \tfrac{13}{12}\left(\varphi_{i-1} - 2\varphi_i + \varphi_{i+1}\right)^2 + \tfrac{1}{4}\left(\varphi_{i-1} - \varphi_{i+1}\right)^2, \\ IS_3 &= \tfrac{13}{12}\left(\varphi_i - 2\varphi_{i+1} + \varphi_{i+2}\right)^2 + \tfrac{1}{4}\left(3\varphi_i - 4\varphi_{i+1} + \varphi_{i+2}\right)^2, \qquad (4.2)\end{aligned}$$

$$\omega_k = \frac{\alpha_k}{\alpha_1 + \alpha_2 + \alpha_3}, \quad \alpha_k = \frac{C_k}{(\varepsilon + IS_k)^2}, \quad (C_1, C_2, C_3) = (0.1, 0.6, 0.3), \quad \varepsilon = 10^{-6}, \qquad (4.3)$$

where IS_k are the smoothness indicators of the stencils. For $u_{i+1/2} < 0$ the stencil is mirrored about the face. Eqs. (4.1)–(4.3) are the uniform-grid form. The standard options use a non-uniform grid version, in which the stencil coefficients, the C_k and the coefficients of IS_k are computed once from the local cell sizes [Shu, 1997]; on a uniform grid they reduce to the values above.

There are two forms of the scheme. The conservative **WENO FLUX** form discretizes the convection term as $-(u_{i+1/2}\varphi_{i+1/2} - u_{i-1/2}\varphi_{i-1/2})/\Delta x_i$, with the face velocities interpolated according to D 11 (FOU or CDS2). The non-conservative **WENO HJ** (Hamilton–Jacobi) form [Jiang and Peng, 2000] discretizes $u\,\partial\varphi/\partial x$. It applies the same weights to the one-sided differences $(\varphi_{i-2} - \varphi_{i-3})/\Delta x, \ldots, (\varphi_{i+2} - \varphi_{i+1})/\Delta x$ in place of $\varphi_{i-2}, \ldots, \varphi_{i+2}$, which gives the upwind derivative $(\partial\varphi/\partial x)^-$ for $u > 0$ (mirrored for $u < 0$). In CFD, D 10 (momentum), F 35 (level set), C 15 (concentration) and H 15 (heat) select 4 WENO5 FLUX or 5 WENO5 HJ. The other options are the third-order variants WENO3 FLUX/HJ (7, 8) and lower-order schemes: first-order upwind FOU (1), central CDS2 (2), QUICK (3), CDS4 (6), TVD limiter schemes (10–16) and CDS6 (60). In NHFLOW (A 514 4, 5) and SFLOW (A 211 4), WENO reconstructs the left and right face states for the HLL-type fluxes. In FNPF, A 311 4/5 applies WENO5 to the horizontal gradients in the free surface boundary conditions.

4.1.3 Time integration and adaptive time stepping

The semi-discrete equations $\partial\varphi/\partial t = \mathcal{L}(\varphi)$ are advanced with the total variation diminishing (TVD) Runge–Kutta schemes of Shu and Osher [1988], in third order

$$\begin{aligned}\varphi^{(1)} &= \varphi^n + \Delta t\,\mathcal{L}(\varphi^n), \qquad \varphi^{(2)} = \tfrac{3}{4}\varphi^n + \tfrac{1}{4}\varphi^{(1)} + \tfrac{1}{4}\Delta t\,\mathcal{L}(\varphi^{(1)}), \\ \varphi^{n+1} &= \tfrac{1}{3}\varphi^n + \tfrac{2}{3}\varphi^{(2)} + \tfrac{2}{3}\Delta t\,\mathcal{L}(\varphi^{(2)}), \qquad (4.4)\end{aligned}$$

or in second order, $\varphi^{n+1} = \frac{1}{2}\varphi^n + \frac{1}{2}\varphi^{(1)} + \frac{1}{2}\Delta t\,\mathcal{L}(\varphi^{(1)})$ with the same first stage. The scheme is selected with N 40 (CFD), A 210 (SFLOW), A 310 (FNPF; option 4 is a classical fourth-order Runge–Kutta scheme) and A 510 (NHFLOW). In CFD and NHFLOW the pressure projection is carried out in every stage. The relaxation zones are applied after every stage.

With N 48 1 the time step follows from a stability criterion multiplied by $c =$ N 47. In CFD (N 50 1), convection, diffusion and source terms are included:

$$\Delta t = c \min_{\text{cells}} \frac{2}{\left(\dfrac{|\mathbf{u}|_{\max}}{\Delta x} + \dfrac{6\,\nu_{\text{eff}}}{\Delta x^2}\right) + \sqrt{\left(\dfrac{|\mathbf{u}|_{\max}}{\Delta x} + \nu_{\text{eff}}\right)^2 + \dfrac{4\,|S_{\max}|}{\Delta x}}}. \tag{4.5}$$

Here $|\mathbf{u}|_{\max} = \sqrt{u_{\max}^2 + v_{\max}^2 + w_{\max}^2}$ is formed from the global maxima of the velocity components, $\Delta x = \min(\Delta x_i, \Delta y_j, \Delta z_k)$ is the local cell size, $\nu_{\text{eff}} = \nu + \nu_t$, and $S_{\max}$ is the largest source term of the momentum equations, including gravity. N 50 2 evaluates Eq. (4.5) for each direction separately, with $u_{\max}$ and Δx_i, $v_{\max}$ and Δy_j, and $w_{\max}$ and Δz_k. The other modules use wave celerity based criteria:

- NHFLOW: $\Delta t = c \min\Big(\Delta x/(u_{\max} + \sqrt{g h_{\max}}),\ \Delta x/w_{\max}\Big)$, with the maximum total water depth $h_{\max}$. A 533 1 (and porous media, B 200) adds a vertical Courant number and a source term criterion.
- FNPF: $\Delta t = c \min\Big(\Delta x/\max(u_{\max}, \sqrt{g d_{\max}})\Big)$, where Δx is the horizontal cell size.
- SFLOW: $\Delta t = 2c \min\Big(\Delta x/(|U| + \sqrt{gh}),\ \Delta y/(|V| + \sqrt{gh})\Big)$, taken over the wet cells, so N 47 is half the Courant number.

In FNPF and NHFLOW the adaptive time step is limited by N 49. N 48 0 selects the fixed time step N 49.

4.1.4 Linear solvers

Each module solves an elliptic equation per time step or stage: the pressure Poisson equation of the projection method in CFD (D 30), the Poisson equation for the non-hydrostatic pressure in NHFLOW, the Laplace equation for Φ in FNPF and the equation for the non-hydrostatic (dynamic) pressure in SFLOW. N 10 selects the solver and N 11 the preconditioner, N 44 sets the tolerance and N 46 the maximum number of iterations. The HYPRE library [Falgout et al., 2006, Cen, 2015] provides the following solvers. On the structured interface (Struct, N 10 11–19; SFLOW uses its 2D version) there are Krylov solvers with the geometric multigrid preconditioners PFMG and SMG; the default is BiCGStab with PFMG. The matrix interface (AIJ, N 10 21–25) provides BoomerAMG, and the semi-structured interface (SStruct, N 10 31–38) provides SysPFMG. The in-house solver REEFMG (N 10 1) is a BiCGStab iteration preconditioned by a multigrid that coarsens only in x and y and solves each vertical grid line exactly (tridiagonal). It is therefore insensitive to strong vertical stretching. REEFMG is designed for the FNPF Laplace and NHFLOW pressure equations and also handles the CFD pressure (N 12–N 15).

4.1.5 Wave generation and absorption

B 90 1 switches on the numerical wave tank. The wave theory is set with B 92, the wave parameters with B 91 or B 93, and the water depth for the theory with B 94. Waves are generated with relaxation zones or Dirichlet conditions (B 98) and absorbed with a numerical beach or active absorption (B 99), see Fig. 4.1.

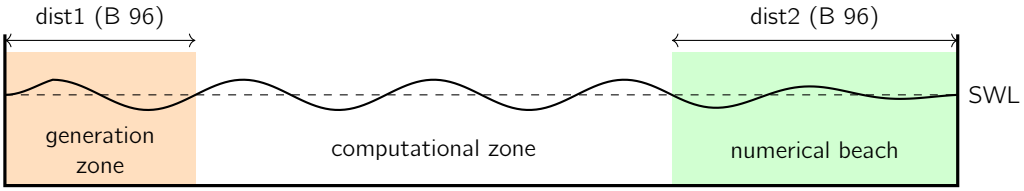

Figure 4.1: Numerical wave tank with relaxation zones for wave generation (B 98 2) and absorption (B 99 1, 2). B 108 and B 107 place the zones freely.

Relaxation method (B 98 2, B 99 1/2). Inside the zones, the solution is blended with a target solution after every Runge–Kutta stage [Jacobsen et al., 2012]:

$$\varphi = \Gamma(\tilde{x})\,\varphi_{\text{computed}} + (1-\Gamma(\tilde{x}))\,\varphi_{\text{target}}, \qquad \Gamma(\tilde{x}) = 1 - \frac{e^{\tilde{x}^{3.5}}-1}{e-1}, \qquad \tilde{x}\in[0,1], \tag{4.6}$$

so $\Gamma = 1$ keeps the computed solution and $\Gamma = 0$ imposes the target. In the generation zone, $\tilde{x} = 1 - x_g/L_g$, where x_g is the distance from the inlet (or from the B 108 line) and L_g is dist1 of B 96 (or d of B 108). The target is the wave theory multiplied by the ramp function B 101. The theory is thus fully imposed at the boundary and hands over smoothly to the computed solution. Waves reflected back towards the inlet are absorbed in the same zone. In the numerical beach, the target is still water: zero velocity, the still water level and hydrostatic pressure. Here $\tilde{x} = (L_b - x_b)/(a\,L_b)$, where x_b is the distance from the outlet (or from the B 107 line), L_b is dist2 of B 96 (or d of B 107), $a = 1$ for B 99 2 and $a = 2$ for the softer B 99 1. Each module relaxes the quantities it advances in time: in CFD u_i, ϕ and p; in NHFLOW the water level, the velocities and p; in SFLOW η and the depth-averaged velocities; in FNPF η and $\tilde{\Phi}$. Typical zone lengths are one wavelength for generation and two wavelengths for the beach.

Dirichlet generation and active absorption (B 98 3/4, B 99 3–5). With Dirichlet generation, the wave theory sets η and the velocities in the inflow ghost cells, and no extra domain length is needed. With active absorption at the inlet (B 98 4), the inflow velocity is corrected with the difference between the target elevation η_T and the computed elevation η_M at the boundary, using shallow water theory: $U_c = (\eta_T - \eta_M)\sqrt{g/d}$. The active beach sets the outflow velocity to $U_c = \eta_M\sqrt{g/d}$ (B 99 3), or to $U_c = \eta_M\,\omega\cosh(k(d+z))/\sinh(kd)$ with the frequency and wave number of the input wave (B 99 4). B 99 5 distributes U_c over the depth like a flap hinged at z = B 123. The active beach also needs the matching boundary condition in DIVEMesh.

Wave theories (B 92, full list and compatibility table in Section 6.2): shallow water, linear and deep water waves, Stokes waves of second and fifth order, cnoidal and solitary waves, piston and flap wavemakers, first- and second-order irregular and focused waves from a spectrum (B 85, e.g. Pierson–Moskowitz or JONSWAP), wave reconstruction, steady surface gravity waves (FNPF) and hydrodynamic coupling.

Hydrodynamic coupling (HDC) (B 92 61). Instead of a wave theory, the input comes from a previous large-scale simulation, typically FNPF (or SFLOW) driving a local CFD model. The large-scale run writes state files (P 40, area restricted with P 43), which DIVEMesh converts into the HDC input of the receiving model. There, η, u, v and w are interpolated linearly in time between the stored steps and linearly in space. The vertical positions are rebuilt from the stored σ levels, $z = \sigma(\eta + d - z_b) + z_b$ with the bed elevation z_b. The interpolated fields serve as the target of the relaxation zone (B 98 2) or of the Dirichlet inflow. I 231–I 233 shift the input, and B 125 and B 127 allow 2D input for a 3D domain.

4.1.6 Key input commands

M 10	number of MPI processes/subdomains, as in DIVEMesh
D 10, F 35, C 15, H 15	CFD convection (4 WENO5 FLUX, 5 WENO5 HJ), D 11
A 211, A 311, A 514	WENO reconstruction/gradients in SFLOW, FNPF, NHFLOW
N 40, A 210, A 310, A 510	time schemes of CFD, SFLOW, FNPF, NHFLOW
N 47, N 48, N 49, N 50	CFL factor, adaptive on/off, fixed/maximum Δt, CFD criterion
N 10, N 11, N 44, N 46	solver, preconditioner, tolerance, max. iterations; N 12–15 REEFMG
B 90, B 92, B 91/93, B 94	wave tank, wave theory, wave parameters, depth for the theory
B 96, B 107, B 108	zone lengths, free placement of beach and generation zones
B 98, B 99, B 101/102	wave generation, numerical beach, ramp function
B 92 61, I 231–233, B 125	hydrodynamic coupling, input offsets, 2D slice

4.2 REEF3D::CFD

REEF3D::CFD (A 10 6) solves the incompressible Reynolds-averaged Navier-Stokes (RANS) equations for two immiscible phases, water and air, on a staggered rectilinear grid. The free surface is captured with the level set method, and solids and the bed enter through immersed boundary methods. The model and its validation as a numerical wave tank are described in Bihs et al. [2016a]. The shared building blocks (WENO, TVD Runge-Kutta, adaptive time stepping, linear solvers, wave generation and absorption) are described in Section 4.1.

4.2.1 Governing equations

Both phases are described by one set of equations with variable density ρ and kinematic viscosity ν. Continuity and momentum read

$$\frac{\partial u_i}{\partial x_i} = 0, \tag{4.7}$$

$$\frac{\partial u_i}{\partial t} + u_j \frac{\partial u_i}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + g_i + f_i, \tag{4.8}$$

with the velocities u_i, the pressure p, the eddy viscosity ν_t from the turbulence model, the gravity vector g_i = (W 20, W 21, W 22) and additional source terms f_i: a constant pressure gradient (W 29), the direct forcing of solids and floating bodies, porous media resistance, wall functions and, if W 5 > 0, surface tension with the continuum surface force approach

[Brackbill et al., 1992]. Without a turbulence model (T 10 0), $\nu_t = 0$ and the equations are the Navier-Stokes equations.

The phases are distinguished by the level set function ϕ, which is the signed distance to the interface, positive in water (phase 1) and negative in air (phase 2). The material properties are blended across the interface with a smoothed Heaviside function:

$$\rho = \rho_1 H(\phi) + \rho_2 (1 - H(\phi)), \qquad \nu = \nu_1 H(\phi) + \nu_2 (1 - H(\phi)), \tag{4.9}$$

$$H(\phi) = \begin{cases} 0 & \phi < -\epsilon, \\ \frac{1}{2}\left(1 + \frac{\phi}{\epsilon} + \frac{1}{\pi}\sin\left(\frac{\pi\phi}{\epsilon}\right)\right) & |\phi| \le \epsilon, \\ 1 & \phi > \epsilon, \end{cases} \tag{4.10}$$

with ρ_1 = W 1, ν_1 = W 2, ρ_2 = W 3 and ν_2 = W 4. The half-thickness of the interface is $\epsilon = \mathrm{F}45 \cdot \overline{\Delta x}$ (default F 45 = 2.1). By default, $\overline{\Delta x}$ is the average of the mean cell sizes in the coordinate directions (x and z only in 2D). For wave and current simulations (B 90 > 0 or B 60 > 0), $\overline{\Delta x}$ is instead the mean vertical cell size of the cells near the initial free surface, so that vertical grid stretching towards the free surface is taken into account. Where the density is needed at a cell face (pressure gradient and pressure Poisson equation), the level set is first averaged to the face and H is evaluated there. This avoids averaging the density itself across the density jump [Bihs et al., 2016a].

4.2.2 Pressure-velocity coupling

The velocity components are stored at the cell faces, and the pressure, the level set and the turbulence quantities at the cell centres (staggered grid). Velocity and pressure are coupled with Chorin's projection method [Chorin, 1968] in incremental pressure-correction form (D 30 1). In every Runge-Kutta stage with coefficient α ($\alpha = 1, \frac{1}{4}, \frac{2}{3}$ for the three stages of the third-order scheme), an intermediate velocity u_i^* is computed from Eq. (4.8) with the pressure p^{old} of the previous stage. The pressure correction p' then follows from a Poisson equation with variable density:

$$\frac{\partial}{\partial x_i}\left(\frac{1}{\rho}\frac{\partial p'}{\partial x_i}\right) = \frac{1}{\alpha \Delta t}\frac{\partial u_i^*}{\partial x_i}. \tag{4.11}$$

Pressure and velocity are corrected as

$$p = p^{\mathrm{old}} + p', \qquad u_i = u_i^* - \frac{\alpha \Delta t}{\rho}\frac{\partial p'}{\partial x_i}, \tag{4.12}$$

which makes u_i divergence-free. Eq. (4.11) is solved with the solver selected in N 10 and N 11 (Section 4.1), and the pressure level is fixed with B 30 – B 33. With D 30 10, no Poisson equation is solved: the pressure is computed hydrostatically from the level set function, and only this pressure gradient enters the velocity update.

4.2.3 Free surface: level set method

The free surface is the zero contour of the level set function ϕ [Osher and Sethian, 1988], which is advected with the flow:

$$\frac{\partial \phi}{\partial t} + u_j \frac{\partial \phi}{\partial x_j} = 0. \tag{4.13}$$

The convection term is discretized with the fifth-order WENO scheme in Hamilton-Jacobi form (F 35 5, default). In the coupled momentum time schemes (N 40 = 2, 3, 4, 22, 23, 24, 33), Eq. (4.13) is advanced inside every Runge-Kutta stage of the momentum equations, followed by an update of density and viscosity. In the separate schemes (N 40 = 12, 13, 14, 44), the level set has its own time scheme F 30. In both cases F 30 > 0 activates the two-phase properties.

Since advection destroys the signed distance property on which Eq. (4.10) relies, ϕ is reinitialized with the PDE

$$\frac{\partial \phi}{\partial \tau} + S(\phi)\left(\left|\frac{\partial \phi}{\partial x_j}\right| - 1\right) = 0, \qquad S(\phi) = \frac{\phi}{\sqrt{\phi^2 + \left|\partial\phi/\partial x_j\right|^2 \overline{\Delta x}^2}}, \tag{4.14}$$

in pseudo time τ [Sussman et al., 1994]. The smoothed sign function $S(\phi)$ follows Peng et al. [1999]; here $\overline{\Delta x}$ is the local average cell size. The gradient is discretized with a Godunov-type upwind selection of fifth-order WENO derivatives, and Eq. (4.14) is integrated with the third-order TVD Runge-Kutta scheme (F 40 3). The pseudo time step is $\Delta\tau = \mathrm{F}43 \cdot \min(\Delta x, \Delta y, \Delta z)$ (default F 43 = 0.55). F 44 reinitialization steps are carried out per time step (default 3). In the coupled scheme N 40 3 the number is set internally to two, two and three steps in the three Runge-Kutta stages. At the start, ϕ is initialized from F 51 – F 72 and reinitialized with $2L/(\mathrm{F}43\,\overline{\Delta x})$ steps, where L is the maximum domain length or F 42. Optionally, the level set is shifted by a constant after advection and reinitialization so that the water volume is conserved (F 46, F 47).

As an alternative, the volume of fluid (VOF) method can be activated with F 80, with an algebraic compressive scheme (F 84, F 85) or the geometric PLIC reconstruction (F 80 4). The VOF implementation is in beta; the open channel flow and wave generation algorithms are designed for the level set method.

4.2.4 Turbulence modelling

The turbulence model is selected with T 10. The RANS model mainly used for wave and coastal applications is the k-ω model [Wilcox, 1994] (T 10 2):

$$\frac{\partial k}{\partial t} + u_j \frac{\partial k}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_k}\right)\frac{\partial k}{\partial x_j}\right] + P_k - \beta_k\, k\,\omega, \tag{4.15}$$

$$\frac{\partial \omega}{\partial t} + u_j \frac{\partial \omega}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\nu_t}{\sigma_\omega}\right)\frac{\partial \omega}{\partial x_j}\right] + \alpha\,\frac{\omega}{k}\,P_k - \beta\,\omega^2, \tag{4.16}$$

with the production $P_k = \nu_t\, 2S_{ij}S_{ij}$, where $S_{ij} = \frac{1}{2}\left(\partial u_i/\partial x_j + \partial u_j/\partial x_i\right)$. The closure coefficients in the code are $\alpha = 5/9$, $\beta = 3/40$, $\beta_k = c_\mu = 0.09$ and $\sigma_k = \sigma_\omega = 2$. Both equations are solved implicitly in time, and their convection terms are discretized with T 12.

The eddy viscosity is $\nu_t = k/\omega$. With T 34 1 (default), it is bounded by a strain-based limiter [Durbin, 2009]:

$$\nu_t = \max\left(\min\left(\frac{k}{\omega},\ \mathrm{T}31\,\frac{k}{|S|}\right),\ 10^{-4}\,\nu\right), \qquad |S| = \sqrt{2S_{ij}S_{ij}}, \tag{4.17}$$

with T 31 = $1/\sqrt{3}$ by default. This prevents the over-production of turbulence under propagating waves. T 41 activates the stabilization of Larsen and Fuhrman [2018] for nearly potential flow regions, $\nu_t \leq (k/\omega)\, c_\mu \alpha\, (2\Omega_{ij}\Omega_{ij})/(\lambda_2 \beta 2 S_{ij} S_{ij})$, where Ω_{ij} is the rotation tensor and λ_2 = T 42.

To avoid spurious turbulence production at the density jump of the free surface, T 36 > 0 overwrites ω in a band $|\phi| < \epsilon_t$ around the interface with a free surface value based on Naot and Rodi [1982]:

$$\omega = \delta(\phi)\,\frac{c_\mu^{-1/4}\sqrt{k}}{\kappa\, y'}, \qquad \delta(\phi) = \frac{1}{2\epsilon_t}\left(1 + \cos\left(\frac{\pi\phi}{\epsilon_t}\right)\right), \tag{4.18}$$

with $\kappa = 0.4$, $\epsilon_t = \mathrm{T}38 \cdot \overline{\Delta x}$ and the virtual origin y' = T 37. T 36 2 adds the inverse wall distance to $1/y'$, and with T 36 3, y' is T 37 times the local water depth. T 39 bounds ν_t near the interface from below by a Smagorinsky-type eddy viscosity. T 45 adds a buoyancy term to the k-equation, and T 33 adds the isotropic part of the Reynolds stress, $\frac{2}{3}\partial k/\partial x_i$, to the momentum equations.

Further options of T 10 are the k-ϵ model (1; $c_{1\epsilon} = 1.44$, $c_{2\epsilon} = 1.92$, $\sigma_k = 1.0$, $\sigma_\epsilon = 1.3$), an explicit algebraic Reynolds stress model based on k-ω (12) and URANS variants (21, 22). For T 10 22, ν_t is multiplied by $\min\left(1,\ \mathrm{T}23\, c_\mu\, \Delta\omega/\sqrt{k}\right)$ with $\Delta = (\Delta x \Delta y \Delta z)^{1/3}$. For large eddy simulation, T 10 31 selects the Smagorinsky model [Smagorinsky, 1963], $\nu_t = (C_s\Delta)^2|\tilde{S}|$ with $C_s = 0.2$ and the filtered velocity field (filter T 21), and T 10 33 the WALE model [Nicoud and Ducros, 1999] with $C_w = 0.6$.

4.2.5 Boundaries, solids and topography

Ghost cells. The outer boundaries of the domain and the cells that DIVEMesh excludes from the grid are treated with the ghost cell method. The ghost cell values are extrapolated along the grid line normal to the boundary, following the local directional ghost cell approach of Berthelsen and Faltinsen [2008]. The extrapolation uses Lagrange polynomials through the interior values and the boundary value at the wall location. B 20 selects slip or no-slip conditions for the velocities, B 23 extrapolation or reflection, and B 29 the image point distance for the Dirichlet extrapolation of B 20 3.

Direct forcing. Solids and the bed topography from DIVEMesh are described by their own signed distance functions, ϕ_s for solids and ϕ_b for the topography. These are reinitialized with the same PDE as the free surface level set (G 40). The no-slip condition is imposed with a continuous direct forcing immersed boundary method [cf. Fadlun et al., 2000]. After the predictor step of every Runge-Kutta stage, the forcing

$$f_i = H_s\,\frac{u_{s,i} - u_i^*}{\alpha \Delta t} \tag{4.19}$$

is added to the intermediate velocity before the projection step. H_s is the smoothed Heaviside function of Eq. (4.10), evaluated for $-\min(\phi_s, \phi_b)$ at the velocity location with the half-width $\mathrm{X\,41} \cdot \overline{\Delta x}$ (default X 41 = 0.6), and $u_{s,i}$ is the velocity of the solid (zero for fixed solids). With B 20 2 and B 21 1 (defaults), the forcing in the transition layer outside the body is weighted with the components of the surface normal, which preserves the tangential velocity. D 22 selects how the implicit diffusion treats neighbour cells inside direct-forcing regions, and X 48 extrapolates ϕ, k and ω into the solid.

Wall functions. With B 10 1, the wall shear stress from the rough-wall logarithmic law is applied as a momentum sink in the cells next to ghost cell walls and direct-forcing surfaces:

$$f_i = -\frac{|u_i|\, u_i}{(u^+)^2\, \Delta n}, \qquad u^+ = \frac{1}{\kappa} \ln\!\left(\frac{30\, y}{k_s}\right), \qquad y = \max\!\left(\frac{\Delta n}{2},\ \frac{k_s}{30}\right), \tag{4.20}$$

where u_i is a velocity component tangential to the wall, Δn the cell size normal to the wall, $\kappa = 0.4$ and k_s the equivalent sand-grain roughness (B 50 – B 56). For the k-ω model, the wall value $\omega = \sqrt{k}/(\kappa\, y\, c_\mu^{1/4})$ is used, and the production of k in the wall cells is computed from the log-law wall shear stress.

4.2.6 Discretization summary

Momentum convection uses the conservative fifth-order WENO scheme (D 10 4, default) of Jiang and Shu [1996], with the convecting velocities interpolated to the faces according to D 11; lower-order, TVD and central schemes are alternatives. Diffusion is implicit by default (D 20 2), which removes the viscous time step restriction. Only the part $\partial/\partial x_j[(\nu + \nu_t)\partial u_i/\partial x_j]$ is implicit; the transposed part is explicit. D 20 1 is explicit and D 20 3 Crank-Nicolson (3D). The time scheme N 40 defaults to the third-order TVD Runge-Kutta scheme with the level set coupled into each stage (3). Alternatives are a separate level set step (13), low-storage variants (4, 14, 44) and a conservative formulation with face densities (33). The time step follows the CFL criterion with N 48 1 and N 47 (N 50), see Section 4.1.

4.2.7 Multiphysics coupling

Floating bodies (X). With X 10 1, rigid bodies move in six degrees of freedom (X 11). Each body is represented by a level set function from an STL file or primitives. Force and moment are integrated from the pressure and viscous stresses over the triangulated body surface, and the Newton-Euler equations are advanced within the Runge-Kutta stages of the fluid solver. The body velocity is imposed with the direct forcing of Eq. (4.19) (X 41, X 14). With X 10 > 0, N 40 3 is changed to 4, N 40 13 to 14, and D 22 is set to 2.

Sediment transport (S, Q). With S 10 1, the bed is the zero contour of the topography level set $\phi_b = z - z_b(x, y)$. It enters the flow through the direct forcing described above. The bed level z_b is updated with the Exner equation

$$(1 - n)\frac{\partial z_b}{\partial t} + \frac{\partial q_{b,x}}{\partial x} + \frac{\partial q_{b,y}}{\partial y} = 0, \tag{4.21}$$

where n is the bed porosity (S 24) and q_b the bedload transport from the formula selected in S 11 with the bed shear stress of S 16. With suspended load (S 12), the erosion and deposition fluxes are added. S 10 2 treats the bed as a porous VRANS region, and Q 10 1 replaces the Exner model by the particle model of Section 6.12.

Porous media (B 270 – B 295). In porous zones (porosity n, grain size d_{50}), the volume-averaged (VRANS) form of the equations is solved. Its main terms are a linear and a nonlinear resistance force [see Jensen et al., 2014]:

$$f_i = -a\,u_i - b|u_i|\,u_i, \qquad a = \alpha_p \frac{(1-n)^2}{n^3}\frac{\nu}{d_{50}^2}, \qquad b = \beta_p\left(1+\frac{7.5}{KC}\right)\frac{1-n}{n^3}\frac{1}{d_{50}}, \tag{4.22}$$

where the coefficients α_p and β_p are given with the porous objects, KC is set with B 264 (no correction by default), and the added mass coefficient B 260 scales the time derivative inside the porous media. VRANS is switched on automatically when porous objects are defined. B 295 controls the turbulence source terms in porous zones.

Heat and concentration (H, C). H 10 solves an advection-diffusion equation for the temperature, and the fluid properties then follow from H 3 (tabulated properties or the Boussinesq approximation). C 10 solves a transport equation for a concentration c. The concentration changes density and viscosity, $\rho = (\rho_1 + c\,\mathrm{C}1)H + (\rho_2 + c\,\mathrm{C}3)(1-H)$, and correspondingly for ν with C 2 and C 4.

4.2.8 Solution procedure

Figure 4.2 summarizes the solution procedure of REEF3D::CFD as it is executed by the code for the default time scheme N 40 3. After the control file and the grid have been read and all solver objects have been initialized, every time step first updates the inflow and wave generation, turbulence and sediment transport, and then advances the level set function and the velocities together in the three stages of the TVD Runge-Kutta scheme. In each stage, the pressure projection of Eqs. (4.11) and (4.12) follows the momentum predictor and the direct forcing of solid and floating bodies, before the fluid properties are updated with the new level set. The adaptive time step for the next step is computed at the end of each time step, after which the output is written.

4.2.9 Key input commands

A 10 6	select REEF3D::CFD
W 1 – W 5, W 20 – W 22	fluid properties, surface tension, gravity
D 10, D 11, D 20	momentum convection, convecting velocities, diffusion
D 30, B 30 – B 33	pressure algorithm, pressure reference
N 10, N 11	Poisson solver and preconditioner
N 40, N 47, N 48, N 50	momentum time scheme, adaptive time step
F 30, F 35, F 45	level set time and convection scheme, interface half-thickness ϵ

F 40, F 42 – F 44	reinitialization
F 46, F 47	level set volume correction
F 80 – F 85	VOF (beta)
T 10, T 12	turbulence model, convection scheme of k and ω
T 31, T 34, T 41, T 42	eddy viscosity limiter, k-ω stabilization
T 36 – T 39	free surface turbulence damping
B 10, B 50 – B 56	wall functions, roughness
B 20 – B 29, X 41, G 40	ghost cells, direct forcing, solid and bed level sets
X 10	6DOF floating bodies
S 10, Q 10	sediment transport, particle model
L 10 – L 62	Lagrangian particle tracking (Section 6.8)
B 260 – B 295	VRANS porous media
H 10, C 10	heat transfer, concentration

4.3 REEF3D::NHFLOW

REEF3D::NHFLOW (A 10 5) solves the incompressible Navier-Stokes equations for a single-valued free surface on a three-dimensional σ-grid. The hydrostatic part of the pressure is treated together with the free surface in a shock-capturing Godunov-type finite volume scheme, while the non-hydrostatic (dynamic) pressure is obtained from a Poisson equation. The method is described in Bihs and Wang [2025] and Bihs et al. [2026]; this section summarises the formulation as it is implemented in the `nhflow_*` classes.

4.3.1 Governing equations

The flow is governed by continuity and momentum conservation,

$$\frac{\partial u_i}{\partial x_i} = 0, \tag{4.23}$$

$$\frac{\partial u_i}{\partial t} + \frac{\partial (u_i u_j)}{\partial x_j} = -\frac{1}{\rho}\frac{\partial p}{\partial x_i} + \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right] + g_i. \tag{4.24}$$

The free surface $z = \eta(x, y, t)$ is single-valued. The still water level is F 60, the bed lies at $z = -d(x, y)$ below it, and $h = \eta + d$ is the total water depth. The pressure is split into a hydrostatic and a non-hydrostatic part,

$$p = \rho g (\eta - z) + p_{nh}, \tag{4.25}$$

where $g = |\text{W } 22|$ and ρ is W 1. In the following, p stands for p_{nh} only. Integration of (4.23) over the water column together with the kinematic conditions at the free surface and at the bed,

$$w\big|_{\eta} = \frac{\partial \eta}{\partial t} + u\frac{\partial \eta}{\partial x} + v\frac{\partial \eta}{\partial y}, \qquad w\big|_{-d} = -u\frac{\partial d}{\partial x} - v\frac{\partial d}{\partial y}, \tag{4.26}$$

gives the free surface equation

$$\frac{\partial \eta}{\partial t} + \frac{\partial}{\partial x}\int_{-d}^{\eta} u\, dz + \frac{\partial}{\partial y}\int_{-d}^{\eta} v\, dz = 0. \tag{4.27}$$

Since d does not change in time (except with sediment transport), the code advances h with (4.27) and sets $\eta = h - d$.

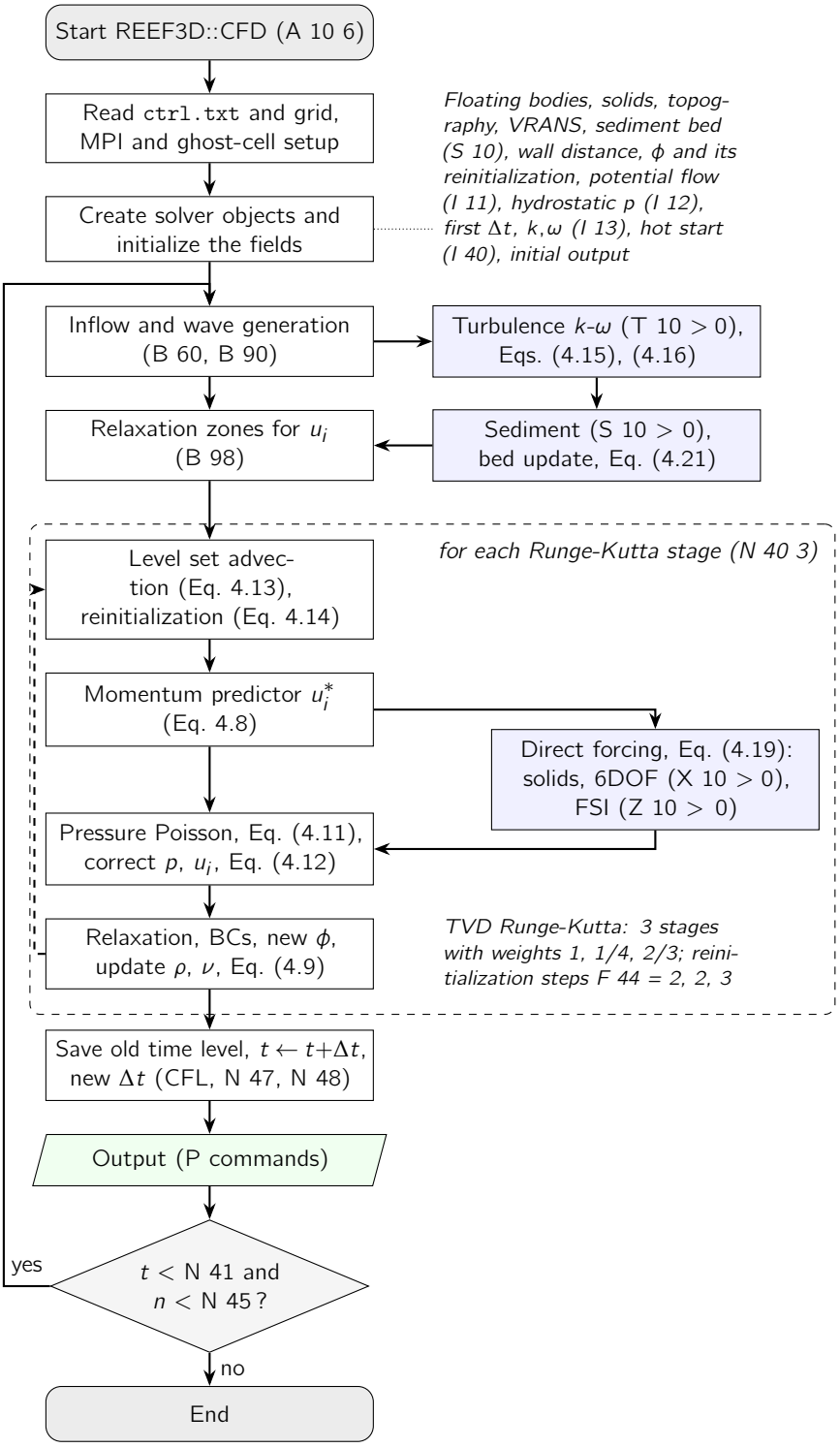

Figure 4.2: Solution procedure of REEF3D::CFD for the default time scheme N 40 3, in which the level set equation is advanced within each Runge-Kutta stage of the momentum equations (the dashed arrow marks the next stage), and optional models are drawn as shaded boxes. With X 10 > 0 the scheme N 40 3 is replaced by its low-storage variant N 40 4 with the same stage order; for the separate schemes N 40 12, 13, 14 and 44 the level set is instead advanced with its own time scheme (F 30) and reinitialized before the turbulence step, and N 40 14 additionally uses dedicated main loops for fixed and floating bodies with low-storage Runge-Kutta stages.

4.3.2 Sigma-coordinate transformation

The vertical coordinate is mapped onto $\sigma \in [0,1]$ from the bed to the free surface,

$$\sigma = \frac{z + d(x,y)}{h(x,y,t)}, \tag{4.28}$$

so that every water column has the same number of cells, independent of the local depth (Figure 4.3). The vertical node distribution is taken from the DIVEMesh grid and normalised to $[0,1]$, so vertical stretching towards the free surface is set in DIVEMesh (B 103, B 113, B 116); the number of vertical cells is the number of cells of the DIVEMesh grid in z-direction. Derivatives are transformed with the chain rule,

$$\frac{\partial}{\partial x} \to \frac{\partial}{\partial x} + \sigma_x \frac{\partial}{\partial \sigma}, \quad \frac{\partial}{\partial y} \to \frac{\partial}{\partial y} + \sigma_y \frac{\partial}{\partial \sigma}, \quad \frac{\partial}{\partial z} \to \sigma_z \frac{\partial}{\partial \sigma}, \quad \frac{\partial}{\partial t} \to \frac{\partial}{\partial t} + \sigma_t \frac{\partial}{\partial \sigma}, \tag{4.29}$$

with the metric coefficients

$$\sigma_x = \frac{1-\sigma}{h}\frac{\partial d}{\partial x} - \frac{\sigma}{h}\frac{\partial \eta}{\partial x}, \quad \sigma_y = \frac{1-\sigma}{h}\frac{\partial d}{\partial y} - \frac{\sigma}{h}\frac{\partial \eta}{\partial y}, \quad \sigma_z = \frac{1}{h}, \quad \sigma_t = -\frac{\sigma}{h}\frac{\partial \eta}{\partial t}. \tag{4.30}$$

The horizontal free surface gradients in (4.30) are discretised as selected with A 524. The metric terms are updated at every Runge-Kutta stage and are set to zero in dry columns.

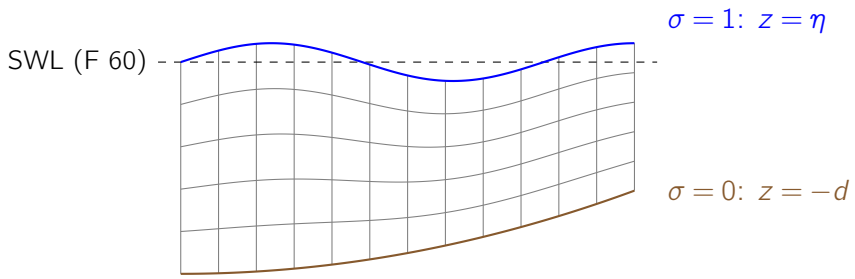

Figure 4.3: Sketch of the NHFLOW σ-grid: each water column is divided into the same number of layers between the bed and the free surface.

Applying the transformation to (4.23)–(4.24) and multiplying by h gives the conservative form solved by NHFLOW. Local continuity in the σ-layer reads

$$\frac{\partial h}{\partial t} + \frac{\partial (hu)}{\partial x} + \frac{\partial (hv)}{\partial y} + \frac{\partial \omega}{\partial \sigma} = 0, \tag{4.31}$$

where ω is the vertical velocity relative to the moving σ-surfaces (scaled with h). It is not a prognostic variable: the code integrates (4.31) from the bed ($\omega = 0$) upwards layer by layer, using the same numerical fluxes of hu and hv as the free surface equation; ω is also zero at the free surface. Summing (4.31) over all layers gives the discrete form of (4.27), $\partial h/\partial t = -\sum_k \Delta\sigma_k \left[\partial (hu)_k/\partial x + \partial (hv)_k/\partial y\right]$.

The momentum equations are written for the conserved variables $\mathbf{U} = (hu, hv, hw)^T$, located at the cell centres,

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} + \frac{\partial \mathbf{H}}{\partial \sigma} = \mathbf{S}, \tag{4.32}$$

$$\mathbf{F} = \begin{pmatrix} hu^2 + \frac{1}{2}g\eta^2 + g\eta d \\ huv \\ huw \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} hvu \\ hv^2 + \frac{1}{2}g\eta^2 + g\eta d \\ hvw \end{pmatrix}, \quad \mathbf{H} = \begin{pmatrix} \omega u \\ \omega v \\ \omega w \end{pmatrix}, \tag{4.33}$$

$$\mathbf{S} = \begin{pmatrix} g\eta \dfrac{\partial d}{\partial x} - \dfrac{h}{\rho}\left(\dfrac{\partial p}{\partial x} + \sigma_x \dfrac{\partial p}{\partial \sigma}\right) \\ g\eta \dfrac{\partial d}{\partial y} - \dfrac{h}{\rho}\left(\dfrac{\partial p}{\partial y} + \sigma_y \dfrac{\partial p}{\partial \sigma}\right) \\ -\dfrac{1}{\rho}\dfrac{\partial p}{\partial \sigma} \end{pmatrix} + \mathbf{S}_D + \mathbf{S}_{ext}. \tag{4.34}$$

The hydrostatic pressure gradient $gh\partial\eta/\partial x$ is split into the flux part $\partial(\frac{1}{2}g\eta^2 + g\eta d)/\partial x$ and the bed slope source $g\eta\,\partial d/\partial x$. In the code the bed slope term is evaluated with the same face depths $d_{i\pm1/2}$ (arithmetic mean of the neighbouring columns) that enter the flux, so that for a lake at rest (η = const, $u_i = 0$) the flux difference and the source cancel exactly. $\mathbf{S}_D$ contains the viscous and turbulent diffusion of hu_i in σ-coordinates (below), and $\mathbf{S}_{ext}$ collects the optional sources: direct forcing of solids and floating bodies, VRANS porous media (B 200 ff.), wind forcing (A 570 ff.), bed and wall friction (A 519) and the relaxation zones of the wave generation.

4.3.3 Shock-capturing finite volume scheme

Reconstruction. The face values left and right of each cell face are reconstructed from the cell averages of η, u, v, w, hu, hv and hw. The free surface is reconstructed through η (not h); the face depth is then $D_{L,R} = \max(\eta_{L,R} + d_{i+1/2}$, A 544). The scheme is chosen with A 514: options 1–3 use a limited linear (MUSCL-type) reconstruction with the van Leer, Superbee or van Albada limiter; with A 514 4 (default) the horizontal reconstruction is the fifth-order WENO scheme [Jiang and Shu, 1996] in its formulation for non-uniform grids (see Section 4.1) and the vertical one is linear with the Superbee limiter; A 514 5 applies WENO in all three directions. The limited slopes are set to zero next to dry cells.

Riemann solver. The horizontal numerical fluxes are computed with the HLL approximate Riemann solver (A 511 1) [Harten et al., 1983, Toro, 2009],

$$\mathbf{F}_{i+1/2} = \begin{cases} \mathbf{F}_L & \text{if } S_L \geq 0, \\ \dfrac{S_R\mathbf{F}_L - S_L\mathbf{F}_R + S_L S_R(\mathbf{U}_R - \mathbf{U}_L)}{S_R - S_L} & \text{if } S_L < 0 < S_R, \\ \mathbf{F}_R & \text{if } S_R \leq 0, \end{cases} \tag{4.35}$$

where $\mathbf{F}_{L,R}$ are the physical fluxes evaluated with the reconstructed states. The same formula is used for the fluxes of the free surface equation (flux hu, jump $D_R - D_L$). The wave speed estimates are those of the two-rarefaction approximation for shallow water [Toro, 2009], with $c_{L,R} = \sqrt{gD_{L,R}}$:

$$\begin{aligned} u^* &= \tfrac{1}{2}(u_L + u_R) + c_L - c_R, & c^* &= \tfrac{1}{2}(c_L + c_R) + \tfrac{1}{4}(u_L - u_R), \\ S_L &= \min(u_L - c_L,\, u^* - c^*), & S_R &= \max(u_R + c_R,\, u^* + c^*). \end{aligned} \tag{4.36}$$

At a wet-dry face the dry-bed estimates are used, e.g. $S_L = u_R - 2c_R$, $S_R = u_R + c_R$ for a dry left cell. With A 511 2 the HLLC solver is used instead, which restores the contact wave with the speed $S^* = u^*$ and the star states $D_K(S_K - u_K)/(S_K - S^*)$ times the transported velocity ($K = L, R$); next to dry cells it falls back to HLL. The vertical fluxes **H** are upwinded with the sign of ω at the σ-face, using the vertically reconstructed velocities. Because the method is conservative and shock-capturing, bores and broken waves are represented as moving discontinuities without additional treatment.

Diffusion. Viscous and turbulent diffusion (A 512) is applied to hu_i with the Laplacian transformed to σ-coordinates, including the metric terms $(\sigma_x^2 + \sigma_y^2 + \sigma_z^2)\partial^2/\partial\sigma^2$ and the mixed derivatives. The implicit option (A 512 2) solves the resulting system for each velocity component with BiCGStab; A 513 sets the treatment next to solid walls. A 512 is set to 2 automatically when a turbulence model is used.

4.3.4 Non-hydrostatic pressure

After the momentum predictor, the non-hydrostatic pressure enforces the continuity equation (4.23). The pressure is stored at the vertical cell faces (the σ-nodes), the velocities at the cell centres. A 520 selects the scheme: 0 is hydrostatic (no pressure solution), 1 is a projection method (PJM) which solves for the full p each Runge-Kutta stage from a predictor without non-hydrostatic pressure gradient, and 2 (default) is an incremental pressure-correction method [Zijlema et al., 2005]: the predictor includes the gradient of the old pressure p^n, a correction p' is solved, and $p^{n+1} = p^n + p'$. For option 2, the Poisson equation in σ-coordinates reads

$$\frac{\partial^2 p'}{\partial x^2} + \frac{\partial^2 p'}{\partial y^2} + (\sigma_x^2 + \sigma_y^2 + \sigma_z^2)\frac{\partial^2 p'}{\partial \sigma^2} + 2\sigma_x \frac{\partial^2 p'}{\partial x\,\partial\sigma} + 2\sigma_y \frac{\partial^2 p'}{\partial y\,\partial\sigma} + (\sigma_{xx} + \sigma_{yy})\frac{\partial p'}{\partial \sigma} = \frac{\rho}{\alpha \Delta t}\left(\frac{\partial u^*}{\partial x} + \sigma_x \frac{\partial u^*}{\partial \sigma} + \frac{\partial v^*}{\partial y} + \sigma_y \frac{\partial v^*}{\partial \sigma} + \sigma_z \frac{\partial w^*}{\partial \sigma}\right), \quad (4.37)$$

where u_i^* is the predicted velocity and α the weight of the current Runge-Kutta stage. The mixed derivatives are evaluated explicitly on the right-hand side (deferred correction), so that the matrix has a seven-point stencil. In the bottom layer, the vertical interpolation of the horizontal velocities in the divergence is reduced over steep slopes with A 522. The velocities are then corrected with

$$(hu)^{n+1} = (hu)^* - \alpha \Delta t \frac{h}{\rho}\left(\frac{\partial p'}{\partial x} + \sigma_x \frac{\partial p'}{\partial \sigma}\right), \quad (hv)^{n+1} = (hv)^* - \alpha \Delta t \frac{h}{\rho}\left(\frac{\partial p'}{\partial y} + \sigma_y \frac{\partial p'}{\partial \sigma}\right),$$
$$(hw)^{n+1} = (hw)^* - \frac{\alpha \Delta t}{\rho}\frac{\partial p'}{\partial \sigma}. \quad (4.38)$$

At the free surface the Dirichlet condition $p = 0$ is imposed. At the bed, at closed side walls and next to solid or dry cells the coupling to the neighbouring ghost node is removed from the matrix; with A 520 1 the ghost value is replaced by the pressure of the current cell (zero gradient). At the outflow boundary $p = 0$ is prescribed, at the inflow either $p = 0$ (wave generation) or a zero gradient (B 76). The pressure is only solved in "deep" cells: wet columns that are deeper than A 545·A 544 and not directly next to dry cells. In the shallow swash

zone the model therefore reduces to the hydrostatic, shock-capturing formulation. The linear system is solved with the solver selected by N 10, either HYPRE (N 10 11–19 Struct, 21–25 AIJ, 31–38 SStruct, preconditioner N 11) or REEF3D's semicoarsening multigrid REEFMG (N 10 1), see Section 4.1. Option A 520 3 is an alternative non-hydrostatic scheme with its own flux formulation.

4.3.5 Wetting and drying, breaking waves and turbulence

Wetting and drying. A column is dry when $h \leq$ A 544 (default 0.001 m); its depth is then kept at A 544 and its velocities are set to zero. With A 540 1 (default) a dry cell only becomes wet when a wet neighbour has a higher free surface, following the approach of Roeber et al. [2010]; A 540 2 decides from the depth threshold alone. Isolated wet cells are dried. With A 540 1, the free surface and depth of the wet cell are used on both sides of a wet-dry face and the face velocities are set to zero; in the Riemann solver the dry-bed wave speeds are used. The Froude number limiter $|hu_i| \leq$ A 531$\cdot h\sqrt{gh}$ (area A 532) keeps the velocities bounded in very shallow water.

Wave breaking. Breaking bores are captured by the Godunov-type scheme itself, which dissipates energy at the wave front without a breaking criterion. In addition, A 550 1 activates a breaking-wave viscosity. Breaking is detected (A 551) either in shallow water from the rise rate of the free surface, $\partial\eta/\partial t >$ A 554$\cdot\sqrt{gh}$, or in deep water from the free-surface slope $|\partial\eta/\partial x|$ or $|\partial\eta/\partial y| >$ A 555. In the flagged columns and their direct neighbours the artificial viscosity A 557 is added to the diffusion coefficient (half the value in the second ring of neighbours); A 552 additionally filters the water depth in breaking columns.

Turbulence. A 560 selects the turbulence closure: the k-ϵ (1) or k-ω (2) model, their URANS-filtered variants (21, 22, filter coefficient A 568), or the Smagorinsky LES model (31). The transport equations of the two-equation models are solved implicitly on the σ-grid, with the same closure coefficients as in CFD (Section 4.2); options A 564–A 567 control the eddy viscosity limiter, the stabilisation in nearly potential flow regions, buoyancy and the free surface condition for ϵ/ω. The eddy viscosity ν_t enters the implicit diffusion of (4.32).

4.3.6 Time stepping

The momentum equations and the free surface equation are integrated together with the second- (A 510 2, default) or third-order (A 510 3) TVD Runge-Kutta scheme (Section 4.1). Each stage consists of: update of the σ-metrics, reconstruction and wave speeds, free surface update with (4.27) and wetting-drying, computation of ω, momentum predictor with the HLL fluxes and sources, implicit diffusion, direct forcing, pressure solution and velocity correction, and finally the relaxation zones of the wave generation and absorption (B 98, B 99; Section 4.1). With adaptive time stepping (N 48 1) the time step is

$$\Delta t = \text{N } 47 \cdot \min\left(\frac{\Delta x}{|u|_{max} + \sqrt{g\, h_{max}}},\ \frac{\Delta x}{|w|_{max}} \right), \tag{4.39}$$

with Δx the smallest horizontal cell size, limited from above by N 49. A 533 1 adds the vertical Courant number $h\Delta\sigma/|w|_{max}$ and a source-term criterion.

4.3.7 Floating bodies and sediment transport

Fixed structures (A 581–A 590) and floating bodies (X group) are represented by a direct forcing immersed boundary method: a smoothed Heaviside function H of the body's signed distance, with an interface width of A 526 times the horizontal cell size, gives the forcing term $f_i = H(u_{i,body} - u_i)/(\alpha \Delta t)$, which is added to hu_i before and after the pressure solution. With A 517 1 only the normal velocity is enforced in the interface band, so that the tangential velocity is preserved. For floating bodies (X 10 1), the pressure and viscous forces are integrated over the triangulated body surface and the rigid body motion is integrated with the scheme of A 510; A 516 sets the treatment of the free surface inside the body. Alternatively, A 599 1 uses a distributed Lagrange multiplier method. Sediment transport (S 10 1, or S 10 12 for the fully coupled RK2 version) uses the bed shear stress from the NHFLOW velocity field (wall law B 22 with S 16 1) in the Exner-based morphology model of REEF3D; the updated bed changes d, and thereby the σ-grid, at every time step.

4.3.8 Solution procedure

The solution procedure of REEF3D::NHFLOW, as implemented in the NHFLOW driver and the Runge-Kutta momentum classes, is shown in Fig. 4.4. The free surface and the momentum equations are advanced together in each Runge-Kutta stage: the free surface is updated first with the HLL fluxes of the reconstructed state, then the momentum predictor is computed on the new water depth and projected with the non-hydrostatic pressure. Turbulence and sediment transport are solved once per time step before the Runge-Kutta stages, and the new time step size is computed at the end of the step.

4.3.9 Key input commands

Command	Meaning
A 10 5	select REEF3D::NHFLOW
A 510	time scheme: 2nd- or 3rd-order TVD Runge-Kutta
A 511	Riemann solver: 1 HLL, 2 HLLC
A 512, A 513	diffusion: off / explicit / implicit; wall treatment
A 514	reconstruction: 1–3 TVD (van Leer, Superbee, van Albada), 4 WENO horizontal, 5 WENO 3D
A 518, A 519	bed boundary condition (slip/no-slip), bed and wall friction
A 520	pressure: 0 hydrostatic, 1 projection, 2 pressure correction, 3 alternative scheme
A 522, A 524	bed slope factor in the Poisson right-hand side; discretisation of σ-metrics
A 531, A 532	Froude number limiter and its area
A 540, A 544, A 545	wetting-drying scheme, dry depth threshold, shallow (hydrostatic) cell criterion
A 550–A 557	breaking detection and breaking viscosity
A 560–A 568	turbulence model and its options

A 570–A 578	wind forcing
A 581–A 599	direct forcing solids, STL geometry, DLM
B 98, B 99	wave generation and absorption (Section 4.1)
B 103, B 113, B 116	vertical grid stretching (DIVEMesh)
F 60	still water level
N 10, N 11	linear solver and preconditioner for the pressure
N 47, N 48, N 49	CFL number, adaptive time stepping, maximum/fixed time step
X 10	floating bodies (6DOF)
S 10	sediment transport
L 10 – L 62	Lagrangian particle tracking (Section 6.8)

4.4 REEF3D::FNPF

REEF3D::FNPF (A 10 3) solves the fully nonlinear potential flow equations on a σ-coordinate grid that follows the bed and the free surface [Bihs et al., 2020]. The breaking wave detection and the coastline algorithm for complex topographies are described in Wang et al. [2022]. The flow is assumed inviscid and irrotational, and the free surface single-valued. The unknowns are the free surface elevation $\eta(x,y,t)$ and the velocity potential at the free surface $\tilde{\Phi}(x,y,t)$, both stored on the horizontal (2D) grid, and the velocity potential $\Phi(x,y,\sigma,t)$ in the water column, which is obtained from a Laplace equation in every Runge-Kutta stage.

4.4.1 Governing equations

With z measured from the still water level F 60 and the still water depth $d(x,y)$ (bed at $z=-d$), the velocity potential satisfies the Laplace equation in the fluid domain $-d \le z \le \eta$:

$$\frac{\partial^2\Phi}{\partial x^2}+\frac{\partial^2\Phi}{\partial y^2}+\frac{\partial^2\Phi}{\partial z^2}=0. \tag{4.40}$$

At the bed, the normal velocity vanishes (Neumann condition including the bed slope):

$$\frac{\partial\Phi}{\partial z}+\frac{\partial d}{\partial x}\frac{\partial\Phi}{\partial x}+\frac{\partial d}{\partial y}\frac{\partial\Phi}{\partial y}=0 \qquad \text{at } z=-d. \tag{4.41}$$

At the free surface, the kinematic and dynamic boundary conditions are written in terms of η, the surface potential $\tilde{\Phi}=\Phi(x,y,\eta,t)$ and the vertical velocity at the surface $\tilde{w}=\partial\Phi/\partial z|_{z=\eta}$ (Zakharov form):

$$\frac{\partial\eta}{\partial t}=-\frac{\partial\eta}{\partial x}\frac{\partial\tilde{\Phi}}{\partial x}-\frac{\partial\eta}{\partial y}\frac{\partial\tilde{\Phi}}{\partial y}+\tilde{w}\left(1+\left(\frac{\partial\eta}{\partial x}\right)^2+\left(\frac{\partial\eta}{\partial y}\right)^2\right), \tag{4.42}$$

$$\frac{\partial\tilde{\Phi}}{\partial t}=-\frac{1}{2}\left(\left(\frac{\partial\tilde{\Phi}}{\partial x}\right)^2+\left(\frac{\partial\tilde{\Phi}}{\partial y}\right)^2\right)+\frac{1}{2}\tilde{w}^2\left(1+\left(\frac{\partial\eta}{\partial x}\right)^2+\left(\frac{\partial\eta}{\partial y}\right)^2\right)-g\eta. \tag{4.43}$$

The gravitational acceleration g is taken from W 22. With A 314 1 the linearised conditions $\partial\eta/\partial t=\tilde{w}$ and $\partial\tilde{\Phi}/\partial t=-g\eta$ are used instead (default A 314 2: fully nonlinear). Optional wind forcing (A 370–A 373) adds a surface pressure or stress term to Eq. (4.43). The velocities follow from the potential, $u_i=\partial\Phi/\partial x_i$; they are needed for output, wave kinematics and hydrodynamic coupling, but not for the time integration itself.

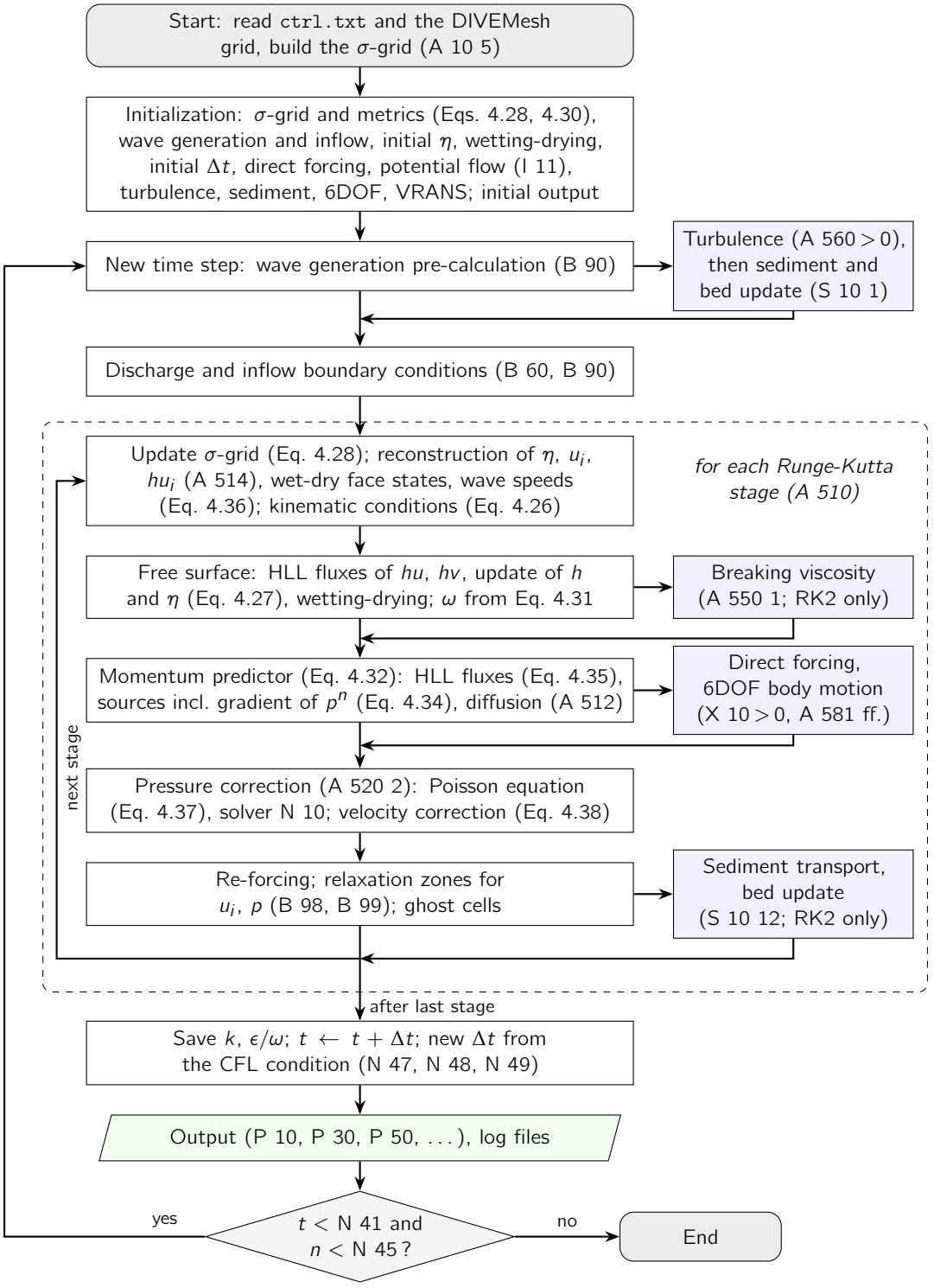

Figure 4.4: Solution procedure of REEF3D::NHFLOW for the default second-order Runge-Kutta scheme (A 510 2) and pressure-correction method (A 520 2); with A 510 3 there are three stages and neither the breaking viscosity nor the stage-wise sediment update (S 10 12) is called, and with A 520 0 or 1 the pressure step is skipped or replaced by the projection method. The loop also ends when the sediment time exceeds S 19.

4.4.2 σ-coordinate transformation and Laplace solver

The vertical coordinate is mapped onto $\sigma \in [0,1]$, with $\sigma = 0$ at the bed and $\sigma = 1$ at the free surface:

$$\sigma = \frac{z + d(x,y)}{\eta(x,y,t) + d(x,y)} = \frac{z+d}{h}, \qquad h = \eta + d. \tag{4.44}$$

The σ-levels are the normalised vertical grid of DIVEMesh, so the number of vertical cells and the vertical stretching are controlled by B 2 and B 103, B 113, B 116 (typically a refinement towards the free surface); the vertical extent given in B 10 is irrelevant. The grid moves with η and is updated after every stage. Applying the chain rule to Eq. (4.40) gives, with derivatives now taken at constant σ,

$$\frac{\partial^2\Phi}{\partial x^2} + \frac{\partial^2\Phi}{\partial y^2} + \left(\sigma_x^2 + \sigma_y^2 + \sigma_z^2\right)\frac{\partial^2\Phi}{\partial\sigma^2} + 2\sigma_x\frac{\partial^2\Phi}{\partial x\,\partial\sigma} + 2\sigma_y\frac{\partial^2\Phi}{\partial y\,\partial\sigma} + \left(\sigma_{xx} + \sigma_{yy}\right)\frac{\partial\Phi}{\partial\sigma} = 0, \tag{4.45}$$

with the metric terms

$$\sigma_x = \frac{(1-\sigma)\,d_x - \sigma\,\eta_x}{h}, \qquad \sigma_z = \frac{1}{h}, \qquad \sigma_{xx} = \frac{(1-\sigma)\,d_{xx} - \sigma\,\eta_{xx} - 2\sigma_x\,(d_x + \eta_x)}{h}, \tag{4.46}$$

and the corresponding expressions in y ($d_x = \partial d/\partial x$, $\eta_x = \partial\eta/\partial x$ etc.). The first derivatives η_x, d_x are computed with the scheme of A 311 (Section 4.4.3), the second derivatives η_{xx}, d_{xx} with second-order (A 312 2) or fourth-order (A 312 3) central differences. The bed metrics can be smoothed with A 309 passes of a 1-2-1 filter applied to the bed elevation, and A 317 1 (default) uses a symmetric bed gradient including the bed curvature terms d_{xx}, d_{yy}.

The boundary conditions become $\Phi = \tilde{\Phi}$ at $\sigma = 1$ (Dirichlet) and, from Eq. (4.41),

$$\frac{\partial\Phi}{\partial\sigma} = -\frac{d_x\,\partial\Phi/\partial x + d_y\,\partial\Phi/\partial y}{\sigma_z + d_x\sigma_x + d_y\sigma_y} \qquad \text{at } \sigma = 0, \tag{4.47}$$

which is imposed through ghost points below the bed, with second-order central differences for $\partial\Phi/\partial x$ and $\partial\Phi/\partial y$.

Eq. (4.45) is discretised with second-order central differences on the non-uniform σ grid (7-point stencil, fnpf_laplace_cds2). The mixed derivatives $\partial^2\Phi/\partial x\partial\sigma$ and $\partial^2\Phi/\partial y\partial\sigma$ are evaluated explicitly from the current values of Φ and moved to the right-hand side; the $\sigma_{xx}\,\partial\Phi/\partial\sigma$ term and the bed condition enter the matrix, which is therefore not symmetric. At closed lateral boundaries and next to dry cells, $\partial\Phi/\partial n = 0$ is used; at Dirichlet wave generation or active absorption boundaries (B 98 3, 4, B 99 3, 4) the prescribed horizontal velocity enters as a Neumann condition, extrapolated to first or second order (A 329). Dry columns are removed from the system (identity rows). The linear system is solved with the solver chosen in N 10 and N 11 (Section 4.1): by default HYPRE's Struct BiCGStab with the PFMG geometric multigrid preconditioner (N 10 14, N 11 11), or REEF3D's own semicoarsening multigrid-preconditioned BiCGStab REEFMG (N 10 1, controlled by N 12–N 15). The solution of the previous stage is the initial guess.

After the last stage, the velocities in the water column are computed from Φ: $u = \partial\Phi/\partial x + \sigma_x\,\partial\Phi/\partial\sigma$, $v = \partial\Phi/\partial y + \sigma_y\,\partial\Phi/\partial\sigma$, $w = \sigma_z\,\partial\Phi/\partial\sigma$, with fourth-order central differences in the interior (second order in σ for the metric terms and at the free surface, where $w = \tilde{w}$). Velocities in columns next to a dry cell are set to zero.

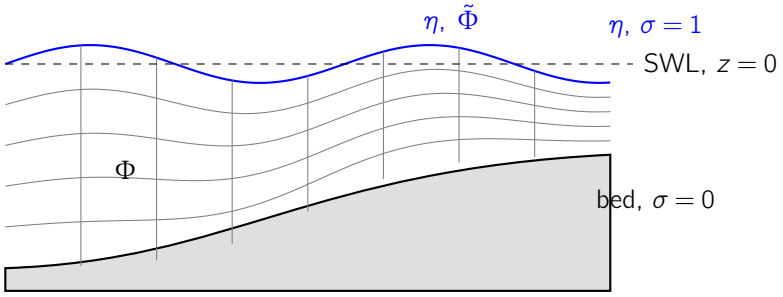

Figure 4.5: Sketch of the σ grid of REEF3D::FNPF: the vertical grid lines connect bed and free surface; η and $\tilde{\Phi}$ live on the horizontal grid at $\sigma = 1$, Φ in the water column.

4.4.3 Time integration of the free surface boundary conditions

Eqs. (4.42) and (4.43) are integrated with the third-order TVD Runge-Kutta scheme (A 310 3, default, Section 4.1); a classical fourth-order Runge-Kutta scheme is available with A 310 4. In every stage, the right-hand sides of Eqs. (4.42) and (4.43) are evaluated, the breaking and coastline damping (Section 4.4.4) is applied and η, $\tilde{\Phi}$ are updated; then the wave generation and beach relaxation, the wetting-drying and the coastline relaxation are applied, breaking is detected, the horizontal gradients and the σ grid are updated, and finally the Laplace equation is solved with $\Phi = \tilde{\Phi}$ at $\sigma = 1$ to obtain $\tilde{w}$ for the next stage. The horizontal gradients $\partial\eta/\partial x_i$ and $\partial\tilde{\Phi}/\partial x_i$ are discretised with the scheme selected in A 311: second-, fourth- or sixth-order central differences (1, 2, 6), third-order WENO (3) or the default fifth-order WENO (4, 5), written for non-uniform grids. The WENO stencil for $\tilde{\Phi}_x$ is upwinded with the sign of a central estimate of $\tilde{\Phi}_x$. In the kinematic condition, η_x is upwinded with the characteristic speed $\tilde{\Phi}_x - 2\tilde{w}\eta_x$ of Eq. (4.42) (A 315 1, default); A 315 2 additionally uses symmetric η gradients for the geometric terms (metrics, breaking, wind, dynamic condition), and A 315 0 is the legacy treatment. The wetting-drying versions of the schemes are selected automatically with A 343 ≥ 1 (A 316 controls the WENO5 fallback at the shoreline).

The vertical velocity at the free surface is obtained from the Laplace solution as $\tilde{w} = \sigma_z \partial\Phi/\partial\sigma|_{\sigma=1}$, using a one-sided backward difference in σ whose order follows A 311: 3 points (second order) for CDS2, 5 points (fourth order) for CDS4 and WENO5 (with a reduction of the stencil where not enough wet points are available), 7 points for CDS6 and 5-point finite difference weights on the actual grid for WENO3. The one-sided stencil is applied to Φ and to the σ coordinates of the grid points, so it remains consistent on stretched grids.

The time step is adaptive (N 47, N 48, Section 4.1), based on the maximum of the surface velocity and the shallow water celerity:

$$\Delta t = \text{N } 47 \cdot \min\left(\frac{\Delta x}{\max\left(|u|_{max},\ \sqrt{g\, d_{max}}\right)}\right), \tag{4.48}$$

limited by N 49. Since the phase speed of the waves determines the time step, FNPF is typically run with larger CFL numbers than CFD (e.g. N 47 1.0).

4.4.4 Wave breaking

Potential flow cannot represent the overturning of a breaker, so breaking is modelled by detecting breaking cells and dissipating energy there (A 350). The breaking detection is controlled by A 351:

- shallow water (depth-induced) breaking, A 351 1 or 3: the surface rises faster than a fraction of the shallow water celerity, $\partial\eta/\partial t > \alpha\sqrt{gh}$ with α = A 354 (default 0.6), where $\partial\eta/\partial t$ is evaluated from the change of η over the current stage;
- deep water (steepness-induced) breaking, A 351 2 or 3: the free surface slope exceeds a threshold, $|\partial\eta/\partial x| >$ A 355 or $|\partial\eta/\partial y| >$ A 355 (default 1.25).

With the base algorithm (A 350 1), the flagged region is widened by up to two cells, the flags are collected over the stages of one time step, and an artificial viscosity $\nu_b =$ A 365 (default $1.86\,\mathrm{m}^2/\mathrm{s}$) is assigned to the breaking cells and their direct neighbours, with $0.5\,\nu_b$ in the next ring of cells. The dissipation is applied as an implicit horizontal diffusion step to $\tilde{\Phi}$ and/or η (A 357) before the Runge-Kutta update of each stage:

$$\frac{f^* - f}{\alpha_{RK}\,\Delta t} = \nu_f \left(\frac{\partial^2 f^*}{\partial x^2} + \frac{\partial^2 f^*}{\partial y^2} \right), \qquad f \in \{\eta, \tilde{\Phi}\}, \quad \nu_\eta = \text{A 366} \cdot \nu_b, \quad \nu_{\tilde{\Phi}} = \nu_b, \tag{4.49}$$

where α_{RK} is the weight of the stage (1, 1/4, 2/3 for RK3). The 2D system is solved with a BiCGStab solver and skipped when $\nu_b = 0$ everywhere. Optionally (A 352), η and $\tilde{\Phi}$ are additionally smoothed in the breaking cells with a local 1-2-1 filter with predictor-corrector iterations (A 361 outer, A 362 inner iterations).

With A 350 2, an eddy viscosity in the spirit of Kennedy et al. [2000] is used: depth-induced breaking starts when $\partial\eta/\partial t > 0.65\sqrt{gh}$ and stops when $\partial\eta/\partial t < 0.15\sqrt{gh}$ (fixed values in the code, A 354 is not used), and the viscosity is $\nu_b = \delta_b^2\, h\,|\partial\eta/\partial t|$ with $\delta_b =$ A 365; for steepness-induced breaking (onset at A 355, cessation below A 356·A 355) $\nu_b = \delta_b^2 h$. The viscosity is limited to $10\,\mathrm{m}^2/\mathrm{s}$ and applied with Eq. (4.49).

A 350 3 implements an energy-dissipation-based model following Romero et al. [2012]: from a local wavenumber k (estimated from the curvature of η), a local steepness $S = kA$ and the linear phase speed c, breaking occurs for $S > S_0$ and ceases for $S < 0.5S_0$; the viscosity is obtained by matching the dissipation to $b(S)\,c^5/(gL_b)$ with the breaking strength $b = a(S - S_0)^{5/2}$ and the breaking length $L_b = \alpha_L 2\pi/k$, ramped up over $3\sqrt{h/g}$. The coefficients are fixed in the code ($a = 0.5$, $S_0 = 0.08$, $\alpha_L = 0.35$, $\nu_b \le 10\,\mathrm{m}^2/\mathrm{s}$); A 351–A 356 are not used by this option.

4.4.5 Coastline and wetting-drying

FNPF does not resolve the swash zone like a shallow water model. Instead, a cell is dry when its water depth is below A 344 (default 0.001 m), and the waves are damped before they reach the coastline, so that complex coastlines, islands and skerries can be included directly from the topography without special meshing [Wang et al., 2022]. A 343 selects the treatment:

- A 343 1 (default), static coastline: the wet and dry cells are determined once from the still water depth. In dry cells $\tilde{\Phi} = 0$ and the Laplace equation is not solved. Wet cells whose depth drops below A 344 during a drawdown are clamped to a thin film ($h = 1.1 \cdot$A 344) with $\tilde{\Phi}$ averaged from the wet neighbours; with A 337 1 (default) the clamp volume is redistributed (A 334) and one-sided gradients are used at the coast (A 336).
- A 343 2 and 3, dynamic wetting-drying (runup and rundown, or rundown only): the wet/dry flags are updated once per time step. A cell dries when $h \leq$ A 344 and it has been wet for at least A 330 steps; it rewets when a wet neighbour is higher by at least A 331$\cdot$A 344, taking $\tilde{\Phi}$ from its wet neighbours. A 335 limits the runup elevation, and a viscosity band of A 332 cells with $\nu \geq$ A 346 is applied along the moving front. Option 2 uses no coastline relaxation.

For the static coastline (and A 343 3), the waves are absorbed in a relaxation zone along the coastline. At the start of the simulation, a signed distance function d_c to the coastline is computed from the wet/dry mask by fast sweeping (A 339 1, default) or by PDE reinitialisation (A 339 0); isolated wet areas not connected to the main water body can be dried out first (A 338). Within the zone width L_c = A 341$\cdot\Delta x$ or L_c = A 342 (absolute width, has priority), the free surface quantities are relaxed after every stage,

$$f \leftarrow r(\hat{x})\, f + (1 - r(\hat{x}))\, f_{ref}, \qquad r(\hat{x}) = 1 - \frac{e^{\hat{x}^{3.5}} - 1}{e - 1}, \qquad \hat{x} = \max\left(0,\ 1 - \frac{d_c}{L_c}\right), \quad (4.50)$$

so that $r = 0$ at the coastline and $r = 1$ at the edge of the zone. This is applied to η and/or $\tilde{\Phi}$ (A 347). The target f_{ref} is zero by default; with A 326 > 0 it is a running mean of f over the time scale A 326, which lets wave set-up and the slow drift of $\tilde{\Phi}$ pass through the zone. The velocities and $\tilde{w}$ are damped in the same way over a zone of width $3L_c$ with exponent 1.5. With A 337 1, the coastline viscosity A 346 is added to ν_b inside the zone, rising to A 346 towards the coast, and in thin-film cells; it is applied through Eq. (4.49) and therefore requires A 350 ≥ 1. For a full-domain initialisation (I 30 1) the zone is widened by the factor A 349 in the first time step.

4.4.6 Wave generation, absorption and coupling

Waves are generated and absorbed with the methods of Section 4.1. With the relaxation method (B 98 2), η and $\tilde{\Phi}$ are relaxed towards the wave theory (B 92) in the generation zone, and in the numerical beach (B 99 1, 2) towards zero (A 348 selects η and/or $\tilde{\Phi}$); only wave theories that provide the potential are suitable (marked in the B 92 table). With Dirichlet generation (B 98 3, 4) and active absorption (B 99 3, 4) the horizontal velocity of the wave theory is imposed as a Neumann condition in the Laplace equation.

Because FNPF is cheap compared with CFD, a typical workflow is to propagate the waves over a large domain in FNPF and to write the free surface, the surface potential and the velocities to state files (P 40, P 44 1). After preprocessing with DIVEMesh (H options), these drive a smaller REEF3D::CFD domain around a structure through the hydrodynamic coupling (HDC, B 92 61), with Dirichlet or relaxation wave generation (B 98) at the coupling boundary.

4.4.7 Solution procedure

The solution procedure of REEF3D::FNPF is shown in Fig. 4.6. After the σ grid, the wet/dry mask with the coastline distance and the wave generation have been initialised, every time step advances η and $\tilde{\Phi}$ with the Runge-Kutta scheme of A 310. In each stage the free surface boundary conditions are integrated explicitly, relaxation, wetting-drying and breaking are applied to the new surface values, and the σ grid is updated before the Laplace equation is solved for Φ, which provides $\tilde{w}$ for the next stage.

4.4.8 Key input commands

Command	Meaning
A 10 3	select REEF3D::FNPF
A 309	number of smoothing passes for the bed
A 310	time scheme: 3 TVD-RK3 (default), 4 RK4
A 311	horizontal gradients in the FSBCs: CDS2/4/6, WENO3, WENO5 (default 5)
A 312	second derivatives of η, d in the σ metrics: CDS2 (default), CDS4
A 314	linear (1) or fully nonlinear (2, default) free surface boundary conditions
A 315–A 317	gradient treatment of η, WENO5 at the shoreline, bed metrics
A 329	extrapolation order for Dirichlet wave generation and active absorption
A 326, A 341, A 342, A 347, A 349	coastline relaxation: reference time scale, zone width (relative, absolute), variables, initialisation factor
A 330–A 339, A 343, A 344	wetting-drying: method (A 343), dry threshold (A 344), dynamic front parameters, coastline distance
A 346	coastline damping viscosity
A 348	variables relaxed in the numerical beach
A 350, A 351	breaking model (0 off, 1 base, 2 Kennedy, 3 Romero) and detection (shallow/deep)
A 352, A 361, A 362	additional filtering in breaking cells
A 354, A 355, A 356	breaking thresholds: $\partial\eta/\partial t$ factor, slope onset, slope cessation
A 357, A 365, A 366	breaking viscosity: applied to $\eta/\tilde{\Phi}$, value, factor for η
A 370–A 373	wind forcing
B 2, B 103, B 113, B 116	number of vertical cells and vertical stretching of the σ grid (DIVEMesh)
B 92, B 98, B 99	wave theory, generation and absorption method
F 60	still water level
N 10, N 11	solver and preconditioner for the Laplace equation (HYPRE, REEFMG N 10 1)
N 47, N 48, N 49	CFL number, adaptive time stepping, maximum time step
P 40, P 44	state files for hydrodynamic coupling and nesting

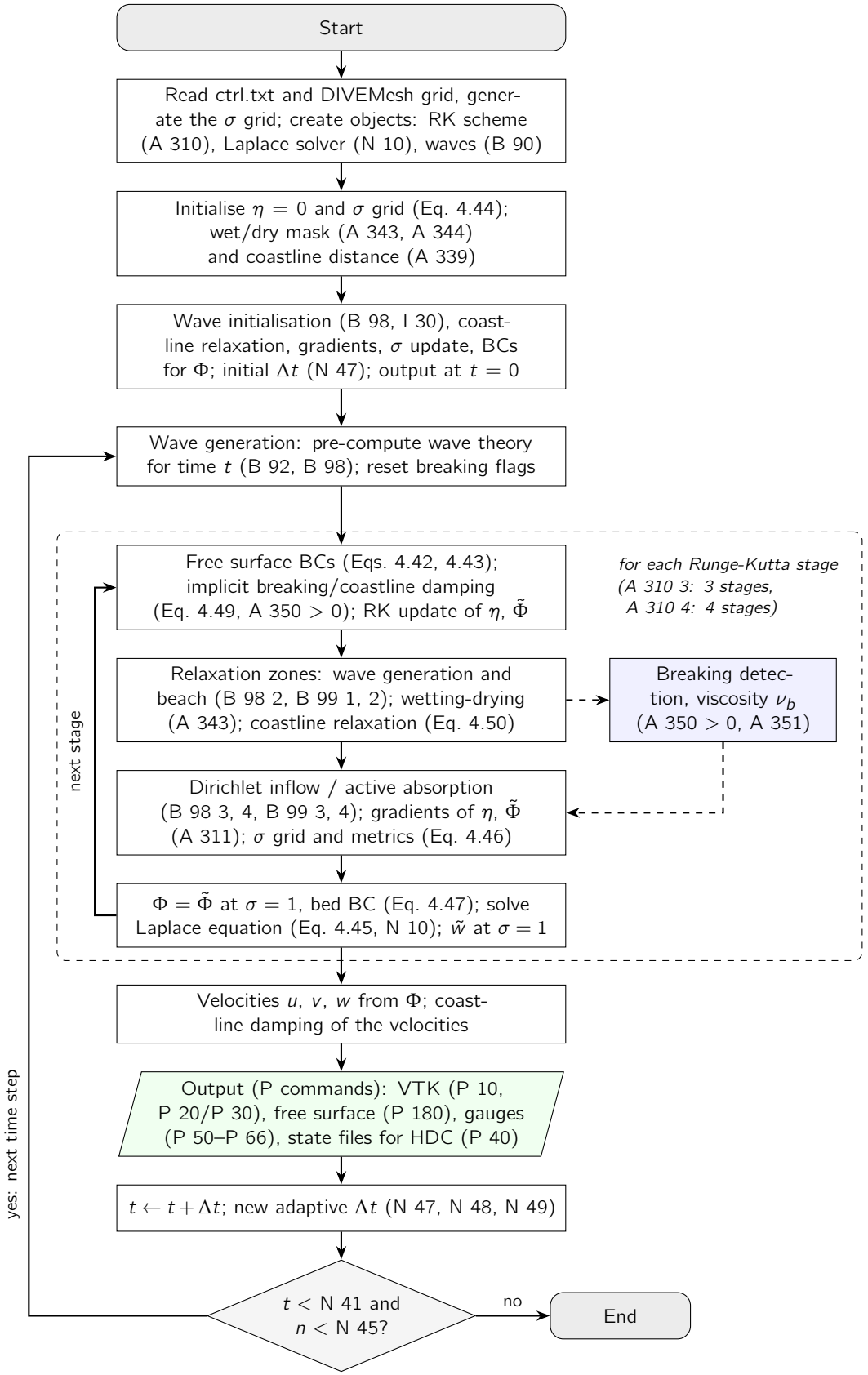

Figure 4.6: Solution procedure of REEF3D::FNPF. Shown is the default RK3 path (A 310 3); with A 310 4 the same steps are carried out in four stages, but the wetting-drying is applied after the breaking detection.

4.5 REEF3D::SFLOW

REEF3D::SFLOW (A 10 2) solves the depth-averaged shallow water equations (SWE) on a two-dimensional horizontal grid. A depth-averaged non-hydrostatic pressure can be added to represent frequency dispersion, so that waves from shallow to intermediate water depth can be computed [Stelling and Zijlema, 2003, Jeschke et al., 2017, Wang et al., 2020]. Bed friction, depth-averaged turbulence models, wetting and drying, wave breaking and bedload sediment transport are included.

Note. The numerical scheme of SFLOW is under active development. This section describes the current source code: a cell-centred finite volume scheme with HLL fluxes. It differs from the staggered finite difference discretization described by Wang et al. [2020]. The Boussinesq option A 220 4 is new. Details may change in future versions.

4.5.1 Governing equations

The bed is at $z = -d(x, y)$ below the still water level, η is the free surface elevation and $h = \eta + d$ the total water depth. u, v are the depth-averaged horizontal velocities and $\bar{w}$ is the depth-averaged vertical velocity. Continuity and horizontal momentum read

$$\frac{\partial h}{\partial t} + \frac{\partial (hu)}{\partial x} + \frac{\partial (hv)}{\partial y} = 0, \tag{4.51}$$

$$\frac{\partial (hu)}{\partial t} + \frac{\partial}{\partial x}\Big(hu^2 + \tfrac{1}{2} g\eta^2 + g\eta d\Big) + \frac{\partial (huv)}{\partial y} = g\eta \frac{\partial d}{\partial x} - \frac{1}{\rho}\left(\frac{\partial (h\bar{p})}{\partial x} - p_b \frac{\partial d}{\partial x}\right) - \frac{\tau_{bx}}{\rho} + D_x, \tag{4.52}$$

and correspondingly for hv. The hydrostatic pressure is written in the pre-balanced form $\frac{1}{2} g\eta^2 + g\eta d$ with the bed slope source $g\eta\, \partial d/\partial x$. Together they equal $gh\, \partial \eta/\partial x$, and still water over an uneven bed stays exactly at rest in the discrete equations (well-balanced scheme). $\bar{p}$ is the depth-averaged non-hydrostatic pressure and p_b its value at the bed (Section 4.5.2). With A 220 0 both are zero and the equations are the nonlinear SWE.

Friction, diffusion and turbulence. With A 218 1, the Manning bed shear stress

$$\frac{\tau_{b,i}}{\rho} = \frac{g n^2}{h^{1/3}}\, u_i \sqrt{u^2 + v^2}, \qquad n = \frac{k_s^{1/6}}{20}, \tag{4.53}$$

is applied, with the roughness k_s from B 50. W 90 1 adds a Bingham-type bed stress for non-Newtonian flows. Horizontal diffusion is switched on with A 212 (1 explicit, 2 implicit):

$$D_x = h\nu_e \left(2\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 v}{\partial x \partial y}\right), \qquad \nu_e = \nu + \nu_t \text{ (+A 250 in breaking cells with A 246 2).} \tag{4.54}$$

The eddy viscosity ν_t comes from the depth-averaged turbulence model A 260: k-ϵ (1), k-ω (2, and an alternative implementation 5), Prandtl mixing length (3) with $l = 0.4 \cdot$A 261$\cdot h$, or a parabolic profile (4) $\nu_t =$ A 262$\cdot u_* h$. It acts on the velocities only when A 212 is on.

4.5.2 Non-hydrostatic extension

The non-hydrostatic pressure vanishes at the free surface, and the vertical velocity varies linearly over the depth, $\bar{w} = (w_s + w_b)/2$, with the surface value w_s and the kinematic bed value $w_b = -u\,\partial d/\partial x - v\,\partial d/\partial y$. The depth-integrated continuity equation then gives the constraint

$$h\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + 2\left(\bar{w} + u\frac{\partial d}{\partial x} + v\frac{\partial d}{\partial y}\right) = 0, \tag{4.55}$$

and the depth-averaged vertical momentum equation is

$$\frac{\partial (h\bar{w})}{\partial t} + \frac{\partial (hu\bar{w})}{\partial x} + \frac{\partial (hv\bar{w})}{\partial y} = \frac{p_b}{\rho} + D_w. \tag{4.56}$$

The convective terms are included with A 214 1 and omitted with A 214 0. The bed pressure depends on the vertical pressure profile (A 220):

- **Linear (A 220 1)**, as in the one-layer Keller-box approach of Stelling and Zijlema [2003]: $p_b = 2\bar{p}$.
- **Quadratic (A 220 2)** following Jeschke et al. [2017]:

  $$p_b = \tfrac{3}{2}\bar{p} + \frac{\rho h}{4}\varphi, \qquad \varphi = \frac{Dw_b}{Dt} = -\frac{Du}{Dt}\frac{\partial d}{\partial x} - \frac{Dv}{Dt}\frac{\partial d}{\partial y} - u^2\frac{\partial^2 d}{\partial x^2} - v^2\frac{\partial^2 d}{\partial y^2} - 2uv\frac{\partial^2 d}{\partial x \partial y}. \tag{4.57}$$

  The bed acceleration φ is evaluated explicitly in each Runge-Kutta stage. On a flat bed, $\varphi = 0$.
- **Quadratic with improved dispersion (A 220 3)**, with the parameter $\alpha \geq 1$ (A 224, default 1.159). The pressure source in Eq. (4.56) becomes $\frac{3}{2\alpha}\bar{p}/\rho + h\varphi/4$. In the horizontal momentum, φ is multiplied by α and the explicit term $Bg\nabla(h^3\nabla^2\eta)$, $B = (\alpha - 1)/3$, is added. $\alpha = 1$ recovers option 2.

Pressure projection. In every Runge-Kutta stage, provisional values $(hu)^*$, $(hv)^*$, $(h\bar{w})^*$ are computed without $\bar{p}$ and then corrected:

$$(hu)^{s+1} = (hu)^* - \frac{\Delta t_s}{\rho}\left(\frac{\partial (h\bar{p})}{\partial x} - p_b\frac{\partial d}{\partial x}\right), \qquad (h\bar{w})^{s+1} = (h\bar{w})^* + \Delta t_s\left(\frac{c_w\bar{p}}{\rho} + \delta\,\frac{h\varphi}{4}\right), \tag{4.58}$$

with $c_w = 2$, $\delta = 0$ (linear) or $c_w = 3/(2\alpha)$, $\delta = 1$ (quadratic). Δt_s is the effective time step of the stage. Inserting Eq. (4.58) into the constraint (4.55) gives the Poisson-type equation

$$-\frac{h}{\rho}\left(\frac{\partial^2\bar{p}}{\partial x^2} + \frac{\partial^2\bar{p}}{\partial y^2}\right) + \frac{2c_w}{\rho h}\bar{p} = -\frac{1}{\Delta t_s}\left[h\left(\frac{\partial u^*}{\partial x} + \frac{\partial v^*}{\partial y}\right) + 2\left(\bar{w}^* + u^*\frac{\partial d}{\partial x} + v^*\frac{\partial d}{\partial y}\right)\right] - \delta\,\frac{\varphi}{2}. \tag{4.59}$$

The left-hand side is discretized as a five-point stencil with the cell depth h, the gradients with central differences. The equation is solved with HYPRE Struct (N 10 11–19, preconditioner N 11, stopping criterion N 44), with $\bar{p} = 0$ at in- and outflow boundaries and $\partial\bar{p}/\partial n = 0$ at walls. $\bar{p}$ is only solved in wet cells with wet neighbours that are not breaking and not excluded by A 221 0 (cells close to the shoreline or with $h \leq 10\cdot$A 244). Elsewhere $\bar{p} = 0$ and the flow is hydrostatic.

Boussinesq equations (A 220 4, new). Instead of a pressure equation, this option solves the fully nonlinear Boussinesq equations of FUNWAVE-TVD in the conservative form of Shi et al. [2012]:

$$\frac{\partial \eta}{\partial t}+\frac{\partial M_j}{\partial x_j}=0, \qquad \frac{\partial V_i}{\partial t}+\frac{\partial}{\partial x_j}\left(\frac{M_i M_j}{h}\right)+\frac{\partial}{\partial x_i}\left(\tfrac{1}{2}g\eta^2+g\eta d\right)=g\eta\frac{\partial d}{\partial x_i}+\psi_i-\frac{\tau_{b,i}}{\rho}+D_i, \quad (4.60)$$

with $M_i = h(u_{\alpha,i}+u_{2,i})$ and $V_i = h(u_{\alpha,i}+V'_{1,i})$. $u_{\alpha,i}$ is the velocity at the reference level $z_\alpha = \zeta d+(1+\zeta)\eta$, $\zeta=-0.53$. The dispersive velocities $u_{2,i}$ and $V'_{1,i}$ depend on $\nabla(\nabla\cdot\mathbf{u}_\alpha)$ and $\nabla(\nabla\cdot(d\,\mathbf{u}_\alpha))$, and ψ_i collects the remaining dispersive terms, all as given by Shi et al. [2012]. ψ_i is explicit. After each stage, $u_{\alpha,i}$ is recovered from V_i by a line-implicit solve per component, with lagged cross derivatives. M_i enters the HLL fluxes, while the dissipative part of the HLL flux uses V_i, as in FUNWAVE-TVD. The dispersive terms are switched off smoothly in breaking cells (A 246), where $|\eta|/d \geq 0.8$, next to the shoreline and next to in- and outflow boundaries. N 47 is limited to 0.25.

4.5.3 Discretization

Finite volume scheme. SFLOW uses a cell-centred finite volume method on the horizontal, possibly non-uniform grid. The prognostic variables h, hu, hv, $h\bar{w}$ (or V_i for A 220 4) and the pressure $\bar{p}$ are stored at the cell centres. Face velocities are only computed as diagnostic quantities for the sediment, turbulence and output modules. At each face, left and right states of η, u, v and the conserved variables are reconstructed as selected with A 211: first order (0), TVD with the van Leer (1), Superbee (2) or van Albada (3) limiter, or fifth-order WENO (4, default; Section 4.1). The limiters fall back to first order next to dry cells. The face depth is $D=\max(\eta_{face}+d_{face}$, A 244), with d_{face} the mean of the two adjacent cells. The HLL flux [Harten et al., 1983] is

$$F=\begin{cases} F_L, & S_L\geq 0,\\ \big(S_R F_L - S_L F_R + S_L S_R (q_R-q_L)\big)/(S_R-S_L), & S_L<0<S_R,\\ F_R, & S_R\leq 0, \end{cases} \quad (4.61)$$

with the conserved variable q and the physical flux F. Between wet cells, the signal speeds are $S_L=\min(u_L-\sqrt{gD_L}, u_*-c_*)$ and $S_R=\max(u_R+\sqrt{gD_R}, u_*+c_*)$, with $u_*=\frac{1}{2}(u_L+u_R)+\sqrt{gD_L}-\sqrt{gD_R}$ and $c_*=\frac{1}{2}(\sqrt{gD_L}+\sqrt{gD_R})+\frac{1}{4}(u_L-u_R)$. Next to a dry cell, dry-bed speeds are used, e.g. $S_L=u_R-2\sqrt{gD_R}$ for a dry left side. The bed slope source uses the same d_{face} as the flux, which keeps the scheme well balanced. At walls, the mass flux is zero and the momentum flux is $\frac{1}{2}g\eta^2+g\eta d$ of the wall cell. At in- and outflow boundaries, the flux of the ghost cell state is used. The discharge is limited to $|hu|\leq$ A 531$\cdot h\sqrt{gh}$.

Time stepping. The SSP Runge-Kutta schemes of Section 4.1 are used, third order (A 210 3, default) or second order (A 210 2). Each stage computes $q^{(s+1)}=a\,q^n+(1-a)\big(q^{(s)}+\Delta t\,L(q^{(s)})\big)$, with $a=0,\frac{3}{4},\frac{1}{3}$ (RK3) or $a=0,\frac{1}{2}$ (RK2), and includes the pressure projection with $\Delta t_s=(1-a)\Delta t$. The adaptive time step (N 48 1) is

$$\Delta t = 2\cdot \text{N } 47\cdot\min\left(\frac{\Delta x}{|u|+\sqrt{gh}},\frac{\Delta y}{|v|+\sqrt{gh}}\right) \quad (4.62)$$

over all wet cells. A 219 2 adds the limit $\Delta x/|u|$, and for A 220 3 the explicit dispersion term requires $\Delta t \leq 1.8/\left[(\Delta x^{-2} + \Delta y^{-2})\sqrt{gBh^3}\right]$. N 48 0 uses the fixed time step N 49.

Wetting and drying. A cell is dry when h falls below A 244. Its depth is then set to A 244 and its momentum to zero. With A 243 2 (default), a dry cell becomes wet again only when a wet neighbour has a higher free surface. At wet-dry faces, the face states are taken from the wet cell with zero velocity, and no mass flux enters a dry cell.

Wave breaking. With A 246 1, a cell breaks when $\partial\eta/\partial t >$ A 247$\cdot\sqrt{gh}$. A 246 2 also flags free surface slopes $|\partial\eta/\partial x| >$ A 355 and adds the breaking viscosity A 250. With A 248 1, breaking continues while $\partial\eta/\partial t >$ A 249$\cdot\sqrt{gh}$. A 242 flags further shallow cells near the shoreline. In breaking cells, the non-hydrostatic pressure (or the Boussinesq dispersion) is switched off. The breaker then travels as a bore of the SWE and loses energy through the shock-capturing HLL flux.

Waves are generated and absorbed with relaxation zones or Dirichlet boundaries (B 98, B 99; Section 4.1), with the input velocities averaged over B 160 + 1 points over the depth.

4.5.4 Sediment transport

With S 10 1, the bed elevation z_b follows the Exner equation for bedload transport,

$$\frac{\partial z_b}{\partial t} = -\frac{\text{S 35}}{1-n}\left(\frac{\partial q_{b,x}}{\partial x} + \frac{\partial q_{b,y}}{\partial y}\right), \tag{4.63}$$

with the porosity n (S 24) and the morphological factor S 35. The sediment module is shared with NHFLOW and CFD (Section 6.13): bedload formula S 11, slope reduction of the critical shear stress S 80, Exner formulation S 31, non-equilibrium transport S 33 and sandslide S 90. In SFLOW, the bed shear stress is computed from the depth-averaged velocity: S 16 1 uses the Manning form $\tau_b = \rho g n^2 |\mathbf{u}|^2/h^{1/3}$, S 16 2 and 3 the depth-averaged log-law $\tau_b = \rho g |\mathbf{u}|^2/C^2$ with $C = 2.5\ln(12h/k_s)$, and S 16 4 $\tau_b = 0.3\rho k$. After each bed update, d is recomputed while h is kept, so that η follows the bed.

4.5.5 Solution procedure

The solution procedure of REEF3D::SFLOW is shown in Fig. 4.7. After the initialization, each time step starts with the wave generation and the in- and outflow boundary values. The Runge-Kutta stages then compute the HLL fluxes, update the free surface and the momentum and solve the non-hydrostatic pressure. Turbulence and sediment transport follow once per time step, before the new time step size is computed and the results are written.

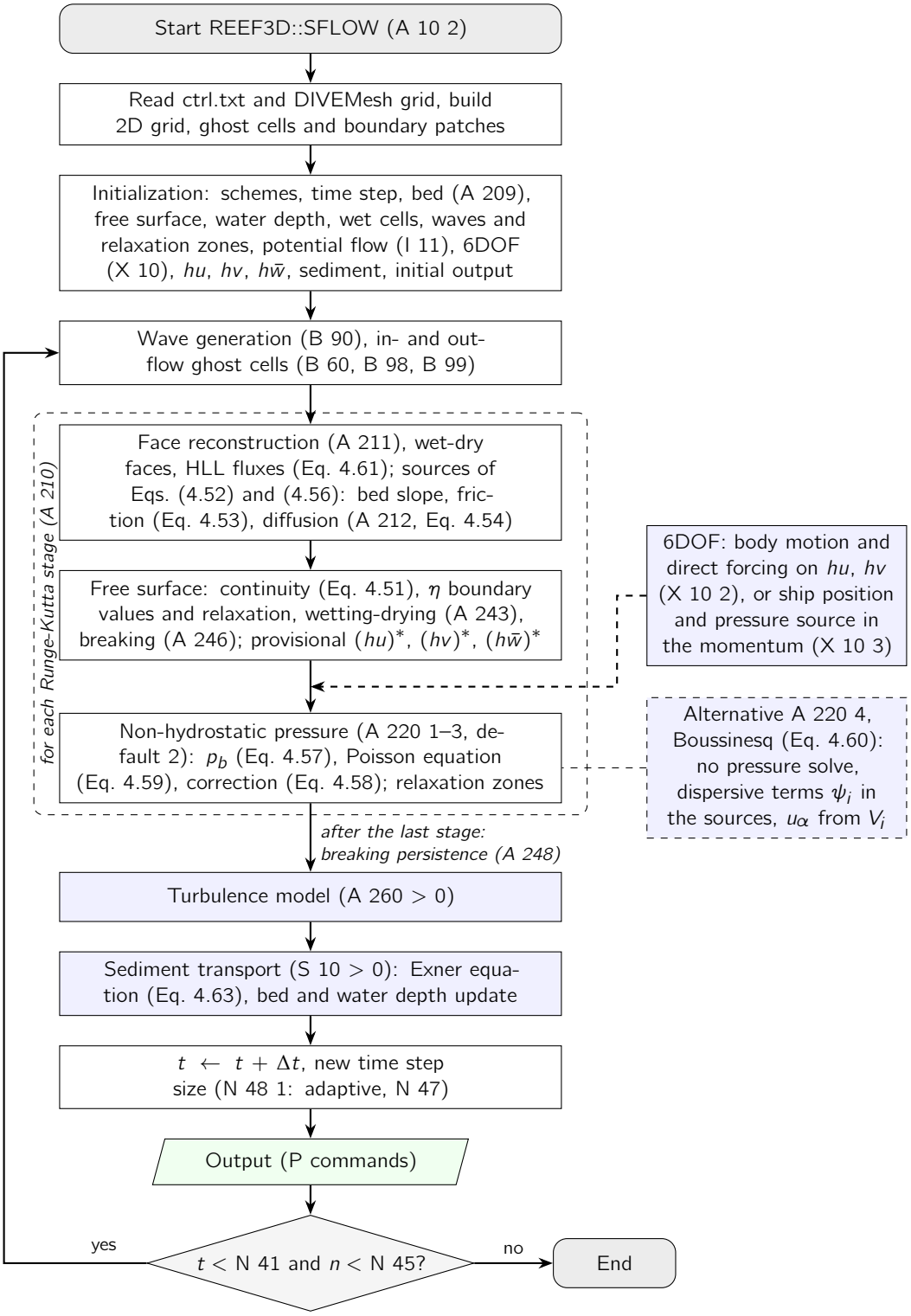

Figure 4.7: Solution procedure of REEF3D::SFLOW with the default quadratic non-hydrostatic pressure (A 220 2); with A 220 0 the pressure step is skipped, and with the Boussinesq option A 220 4 (dashed box) the dispersion enters through the momentum sources and the recovery of u_α instead of a pressure equation. The time step size is computed at the end of each step (initially during the initialization), and with sediment transport the loop also stops at the sediment time S 19.

4.5.6 Key input commands

Command	Meaning
A 10 2	select SFLOW
A 210	Runge-Kutta scheme: 2 (RK2) or 3 (RK3, default)
A 211	face reconstruction: 0 first order, 1–3 TVD limiters, 4 WENO
A 212	horizontal diffusion: 0 off, 1 explicit, 2 implicit
A 214	convection of $\bar{w}$ in the non-hydrostatic schemes
A 218, B 50	Manning bed friction and roughness k_s
A 220	0 hydrostatic, 1 linear, 2 quadratic, 3 quadratic with A 224, 4 Boussinesq (new)
A 221, A 242	hydrostatic treatment of cells near the shoreline
A 243, A 244	wetting and drying algorithm and threshold
A 246–A 250	breaking criterion, persistence and breaking viscosity
A 260–A 264	turbulence model and coefficients
A 531	discharge (Froude number) limiter
B 98, B 99, B 160	wave generation, absorption, depth averaging of the input wave
N 10, N 11, N 44	solver, preconditioner and stopping criterion for $\bar{p}$
N 47, N 48, N 49	time step factor, adaptive or fixed time step
S 10, S 11, S 16	sediment transport, bedload formula, bed shear stress

5. Multiphysics

Besides the flow of water and air, REEF3D contains a number of models for the physical processes that interact with the flow: rigid floating bodies with mooring lines and aquaculture nets, flexible structures, sediment transport and scour, particle-laden flows, porous media and vegetation, heat and concentration transport, non-Newtonian rheology, Lagrangian particle tracking, sea ice and wind forcing. These multiphysics models are not separate solvers. They are coupled to the hydrodynamic modules of Chapter 4, receive the velocities, the pressure and the free surface from them and act back on the flow through source terms, boundary conditions or a change of the geometry. This chapter summarizes the governing equations and the numerical methods of these models in the same compact form as Chapter 4; the shared numerics (grid, WENO schemes, time stepping, linear solvers) are described there and are not repeated. The commands that activate a model are given in parentheses and are described in Chapter 6.

5.1 Overview

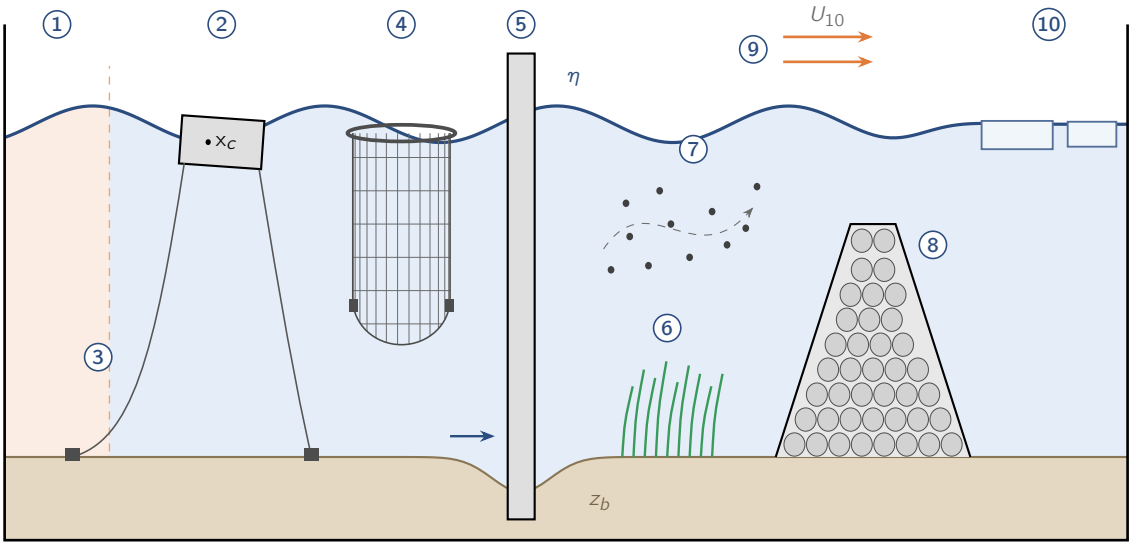

Figure 5.1: Multiphysics models of REEF3D in a schematic coastal wave tank: (1) wave generation zone (Section 4.1.5), (2) floating body with the centre of gravity $\mathbf{x}_c$ (Section 5.2), (3) mooring lines and (4) aquaculture net cage (Section 5.3), (5) pile with a scour hole in the sand bed z_b (Section 5.5; particle-based sediment: Section 5.6), (6) vegetation patch (Section 5.7; flexible vegetation as rod trees: Section 5.4), (7) drifting Lagrangian particles (Section 5.9), (8) porous rubble-mound breakwater (Section 5.7), (9) wind forcing (Section 5.11) and (10) ice floes (Section 5.10).

Figure 5.1 shows the multiphysics models in a schematic coastal numerical wave tank. The models are coupled to the hydrodynamics in different ways:

- In NHFLOW and CFD, rigid bodies, nets, flexible strips and rod trees are represented on the fixed grid and exchange forces with the flow through forcing terms in the momentum equations; in FNPF, a floating body enters through the boundary condition on its wetted surface. The body or structure is then moved with its own equation of motion (Sections 5.2 to 5.4).

- Sediment transport changes the bed level z_b and thus the geometry of the flow domain (Section 5.5); the MP-PIC model instead represents the sediment as computational parcels (Section 5.6).
- Porous media and vegetation modify the governing equations through the porosity and additional resistance terms (VRANS, Section 5.7).
- Heat and concentration are transported scalars that change the density and viscosity of the fluid (Section 5.8).
- Lagrangian particles are passive: they are moved by the flow but do not act back on it (Section 5.9).
- Sea ice floes and wind act on the free surface through an additional surface pressure or stress (Sections 5.10 and 5.11).

Table 5.1 lists in which hydrodynamic modules each model is available and which command activates it. Not every model is implemented in every module: the σ-grid and depth-averaged modules do not resolve the air phase, and some models have so far only been implemented in one module. The validity of each command is also given in the reference entries of Chapter 6.

Table 5.1: Availability of the multiphysics models in the hydrodynamic modules (●available, ○not available) and the command that activates the model. Letters refer to the notes below the table.

Model	SFLOW	FNPF	NHFLOW	CFD	Commands	Sec.
Floating bodies (6DOF)	●[a]	●[b]	●	●	X 10	5.2
Mooring lines	○	●	●	●	X 310	5.3
Nets	○	○[c]	●	●	X 320	5.3
FSI: flexible strips	○	○	○	●	Z 10	5.4
FSI: rod trees	○	○	●	●	Z 20 (`rodtree.dat`)	5.4
Sediment: bedload	●	○	●	●	S 10, S 11	5.5
Sediment: suspended load	○	○	●	●	S 12	5.5
MP-PIC sediment particles	○	○	○	●	Q 10 1, Q 11	5.6
Porous media (VRANS)	○	○	●	●	B 201, B 210–B 230 (NHFLOW); B 270–B 291 (CFD); B 260 (both)	5.7
Vegetation	○	○	○	●	B 308, B 310–B 322	5.7
Heat, concentration, stratification	○	○	○	●	H 10, C 10	5.8
Non-Newtonian rheology	●	○	○	●	W 90, W 110	5.8
Lagrangian particles	○	○	●	○	L 10	5.9
Sea ice floes	○	●	○	○	A 380 (`ice_floes.dat`)	5.10
Wind forcing	○	●	●	○	A 370 (FNPF), A 570 (NHFLOW)	5.11

[a] SFLOW: one-way coupling only, the body moves with the prescribed or initial velocities without hydrodynamic feedback (X 10 2), or acts as a moving pressure field on the free surface for ship waves (X 10 3).
[b] FNPF: triangulated bodies only (box, cylinder, wedge, hexahedron, STL); the level-set defined objects are not supported.
[c] FNPF: nets (X 320) are ignored with a warning.

The main restrictions follow from the governing equations of the modules. FNPF solves the Laplace equation for the potential Φ and therefore has no viscous stresses, turbulence or transport equations; its multiphysics models (floating bodies, moorings, ice floes and wind)

act through the boundary conditions on the body surface and on the free surface. SFLOW is depth-averaged, so that bedload transport, the rheology of the depth-averaged flow and one-way body motion are available, but not the three-dimensional models. NHFLOW and CFD resolve the vertical flow structure and provide most of the models; the two-phase CFD module is required for heat and concentration transport, flexible strips, vegetation and the particle-based sediment model.

5.2 Floating Bodies (6DOF)

Floating rigid bodies are handled by one kernel shared by all modules (body state, surface triangulation, moorings, nets, time integration); only the body representation in the fluid and the loads are module specific. The module is switched on with X 10. **CFD** supports two-way coupling by direct forcing (X 10 1), with several bodies (X 20). **NHFLOW** supports two-way (1) and one-way coupling (2), a moving pressure field for ship waves (3) and prescribed wavemakers (4). **FNPF** (X 10 >0) resolves the body in the Laplace problem with implicit added mass, and **SFLOW** supports one-way coupling (2) and the moving pressure field (3). NHFLOW, FNPF and SFLOW use one body. The body is a closed surface triangulation from primitives or an STL file (Section 6.16); mass, centre of gravity and inertia follow from it and the density X 21, unless given with X 22–X 24 (Fig. 5.2).

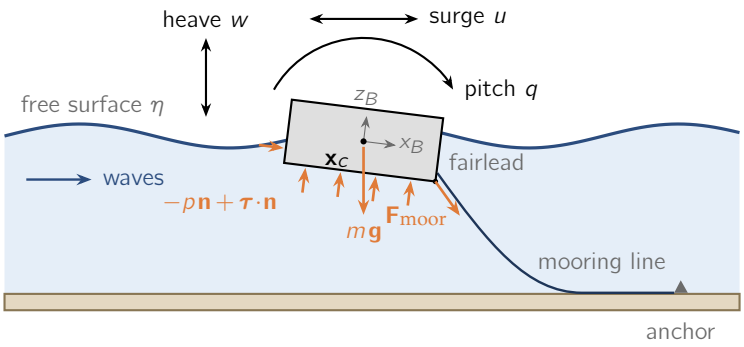

Figure 5.2: Floating body in waves: centre of gravity $\mathbf{x}_C$ with body axes, surface loads, gravity and a mooring line at the fairlead. Surge, heave and pitch are the 2D degrees of freedom; 3D adds sway v, roll p and yaw r (X 11).

5.2.1 Rigid-body equations

The orientation is represented by the unit quaternion $\mathbf{e} = (e_0, e_1, e_2, e_3)$, which avoids the singularities of Euler angles (these are only used for the initial rotation X 101 and the output). With

$$\mathbf{E} = \begin{pmatrix} -e_1 & e_0 & -e_3 & e_2 \\ -e_2 & e_3 & e_0 & -e_1 \\ -e_3 & -e_2 & e_1 & e_0 \end{pmatrix}, \qquad \mathbf{G} = \begin{pmatrix} -e_1 & e_0 & e_3 & -e_2 \\ -e_2 & -e_3 & e_0 & e_1 \\ -e_3 & e_2 & -e_1 & e_0 \end{pmatrix}, \qquad \mathbf{R} = \mathbf{E}\mathbf{G}^T, \quad (5.1)$$

$\mathbf{R}$ rotates body-frame vectors into the inertial frame. The linear momentum $\mathbf{P} = m\mathbf{u}_c$ and the position $\mathbf{x}_c$ (inertial frame) and the angular momentum $\mathbf{h}$ and the quaternion $\mathbf{e}$ (body

frame) are integrated in the Hamiltonian form

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}, \qquad \frac{d\mathbf{x}_c}{dt} = \frac{\mathbf{P}}{m}, \qquad \frac{d\mathbf{e}}{dt} = \frac{1}{2}\mathbf{G}^T \mathbf{I}_B^{-1}\mathbf{h}, \qquad \frac{d\mathbf{h}}{dt} = 2\dot{\mathbf{G}}\mathbf{G}^T\mathbf{h} + \mathbf{R}^T\mathbf{M}, \tag{5.2}$$

with the body-frame inertia tensor $\mathbf{I}_B$ about $\mathbf{x}_c$. With $\boldsymbol{\omega} = \mathbf{R}\mathbf{I}_B^{-1}\mathbf{h}$, a body point moves with $\mathbf{u}_b = \mathbf{u}_c + \boldsymbol{\omega} \times (\mathbf{x} - \mathbf{x}_c)$, and the surface points are moved rigidly, $\mathbf{x} = \mathbf{x}_c + \mathbf{R}\mathbf{x}_0$. Force and moment (about $\mathbf{x}_c$) are collected in the inertial frame:

$$\begin{aligned} \mathbf{F} &= \mathbf{F}_{\text{hyd}} + m\mathbf{g} + \sum \mathbf{F}_{\text{moor}} + \sum \mathbf{F}_{\text{net}} - \mathbf{C}_t\,\mathbf{u}_c, \\ \mathbf{M} &= \mathbf{M}_{\text{hyd}} + \sum (\mathbf{x}_f - \mathbf{x}_c) \times \mathbf{F}_{\text{moor}} + \sum \mathbf{M}_{\text{net}} - \mathbf{C}_r\,\boldsymbol{\omega}, \end{aligned} \tag{5.3}$$

with $\mathbf{g}$ = (W 20, W 21, W 22), the linear damping coefficients $\mathbf{C}_t$ (X 26) and $\mathbf{C}_r$ (X 25), the fairlead positions $\mathbf{x}_f$ and the mooring and net loads of Section 5.3. X 11 selects for each degree of freedom whether it is solved (1), fixed (0) or prescribed (2); in 2D, sway, roll and yaw are switched off.

Prescribed motion. Prescribed degrees of freedom follow constant velocities (X 210, X 211) or a motion file (X 240, X 241) with a ramp (X 205, X 206). In NHFLOW, X 10 2 imposes the motion on the flow without feedback, and X 10 4 moves a piston (X 170) or flap (X 172) wavemaker through the same direct forcing.

Time integration. Eq. (5.2) is advanced stage by stage with the Runge–Kutta scheme of the fluid solver (CFD: X 10 >0 changes N 40 3 to N 40 4; NHFLOW: A 510, Eq. 4.4; FNPF: A 310; SFLOW: A 210), normalizing the quaternion after every stage. The coupling is explicit in CFD and NHFLOW; the direct forcing uses the stage weight α_s.

5.2.2 Hydrodynamic loads

With X 60 1, force and moment are integrated over the surface triangles,

$$\mathbf{F}_{\text{hyd}} = \sum_{\triangle} (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, A_{\triangle}, \qquad \mathbf{M}_{\text{hyd}} = \sum_{\triangle} (\mathbf{x}_{\triangle} - \mathbf{x}_c) \times (-p\mathbf{n} + \boldsymbol{\tau}\cdot\mathbf{n})\, A_{\triangle}, \tag{5.4}$$

with the normal $\mathbf{n}$ into the fluid and the area $A_{\triangle}$ (X 60 2: triangulation of the body level set, CFD and NHFLOW). In **CFD**, p is sampled X 42 cell sizes off the wall, and the viscous stress (X 38, Section 6.16) is computed from the velocity X 43 cell sizes off the wall. In **NHFLOW** and **FNPF**, each triangle is clipped at the local free surface $z = d + \eta(x, y)$, and the wetted part is integrated with the edge midpoints, which is exact for the hydrostatic pressure on a planar facet. NHFLOW uses $p = p_{nh} + \rho g \max(0,\, d + \eta - z)$ (plus a skin friction with X 38 1), FNPF the Bernoulli pressure of Eq. (5.7).

5.2.3 Coupling with the hydrodynamic modules

CFD: level set of the body and direct forcing. After every position update, the signed distance ϕ_b of the body (negative inside) is rebuilt from the triangulation by ray casting (X 188). The body velocity is imposed on the predicted velocity $\mathbf{u}^*$ of each stage by the continuous direct forcing

$$f_i = H(\phi_b)\,\frac{u_{b,i} - u_i^*}{\alpha_s \Delta t}, \qquad H = \begin{cases} 1 & -\phi_b > \epsilon \\ 0 & -\phi_b < -\epsilon \\ \frac{1}{2}\left(1 - \dfrac{\phi_b}{\epsilon} - \dfrac{1}{\pi}\sin\dfrac{\pi\phi_b}{\epsilon}\right) & \text{else} \end{cases}, \tag{5.5}$$

at the staggered velocity positions, with ϵ = X 41 times the mean cell size. With X 14 1, the forcing outside the body is weighted with $|n_i|$ of the level set normal.

NHFLOW: ray casting on the σ-grid. The level set of the body is built on the moving σ-grid (Fig. 5.3): vertical rays through the triangles flag the cell centres of each column as inside or outside, and the signed distance is computed in a band around the surface. Eq. (5.5) is applied to the cell-centred velocities with ϵ = A 526 cell sizes, as no-slip (X 15 0), $|n_i|$-weighted (1) or free-slip forcing (2). With X 16, the body is a moving porous medium with a Darcy–Forchheimer resistance on the velocity relative to $\mathbf{u}_b$.

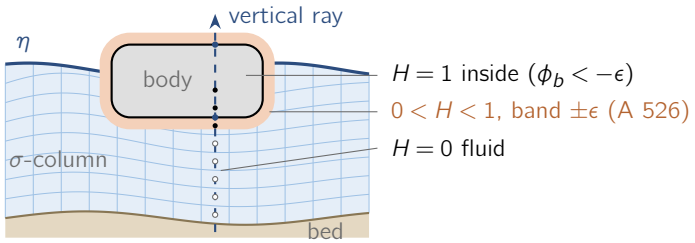

Figure 5.3: NHFLOW body on the σ-grid: vertical rays flag the cell centres inside the body (filled); the signed distance in a band defines the Heaviside function H of the forcing.

FNPF: resolved body with implicit added mass. The body is a boundary of the Laplace problem for Φ (Section 4.4). A vertical ray per σ-column flags the σ-nodes inside the body; these are decoupled from the Laplace matrix, and every fluid/body face carries the staircase Neumann condition

$$\left.\frac{\partial \Phi}{\partial x_i}\right|_{\text{face}} = \mathbf{u}_b \cdot \mathbf{e}_i, \qquad \mathbf{u}_b = \mathbf{u}_c + \boldsymbol{\omega} \times \mathbf{r}, \quad \mathbf{r} = \mathbf{x} - \mathbf{x}_c, \tag{5.6}$$

with the face direction $\mathbf{e}_i$. In the body footprint, η and $\tilde{\Phi}$ are harmonically extended from the surrounding free surface. The pressure follows from the Bernoulli equation,

$$p = -\rho\left(\psi + \tfrac{1}{2}|\nabla\Phi|^2\right) + \rho g\,(d - z), \qquad \psi = \frac{\partial \Phi}{\partial t}, \tag{5.7}$$

sampled half a mean cell off the hull. ψ is solved with the same Laplace operator; its hull condition contains the unknown body acceleration $\mathbf{a} = (\dot{\mathbf{u}}_c, \dot{\boldsymbol{\omega}})$, so ψ is split into

$$\begin{aligned}
&\psi = \psi_0 + \sum_{j=1}^{6} a_j \psi_j, \qquad \mathbf{N} = (\mathbf{n}, \mathbf{r}\times\mathbf{n}),\\
&\psi_0 = \frac{\partial\tilde{\Phi}}{\partial t} - \frac{\partial\Phi}{\partial z}\frac{\partial\eta}{\partial t} \text{ (free surface)}, \quad \frac{\partial\psi_0}{\partial n} = \Big(\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r})\Big)\cdot\mathbf{n} \text{ (hull)},\\
&\psi_j = 0 \text{ (free surface)}, \quad \frac{\partial\psi_j}{\partial n} = N_j \text{ (hull)},
\end{aligned} \tag{5.8}$$

where the m-terms of the hull condition are neglected. ψ_0 gives the load $\mathbf{F}_0$, and the unit modes give the instantaneous added-mass matrix $A_{ij} = -\rho\int_S \psi_j N_i\, dS$ (symmetrized) over the wetted hull. The body is advanced with

$$\begin{pmatrix} m\mathbf{I} + \mathbf{A}_{tt} & \mathbf{A}_{tr} \\ \mathbf{A}_{rt} & \mathbf{I}_I + \mathbf{A}_{rr} \end{pmatrix} \begin{pmatrix} \dot{\mathbf{u}}_c \\ \dot{\boldsymbol{\omega}} \end{pmatrix} = \begin{pmatrix} \mathbf{F} \\ \mathbf{M} - \boldsymbol{\omega}\times(\mathbf{I}_I\boldsymbol{\omega}) \end{pmatrix}, \tag{5.9}$$

with $\mathbf{I}_I = \mathbf{R}\mathbf{I}_B\mathbf{R}^T$ and $\mathbf{F}$, $\mathbf{M}$ from Eq. (5.3) with $\mathbf{F}_{\mathrm{hyd}} = \mathbf{F}_0$; the implicit added mass keeps the coupling stable for light bodies. Only triangulated bodies are supported (X 110, X 163, X 164, X 180), and nets are ignored.

SFLOW: one-way forcing and moving pressure fields. With X 10 2, the depth-averaged velocities under the body footprint are forced to the body velocity as in Eq. (5.5), with a 2D footprint level set. With X 10 3 (SFLOW and NHFLOW, ship waves), the body is a moving surface pressure $p_s(x, y, t)$ (X 400–X 410) with the momentum source $-(h/\rho)\nabla p_s$, moving with the prescribed velocities.

5.2.4 Key input commands

X 10, X 11, X 20	6DOF mode, degrees of freedom, number of bodies (CFD)
X 21–X 24	density, mass, centre of gravity, inertia
X 25, X 26	rotational and translational damping
X 110–X 164, X 180–X 186	primitives, STL geometry and surface refinement
X 101–X 103	initial rotation and velocities
X 38, X 42, X 43, X 60	viscous force model, sampling distances, STL/LSM integration
X 14, X 41, X 188	CFD forcing: tangential velocity, layer width, ray casting
X 15, X 16, A 526	NHFLOW forcing type, porous body, layer width
X 205–X 241	ramps, prescribed velocities, motion files
X 170, X 172	NHFLOW wavemakers (X 10 4)
X 400–X 410	moving pressure field (X 10 3)
N 40, A 510, A 310, A 210	time schemes of CFD, NHFLOW, FNPF, SFLOW
X 19, X 50	output interval, VTP/STL output

5.3 Moorings and Nets

Mooring lines and nets add their loads to Eq. (5.3), evaluated in every Runge–Kutta stage. Moorings (X 310) are available in CFD and NHFLOW (X 10 1) and in FNPF, nets (X 320) in CFD and NHFLOW, where they are fixed structures with X 10 0. SFLOW supports neither.

5.3.1 Mooring models

A line (X 311) is defined by the anchor $\mathbf{x}_a$, the fairlead $\mathbf{x}_f$, the mass per length in air w_a and density ρ_c (submerged weight $w = w_a\, g\,(\rho_c - \rho)/\rho_c$), the axial stiffness EA, the diameter d_c, the length L and N_e elements. It breaks at the tension X 314 or the time X 315.

Quasi-static catenary (X 310 1). The elastic catenary with a part resting on the flat seabed (no friction) gives the fairlead forces H and V (Fig. 5.4). With the span $l_h = |\mathbf{x}_f - \mathbf{x}_a|_{xy}$ and the height $\Delta z = z_f - z_a$,

$$l_h = L - \frac{V}{w} + \frac{H}{w}\operatorname{asinh}\!\left(\frac{V}{H}\right) + \frac{HV}{EAw}, \tag{5.10}$$

$$\Delta z = \frac{H}{w}\left(\sqrt{1 + \left(\frac{V}{H}\right)^2} - 1\right) + \frac{V^2}{2EAw}. \tag{5.11}$$

solved with Newton's method; $\mathbf{F}_{\text{moor}} = -(H\,\mathbf{e}_h + V\,\mathbf{e}_z)$ with the horizontal unit vector $\mathbf{e}_h$ from anchor to fairlead.

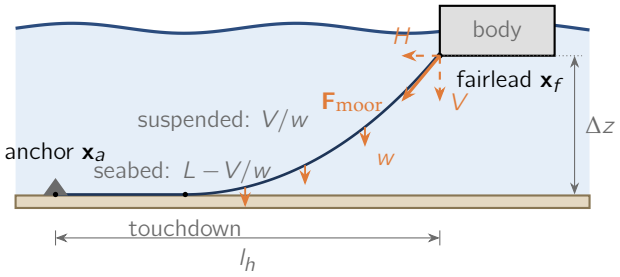

Figure 5.4: Quasi-static catenary (X 310 1) with seabed part, submerged weight w and fairlead forces H, V.

Spring (X 310 4). A spring (X 312) carries $T = \max(T_0 + k\,(|\mathbf{x}_f - \mathbf{x}_a| - L_0), 0)$ (stiffness k, pre-tension T_0, initial length L_0).

Quasi-static bar model (X 310 2). N_e bars of the unstretched length $l_0 = L/N_e$ with the directions $\mathbf{f}_j$ and tensions T_j satisfy knot equilibrium and geometric closure,

$$T_{j+1}\mathbf{f}_{j+1} - T_j\mathbf{f}_j = w\,\bar{l}_j\,\mathbf{e}_z - \mathbf{R}_j, \quad \sum_j l_j\,\mathbf{f}_j = \mathbf{x}_f - \mathbf{x}_a, \quad |\mathbf{f}_j| = 1, \quad l_j = l_0\left(1 + \frac{T_j}{EA}\right), \tag{5.12}$$

with the mean bar length $\bar{l}_j$ at knot j and a Morison-type drag $\mathbf{R}_j$, solved iteratively; the body force is $-T\mathbf{f}$ of the last bar.

Dynamic model (X 310 3). The line is a geometrically exact rod ($A = w_a/\rho_c$, no bending stiffness) solved with the rod solver of Section 5.4 and RADAU5 [Hairer and Wanner, 1996] once per time step. The loads per unit length are

$$\mathbf{f} = \tfrac{1}{2}\rho d_c\,(C_{dt}|u_t|u_t\,\mathbf{t} + C_{dn}|\mathbf{u}_n|\mathbf{u}_n) + \rho\frac{\pi d_c^2}{4}\,(\dot{\mathbf{u}}_f + C_{mt}a_t\,\mathbf{t} + C_{mn}\mathbf{a}_n) - w\,\mathbf{e}_z + \mathbf{f}_{\text{bed}}, \tag{5.13}$$

with the tangent $\mathbf{t}$, the relative velocity $\mathbf{u}_f - \dot{\mathbf{c}}$ and acceleration, $C_{dt} = 0.5$, $C_{dn} = 2.5$, $C_{mt} = 0$, $C_{mn} = 3.8$ [Morison et al., 1950], and a spring–damper seabed.

5.3.2 Nets

Nets (X 320–X 322: solidity S_n, twine diameter d_t and length l_t, knot diameter d_k) are meshes of knots and bars (twines). The load is computed for screens spanned by neighbouring bars (Fig. 5.5), with area A_s, normal $\mathbf{n}_s$ and inflow $\mathbf{u}_{\mathrm{rel}}$ at the angle θ to $\mathbf{n}_s$ [Kristiansen and Faltinsen, 2012]:

$$\mathbf{F}_s = \tfrac{1}{2}\rho A_s |\mathbf{u}_\infty|^2 (C_D\,\mathbf{n}_d + C_L\,\mathbf{n}_l)\,, \qquad \mathbf{n}_d = \frac{\mathbf{u}_{\mathrm{rel}}}{|\mathbf{u}_{\mathrm{rel}}|}, \quad \mathbf{n}_l = \frac{(\mathbf{n}_d \times \mathbf{n}_s)\times \mathbf{n}_d}{|(\mathbf{n}_d \times \mathbf{n}_s)\times \mathbf{n}_d|}, \tag{5.14}$$

$$C_D = C_{D0}\left[(1-a_3-a_5)\cos\theta + a_3\cos 3\theta + a_5\cos 5\theta\right],$$
$$C_L = C_{L0}\left[b_2\sin 2\theta + b_4\sin 4\theta + b_6\sin 6\theta\right], \tag{5.15}$$

with $a_3 = 0.0129$, $a_5 = 0.0151$, $b_2 = 1.2044$, $b_4 = 0.1466$, $b_6 = 0.135$,

$$C_{D0} = C_{\mathrm{cyl}}\frac{S_n(2-S_n)}{2(1-S_n)^2}, \quad C_{L0} = \frac{C_{D0}/2 - C_{T,\pi/4}}{\sqrt{2}}, \quad C_{T,\pi/4} = \frac{\pi C_{D0}\cos^2(\pi/4)}{8 + C_{D0}\cos^2(\pi/4)}. \tag{5.16}$$

where $C_{\mathrm{cyl}} = 1.1 + 4\,Re^{-1/2}$, $Re = d_t|\mathbf{u}|/(\nu(1-S_n))$, and $|\mathbf{u}_\infty|$ follows from momentum theory. Y 1 selects the model of Martin et al. [2022].

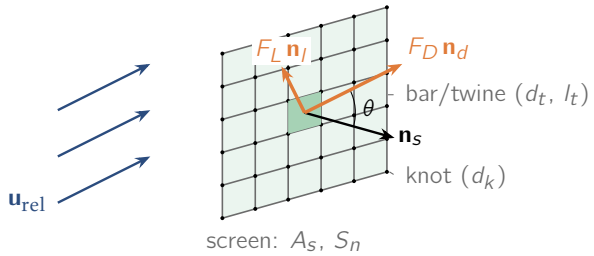

Figure 5.5: Net panel of knots and bars; screen (shaded) load from θ and Eq. (5.15).

Net structural models.

- **Quasi-static bar net** (X 320 2, 3): static knot equilibrium under weight, screen and twine loads, with the bar tensions as unknowns [Martin et al., 2020]; no load is returned to the body.
- **Dynamic bar net** (X 320 12–14): the knots carry the lumped twine mass plus added mass and move under

$$(m_k + m_{a,k})\,\ddot{\mathbf{x}}_k = \textstyle\sum_b T_b\,\mathbf{e}_b + \mathbf{F}_k,$$
$$\mathbf{F}_k = -g\,(m_k - \rho V_k)\,\mathbf{e}_z + (1+C_a)\rho V_k\dot{\mathbf{u}}_f + \textstyle\sum_s \mathbf{F}_s + \mathbf{F}_{d,k} - \mathbf{r}\,\dot{\mathbf{x}}_k, \tag{5.17}$$

  with the bar directions $\mathbf{e}_b$, $C_a = 1$, knot and sinker drag $\mathbf{F}_{d,k}$ (X 323) and relaxation $\mathbf{r}$ (X 325). The tensions T_b of the inextensible bars follow from an accelerated Newton method [Martin and Bihs, 2021a]; the top-bar tensions give $\mathbf{F}_{\mathrm{net}}$, $\mathbf{M}_{\mathrm{net}}$.
- **Rigid sheets** (X 320 4): moving with the body, loads added to the body.

Coupling to the fluid. The screen load $\mathbf{F}_k$ at the centroid $\mathbf{X}_k$ of each net triangle [Martin et al., 2020] is spread onto the grid as a volume force with the four-point cosine kernel [Peskin, 2002]:

$$\mathbf{f}(\mathbf{x}) = -\sum_k \frac{\mathbf{F}_k}{\rho}\,\frac{\delta_h(\mathbf{x}-\mathbf{X}_k)}{\Delta x\,\Delta y\,\Delta z}, \qquad \varphi(r) = \begin{cases} \frac{1}{4}\left(1+\cos\frac{\pi r}{2}\right) & |r| \le 2 \\ 0 & \text{else} \end{cases},$$
$$\delta_h = \varphi\!\left(\frac{x-X_k}{\Delta x}\right)\varphi\!\left(\frac{y-Y_k}{\Delta y}\right)\varphi\!\left(\frac{z-Z_k}{\Delta z}\right). \tag{5.18}$$

It enters the momentum equations in the next predictor step: the net acts as a porous, momentum-absorbing surface not resolved by the grid.

5.3.3 Key input commands

X 310, X 311, X 312	mooring model, line definition, spring definition
X 313	rotate the fairleads with the initial body rotation
X 314, X 315	breaking tension and breaking time of the lines
X 320, X 321, X 322	net type, net material, net dimensions and position
X 323, X 324, X 325	sinkers, knot probes, net time step and relaxation (dynamic nets)
Y 1	simulation-based screen force model

5.4 Flexible Structures (FSI)

REEF3D has two models for flexible, slender structures, both treated as rods and coupled to the fluid through Lagrangian points and a regularized delta function: **flexible strips** (Z 10 1, Z 11, Z 12), thin blades resolved by the grid, modelled as a geometrically exact Cosserat rod with direct-forcing immersed boundary coupling (REEF3D::CFD only); and **rod trees** (Z 20 1, Z 21, input file `rodtree.dat`), branched soft corals, gorgonians or flexible vegetation thinner than the cells, with Morison-type loads spread as a body force (unresolved, actuator-line type coupling; REEF3D::CFD and REEF3D::NHFLOW). Neither is available in REEF3D::FNPF or REEF3D::SFLOW.

5.4.1 Flexible strips: geometrically exact rod

Each strip (Z 11) is a rod of length L, width W and thickness T, clamped at its bottom point and initially pointing in $+z$, with the width in y. It is described by the centre line $\mathbf{c}(s,t)$, $s \in [0, L]$, and a unit quaternion $\mathbf{q}(s,t)$ for the orientation of the cross-section (rotation matrix $\mathbf{R}(\mathbf{q})$), which avoids the singularities of Euler angles at large rotations [Martin and Bihs, 2021b]. The translational balance is

$$\rho_s A\,\frac{\partial^2 \mathbf{c}}{\partial t^2} = \frac{\partial \mathbf{n}}{\partial s} + \rho_s A\,\mathbf{f}_{\text{ext}}, \qquad \mathbf{n} = \mathbf{R}(\mathbf{q})\,\mathbf{n}^0, \tag{5.19}$$

with the material density ρ_s, $A = WT$, the internal force $\mathbf{n}$ and the external load per unit mass $\mathbf{f}_{\text{ext}}$. Body-frame force and moment follow from a linear visco-elastic law,

$$\mathbf{n}^0 = \mathbf{C}_\varepsilon\left(\mathbf{R}^T\frac{\partial \mathbf{c}}{\partial s} - \mathbf{R}_0^T\frac{\partial \mathbf{c}_0}{\partial s}\right) + \mathbf{D}_\varepsilon\dot{\boldsymbol{\varepsilon}}, \qquad \mathbf{C}_\varepsilon = \operatorname{diag}(EA,\ GA,\ GA), \tag{5.20}$$

$$\mathbf{m}^0 = \mathbf{C}_\kappa(\boldsymbol{\kappa} - \boldsymbol{\kappa}_0) + \mathbf{D}_\kappa\dot{\boldsymbol{\kappa}}, \qquad \mathbf{C}_\kappa = \operatorname{diag}(GI_x,\ EI_y,\ EI_z), \tag{5.21}$$

with E, $G = E/(2(1+\nu_s))$, I_x, I_y, I_z from Z 11, the damping matrices $\mathbf{D}_\varepsilon$, $\mathbf{D}_\kappa$ from Z 12, the reference configuration (index 0) and the curvature $\boldsymbol{\kappa}$ from the quaternion product of neighbouring cross-sections. The rotational balance is a second-order equation for $\mathbf{q}$ whose term $-|\dot{\mathbf{q}}|^2\mathbf{q}$ keeps $|\mathbf{q}| = 1$. The rod is discretized with finite differences on N elements (positions at the nodes, quaternions at the element centres), clamped at the bottom and free at the top, and integrated with the implicit Runge-Kutta method RADAU5 with step size control [Hairer and Wanner, 1996].

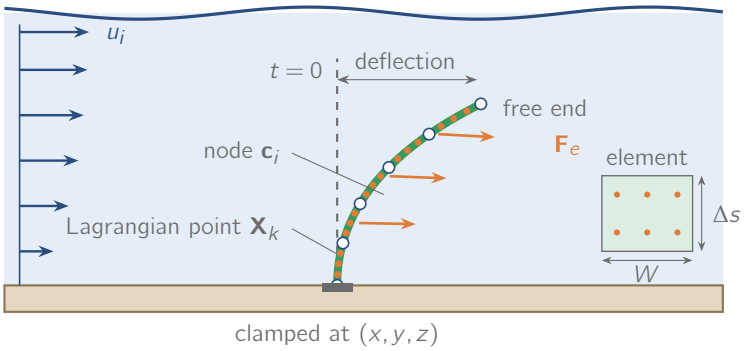

Figure 5.6: Flexible strip (Z 11) in a current: rod nodes $\mathbf{c}_i$ (white), Lagrangian points (orange) over the length and width of each element (inset), and the fluid load $\mathbf{F}_e$ bending the strip from its initial position (dashed).

5.4.2 Flexible strips: immersed boundary coupling

Each element carries $n_l \times n_w$ Lagrangian points $\mathbf{X}_k$ over its length and width, spaced close to the local cell size Δx_b, with the area ΔA_k and the fluid volume $\Delta V_k = \Delta A_k\,\Delta x_b$ (Fig. 5.6); they move rigidly with the element. Interpolation and spreading use the three-point kernel of Roma et al. [1999] at the staggered velocity positions,

$$\varphi(r) = \begin{cases} \frac{1}{3}\left(1 + \sqrt{1-3r^2}\right), & |r| \le \frac{1}{2}, \\ \frac{1}{6}\left(5 - 3|r| - \sqrt{1-3(1-|r|)^2}\right), & \frac{1}{2} < |r| \le \frac{3}{2}, \\ 0, & \text{otherwise}, \end{cases} \qquad \delta_h(\mathbf{x}-\mathbf{X}) = \frac{\varphi\left(\frac{x-X}{\Delta x}\right)\varphi\left(\frac{y-Y}{\Delta y}\right)\varphi\left(\frac{z-Z}{\Delta z}\right)}{\Delta x\,\Delta y\,\Delta z}. \tag{5.22}$$

In every Runge-Kutta stage (coefficient α) of the CFD momentum equations, the fluid velocity $\mathbf{U}_k$ is interpolated to the points and the rod is advanced over $\alpha\Delta t$. Its external load contains the buoyancy-reduced gravity $(1-\rho_f/\rho_s)\mathbf{g}$ and the fluid load from the momentum balance of the fluid carried by the points of element e: with $m_e = \sum_k \rho_f \Delta V_k$, the momentum $\mathbf{P}_e = m_e\bar{\mathbf{u}}_e + \boldsymbol{\omega}_e \times \mathbf{R}\mathbf{s}_0$ in the structural motion ($\mathbf{s}_0$ static moment of the points) and $\mathbf{P}_e^* = \sum_k \rho_f \Delta V_k \mathbf{U}_k$,

$$\mathbf{F}_e = -\frac{\mathbf{P}_e - \mathbf{P}_e^n}{t - t^n} - \frac{\mathbf{P}_e^n - \mathbf{P}_e^*}{\alpha\Delta t}, \tag{5.23}$$

with t the time within the RADAU5 integration; the moment follows analogously (switched off with Y 2 1). The structural point velocity $\mathbf{U}_k^s$ then gives the direct-forcing term $\mathbf{f}_k = (\mathbf{U}_k^s - \mathbf{U}_k)/(\alpha\Delta t)$, which is spread to the grid, $f_i(\mathbf{x}) = \sum_k f_{k,i}\,\delta_h(\mathbf{x}-\mathbf{X}_k)\,\Delta V_k$, and added as source term in Eq. (4.8). With the k-ω model (T 10 2), ω is set in the kernel footprint to represent the wall friction of the strip. STL output follows P 20/P 30, log files X 19.

5.4.3 Rod trees for soft corals

With Z 20 1, colonies are read from `rodtree.dat`: blocks `colony ... end` with `node` (id x y z r), `edge` (parent, child), `clamp`, `material` (E ν_s ρ_s), `hydro` (C_{dn} C_{dt} C_a [C_m], default $C_m = 1 + C_a$), `damping` (β), `refine` and optionally `instance`, `polyps`, `polyp_response`; global keywords `integrator`, `substeps`, `reaction`.

The tree is a discrete Cosserat rod of rigid, slightly conical cylinder elements (length l, mean radius r, mass m, centre $\mathbf{c}_e$, orientation $\mathbf{q}_e$), connected by elastic 6-DOF joints at the shared nodes; branch points and the root (joint to the fixed bed) are ordinary joints (Fig. 5.7). With the gap $\mathbf{g}$ between the attachment points and the rotation vector $\boldsymbol{\theta}$ of the relative rotation with respect to the rest state,

$$\mathbf{F}_j = -k\,(\mathbf{g} + \beta\dot{\mathbf{g}})\,, \qquad \mathbf{T}_j = -\mathbf{K}\,(\boldsymbol{\theta} + \beta\,\boldsymbol{\omega}_{\mathrm{rel}})\,, \qquad \mathbf{K} = \frac{EI}{l_j}\left(\mathbf{I} - \mathbf{t}\mathbf{t}^T\right) + \frac{G\,2I}{l_j}\,\mathbf{t}\mathbf{t}^T, \tag{5.24}$$

with $k = EA/l_j$, $A = \pi r_j^2$, $I = \pi r_j^4/4$ at the joint radius r_j, the rest axis $\mathbf{t}$ and $l_j = (l_a + l_b)/2$ ($l/2$ at a root). Each element (volume V, $D = 2r$) is loaded with the relative velocity $\mathbf{u}_r = \mathbf{u}_f - \dot{\mathbf{c}}_e$, split into the normal part $\mathbf{u}_n = (\mathbf{I} - \mathbf{t}\mathbf{t}^T)\mathbf{u}_r$ and the tangential part $\mathbf{u}_t$ [Morison et al., 1950]:

$$\mathbf{F}_e = (m - \chi\rho_f V)\,\mathbf{g} + c_n|\mathbf{u}_n|\mathbf{u}_n + c_t|\mathbf{u}_t|\mathbf{u}_t + \chi\rho_f V\,(\mathbf{I} - \mathbf{t}\mathbf{t}^T)\,(C_m\mathbf{a}_f - C_a\ddot{\mathbf{c}}_e)\,, \tag{5.25}$$

$$c_n = \tfrac{1}{2}\rho_f\,l\,\chi\Big(C_{dn}D + C_{dp}b_p\Big), \qquad c_t = \tfrac{1}{2}\rho_f\,l\,\chi\Big(C_{dt}\pi D + k_t C_{dp} b_p\Big), \tag{5.26}$$

with the submerged fraction χ. The optional polyp layer (height h_p, frontal solidity ϕ_p) widens the element by $b_p = 2\,e_p h_p \phi_p$; with `polyp_response` U_r ΔU_r τ_p its extension e_p relaxes with time scale τ_p to a target that decreases linearly from 1 at $|\mathbf{u}_n| = U_r$ to 0 at $U_r + \Delta U_r$. By default the tree is advanced with a linearly implicit Euler step (added mass in the mass matrix, drag and gyroscopic terms linearized) with adaptive sub-steps; `integrator explicit` selects symplectic Euler with sub-cycling.

5.4.4 Rod trees: coupling to CFD and NHFLOW

Each element carries $n_q = \lceil l/h \rceil$ Lagrangian points along its axis (h smallest horizontal cell size, $1 \le n_q \le 64$); the structure is replicated on all MPI ranks. In every Runge-Kutta stage, the fluid velocity is sampled at the points (CFD: trilinear, points with $\phi \ge 0$; NHFLOW: bilinear horizontally and linear in the σ column, wet points only); $\mathbf{u}_f$ is the mean over the submerged points and χ their fraction. The reaction $-$(drag + inertia) (`reaction full`, default), $-$drag or zero (one-way coupling) is distributed over the submerged points and spread onto the grid as an acceleration with the cosine kernel [Peskin, 2002]

$$\varphi(r) = \tfrac{1}{4}\Big(1 + \cos\tfrac{\pi r}{2}\Big) \quad \text{for } |r| < 2, \qquad \varphi(r) = 0 \quad \text{otherwise.} \tag{5.27}$$

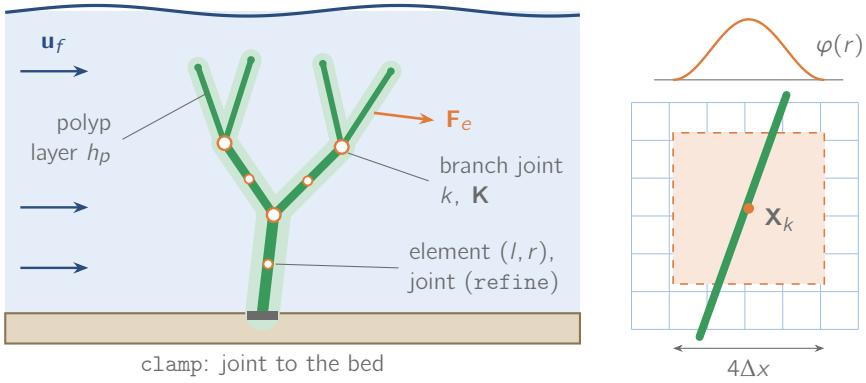

Figure 5.7: Rod tree (Z 20 1). Left: chains of rigid cylinder elements (green) connected by elastic joints (orange), root clamped to the bed, optional polyp layer (light green). Right: the reaction at a Lagrangian point $\mathbf{X}_k$ is spread over the 4×4 cell support of the cosine kernel $\varphi(r)$.

In REEF3D::CFD it enters the forcing fields f_i; in REEF3D::NHFLOW it is added to U,V,W and UH,VH,WH, with the kernel renormalized over the wet cells of the σ column so that no force is lost at the bed or the free surface. The fluid-side drag is treated point-implicitly, which keeps dense canopies stable. The structure is advanced once per time step, in the final stage, with the fluid acceleration $\mathbf{a}_f$ from a backward difference (staggered explicit coupling). Tip displacement, forces, submerged elements and polyp extension per colony are written to `REEF3D_RodTree/`, VTP files every Z 21 seconds.

5.4.5 Key input commands

Z 10 1	flexible strips, resolved immersed boundary coupling (CFD)
Z 11	strip definition: bottom point, L, W, T, ρ_s, E, I_x, I_y, I_z, ν_s, N
Z 12	structural damping of the strips, translational and rotational
Y 2	1: coupling moments on the strips off
Z 20 1	rod trees (soft corals, vegetation), unresolved coupling, input `rodtree.dat` (CFD, NHFLOW)
Z 21	rod-tree VTP print interval [s]
T 10 2	k-ω model; ω forcing at the strips
X 19; P 20, P 30	strip log file interval; STL output interval

5.5 Sediment Transport

The morphodynamic model computes the bed level change due to bedload and suspended load. It is shared by REEF3D::SFLOW, REEF3D::NHFLOW and REEF3D::CFD: τ_b comes from the respective flow solution, while the transport formulas, the Exner equation, the sandslide and the time stepping are common code. REEF3D::FNPF has no sediment transport. The model is based on Bihs [2011] and Bihs and Olsen [2011]; scour applications are given e.g. by Afzal et al. [2015]. The sediment is characterized by d_{50} (S 20), ρ_s (S 22) and the bed porosity n (S 24), with $s = \rho_s/\rho$ (Fig. 5.8). All options are listed in Section 6.13.

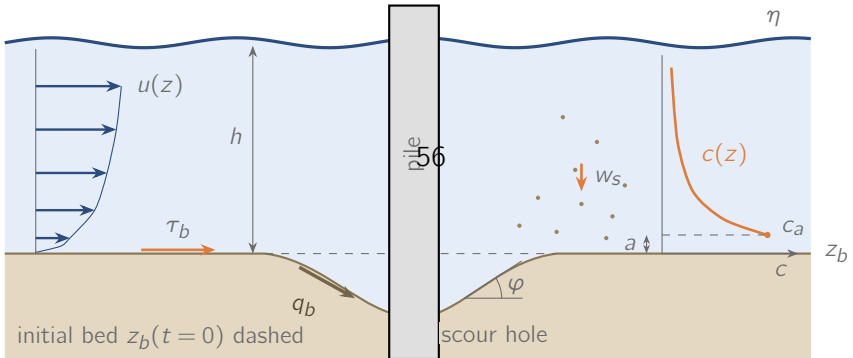

Table 5.3: Sediment transport options per module.

	SFLOW	NHFLOW	CFD
Exner model with bedload (S 10 1)	●	●	●
bed update within the RK2 stages (S 10 12, with A 510 2)	○	●	○
porous VRANS sediment bed (S 10 2)	○	○	●
bed shear stress options S 16	1–4	1–4, 6	1–8
inertia and shallow water blending of τ_b (S 37, S 38)	○	●	○
suspended load (S 12)	○	●	●
Exner sub-iterations (S 27)	○	○	●
graded sediment (S 51)	●	●	●
particle model MP-PIC (Q 10 1, Section 5.6)	○	○	●

5.5.1 Bed representation and module support

The bed level $z_b(x,y)$ is a two-dimensional field on the horizontal grid. In **CFD** the bed is the zero contour of the level set $\phi_b = z - z_b(x,y)$, which enters the flow through the direct forcing of Section 4.2 and is recomputed and reinitialized after each bed update; with S 10 2 the sediment is additionally a porous VRANS region (S 24, S 26). In **NHFLOW** z_b becomes the bed of the σ-grid, and in **SFLOW** d is recomputed while h is kept, so that η follows the bed. Table 5.3 lists the options per module.

5.5.2 Bed shear stress

The default (S 16 1) in CFD and NHFLOW is the rough-wall logarithmic law

$$\tau_b = \rho\left(\frac{\kappa\,|\mathbf{u}_p|}{\ln\!\left(30\,\Delta z_p/k_s\right)}\right)^2, \qquad \kappa = 0.4, \qquad k_s = \text{S 21}\cdot d_{50}, \tag{5.28}$$

with the near-bed velocity $\mathbf{u}_p$ at the height Δz_p above the bed (CFD: interpolated at $z_b{+}1.6\,\Delta z$, $\Delta z_p = \Delta z$; NHFLOW: lowest σ-cell, Δz_p half its height, smooth-wall law with B 22 1). In SFLOW, S 16 1 is the Manning form $\tau_b = \rho g\, n_M^2 |\mathbf{u}|^2/h^{1/3}$, $n_M = k_s^{1/6}/20$. The other S 16 options (k-based, viscous, depth-averaged Chezy/Manning, transitional roughness) are listed in Section 6.13. S 36 1 adds the bedform roughness of van Rijn [1984b] to k_s, giving $k_{s,eff}$; in NHFLOW, S 37 adds an inertia contribution and S 38 blends with the Manning form in very shallow water.

5.5.3 Critical shear stress on sloping beds

The Shields parameter and the critical shear stress are

$$\theta = \frac{\tau_b}{(\rho_s - \rho)\, g\, d_{50}}, \qquad \tau_{cr} = r\,\theta_{cr}\,(\rho_s - \rho)\, g\, d_{50}, \tag{5.29}$$

with θ_{cr} of a flat bed (S 30, default 0.047) [Shields, 1936]. The factor r (S 80) accounts for gravity on a sloping bed [Bihs and Olsen, 2011]; with the longitudinal slope angle θ_l (positive

for a bed descending in the flow direction), the transverse slope angle α and the angle of repose φ, S 80 4 reads [Fredsøe and Deigaard, 1992]

$$r = \cos\theta_l \left(1 - \frac{\tan\theta_l}{\tan\varphi}\right) \cos\alpha \sqrt{1 - \frac{\tan^2\alpha}{\tan^2\varphi}}. \tag{5.30}$$

S 80 1 follows Kovacs and Parker [1994], S 80 2 and 3 Dey [2001, 2003]. r is limited to $0.01 \le r \le 1.5$ (S 84 beyond φ); a descending bed gives $r < 1$, an adverse slope $r > 1$. φ varies around S 81 by up to S 82 with the slope direction.

5.5.4 Bedload transport

The equilibrium bedload capacity $q_{b,e}$ (S 11) of van Rijn [1984a] (S 11 1) and Meyer-Peter and Müller [1948] (S 11 2) reads

$$q_{b,e} = 0.053\sqrt{(s-1)g}\, d_{50}^{1.5} \frac{T^{2.1}}{D_*^{0.3}}, \qquad T = \frac{\theta - \theta_{cr} r}{\theta_{cr} r}, \qquad D_* = d_{50}\left(\frac{(s-1)g}{\nu^2}\right)^{1/3}, \tag{5.31}$$

$$q_{b,e} = 8\,(\theta - r\theta_{cr})^{1.5} \sqrt{(s-1)g\, d_{50}^3}, \tag{5.32}$$

and $q_{b,e} = 0$ for $\theta < r\theta_{cr}$. S 11 3 is Engelund and Fredsøe [1976] in the form of Fredsøe and Deigaard [1992], S 11 4 Engelund-Hansen and S 11 5 Einstein-Brown. The bedload moves in the near-bed flow direction $\hat{\mathbf{s}} = \mathbf{u}_p/|\mathbf{u}_p|$; S 85 1 scales its magnitude with $1 - \min(1,\, 1.3\hat{\mathbf{s}}\cdot\nabla z_b)$. With S 33 > 0 (non-equilibrium), q_b adapts to the capacity over the length L_s,

$$q_b + L_s\, \hat{\mathbf{s}} \cdot \nabla q_b = q_{b,e}, \tag{5.33}$$

with L_s from S 40 or the saltation length of van Rijn [1984a]; S 33 0 gives $q_b = q_{b,e}$.

5.5.5 Suspended load

With S 12 1 (CFD and NHFLOW), the volumetric concentration c follows

$$\frac{\partial c}{\partial t} + \frac{\partial}{\partial x_j}\Big(\big(u_j - w_{s,eff}\,\delta_{j3}\big)c\Big) = \frac{\partial}{\partial x_j}\left((\nu + \nu_t)\frac{\partial c}{\partial x_j}\right), \qquad w_{s,eff} = w_s\left(1 - \frac{c}{0.635}\right)^{m_s}, \tag{5.34}$$

with hindered settling (m_s depends on $Re_p = w_s d_{50}/\nu$) and w_s from S 23 or computed (S 25 1, including van Rijn [1984b]). It is solved implicitly with first-order upwind convection; in the cell above the bed the exchange flux $w_s(c_{b,e} - c)$ is an implicit source. The equilibrium concentration $c_{b,e}$ follows from the reference concentration of van Rijn [1984b] at the level a,

$$c_a = \min\left(0.015\,\frac{d_{50}}{a}\,\frac{T^{1.5}}{D_*^{0.3}},\, 0.05\right), \tag{5.35}$$

transferred with the Rouse profile to the first cell centre. S 61 selects the near-bed concentration c_b, and S 62 the coupling to the bed (Exner source $E - D$, or in NHFLOW the depth-integrated suspended flux added to q_b).

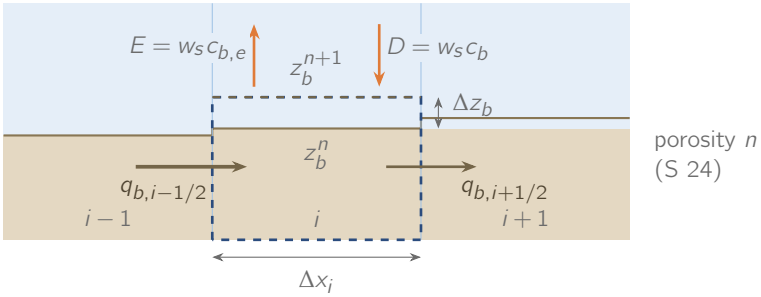

Figure 5.9: Exner balance of the bed column i: the difference of the bedload fluxes through the faces and the net erosion $E - D$ change the sediment volume $(1-n)\,\Delta z_b\,\Delta x_i$ of the column.

5.5.6 Exner equation

The bed level follows from the sediment mass balance of a bed column (Fig. 5.9),

$$(1-n)\,\frac{\partial z_b}{\partial t} = -f_m\left[\frac{\partial (q_b s_x)}{\partial x} + \frac{\partial (q_b s_y)}{\partial y} + w_s\,(c_{b,e} - c_b)\right], \tag{5.36}$$

with the morphological factor f_m (S 35) and $(s_x, s_y) = \hat{\mathbf{s}}$. The right-hand side is zero outside S 71 $\le x \le$ S 72 and in non-erodible zones (S 74), and it is multiplied by the relaxation function (S 73, S 75). S 31 and S 32 select the divergence form and its reconstruction (Section 6.13). The bed is updated explicitly, $z_b^{n+1} = z_b^n + \Delta t_{sed}\,(\partial z_b/\partial t)^n$, or with S 10 12 (NHFLOW) by a two-stage Runge-Kutta scheme within the flow stages.

Sandslide. Where the slope between neighbouring columns exceeds the angle of repose, $\Delta z_b > \tan\varphi\,\Delta x$, the sandslide S 90 moves sediment downslope conserving its volume, repeated up to S 91 times until no slope exceeds φ.

5.5.7 Morphological time stepping

The sediment module is decoupled in time from the flow: it starts after S 41 (S 43, S 45, S 47) and is called at the interval S 42 (S 44, S 46, S 48). Each call advances the bed by

$$\Delta t_{sed} = \text{S 17} \cdot \min\left(\text{S 13},\ \frac{\text{S 14}\,\Delta x}{\max|\partial z_b/\partial t|}\right) \qquad \text{(S 15 0)}, \tag{5.37}$$

and the run stops when the morphological time $t_{sed} = \sum \Delta t_{sed}$ reaches S 19. **Morphological acceleration** is obtained with f_m (S 35) in Eq. (5.36), or by advancing the bed by S 27·Δt_{sed} per call (CFD) while the flow advances only by the call interval. In both cases the flow must stay close to equilibrium with the bed.

5.5.8 Graded sediment

With S 51, the bed has up to 20 size fractions d_k in an active layer over a substrate. The capacity of fraction k is $q_{b,e,k} = F_k\,q_{b,e}(d_k)$ with the fraction F_k in the active layer, $\theta_k = \theta\,d_{50}/d_k$ and $\theta_{cr,k} = \xi_k\,r\,\theta_{cr}$, with the hiding factor ξ_k of Wu et al. [2000] (S 54 1, S 55). Eq. (5.36) is solved per fraction and updates the active layer composition. Not available with S 10 12.

5.5.9 Key input commands

S 10	on (1), porous VRANS bed (2, CFD), RK2 coupled (12, NHFLOW)
S 20, S 22, S 24	grain size d_{50}, sediment density, bed porosity
S 11, S 18	bedload formula, dynamic friction coefficient for S 11 3
S 16, S 21, S 36	bed shear stress, roughness factor, bedform roughness
S 30, S 80–S 84	critical Shields parameter, slope reduction, angle of repose
S 33, S 40, S 85	non-equilibrium bedload, adaptation length, slope correction
S 12, S 23, S 25, S 61, S 62	suspended load, settling velocity, c_b, coupling
S 31, S 32, S 100–S 102	Exner formulation and discretization, bed filter
S 13–S 15, S 17, S 19, S 29	sediment time step, factor, end time, ramp
S 35, S 27, S 103	morphological factor, Exner sub-iterations, celerity limit
S 41–S 48	start and interval of the sediment calls
S 90–S 94	sandslide
S 50, S 71–S 78	bed boundary conditions, erodible regions, relaxation zones
S 51–S 56	graded sediment

5.6 Particle Model for Sediment (MP-PIC)

REEF3D::CFD contains a particle-based alternative to the Exner model, the continuum particle method (CPM), activated with S 10 1 (or S 10 2) together with Q 10 1; it is not available in SFLOW, NHFLOW and FNPF. The sediment below the initial bed ($\phi_b < 0$) is represented by Q 24 parcels per cell (Fig. 5.10), each representing $N_p = (1-n)\overline{\Delta x}^3/(\text{Q } 24\,\pi d_{50}^3/6)$ grains of size d_{50} and density ρ_s. The particle volume fraction θ_p follows from distributing the parcel volumes to the cell centres with trilinear weights. All options are listed in Section 6.12.

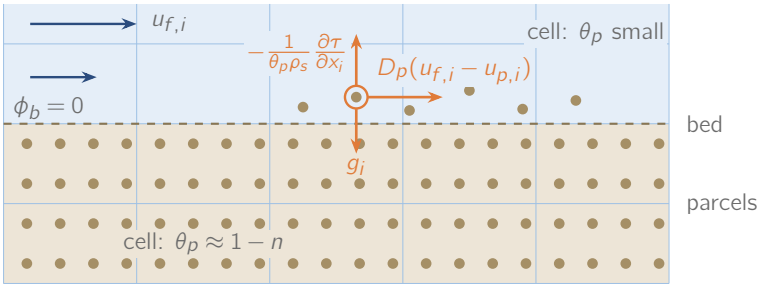

Figure 5.10: Continuum particle method: parcels seeded below the bed $\phi_b = 0$. A parcel is accelerated by the drag of the fluid velocity interpolated to its position, by gravity, by the pressure gradients and by the gradient of the inter-particle stress $\tau(\theta_p)$, which keeps the packing below the close-packing limit.

With Q 11 2, the parcels follow the multiphase particle-in-cell (MP-PIC) formulation of Snider [2001]:

$$\frac{du_{p,i}}{dt} = D_p\left(u_{f,i}-u_{p,i}\right) - \frac{1}{\rho_s}\frac{\partial(p+p_s)}{\partial x_i} + g_i - \frac{1}{\theta_p \rho_s}\frac{\partial \tau}{\partial x_i}, \qquad \frac{dx_{p,i}}{dt} = u_{p,i}, \tag{5.38}$$

with $u_{f,i}$ and p interpolated to the parcel and the lithostatic pressure $p_s = (\rho_s - \rho)\,|g|\,(-\phi_b)$. The drag coefficient is $D_p = \frac{3}{4}C_d(\rho/\rho_s)|u_f|/d_{50}$, with a volume-fraction dependent $C_d(Re_p, \theta_f)$, $\theta_f = 1-\theta_p$, limited to 10. The inter-particle stress [Snider, 2001]

$$\tau = \frac{P_s \theta_p^{\beta}}{\max(\theta_{cp}-\theta_p,\ \epsilon(1-\theta_p))}, \tag{5.39}$$

with $P_s = 100\,\mathrm{Pa}$, $\beta = 5$, $\epsilon = 10^{-7}$ and $\theta_{cp} = 1 - n + 0.05$ fixed in the code, rises steeply towards close packing θ_{cp} and pushes the parcels apart. The equations are integrated with an explicit Euler step and implicit drag in every flow time step, with Δt_{sed} from S 13–S 15. The plain model Q 11 1 uses a two-stage Runge-Kutta scheme without τ: a parcel moves where the drag exceeds a resistance force scaled by Q 30, and the net parcel flux through a water column changes the bed level.

The coupling is one-way in momentum: no drag reaction is added to the fluid. The fluid sees the sediment through ϕ_b (S 10 1) or the VRANS bed (S 10 2), and the density in the pressure solver is blended to ρ_s below the bed. τ_b (S 16) and τ_{cr} (S 80) are computed for output only.

5.6.1 Key input commands

S 10, Q 10	sediment transport on, particle model instead of Exner (CFD only)
Q 11	plain particle transport (1) or MP-PIC with Snider stress (2)
Q 24, Q 29	parcels per cell (must be set), seeding type
S 20, S 22, S 24	grain size, sediment density, porosity (parcel size, packing limit)
Q 30	friction factor of the resistance force (Q 11 1)
S 13–S 15, S 41–S 47	sediment time step, start of the particle transport
Q 73	relaxation zones for the particle forces
Q 180–Q 182	particle output

5.7 Porous Media and Vegetation

Breakwaters, rock filters, porous beds and vegetation contain pores and stems far smaller than a grid cell. REEF3D therefore solves the volume-averaged RANS (VRANS) equations, in which the solid skeleton is described by the porosity n (pore volume per total volume) and its effect on the flow by resistance and added mass terms. VRANS porous media are available in REEF3D::CFD (B 270 – B 291, porous sediment bed S 10 2) and REEF3D::NHFLOW (B 201 – B 230, porous floating bodies X 16), the vegetation model in REEF3D::CFD only (B 307 – B 322); REEF3D::FNPF and REEF3D::SFLOW have neither. Porous regions are defined with input commands, not in the DIVEMesh grid. Flexible vegetation is treated with the rod models of Section 5.4.

5.7.1 VRANS equations in REEF3D::CFD

In the porous zones, u_i is the filtered (Darcy) velocity, so that continuity keeps its form $\partial u_i/\partial x_i = 0$. In cells with $n < 1$, the momentum equations are

$$(1 + c_A)\frac{\partial u_i}{\partial t} + \frac{u_j}{n}\frac{\partial u_i}{\partial x_j} = n\left(-\frac{1}{\rho}\frac{\partial p}{\partial x_i} + g_i + f_i - a\,u_i - b\,|u_i|\,u_i\right) + \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)\right], \tag{5.40}$$

which reduces to Eq. (4.8) for $n = 1$. The pore velocity u_j/n convects the momentum (the convecting face velocities of the flux schemes are divided by n). Pressure gradient, gravity,

external forces f_i and resistance are multiplied by n, which also enters the pressure Poisson equation and the velocity correction:

$$\frac{\partial}{\partial x_i}\left(\frac{n}{(1+c_A)\,\rho}\frac{\partial p'}{\partial x_i}\right)=\frac{1}{\alpha\Delta t}\frac{\partial u_i^*}{\partial x_i},\qquad u_i=u_i^*-\frac{n\alpha\Delta t}{(1+c_A)\,\rho}\frac{\partial p'}{\partial x_i}. \tag{5.41}$$

The whole momentum increment of a Runge-Kutta stage, including the implicit diffusion, is divided by $1+c_A$, with the constant c_A = B 260 (default 0.34, not multiplied by $(1-n)/n$). The resistance coefficients follow van Gent [1995] as used by Jensen et al. [2014]:

$$a=\alpha\,\frac{(1-n)^2}{n^3}\,\frac{\nu}{d_{50}^2},\qquad b=\beta\left(1+\frac{7.5}{KC}\right)\frac{1-n}{n^3}\,\frac{1}{d_{50}}, \tag{5.42}$$

with the dimensionless coefficients α, β, the grain size d_{50}, the local ν and the constant KC = B 264 (default 10^{20}, i.e. no correction). $b|u_i|u_i$ is evaluated component-wise; n, d_{50}, α, β are cell-centred and averaged arithmetically to the faces.

Porous geometries. Each object has its own n, d_{50}, α, β: boxes (B 270), vertical cylinders (B 274), wedges (B 281, B 282) and inclined plates (B 291). A cell belongs to an object if its centre lies inside, and later objects overwrite earlier ones, e.g. a core wedge inside a larger armour-layer wedge (Fig. 5.11). Outside, $n=1$ and $\alpha=\beta=0$. With S 10 2, the cells below the bed become porous (n = S 24, d_{50} = S 20, α,β = S 26) and follow the bed evolution (Section 5.5). B 240 – B 243 give a simple Darcy-Forchheimer box, $f_i=-D\,\nu\,u_i-\frac{1}{2}C\,|u_i|u_i$, without porosity scaling.

Turbulence. With B 295 1 (default), source terms keep the turbulence in the porous zone at the level of the pore flow, with $|u|^2=u_iu_i$:

$$k_\infty=3.7\,(1-n)\,n^{3/2}\,|u|^2,\qquad \epsilon_\infty=39\,(1-n)^{5/2}\,n^2\,\frac{|u|^3}{d_{50}}. \tag{5.43}$$

$n\epsilon_\infty$ is added to the k-equation and $n\,c_{2\epsilon}\epsilon_\infty^2/k_\infty$ ($c_{2\epsilon}=1.92$) to the ϵ-equation, with corresponding terms for k-ω. B 295 2 sets $\nu_t=10^{-3}\nu$ in the porous cells.

5.7.2 VRANS in REEF3D::NHFLOW

All NHFLOW porous objects share n_p, d_{50}, α_p, β_p of B 201: boxes, cylinders, spheres and wedges (B 210 – B 219) and an STL geometry (B 230). As for solid structures, a signed distance function ϕ_p is computed, and with the smoothed Heaviside function $H_p=H(-\phi_p)$ of Eq. (4.10) (half-width B 209 times the horizontal cell size)

$$n=H_p\,n_p+(1-H_p),\qquad \alpha=H_p\,\alpha_p,\qquad \beta=H_p\beta_p. \tag{5.44}$$

In the equations of Section 4.3, the fluxes of the free surface equation (4.27) are divided by n, the convective fluxes hu_iu_j by n^2, and the momentum increment of each stage by $1+c_A(1-n)/n^2$. The resistance acts on the momentum hu_i,

$$f_i=-H_p\,h(a\,u_i+b_{nh}|u_i|\,u_i),\qquad b_{nh}=\beta\left(1+\frac{7.5}{KC}\right)\frac{1-n}{n^4}\,\frac{1}{d_{50}}, \tag{5.45}$$

with a from Eq. (5.42). B 208 1 outputs the total force on the structure (resistance, added mass and the part $(1-n)$ of the pressure gradient). There are no turbulence source terms.

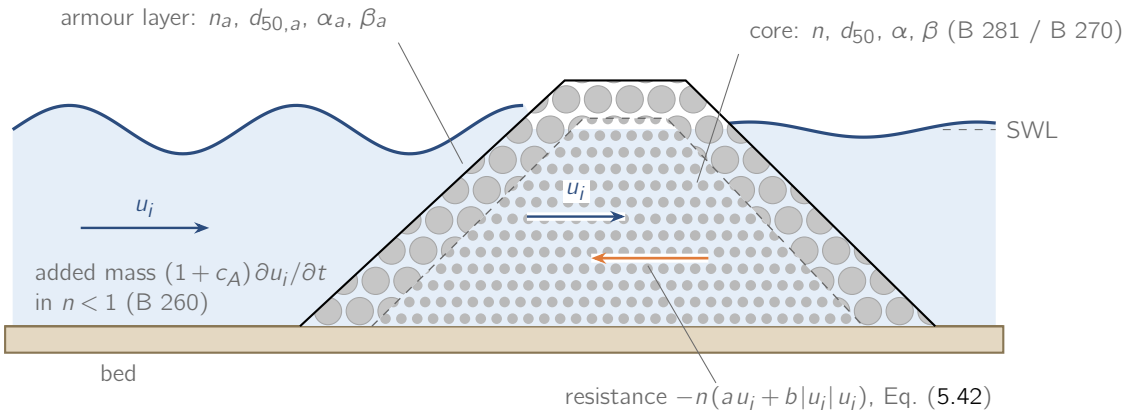

Figure 5.11: Rubble-mound breakwater with armour layer and core as porous VRANS zones in a wave tank. Each zone has its own porosity n, grain size d_{50} and resistance coefficients α, β; inside, the filtered velocity u_i is retarded by the linear and nonlinear resistance and the added mass term.

Porous floating bodies. With X 16 (n_{fb}, d_{50}, α, β), the porosity $n = 1 - H_{fb}(1 - n_{fb})$ follows the moving NHFLOW body, and the resistance of Eq. (5.42) acts on the velocity relative to the body. It is integrated implicitly over the substep, so that large coefficients remain stable, and its reaction is added to the body forces (Section 5.2).

5.7.3 Vegetation

The vegetation model (CFD, activated with B 307 or B 308) represents plants as rigid vertical cylinders with N stems per unit area, diameter D and drag coefficient C_d in boxes (B 310) and wedges (B 321, B 322). The momentum equations receive the Morison-type force per unit mass

$$f_i = -\frac{1}{2} C_d\, N\, D\, |u_i|\, u_i \;-\; C_M \frac{\pi D^2}{4} N \frac{\partial u_i}{\partial t}, \tag{5.46}$$

with C_M = B 309 (default 2.0) and the time derivative from the stored velocities. The drag follows the local velocity and vanishes above the canopy (Fig. 5.12). With B 308 1 (default), the stems block the volume through the porosity $n = 1 - N\pi D^2/4$, which enters Eqs. (5.40) and (5.41) as for porous media. B 295 1 adds turbulence sources proportional to $C_d D N$, B 295 2 sets $\nu_t = 10^{-3}\nu$ in the canopy.

5.7.4 Key input commands

B 260	added mass coefficient c_A (CFD, NHFLOW)
B 264	constant KC number of the nonlinear resistance (CFD, NHFLOW)
B 270, B 274, B 281, B 282, B 291	porous box, vertical cylinder, wedges, plate with n, d_{50}, α, β (CFD)
B 295	turbulence treatment in porous media and vegetation (CFD)
B 240 – B 243	simple Darcy-Forchheimer box (CFD)
S 10 2, S 20, S 24, S 26	porous sediment bed (CFD)
B 201	n, d_{50}, α, β for all NHFLOW porous objects

B 210 – B 219, B 230	NHFLOW porous primitives and porous STL geometry
B 208, B 209	NHFLOW porous force output, Heaviside half-width
X 16	porous floating body (NHFLOW)
B 307, B 308, B 309	vegetation model on, stem porosity, inertia coefficient C_M (CFD)
B 310, B 321, B 322	vegetation box and wedges with N, D, C_d (CFD)

5.8 Heat, Concentration, Stratification and Rheology

REEF3D::CFD can transport a temperature T (H group) or a concentration c (C group) that change the density and viscosity of the fluid, for stratified flows, plumes and density currents, and phase 1 can be a non-Newtonian material (W 90) such as mud or debris. These models require the level set free surface (F 30 >0); REEF3D::SFLOW has only a Bingham-type bed friction, and REEF3D::NHFLOW and REEF3D::FNPF have no heat or concentration transport.

5.8.1 Temperature and concentration

With H 10 >0, the temperature T in °C is advected and diffused,

$$\frac{\partial T}{\partial t} + u_j \frac{\partial T}{\partial x_j} = \frac{\partial}{\partial x_j}\left[\left(\alpha_T + \frac{\nu_t}{\sigma_T}\right)\frac{\partial T}{\partial x_j}\right], \qquad \alpha_T = \alpha_1 H(\phi) + \alpha_2 (1 - H(\phi)), \tag{5.47}$$

with the diffusivities of water and air (H 1, H 2, phase assignment H 9) and $\sigma_T = 0.9$. H 10 selects the time scheme, H 15 the convection scheme; diffusion is implicit. H 3 1 (default) interpolates $\rho(T)$ and $\nu(T)$ of water and air from tables and blends them with Eq. (4.9). With H 3 2, the viscosities stay W 2 and W 4 and the densities follow

$$\rho_k = \rho_{0,k}\,(1 - \beta_k\,(T - T_{0,k}))\,, \qquad k = 1, 2, \tag{5.48}$$

with $\rho_{0,k}$ = W 1, W 3, $T_{0,k}$ from H 50 and β_k = H 4 (default $1/T_{0,k}$ in K).

With C 10 >0, a concentration c is transported with the diffusivity $\nu + \nu_t$ (Schmidt number 1),

$$\frac{\partial c}{\partial t} + u_j \frac{\partial c}{\partial x_j} = \frac{\partial}{\partial x_j}\left[(\nu + \nu_t)\,\frac{\partial c}{\partial x_j}\right], \tag{5.49}$$

with the convection scheme C 15; C 75 defines inclined slabs of c_1 with an initial velocity, e.g. for a turbidity current on a slope. Density and viscosity change linearly in both phases:

$$\begin{aligned} \rho &= (\rho_1 + c\,\mathrm{C}\,1)\,H(\phi) + (\rho_2 + c\,\mathrm{C}\,3)\,(1 - H(\phi))\,, \\ \nu &= (\nu_1 + c\,\mathrm{C}\,2)\,H(\phi) + (\nu_2 + c\,\mathrm{C}\,4)\,(1 - H(\phi))\,. \end{aligned} \tag{5.50}$$

With C 1 = 10 kg/m^3 (default), $c = 1$ gives a salt-water-like excess density of 1 %.

Stratified flow. T and c act on the flow only through the variable density in Eqs. (4.8), (4.11) and (4.12); no separate Boussinesq term is added, and horizontal density gradients drive the motion (Fig. 5.13). The face densities follow from the face-averaged level set and T or c. T 45 adds buoyancy to the k-equation of the k-ω model. A high-order convection scheme (H 15 or C 15 = 4 or 5) keeps the density interface sharp.

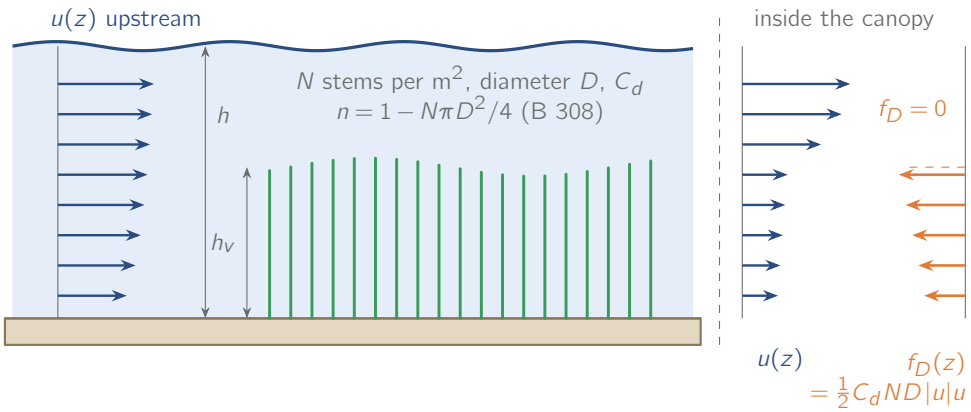

Figure 5.12: Vegetation field of rigid stems (B 310) in a current. Left: approach flow and canopy; right: the velocity inside the canopy is reduced, and the drag force per unit volume, Eq. (5.46), follows the local velocity over the canopy height h_v and vanishes above it.

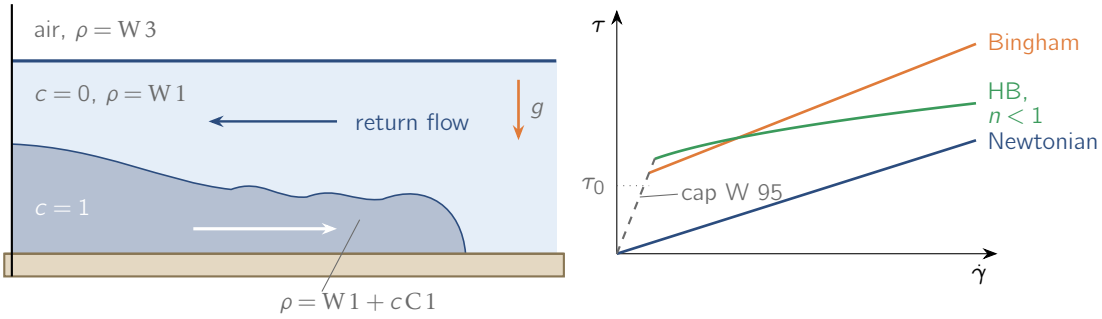

Figure 5.13: Left: a density current in REEF3D::CFD. A dense fluid ($c = 1$, density from Eq. 5.50) spreads along the bed under the lighter water, driven only by the variable density in the momentum and pressure equations; the displaced water returns above it. Right: shear stress over shear rate for a Newtonian fluid, a Bingham fluid (W 98 = 1) and a shear-thinning Herschel-Bulkley fluid with yield stress τ_0; below the grey dashed line, the viscosity is capped at ν_0 (W 95).

5.8.2 Non-Newtonian rheology

With W 90 1, the viscosity of phase 1 follows the Herschel-Bulkley model [Herschel and Bulkley, 1926], $\tau = \tau_0 + K\dot{\gamma}^m$, with the shear rate $\dot{\gamma} = \sqrt{2S_{ij}S_{ij}}$, the yield stress τ_0, K = W 97 and m = W 98, regularized by the cap ν_0 = W 95; phase 2 keeps ν_2 = W 4:

$$\nu_1 = \min\left(\frac{1}{\rho}\left(\frac{\tau_0}{\dot{\gamma}} + K\dot{\gamma}^{m-1}\right),\ \nu_0\right). \tag{5.51}$$

$m = 1$ gives a Bingham fluid, $\tau_0 = 0$ a power-law fluid. In unyielded regions the viscosity reaches ν_0, and the material moves as a very viscous fluid (Fig. 5.13, right). The yield stress is W 96 (W 101 0) or follows a pressure-dependent Mohr-Coulomb law (W 111) with the angle of repose φ and the cohesion c_0 (W 102), for W 101 1 and 2

$$\tau_0 = \max(0,\ p\tan\varphi + c_0), \qquad \tau_0 = (p\tan\varphi + c_0)\left(1 - e^{-m_\tau\dot{\gamma}}\right), \tag{5.52}$$

where m_τ = W 103 gives a Papanastasiou regularization [Papanastasiou, 1987]; W 101 3 – 5 use the buoyant pressure and an excess pore pressure (W 104). W 110 1 (default) puts yield stress and viscous part into the viscosity of Eq. (5.51); W 110 2 – 7 move them into explicit momentum source terms acting in phase 1 only.

For debris flows (W 90 2), $\nu_1 = (a_2\mu_2 + (1-a_2)\mu_3)/\rho_1$ mixes a Herschel-Bulkley slurry, $\mu_2 = k|\dot{\gamma}|^{m-1} + \tau_y/|\dot{\gamma}|$ with $\tau_y = \tau_0 C^2 e^{22CP_1}$ (W 105) and $k = b\tau_y$ (W 106), capped by W 95, and a gravel phase $\mu_3 = \min\left(\mu_{min} + p\sin\delta\,(1 - e^{-m_y\dot{\gamma}})/\dot{\gamma},\ \mu_{max}\right)$ (W 106, W 107), with a_2 = W 108.

SFLOW. In REEF3D::SFLOW, W 90 1 subtracts $\tau_{b,i}/\rho$ with the Bingham-type bed shear stress

$$\tau_{b,i} = \left(1.5\tau_0 + 3K\frac{|\mathbf{U}|}{h}\right)\frac{U_i}{|\mathbf{U}|}, \tag{5.53}$$

from the depth-averaged momentum, with τ_0 from W 96 or W 101 1 with the bed pressure.

5.8.3 Key input commands

H 10, H 15	heat transfer on and time scheme, convection scheme (CFD)
H 1, H 2, H 9	thermal diffusivities of water and air, phase assignment
H 3, H 4	property model (tables or Boussinesq), expansion coefficients
H 50 – H 58, H 61 – H 66	initial temperature field, fixed boundary temperatures
C 10, C 15	concentration on and time scheme, convection scheme (CFD)
C 1 – C 4	density and viscosity change per unit concentration
C 50 – C 58, C 75	initial concentration field, inclined slabs with initial velocity
T 45	buoyancy term in the k-equation
W 90	non-Newtonian rheology of phase 1 (CFD; SFLOW bed friction)
W 95 – W 98	viscosity cap ν_0, yield stress τ_0, consistency K, exponent m
W 101 – W 104, W 111, W 112	pressure-dependent yield stress models
W 105 – W 108	three-phase debris mixture (W 90 2)

5.9 Lagrangian Particle Tracking

Passive Lagrangian particles, e.g. floating or buoyant (macro)plastics, drifters or larvae, can be tracked in the flow field of REEF3D::NHFLOW (L 10 1). They are moved by the resolved flow but do not act back on it. Each release (point L 21, line L 22, box L 23) carries its own rise or settling velocity, windage coefficient and particle mode, so that several particle classes can be simulated together. The options, the seeding and the output are described in Section 6.8.

The particle velocity is the sum of the fluid velocity, the rise velocity w_s ($w_s > 0$ buoyant, $w_s < 0$ settling) and, at the free surface, the windage:

$$\frac{d\mathbf{x}_p}{dt} = \mathbf{v}_p = \mathbf{u}(\mathbf{x}_p, t) + w_s\,\mathbf{e}_z + \delta_s\, c_w U_{10}\,(\cos\alpha_w,\ \sin\alpha_w,\ 0)^T, \tag{5.54}$$

with $\delta_s = 1$ at the free surface and 0 otherwise, the windage coefficient c_w and the wind U_{10}, α_w of L 41 (default: A 571, Section 5.11). $\mathbf{u}$ is interpolated from the cell centres (bilinear in the horizontal, linear along the σ-levels and extrapolated to η), and Eq. (5.54) is integrated with Heun's method with the time step of the flow solver:

$$\mathbf{x}^* = \mathbf{x}_p^n + \Delta t\,\mathbf{v}_p(\mathbf{x}_p^n, t^n), \qquad \mathbf{x}_p^{n+1} = \mathbf{x}_p^n + \frac{\Delta t}{2}\Big(\mathbf{v}_p(\mathbf{x}_p^n, t^n) + \mathbf{v}_p(\mathbf{x}^*, t^{n+1})\Big). \tag{5.55}$$

Since NHFLOW resolves the individual waves, the Stokes drift follows from the orbital motion itself (Fig. 5.14); no Stokes drift parameterisation is added.

Unresolved dispersion is modelled by a random walk added after the Heun step, with the horizontal diffusivity K_h (L 31) and standard normal random numbers R (seed L 11):

$$\Delta x = \sqrt{2K_h\Delta t}\,R_x, \qquad \Delta y = \sqrt{2K_h\Delta t}\,R_y. \tag{5.56}$$

For three-dimensional particles (mode 0) a vertical random walk (L 32) uses either a constant $K_{v,0}$ (L 33) or $K_v(z) = K_{v,0} + \nu_t/Sc_t$ (Sc_t: L 34) with the scheme of Visser [1997], which avoids the artificial accumulation of particles in regions of low diffusivity:

$$\Delta z = K_v'(z)\,\Delta t + \sqrt{2\,K_v\Big(z + \tfrac{1}{2}K_v'(z)\Delta t\Big)\Delta t}\,R_z. \tag{5.57}$$

A particle is in the water column (state 0), at the free surface (1, subject to windage), stranded in a dry cell (2) or deposited on the bed (3) (Fig. 5.14). Surface-trapped particles (mode 1) stay at $z = \eta$. At the bed particles are reflected or, with L 52 1 and $w_s < 0$, deposited (no resuspension); stranded particles refloat when the cell becomes wet again (L 51 0) or stay beached (L 51 1); at the lateral boundaries they are reflected or removed (L 53).

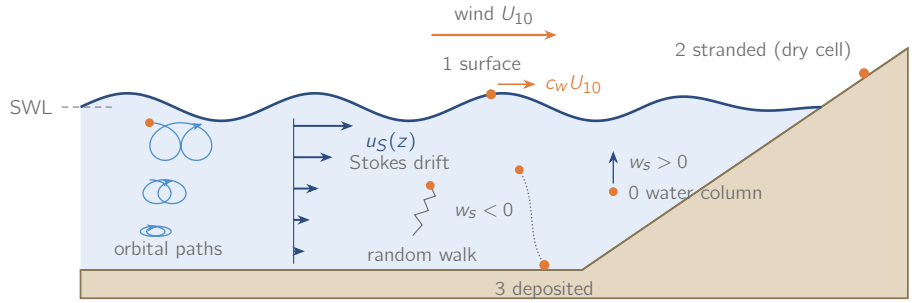

Figure 5.14: Lagrangian particles in NHFLOW: orbital paths with the resulting Stokes drift, windage of floating particles, rise and settling velocity, random walk, and the particle states 0–3.

5.9.1 Key input commands

Command	Meaning
L 10	particle tracking on (NHFLOW, A 10 5)
L 11	random seed of the random walk
L 21, L 22, L 23	point, line and box releases: position, number, release window, w_s, c_w, mode
L 31	horizontal diffusivity K_h
L 32, L 33, L 34	vertical random walk: off / constant $K_{v,0}$ / $K_{v,0} + \nu_t/Sc_t$; $K_{v,0}$; Sc_t
L 41	U_{10} and direction for the windage (default: A 571)
L 51, L 52, L 53	beaching, bed contact, lateral domain boundaries
L 61, L 62	print interval; output format (VTP, CSV track file)

5.10 Sea Ice (FNPF)

REEF3D::FNPF (only) can simulate rigid ice floes that interact with the waves, with each other and with fixed obstacles (A 380 1). The floes are coupled to the potential flow through a compliant lid, contacts are resolved with a non-smooth contact dynamics method, and floes can break by flexural failure or by splitting in contact (Fig. 5.15). A floe is a prism of thickness h_i and density ρ_i (default 917 kg/m^3) over a convex planform; single floes, random floe fields and fixed obstacles are defined in the floe input file.

5.10.1 Compliant lid

Under the footprint of a floe the free surface is loaded with the pressure

$$p_{ice} = \varphi \max\!\left(0,\ \rho_i g h_i + k\,(\eta - z_b) + c\left(\frac{\partial \eta}{\partial t} - \frac{\partial z_b}{\partial t}\right)\right), \tag{5.58}$$

with the lid stiffness $k = \alpha \rho_w g$ (α: A 381, default 20), the damping $c = 2\zeta\sqrt{k\rho_i h_i}$ (ζ: A 382) and the height $z_b(x, y, t)$ of the (tilted) floe bottom. It enters Eq. (4.43) as $-p_{ice}/\rho_w$, like the wind pressure (Section 5.11). The weight term gives the exact static draft; the unilateral spring-dashpot enforces the rigid-body kinematics up to $O(1/\alpha)$ and lets the floe lift off. The cell coverage φ is tapered across each edge over the half width δ (A 386, A 389) with an area-preserving sine ramp, since the single-valued σ-grid surface does not tolerate a step of the size of the draft. The lid stiffens the surface ($g_{eff} = g(1+\alpha)$), so the adaptive time step (N 48) is reduced by $1/\sqrt{1+\alpha}$.

5.10.2 Floe dynamics

The lid pressure integrated over the footprint plus the ice-water skin drag (C_d: A 383) give the force and moment on the floe:

$$\mathbf{F} = \sum_{\text{footprint}} \left[p_{ice} A \left(-\frac{\partial \eta}{\partial x},\ -\frac{\partial \eta}{\partial y},\ 1 \right)^T + \rho_w C_d \varphi A |\Delta \mathbf{u}| \Delta \mathbf{u} \right], \qquad \mathbf{M} = \sum_{\text{footprint}} \mathbf{r} \times d\mathbf{F}, \quad (5.59)$$

with the cell area A, the lever arm $\mathbf{r}$ from the centre of gravity and the horizontal velocity $\Delta\mathbf{u}$ of the surface flow relative to the floe. The slope term is the reaction of the pressure on the fluid, so horizontal momentum is conserved; radiation damping and added mass are provided by the potential flow. The rigid-body equations (orientation as unit quaternion $\mathbf{q}$, $\mathbf{I} = \mathbf{R}\mathbf{I}_b\mathbf{R}^T$) are

$$\frac{d\mathbf{x}_c}{dt} = \mathbf{u}_c, \qquad \frac{d\mathbf{q}}{dt} = \frac{1}{2}(0, \boldsymbol{\omega}) \otimes \mathbf{q}, \qquad m\frac{d\mathbf{u}_c}{dt} = \mathbf{F} - mg\mathbf{e}_z, \qquad \mathbf{I}\frac{d\boldsymbol{\omega}}{dt} = \mathbf{M} - \boldsymbol{\omega} \times \mathbf{I}\boldsymbol{\omega}. \quad (5.60)$$

They are advanced within the same Runge-Kutta stages as the free surface (A 310), which avoids the added-mass instability of a staggered coupling. In a 2D flume only surge, heave and pitch are active.

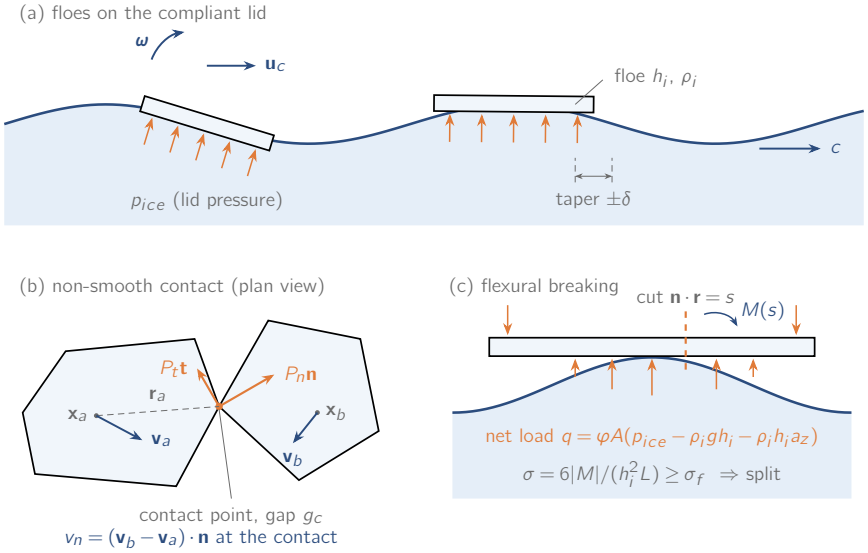

Figure 5.15: Sea ice in REEF3D::FNPF: (a) rigid floes on the compliant lid, loaded by the lid pressure p_{ice} with the tapered footprint edge; (b) planar non-smooth contact between two convex floes with normal and friction impulses; (c) flexural breaking: bending moment of the net load q about a candidate cut line.

5.10.3 Non-smooth contact dynamics

After each time step the contacts between floes, fixed obstacles and, with A 385 1, the domain walls are resolved in the horizontal plane (surge, sway, yaw) with a non-smooth contact dynamics method of Moreau-Jean type [Jean, 1999]. Each contact point has the normal $\mathbf{n}$, the tangent $\mathbf{t}$ and the gap g_c; the relative normal velocity v_n and the normal impulse P_n satisfy the Signorini condition with Coulomb friction (μ, e: A 384),

$$P_n \geq 0, \quad v_n^+ - b \geq 0, \quad P_n(v_n^+ - b) = 0, \qquad |P_t| \leq \mu P_n, \quad (5.61)$$

with the target velocity

$$b = \begin{cases} -g_c/\Delta t & g_c > 0 \text{ (speculative contact)} \\ \max\!\left(-e\, v_n^-,\ \beta \max(0, -g_c - g_s)/\Delta t\right) & g_c \le 0 \end{cases} \tag{5.62}$$

Speculative contacts may close the gap within the step but not more, so that floes do not tunnel through each other; for closed contacts e is the restitution coefficient and the Baumgarte term ($\beta = 0.2$) removes penetrations beyond the slop g_s. The impulses follow from projected Gauss-Seidel iterations (A 388), $P_n \leftarrow \max(0,\ P_n + m_n(b - v_n))$ and P_t clipped to $|P_t| \le \mu P_n$, with the effective masses m_n, m_t of the pair (infinite for obstacles and walls); for obstacles the contact force is logged as ice load.

5.10.4 Breaking

With A 390 a floe can split along a straight cut into two pieces (not narrower than D_{min}, A 393) that keep the rigid-body velocity field of the parent, which conserves mass, momentum and angular momentum. *Flexural failure*: for candidate cut lines $\mathbf{n} \cdot \mathbf{r} = s$ (A 395) the bending moment of the rigid floe follows from the net load on one side of the cut, including the inertia load with the local vertical acceleration a_z,

$$q = \varphi A\,(p_{ice} - \rho_i g h_i - \rho_i h_i a_z)\,, \qquad M(s) = \sum_{\mathrm{n\cdot r} > s} q\,(\mathbf{n}\cdot\mathbf{r} - s), \qquad \sigma = \frac{6\,|M|}{h_i^2\, L}, \tag{5.63}$$

with the chord L of the cut. The floe breaks along the cut with the largest $\sigma \ge \sigma_f$ (A 391); for floes much longer than the flexural length this is an upper bound of the stress. *Contact splitting* (3D only): a floe splits along the contact normal when the pair normal force, averaged over the time t_c (A 396), exceeds the splitting load of linear elastic fracture mechanics [Bhat, 1988, Lu et al., 2015],

$$F_s = C\, K_{IC}\, h_i \sqrt{D}, \qquad D = 2\sqrt{A_f/\pi}, \tag{5.64}$$

with the fracture toughness K_{IC}, a calibration coefficient C (A 392) and the floe area A_f.

5.10.5 Key input commands

Command	Meaning
A 380	ice floes on, read `ice_floes.dat` (FNPF, A 10 3)
A 381, A 382	lid stiffness $\alpha = k/(\rho_w g)$ (default 20); lid damping ratio ζ (default 0.2)
A 383	ice-water drag coefficient C_d (default 0.0055)
A 384	contact friction μ and restitution e (default 0.3, 0.0)
A 385	domain boundaries act as walls for the floes (default 1)
A 386, A 389	footprint sub-samples per cell direction (4); edge taper half width in cells (2.0)
A 387	floe print interval (s, < 0: as the free surface output)
A 388	contact solver iterations (30)
A 390	breaking, bit flags: 1 flexural, 2 contact splitting, 4 spalling (7: all)
A 391, A 392	flexural strength σ_f (0.5 MPa); K_{IC} and C for splitting

A 393, A 394	minimum piece size D_{min} (<0: 4 mean cells); check interval in steps
A 395, A 396	cut directions and cut positions per direction (12, 64); force averaging time t_c (0.05 s)
A 397 – A 402	Young's modulus and Poisson ratio for the flexural stress, crushing strength, Weibull scatter of the strength, cracks per check and per splitting event, spalling (Section 6.1)
A 403	lid time step factor
A 310, N 48	FNPF time scheme (floes use the same RK stages); adaptive time step

5.11 Wind Forcing

Wind forcing is available in REEF3D::FNPF (A 370 – A 378) and REEF3D::NHFLOW (A 570 – A 578). The first command selects the drag coefficient $C_d(U_{10})$ for the wind speed U_{10} at 10 m height and direction α_w (A 371, A 571), which gives the surface stress $\tau = \rho_a C_d U_{10}^2$ and the friction velocity $u_*^2 = C_d U_{10}^2$ (ρ_a: W 3):

$$10^3 C_d = \begin{cases} 0.75 + 0.067\,U_{10} \quad (U_{10} < 41 \text{ m/s}), & 3.5 & \text{Garratt [Garratt, 1977]} \\ 1.2875 \quad (U_{10} < 7.5 \text{ m/s}), & 0.8 + 0.065\,U_{10} & \text{Wu [Wu, 1982]} \\ 0.63 + 0.066\,U_{10} & & \text{Smith and Banke [Smith and Banke, 1975]} \\ 0.55 + 2.97\,\tilde{U} - 1.49\,\tilde{U}^2, & \tilde{U} = U_{10}/31.5 \text{ m/s} & \text{Zijlema [Zijlema et al., 2012]} \end{cases} \tag{5.65}$$

The type of forcing is selected with A 373 / A 573 (option 2 is deprecated and treated as 1).

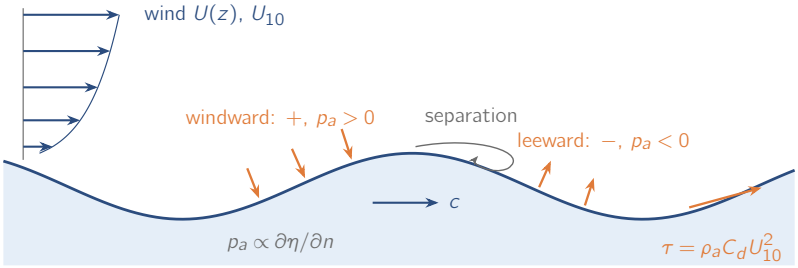

Figure 5.16: Wind forcing: logarithmic wind profile over the waves, surface stress τ, and the air pressure in phase with the slope, positive on the windward and negative on the leeward face (Jeffreys sheltering with flow separation behind steep crests).

Surface stress / wind setup (type 1). In NHFLOW the stress acts on the top σ-layer, so that the depth-integrated momentum receives τ/ρ_w. The potential flow cannot carry a shear stress; in FNPF the stress is spread over the reference depth h_{ref} (A 377) and applied as an along-wind potential, which gives the setup $g\,\partial\eta/\partial s = \tau/(\rho_w h_{ref})$ in the along-wind direction s.

Wave growth by surface pressure (types 3 and 4). A zero-mean air pressure in phase with the along-wind slope $\partial\eta/\partial n$ transfers energy to the waves:

$$p_a = \beta\,\rho_a u_*^2 \frac{\partial\eta}{\partial n} \quad \text{(type 3)}, \qquad p_a = s_J\,\rho_a \max(U_{10} - c, 0)^2\; w\,\frac{\partial\eta}{\partial n} \quad \text{(type 4)}. \tag{5.66}$$

Type 3 follows the shear-flow instability mechanism of Miles [1957] with the energy growth rate $\gamma/\omega = \beta(\rho_a/\rho_w)(u_*/c)^2$; $\beta \approx 32$ (A 375) reproduces the growth rate of Plant [1982].

Type 4 is the sheltering mechanism of Jeffreys [1925] in the modified form of Touboul et al. [2006] and Kharif et al. [2008]: air flow separation on the leeward side of steep crests (Fig. 5.16), so the pressure acts only on faces steeper than the critical slope S_c; the switch w ramps linearly from 0 at $0.8S_c$ to 1 at S_c (A 376: s_J, S_c, c; $c \leq 0$: celerity of the generated wave). Because the input grows like $(u_*/c)^2$, the slope is low-pass filtered (A 378 passes of a 1-2-1 filter) so that grid-scale waves are not forced most strongly. In FNPF p_a enters the dynamic free surface boundary condition as $-p_a/\rho_w$; in NHFLOW it acts in every layer through its horizontal gradient. The forcing can be restricted to a rectangular area (A 372, A 572) with an optional $\cos^2$ downwind decay (A 374, A 574).

5.11.1 Key input commands

Command	Meaning
A 370 / A 570	wind forcing on, C_d: 1 Garratt, 2 Wu, 3 Smith and Banke, 4 Zijlema (FNPF / NHFLOW)
A 371 / A 571	U_{10} and direction
A 372 / A 572	forcing area
A 373 / A 573	type: 1 surface stress (NHFLOW) or wind setup (FNPF), 3 Miles/Plant, 4 modified Jeffreys
A 374 / A 574	$\cos^2$ downwind decay over the forcing area
A 375 / A 575	Miles/Plant coefficient β (default 32)
A 376 / A 576	Jeffreys: sheltering coefficient s_J, critical slope S_c, phase speed c (0.5, 0.3, 0.0)
A 377	reference depth h_{ref} of the FNPF wind setup
A 378 / A 578	1-2-1 filter passes on the slope (default 4)
L 41	wind for the windage of Lagrangian particles (Section 5.9)

6. Functions

REEF3D reads its input from the file `ctrl.txt`, which has to be located in the folder from which REEF3D is started (the simulation folder, see Chapter 8). Each command consists of a capital letter, a number and the corresponding values, separated by blanks, e.g.

```
A 10 6          // CFD module
N 41 30.0       // maximum simulated time
P 51 5.0 0.5    // wave gauge at x = 5.0 m, y = 0.5 m
```

In the command descriptions, the type of each value is given in front of it: for int an integer is expected, for double a floating-point number. The commands can be given in any order. If a command that allows only one instance is given more than once, the last occurrence is used.

How ctrl.txt is read

The file is read token by token, and the following rules follow from the parser:

- **Comments.** A token starting with / makes the parser skip the rest of the line. Comments are therefore written as // and can stand on their own line or after the values of a command. Text that is not preceded by / is not a comment: lower-case letters and digits are skipped, but a capital letter can be interpreted as the start of a command, which usually stops REEF3D (see below). Always start comments with //; then capital letters in comments are no problem.
- **Unknown commands are silently ignored.** A letter or number that does not correspond to an existing command (for example a typo such as `P 03 0.1` instead of `P 30 0.1`) is skipped without a warning, together with its values, and the default value of the intended command is used. Check the command numbers carefully, and check the terminal output and the log files if the simulation does not behave as expected.
- **Missing values stop REEF3D.** If a command is given with fewer values than required, the parser cannot continue and REEF3D stops with the message `!!! missing input parameter in ctrl.txt !!!`. The same happens when a capital letter appears in text that is not preceded by /. Exception: if the value is missing in the very last line of the file, the end of the file is reached and the command silently keeps its default value. A few commands (B 91, B 93, B 96) additionally stop with `wrong input error` if too many values are given.
- **Types.** Give integer values for int commands; a decimal number is not rejected, only its integer part is used.

Some input values are changed automatically after reading, either because an option requires another setting or because two options exclude each other. For example, the level set convection scheme F 35 is set to 0 when VOF is selected with F 80 $>$ 0, and the time scheme N 40 = 3 is changed to 4 when floating bodies are activated with X 10 $>$ 0. Such automatic changes are noted in the descriptions of the affected commands.

Notation

The validity of each command for the four hydrodynamic modules is marked by the circles in front of the command, in the columns S (SFLOW), F (FNPF), N (NHFLOW) and C (CFD): a filled circle ● means that the command is used by that module, an empty circle ○ that it has no effect. For example, ● ○ ○ ● denotes a command that is valid for SFLOW and CFD only. The module itself is selected with A 10.

Commands marked with an asterisk (∗), e.g. **P 51** ∗, can be issued multiple times; each line creates one additional instance (for example one wave gauge per line).

The default value given for each command is used when the command is not specified in `ctrl.txt`.

6.1 A :: Hydrodynamic models

The commands in this section are used to select the hydrodynamic model that is to be activated for the simulation. This will determine the governing equations that will be used for the simulations. The different models available in REEF3D are:

- **S** : SFLOW: A depth-averaged shallow water equations model, that solves the non-hydrostatic pressure.
- **F** : FNPF: This model solves the Laplace equation.
- **N** : NHFLOW: This model solves the Navier-Stokes in three dimensions on a sigma grid, and is suitable for large scale wave modeling with a single-value free surface.
- **C** : CFD : This solves the 3D RANS equations with two-phase flow.

Valid for:
S F N C

6.1.1 SFLOW

● ● ● ●

A 10 int Hydrodynamic model
Selects the governing equations and the numerical grid used for the whole simulation. SFLOW runs on a 2D horizontal grid, FNPF and NHFLOW on a sigma grid, and PTF and CFD on the fixed Cartesian grid.

2 SFLOW

3 FNPF

4 PTF (legacy potential flow solver on the fixed grid)

5 NHFLOW

6 CFD

default: 6

● ○ ○ ○

A 209 int Number of smoothing sweeps for the bed
Number of passes of a 1-2-1 type filter (0.5 centre, 0.125 per neighbour) applied to the bed elevation at initialisation. Use 0 to keep the bed exactly as read.

default: 1

● ○ ○ ○

A 210 int Time scheme for SFLOW momentum equations
Any value other than 2 selects the 3rd-order scheme. The same choice is used for the 6DOF time integration in SFLOW.

2 2nd-order Runge-Kutta

3 3rd-order Runge-Kutta

default: 3

● ○ ○ ○

A 211 int Reconstruction scheme for SFLOW momentum equations
SFLOW uses HLL fluxes. This command selects how the face values for the fluxes are reconstructed. The limiters fall back to first order next to dry cells. Options 0–3 also use first-order upwind convection for the turbulence quantities, option 4 WENO.

0 first-order

1 van Leer TVD

2 Superbee TVD

3 van Albada TVD

4 WENO reconstruction

default: 4

● ○ ○ ○

A 212 int Turn on diffusion for SFLOW velocities

Needed when a turbulence model (A 260) or the breaking viscosity (A 246 2) should act on the velocities.

0 OFF

1 explicit

2 implicit

default: 0

● ○ ○ ○

A 214 int Turn on convection for vertical SFLOW velocity w_s

Only used with the non-hydrostatic pressure schemes (A 220 1–3).

0 OFF

1 ON

default: 1

● ○ ○ ○

A 217 int Boundary conditions at walls

Ghost-cell treatment of the depth-averaged velocities at solid walls.

1 slip

2 no-slip

default: 2

● ○ ○ ○

A 218 int Turn on bed roughness

Adds Manning bed friction $\tau_b/\rho = gn^2|u|u/h^{1/3}$ with $n = k_s^{1/6}/20$. The roughness k_s (m) is taken from B 50.

0 OFF

1 ON

default: 0

● ○ ○ ○

A 219 int Additional time step constraint

With option 2, the time step is also limited by the advective Courant number $\Delta x/|u|$, in addition to the default $\Delta x/(|u|+\sqrt{gh})$ criterion.

1 OFF

2 ON

default: 1

● ○ ○ ○

A 220 int Pressure scheme for SFLOW

Option 3 uses the dispersion parameter A 224 and adds an extra time step limit. Option 4 solves the Boussinesq equations in the FUNWAVE-TVD formulation with hydrostatic pressure. Its dispersive terms are explicit, so N 47 is reduced to 0.25 if set higher. Values of 5 and above fall back to option 1. Non-hydrostatic pressure is only solved in wet, deep (A 221) and non-breaking cells (A 242, A 246).

0 hydrostatic

1 linear non-hydrostatic pressure

2 quadratic non-hydrostatic pressure

3 quadratic non-hydrostatic pressure with improved dispersion (A 224)

4 Boussinesq equations (FUNWAVE-TVD formulation)

default: 2

● ○ ○ ○

A 221 int Non-hydrostatic pressure in very shallow regions

With option 0, cells within two or three cells of a dry cell, and cells with water depth $\leq 10\cdot$A 244, are excluded from the non-hydrostatic pressure solution. They are treated as hydrostatic. With option 1, all wet cells are included.

0 exclude near-dry cells

1 all wet cells

default: 1

● ○ ○ ○

A 223 double Blending of current and previous free surface for the hydrostatic pressure gradient at patch boundaries

Weight θ of the current free surface: $\theta\eta^{n+1}+(1-\theta)\eta^{n}$. Used in the hydrostatic pressure gradient at pressure inflow/outflow patches (patchBC). Range 0–1.

default: 0.5

● ○ ○ ○

A 224 double Dispersion parameter α for A 220 3

Values below 1.0 are set to 1.0. A value of 1.0 gives the standard quadratic pressure. The default 1.159 is the optimised value. For $\alpha > 1$, the time step is also limited by the explicit dispersion correction.

default: 1.159

● ○ ○ ○

A 240 int Free surface scheme for SFLOW

Only option 1 is implemented. The free surface solver is created only for values ≥ 1, so do not set this to 0.

1 ON

default: 1

● ○ ○ ○

A 242 int Hydrostatic pressure for shallow areas

Marks shallow cells as "breaking", which switches off the non-hydrostatic pressure there. Option 1 marks cells within two cells of a dry cell and cells with water depth below 0.01 m. Option 2 marks cells within two cells of a dry cell and cells at steep bed slopes ($|\partial d/\partial x| > 2$, including three neighbours on each side). Option 3 is option 2 plus cells with water depth below 0.02 m.

0 OFF

1 near dry cells and very shallow water

2 near dry cells and steep bed slopes

3 as 2, plus very shallow water

default: 0

● ○ ○ ○

A 243 int Wetting and drying algorithm

Option 1 decides wet or dry from the water depth threshold A 244 alone. Option 2 also requires a dry cell to have a higher, wet neighbour before it can become wet again. Option 2 helps to maintain mass conservation, e.g. in open channel flow. With option 0, the water depth is clipped at A 244 but all cells stay wet.

0 OFF

1 depth criterion

2 depth criterion and wetting from higher wet neighbours

default: 2

● ○ ○ ○

A 244 double Absolute wetting-drying threshold

Water depth (m) below which a cell is dry. It is also the minimum depth used in divisions throughout SFLOW.

default: 0.001

● ○ ○ ○

A 246 int Breaking wave algorithm

Breaking cells are treated hydrostatically. A cell breaks when $\partial\eta/\partial t >$ A 247 $\cdot\sqrt{gh}$. Option 2 also flags cells with free-surface slope $|\partial\eta/\partial x| >$ A 355, and adds the breaking viscosity A 250 in the diffusion terms, which requires A 212 1 or 2.

0 OFF

1 ON

2 ON, with slope criterion and breaking viscosity

default: 1

● ○ ○ ○

A 247 double Breaking wave threshold parameter

Factor α in the onset criterion $\partial\eta/\partial t > \alpha\sqrt{gh}$ (A 246).

default: 0.6

● ○ ○ ○

A 248 int Breaking wave persistence algorithm

With option 0, the breaking flag is reset every time step. With option 1, a breaking cell keeps breaking while $\partial\eta/\partial t >$ A 249 $\cdot\sqrt{gh}$.

0 OFF

1 ON

default: 0

● ○ ○ ○

A 249 double Breaking wave persistence parameter

Factor β in the cessation criterion $\partial\eta/\partial t > \beta\sqrt{gh}$ used with A 248 1.

default: 0.3

● ○ ○ ○

A 250 double Breaking wave viscosity

Additional viscosity (m^2/s) added in breaking cells. Only used with A 246 2 and diffusion turned on (A 212 1 or 2).

default: 1.86

● ○ ○ ○

A 251 double Initial free surface slope in x-direction

Initialises the free surface as $\eta = -s\,(x - x_{min})$ with slope s (m/m). Use it, e.g., to start an open channel flow close to equilibrium. Inactive when not given.

default: not set (flat free surface)

● ○ ○ ○

A 260 int Turbulence model

The eddy viscosity acts on the velocities only when diffusion is turned on (A 212).

0 OFF

1 k-ϵ model

2 k-ω model

3 Prandtl length scale model

4 Parabolic eddy-viscosity model

5 k-ω model, alternative implementation

default: 0

● ○ ○ ○

A 261 double Length scale factor for the Prandtl length scale model

The mixing length is $l = 0.4 \cdot$A 261$\cdot h$. Used with A 260 3.

default: 0.267

● ○ ○ ○

A 262 double Factor for parabolic eddy-viscosity model

Coefficient α_t of the parabolic eddy-viscosity model (A 260 4).

default: 0.0667

● ○ ○ ○

A 263 double Factor for eddy-viscosity limiter

In the k-ϵ model (A 260 1), the eddy viscosity is limited to $\nu_t \leq$ A 263$\cdot k/S$, where S is the strain rate.

default: 2.7

● ○ ○ ○

A 264 double $c_{\epsilon\Gamma}$ coefficient

Coefficient of the bottom-shear production term in the ϵ-equation of the depth-averaged k-ϵ model (A 260 1). Set it explicitly when using A 260 1.

default: not initialised (intended: 2.7)

6.1.2 FNPF

○ ● ○ ○

A 309 int Number of smoothing sweeps for the bed

Number of passes of a 1-2-1 type filter applied to the bed elevation when the sigma grid is set up. Use 0 to keep the bed unfiltered.

default: 1

○ ● ○ ○

A 310 int Time scheme for FNPF algorithm

Floating bodies in FNPF and the ice floes (A 380) are advanced within the same Runge-Kutta stages as the free surface.

3 3rd-order TVD Runge-Kutta

4 4th-order Runge-Kutta

default: 3

○ ● ○ ○

A 311 int Spatial discretization for KFSFBC and DFSFBC

Discretization of the horizontal gradients in the free-surface boundary conditions. Options 4 and 5 are the same scheme. The wetting-and-drying version is selected by A 343.

0 OFF

1 CDS2

2 CDS4

3 WENO3

4 WENO5

5 WENO5

6 CDS6

default: 5

○ ● ○ ○

A 312 int Spatial discretization for second-order derivatives in the sigma coordinate system

2 CDS2

3 CDS4

default: 2

○ ● ○ ○

A 313 int Spatial discretization for the bed boundary condition

Discretization of the bed gradients in the Neumann condition for the velocity potential at the bed.

2 CDS2

3 CDS4

default: 3

○ ● ○ ○

A 314 int Linear or fully nonlinear free-surface boundary conditions

1 linear

2 fully nonlinear

default: 2

○ ● ○ ○

A 315 int Free-surface gradient treatment

Option 0 is the legacy scheme: $\partial\eta/\partial x$ is upwinded by the sign of the potential gradient and used everywhere. Option 1 upwinds the gradient in the kinematic boundary condition by $\partial H/\partial\eta_x$, using a symmetric stencil at zero speed. Option 2 is option 1 plus symmetric gradients for the geometry terms (sigma metrics, breaking, wind, dynamic boundary condition).

0 legacy

1 upwinded kinematic FSBC

2 as 1, plus symmetric geometry gradients

default: 1

○ ● ○ ○

A 316 int WENO5 near the shoreline

Only used with wetting-and-drying (A 343 ≥ 1) and WENO5 (A 311 4, 5). Where the stencil is not fully wet, option 1 falls back to a first-order gradient from wet neighbours. Option 0 is the legacy zero gradient.

0 legacy (zero gradient)

1 first-order fallback from wet neighbours

default: 1

○ ● ○ ○

A 317 int Bed metrics

Option 1 computes a symmetric bed gradient and includes the bed curvature terms. Option 0 is the legacy treatment: a left-biased bed gradient without wetting-drying, and zero bed curvature.

0 legacy

1 symmetric bed gradient and bed curvature

default: 1

○ ● ○ ○

A 318 int Nonlinear weights of the WENO5 free-surface gradients

Only used with WENO5 (A 311 4, 5), with and without wetting-drying. Option 1 uses the WENO-Z weights with $p = 2$ (Borges et al. 2008). Option 2 uses the TENO5 weights (Fu, Hu and Adams 2016): each of the three stencils is either kept with its ideal weight or discarded, depending on the cutoff A 319.

0 WENO-JS

1 WENO-Z

2 TENO5

default: 0

○ ● ○ ○

A 319 double TENO5 cutoff C_T

Used with A 318 2. A stencil is discarded when its normalised smoothness measure falls below C_T. Smaller values discard fewer stencils, i.e. the scheme stays closer to the linear upwind-biased scheme. Dimensionless.

default: $1.0 \cdot 10^{-5}$

○ ● ○ ○

A 324 int Picard iterations for the Laplace cross-derivative terms

The mixed derivatives $2\sigma_x \partial^2\Phi/\partial x \partial\sigma$ (and y) of the sigma-transformed Laplace equation are not in the 7-point matrix, but are evaluated on the right-hand side with the current potential. With 0, they are lagged by one Runge-Kutta stage; this error grows with the wave slope. With A 324 > 0, the assembly and solve are repeated with the new potential at most A 324 times, until the change is below A 325 times the change of the first solve. Each iteration costs one extra Laplace solve.

0 OFF (lagged by one stage)

> 0 maximum number of Picard iterations

default: 0

○ ● ○ ○

A 325 double Picard convergence criterion for the Laplace equation

Used with A 324 > 0. The iterations stop when the maximum change of the potential is below A 325 times the maximum change of the first solve (relative, dimensionless).

default: $1.0 \cdot 10^{-3}$

○ ● ○ ○

A 326 double Coastline relaxation reference time scale

Only used with the static coastline (A 343 1). If > 0, η and ϕ in the coastline relaxation zone are relaxed towards their running mean over this time scale (s) instead of towards zero.

default: 0.0 (relax to zero)

○ ● ○ ○

A 329 int Order of velocity potential extrapolation for Dirichlet wave generation and active wave absorption

1 1st-order

2 2nd-order

default: 1

○ ● ○ ○

A 330 int Dynamic wetting-drying: minimum wet time
Minimum number of time steps a rewetted cell stays wet. Used with A 343 2, 3.

default: 10

○ ● ○ ○

A 331 double Dynamic wetting-drying: rewetting hysteresis
Rewetting margin as a factor of A 344. Used with A 343 2, 3.

default: 1.0

○ ● ○ ○

A 332 int Dynamic wetting-drying: width of the front viscosity band
Cells within this many cells of a dry cell get at least the damping viscosity A 346. Requires A 350 1. Use 0 to turn it off. Used with A 343 2, 3.

default: 2

○ ● ○ ○

A 334 int Wetting-drying: redistribute clamp volume
Redistributes the water volume removed by the wet-dry clamp to wet neighbours. Used with A 343 2, 3, and with A 343 1 if A 337 1.

0 OFF

1 ON

default: 1

○ ● ○ ○

A 335 double Dynamic wetting-drying: maximum runup elevation
Maximum bed elevation above the still water level (m) that can become wet. Used with A 343 2. The default sentinel means no limit.

default: $1.0 \cdot 10^{20}$ (no limit)

○ ● ○ ○

A 336 int Wetting-drying: one-sided free surface gradients at the front
Computes the free surface gradients at the wet-dry front from wet cells only. Used with A 343 2, 3, and with A 343 1 if A 337 1.

0 OFF

1 ON

default: 1

○ ● ○ ○

A 337 int Static coastline: improved treatment

For A 343 1, option 1 averages the ϕ clamp over wet cells only and redistributes the clamp volume (A 334). It also uses wet-only free surface gradients with $\eta_{xx} = 0$ at the coast (A 336). The damping viscosity A 346 is ramped up towards the coastline inside the coastline relaxation zone (A 341, A 342) and applied in full in thin-film cells (water depth $<$ 10·A 344), where a drawdown has exposed the bed. Option 2 is the same, but applies A 346 only in the thin-film cells and leaves the coastline zone undamped (thin coastline). The viscosity requires A 350 1. Option 0 is the legacy treatment.

0 legacy

1 ON

2 ON, damping viscosity only in thin-film cells

default: 1

○ ● ○ ○

A 338 int Remove isolated wet areas

At coastline initialisation, dries out wet areas that are not connected to the main water body.

0 OFF

1 ON

default: 1

○ ● ○ ○

A 339 int Coastline distance initialisation method

0 PDE reinitialisation (legacy)

1 fast sweeping

default: 1

○ ● ○ ○

A 341 double Coastline relaxation zone width as factor of the cell size

The width of the relaxation zone along the coastline is A 341 times the horizontal cell size. The value 0.0 means no coastline relaxation zone. A 342 takes priority when set.

default: 0.0

○ ● ○ ○

A 342 double Coastline relaxation zone absolute width

Width of the coastline relaxation zone (m). When > 0, it overrides A 341.

default: 0.0

○ ● ○ ○

A 343 int Wetting-and-drying

Option 1 keeps the initial coastline fixed and damps η and ϕ in the coastline relaxation zone (A 341, A 342, A 347). Options 2 and 3 move the wet-dry front dynamically (A 330–A 336). Option 2 uses no coastline damping.

0 OFF

1 static coastline

2 dynamic runup and rundown

3 dynamic rundown only

default: 1

○ ● ○ ○

A 344 double Wetting-drying threshold value

Water depth (m) below which a cell is dry.

default: 0.001

○ ● ○ ○

A 346 double Damping viscosity at the coastline

Viscosity (m^2/s) applied within the coastline relaxation zone, modulated by the relaxation function. With dynamic wetting-drying, it is applied in the front band (A 332). It is applied through the breaking viscosity solve, so it requires A 350 1.

default: 1.86

○ ● ○ ○

A 347 int Coastline relaxation is used for

1 η and ϕ

2 η

3 ϕ

default: 1

○ ● ○ ○

A 348 int Numerical beach relaxation is used for

1 η and ϕ

2 η

3 ϕ

default: 1

○ ● ○ ○

A 349 double Coastline relaxation zone factor for full initialisation

In the first time step of a full-domain wave initialisation (I 30 1), the coastline relaxation zone width (A 341, A 342) is multiplied by this factor.

default: 10.0

○ ● ○ ○

A 350 int Breaking wave algorithm

Breaking is modelled with an artificial viscosity (A 365) applied in an implicit damping step. Detection is set by A 351.

0 OFF

1 base version (Baquet)

2 Kennedy

3 Romero (energy dissipation based)

default: 0

○ ● ○ ○

A 351 int Breaking wave detection

Shallow water: $\partial\eta/\partial t >$ A 354$\cdot\sqrt{gh}$. Deep water: free-surface slope $>$ A 355.

0 OFF

1 shallow water

2 deep water

3 shallow and deep water

default: 3

○ ● ○ ○

A 352 int Additional filtering for viscosity-based breaking

Filters η and ϕ in breaking cells for additional robustness (iterations: A 361, A 362), for breaking in:

0 OFF

1 shallow water

2 deep water

3 shallow and deep water

default: 0

○ ● ○ ○

A 354 double Breaking wave threshold parameter shallow water

Factor α in $\partial\eta/\partial t > \alpha\sqrt{gh}$.

default: 0.6

○ ● ○ ○

A 355 double Breaking wave slope threshold deep water α

Breaking starts where the free-surface slope $|\partial\eta/\partial x|$ exceeds this value. SFLOW also uses it with A 246 2.

default: 1.25

○ ● ○ ○

A 356 double Breaking wave slope cessation threshold β

Breaking stops where the slope drops below β·A 355.

default: 0.1

○ ● ○ ○

A 357 int Breaking wave viscosity is applied to

1 η and ϕ

2 ϕ

3 η

default: 1

○ ● ○ ○

A 361 int Breaking wave filter algorithm: outer iterations
Used with A 352.

default: 5

○ ● ○ ○

A 362 int Breaking wave filter algorithm: inner iterations
Used with A 352.

default: 2

○ ● ○ ○

A 365 double Breaking wave viscosity
Artificial viscosity (m^2/s) in breaking cells (A 350 1, 2).

default: 1.86

○ ● ○ ○

A 366 double Breaking wave viscosity factor for η
Factor applied to the breaking viscosity when damping η (A 357 1, 3).

default: 1.0

○ ● ○ ○

A 370 int Wind forcing: drag coefficient formulation
Turns on wind forcing in the dynamic free-surface boundary condition and selects the drag coefficient $C_d(U_{10})$. Wind speed and direction are set with A 371, the type of forcing with A 373.

0 OFF

1 Garratt

2 Wu

3 Smith and Banke

4 Zijlema

default: 0

○ ● ○ ○

A 371 double U_{10}, double direction
Wind speed at 10 m height (m/s) and the direction it blows towards (deg), measured counter-clockwise from the positive x-axis.

default: 0.0 ; 0.0

○ ● ○ ○

A 372 double x_s, double x_e, double y_s, double y_e Wind forcing area
Rectangular area (m) where the wind forcing is applied. When not given, the whole domain is forced.

default: not set (whole domain)

○ ● ○ ○

A 373 int Type of wind forcing
Option 1 gives the wind setup from the surface stress $\tau = \rho_a C_d U_{10}^2$, spread over the reference depth A 377. Option 2 is deprecated and treated as 1. Option 3 applies a surface pressure in phase with the slope, $p_a = \beta \rho_a u_*^2 \, \partial\eta/\partial n$ with β = A 375. Option 4 applies $p_a = s\rho_a (U_{10} - c)^2 \, \partial\eta/\partial n$ only on faces steeper than the critical slope (A 376).

1 wind setup (surface stress)

2 deprecated, same as 1

3 Miles/Plant wave growth

4 modified Jeffreys sheltering

default: 1

○ ● ○ ○

A 374 int Wind decay
Applies a $\cos^2$ downwind decay of the pressure forcing over $[x_s, x_e]$. Requires A 372.

0 OFF

1 ON

default: 0

○ ● ○ ○

A 375 double Miles/Plant wind input coefficient β
Used with A 373 3. $\beta \approx 32$ reproduces the growth rate of Plant (1982).

default: 32.0

○ ● ○ ○

A 376 double s, double slope_c, double c Jeffreys parameters
Used with A 373 4: sheltering coefficient s, critical slope, and wave phase speed c (m/s). For $c \leq 0$, the celerity of the generated wave is used.

default: 0.5 ; 0.3 ; 0.0

○ ● ○ ○

A 377 double Reference depth for wind setup
Depth h_{ref} (m) over which the surface stress is spread for A 373 1. For values ≤ 0, the mean still water depth of the forcing area is used.

default: 0.0

○ ● ○ ○

A 378 int Low-pass filter passes on the wind forcing slope
Number of 1-2-1 filter passes on the free-surface slope used for A 373 3, 4.

default: 4

FNPF sea ice

The commands A 380 – A 403 control the rigid ice floes of REEF3D::FNPF: the compliant-lid coupling to the free surface, the floe contacts and the floe breaking. See Section 5.10 for the model and the symbols used below.

○ ● ○ ○

A 380 int Ice floes
Turns on the ice floes and reads them from the file `ice_floes.dat`. The file has one entry per line; # starts a comment, lengths are in m and angles in deg:

`density` ρ_i ice density (kg/m^3) for the entries that follow (default 917)

`rect` x_c y_c h L_x L_y **yaw** rectangular floe with thickness h

`poly` x_c y_c h D n **yaw** regular n-gon with circumscribed diameter D

`convex` h n x_1 y_1 `...` x_n y_n convex floe from world vertices (the convex hull is taken)

`field` x_s x_e y_s y_e h **conc** D_{min} D_{max} n_{min} n_{max} **seed** random field of convex floes with n_{min} to n_{max} vertices and diameters uniform in $[D_{min}, D_{max}]$, placed without overlap until the area concentration conc is reached

`obstacle` n x_1 y_1 `...` x_n y_n fixed convex polygon, contact only; the ice load on it is logged

`disp` dz **roll pitch** initial offset of the previous floe from its equilibrium

`vel` u v initial drift velocity (m/s) of the previous floe

In a 2D flume, only `rect` is meaningful: y_c and yaw are ignored, and L_y is the floe width the pressure acts on (use $L_y = 1$ for results per unit width).

0 OFF

1 ON

default: 0

○ ● ○ ○

A 381 double Lid stiffness α

Stiffness of the compliant lid, $k = \alpha \rho_w g$. The floe follows the rigid-body kinematics up to $O(1/\alpha)$. Larger values stiffen the free surface under the floes and reduce the time step (A 403).

default: 20.0

○ ● ○ ○

A 382 double Lid damping ratio ζ

Damping of the lid as a fraction of the critical damping of each floe on the lid spring, $c = 2\zeta\sqrt{k\rho_i h}$.

default: 0.2

○ ● ○ ○

A 383 double Ice-water drag coefficient C_d

Skin drag between the floe bottom and the surface flow relative to the floe. Dimensionless. Use 0 to turn it off.

default: 0.0055

○ ● ○ ○

A 384 double μ, double e Contact friction and restitution

Coulomb friction coefficient μ and restitution coefficient e of the contacts between floes, obstacles and walls.

default: 0.3 ; 0.0

○ ● ○ ○

A 385 int Domain boundaries act as walls for the floes

With option 1, the floes are kept inside the domain by contacts with the domain boundaries.

0 OFF

1 ON

default: 1

○ ● ○ ○

A 386 int Footprint supersampling

Number of sub-samples per cell direction used to compute the floe coverage φ of each cell.

default: 4

○ ● ○ ○

A 387 double Floe print interval

Time interval (s) for the floe output. For values < 0, the floes are printed with the free-surface print settings (P 180 – P 182).

default: -1.0 (as the free-surface output)

○ ● ○ ○

A 388 int Contact solver iterations

Number of projected Gauss-Seidel iterations of the non-smooth contact solver.

default: 30

○ ● ○ ○

A 389 double Footprint edge taper half width

Half width δ of the smooth, area-preserving taper of the floe coverage across the floe edge, in multiples of the mean horizontal cell size. A sharp edge (0) gives a step of the size of the draft in the free surface, which the sigma grid does not tolerate once the floe moves.

default: 2.0

○ ● ○ ○

A 390 int Ice breaking

Bit flags for the breaking mechanisms; add the values to combine them (e.g. 7 for all). Contact splitting is only active in 3D. Spalling requires crushing (A 398 > 0). The check interval is set with A 394, the minimum piece size with A 393.

0 OFF

1 flexural failure (A 391, A 395, A 397, A 399, A 400)

2 contact splitting (A 392, A 396, A 401)

4 spalling at crushing contacts (A 398, A 402)

default: 0

○ ● ○ ○

A 391 double Flexural strength σ_f

Flexural strength (Pa) for flexural failure (A 390 1). With A 399, it is the mean of the Weibull distribution for a floe of the reference area.

default: $0.5 \cdot 10^6$

○ ● ○ ○

A 392 double K_{IC}, double C Contact splitting parameters

Fracture toughness K_{IC} (Pa $\mathrm{m}^{0.5}$) and calibration coefficient C of the splitting load $F_s = C\, K_{IC}\, h\sqrt{D}$ (A 390 2).

default: $1.0 \cdot 10^5$; 1.0

○ ● ○ ○

A 393 double Minimum floe size after breaking D_{min}

Pieces narrower than D_{min} (m) are not created. For values ≤ 0, four mean horizontal cell sizes are used.

default: -1.0 (four mean cell sizes)

○ ● ○ ○

A 394 int Breaking check interval

The breaking criteria are checked every A 394 time steps.

default: 1

○ ● ○ ○

A 395 int n_{dir}, int n_{bin} Flexural cut lines

Number of cut directions and number of candidate cut positions (beam bins) per direction for flexural failure. In 2D, one direction is used.

default: 12 ; 64

○ ● ○ ○

A 396 double Contact force averaging time t_c

Averaging time (s) of the pair contact force that is compared with the splitting load (A 390 2).

default: 0.05

○ ● ○ ○

A 397 double E, double ν Elastic ice properties

Young's modulus E (Pa) and Poisson ratio ν for the flexural stress. For $E > 0$, the bending moment follows from a free-free beam on an elastic foundation with the plate stiffness $Eh^3/(12(1-\nu^2))$. For $E \leq 0$, rigid-floe statics are used.

default: $5.0 \cdot 10^9$; 0.3

○ ● ○ ○

A 398 double Ice crushing strength σ_c

Caps the contact force at $\sigma_c hw$ (Pa, with the contact width w). The crushing contacts are logged. Required for spalling (A 390 4). Use 0 to turn crushing off.

default: 0.0 (OFF)

○ ● ○ ○

A 399 double m, double A_{ref}, int seed Weibull flexural strength

Gives each floe a random flexural strength $\sigma_f(A_{ref}/A_f)^{1/m}(-\ln u)^{1/m}/\Gamma(1+1/m)$, with the Weibull modulus m, the reference area A_{ref} (m^2), the floe area A_f, u uniform in $(0,1]$ and σ_f from A 391. For $m \leq 0$, all floes have the strength σ_f, without scatter and size effect.

default: 0.0 ; 1.0 ; 1

○ ● ○ ○

A 400 int Maximum flexural cracks per floe and check

With values > 1, several local stress maxima above the strength can break a floe in one check.

default: 1

○ ● ○ ○

A 401 int Radial cracks per contact splitting event

Number of cracks, fanned around the contact normal, when a floe splits in contact (A 390 2).

default: 1

○ ● ○ ○

A 402 double L_{sp}, double C_{loc}, double T Spalling parameters

Used with A 390 4 and A 398 > 0. When the crushed depth reaches the spall length L_{sp} (m, ≤ 0: floe thickness), the tip of the floe breaks off. Only local chips are created: the chord of the cut must not exceed C_{loc} times the chip depth. Chips narrower than D_{min} (A 393) are removed as rubble, whose lid pressure fades out over the time T (s).

default: -1.0 ; 3.0 ; 1.0

○ ● ○ ○

A 403 double Lid time step factor

The lid stiffens the free surface under the floes. The time step is limited to $\Delta t \leq$ A 403$/\omega_{lid}$, with the frequency ω_{lid} of the fastest lid-stiffened surface mode on the grid (including the lid damping). A 403 1 is about 60 % of the RK3 stability limit. For values ≤ 0, the previous limit is used: the FNPF time step times $1/\sqrt{1+\alpha}$ (A 381). In addition, the floes do not move more than half a cell per time step.

default: 1.0

6.1.3 NHFLOW

○ ○ ● ○

A 509 int Number of smoothing sweeps for the bed

Number of passes of a 1-2-1 type filter applied to the bed elevation at initialisation. Use 0 to keep the bed unfiltered.

default: 1

○ ○ ● ○

A 510 int Time scheme for NHFLOW momentum equations

The same choice is used for the 6DOF time integration in NHFLOW.

2 2nd-order Runge-Kutta

3 3rd-order Runge-Kutta

default: 2

○ ○ ● ○

A 511 int Flux scheme for NHFLOW momentum equations
Ignored with A 520 3, which uses its own flux scheme.

1 HLL
2 HLLC
default: 1

○ ○ ● ○

A 512 int Diffusion treatment
If used, implicit is recommended. If a turbulence model (A 560 >0) or A 550 >1 is used, the code sets A 512 2 automatically.

0 OFF
1 explicit
2 implicit
default: 0

○ ○ ● ○

A 513 int Wall treatment for implicit diffusion
Boundary values used for the implicit diffusion next to solid walls and direct-forcing bodies. Option 1 uses the cell value (zero gradient). Option 2 uses the ghost-cell values.

1 zero gradient
2 ghost-cell values
default: 1

○ ○ ● ○

A 514 int Reconstruction scheme for NHFLOW momentum equations
With option 4, WENO is used in the horizontal and a limited linear reconstruction in the vertical. Option 5 uses WENO in all three directions.

1 van Leer TVD
2 Superbee TVD
3 van Albada TVD
4 WENO reconstruction (horizontal)
5 WENO reconstruction (horizontal and vertical)
default: 4

○ ○ ● ○

A 515 int Free surface treatment at the Dirichlet wave generation boundary

Sets the free surface in the inflow ghost cells and the water depth used for the prescribed discharges UH, VH, WH. Options 1, 2 and 4 copy the interior free surface into the ghost cells. Option 1 computes the discharges with the still water depth, option 2 with the actual water depth. Option 3 sets the theoretical wave elevation and water depth in the ghost cells and uses them for the discharges. Option 4 is mass-conservative: the wave kinematics are sampled at the still-water sigma levels and the wave discharge is taken over the still water depth, so that no net second-order (Stokes) mass flux enters the domain. A current is taken over the actual depth, and the ghost-cell velocity is $U = UH/D$ with the actual depth D. For plain Dirichlet generation without absorption (B 98 5 in the input, see B 98), option 4 also imposes the prescribed discharge exactly as the mass flux across the inflow face.

1 zero gradient, discharge from the still water depth

2 zero gradient, discharge from the actual water depth

3 prescribed wave elevation

4 mass-conservative

default: 1

○ ○ ● ○

A 516 int Free surface treatment at forcing object

Options 2 and 4 are also applied automatically with floating bodies (X 10 4 or X 240 21).

1 free moving

2 fixed inside

3 enforcing zero gradient, free inside

4 enforcing zero gradient, fixed inside

default: 1

○ ○ ● ○

A 517 int Preserve tangential velocities for forcing

0 OFF

1 ON

default: 1

○ ○ ● ○

A 518 int Bed boundary condition

1 slip

2 no-slip

default: 2

○ ○ ● ○

A 519 int Bed roughness for velocities

Option 1 applies log-law bed friction in the bottom layer, with roughness k_s (m) from B 50, or from the sediment settings if sediment transport is active. Option 2 applies wall-function friction in all cells next to the bed (B 50) and next to solid walls and forcing bodies (B 57), including the vertical velocity.

0 OFF

1 bed friction

2 bed and wall friction

default: 0

○ ○ ● ○

A 520 int Pressure scheme for NHFLOW

Option 3 uses its own flux scheme and ignores A 511.

0 hydrostatic

1 PJM for non-hydrostatic pressure

2 Pressure correction for non-hydrostatic pressure

3 alternative non-hydrostatic scheme (YL)

default: 2

○ ○ ● ○

A 521 int Fluxes at direct-forcing interfaces

With option 0, velocities and fluxes are set to zero on cell faces between fluid cells and direct-forcing solids or floating bodies. Any other value turns this off.

0 ON

1 OFF

default: 0

○ ○ ● ○

A 522 double Bed slope factor in the non-hydrostatic pressure

In the bottom layer, the vertical interpolation weight of the horizontal velocities in the pressure solution (A 520 1, 2) is reduced by $\max(1-\text{A } 522|\partial d/\partial x|, 0)$. This improves robustness over steep bed slopes.

default: 5.0

○ ○ ● ○

A 524 int Horizontal gradients for the sigma transformation

Discretization of the free surface and bed gradients in the sigma coordinate metrics.

1 central differences

2 van Leer limited

3 van Leer limited, from the filtered free surface

default: 2

○ ○ ● ○

A 526 double Direct forcing interface width factor

The smoothed direct forcing interface has a width of A 526 times the horizontal cell size. It applies to fixed solids (A 581–A 590) and floating bodies, including porous floating bodies (X 16).

default: 1.1

○ ○ ● ○

A 527 int Nonlinear weights of the WENO reconstruction

Only used with the WENO reconstruction (A 514 4, 5), for all reconstructed quantities. Option 1 uses the WENO-Z weights with $p = 2$ (Borges et al. 2008). Option 2 uses the TENO5 weights (Fu, Hu and Adams 2016): each of the three stencils is either kept with its ideal weight or discarded, depending on the cutoff A 528.

0 WENO-JS

1 WENO-Z

2 TENO5

default: 0

○ ○ ● ○

A 528 double TENO5 cutoff C_T

Used with A 527 2. A stencil is discarded when its normalised smoothness measure falls below C_T. Smaller values discard fewer stencils, i.e. the scheme stays closer to the linear upwind-biased scheme. Dimensionless.

default: $1.0 \cdot 10^{-5}$

● ○ ● ○

A 531 double Froude number limiter

Limits the discharge in each wet cell to $|uh| \leq$ A 531$\cdot h\sqrt{gh}$. Next to dry cells, a factor 0.1 is applied when B 60 is used. In NHFLOW, the area is set by A 532. Also used by SFLOW.

default: 3.0

○ ○ ● ○

A 532 int Froude number limiter area

Option 2 applies the limiter only in shallow cells near dry cells (see A 545).

0 OFF

1 all wet cells

2 shallow cells only

default: 1

○ ○ ● ○

A 533 int Vertical velocity time step control

Adds the vertical Courant number $\Delta z/|w|$ and a source-term criterion to the time step control.

0 OFF

1 ON

default: 0

○ ○ ● ○

A 540 int Wetting-drying scheme

Option 1 lets a dry cell become wet only from a higher, wet neighbour, and uses the wet-dry face flux treatment. Option 2 decides wet or dry from the depth threshold A 544 alone.

1 depth criterion and wetting from higher wet neighbours

2 depth criterion

default: 1

○ ○ ● ○

A 544 double Absolute wetting-drying threshold

Water depth (m) below which a cell is dry. It is also the minimum depth used in divisions throughout NHFLOW.

default: 0.001

○ ○ ● ○

A 545 double Shallow water criterion factor

Cells with water depth $\leq$ A 545·A 544, or within three cells of a dry cell, are "not deep". There, the non-hydrostatic pressure is not solved, and A 532 2 applies the Froude number limiter.

default: 10.0

○ ○ ● ○

A 550 int Breaking wave algorithm

0 OFF

1 base version (Baquet)

default: 0

○ ○ ● ○

A 551 int Breaking wave detection

Shallow water: $\partial\eta/\partial t >$ A 554·$\sqrt{gh}$. Deep water: free-surface slope $>$ A 555.

0 OFF

1 shallow water

2 deep water

3 shallow and deep water

default: 3

○ ○ ● ○

A 552 int Additional filtering for viscosity-based breaking
Filters the water depth in breaking cells for additional robustness, for breaking in:

0 OFF
1 shallow water
2 deep water
3 shallow and deep water
default: 0

○ ○ ● ○

A 554 double Breaking wave threshold parameter shallow water
Factor α in $\partial\eta/\partial t > \alpha\sqrt{gh}$.

default: 0.6

○ ○ ● ○

A 555 double Breaking wave slope threshold deep water
Breaking starts where the free-surface slope exceeds this value.

default: 1.25

○ ○ ● ○

A 557 double Breaking wave viscosity
Artificial viscosity (m^2/s) in breaking cells (A 550 1).

default: 1.86

○ ○ ● ○

A 560 int Turbulence model
Any turbulence model switches on implicit diffusion (A 512 2). The URANS variants scale the eddy viscosity with a filter based on the local cell size and coefficient A 568.

0 OFF
1 k-ϵ model
2 k-ω model
21 k-ϵ model, URANS filtered
22 k-ω model, URANS filtered
31 LES, Smagorinsky
default: 0

○ ○ ● ○

A 564 int Eddy viscosity limiter

Option 1 is the realizability limiter $\nu_t \leq$ T 31·k/S. Option 2 is the SST/Bradshaw limiter $\nu_t \leq a_1 k/(F_2 S)$.

0 OFF

1 strain-based (realizability)

2 SST/Bradshaw

default: 1

○ ○ ● ○

A 565 int Turbulence model stabilization

Stabilization of the eddy viscosity in nearly potential flow regions, following Larsen and Fuhrman (2018). Uses T 42. Applies to the k-ω and k-ϵ models.

0 OFF

1 ON

default: 0

○ ○ ● ○

A 566 int Add buoyancy term to k-equation

0 OFF

1 ON

default: 0

○ ○ ● ○

A 567 int Free surface boundary condition for turbulent dissipation

Sets ϵ or ω in the top layer from a mixing length based on T 37. For k-ϵ, the length is T 37·h, and only option 1 has an effect. For k-ω, the length is T 37 (m) with options 1 and 2, and T 37·h with option 3.

0 OFF

1 ON

2 as 1 (k-ω only)

3 ON, length scaled with water depth (k-ω only)

default: 0

○ ○ ● ○

A 568 double URANS filter coefficient

Coefficient of the URANS eddy viscosity filter for A 560 21 and 22.

default: 1.0

○ ○ ● ○

A 570 int Wind forcing: drag coefficient formulation

Turns on wind forcing and selects the drag coefficient $C_d(U_{10})$. Wind speed and direction are set with A 571, the type of forcing with A 573.

0 OFF

1 Garratt

2 Wu

3 Smith and Banke

4 Zijlema

default: 0

○ ○ ● ○

A 571 double U_{10}, double direction

Wind speed at 10 m height (m/s) and the direction it blows towards (deg), measured counter-clockwise from the positive x-axis. Also used for the windage of Lagrangian particles if L 41 is not given.

default: 0.0 ; 0.0

○ ○ ● ○

A 572 double x_s, double x_e, double y_s, double y_e Wind forcing area

Rectangular area (m) where the wind forcing is applied. When not given, the whole domain is forced.

default: not set (whole domain)

○ ○ ● ○

A 573 int Type of wind forcing

Option 1 applies the surface shear stress $\tau = \rho_a C_d U_{10}^2$ in the top sigma layer. Option 2 is deprecated and treated as 1. Options 3 and 4 apply a surface pressure that acts in all layers: Miles/Plant (A 575) or modified Jeffreys (A 576). The slope used for options 3 and 4 is filtered with A 578 passes.

1 surface shear stress

2 deprecated, same as 1

3 Miles/Plant wave growth

4 modified Jeffreys sheltering

default: 1

○ ○ ● ○

A 574 int Wind decay

Applies a $\cos^2$ downwind decay of the wind forcing over $[x_s, x_e]$ for all forcing types. Requires A 572.

0 OFF

1 ON

default: 0

○ ○ ● ○

A 575 double Miles/Plant wind input coefficient β

Used with A 573 3: $p_a = \beta \rho_a u_*^2 \partial\eta/\partial n$.

default: 32.0

○ ○ ● ○

A 576 double s, double slope_c, double c Jeffreys parameters

Used with A 573 4: sheltering coefficient s, critical slope, and wave phase speed c (m/s). For $c \leq 0$, the celerity of the generated wave is used.

default: 0.5 ; 0.3 ; 0.0

○ ○ ● ○

A 578 int Low-pass filter passes on the wind forcing slope

Number of 1-2-1 filter passes on the free-surface slope used for A 573 3, 4.

default: 4

○ ○ ● ○

A 581 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing rectangular box

Fixed solid box (m) represented by direct forcing. It can be given several times.

default: not set (no object)

○ ○ ● ○

A 583 * double x_c, double z_c, double y_s, double y_e, double r Direct forcing cylinder in y-direction
Horizontal cylinder along y with centre (x_c, z_c), extent y_s to y_e and radius r (m).

default: not set (no object)

○ ○ ● ○

A 584 * double x_c, double y_c, double z_s, double z_e, double r Direct forcing cylinder in z-direction
Vertical cylinder with centre (x_c, y_c), extent z_s to z_e and radius r (m). These cylinders are also used by the DLM method (A 599).

default: not set (no object)

○ ○ ● ○

A 585 * double x_s, double y_s, double z_s, double r_s, double x_e, double y_e, double z_e, double r_e Direct forcing cylinder with flexible orientation
Cylinder or cone between the start and end points (m), with end faces orthogonal to the axis and radii r_s and r_e (m) at the two ends. Suitable, e.g., for jacket members.

default: not set (no object)

○ ○ ● ○

A 586 * double x_c, double y_c, double z_c, double r Direct forcing sphere
Sphere with centre (x_c, y_c, z_c) and radius r (m).

default: not set (no object)

○ ○ ● ○

A 587 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing wedge object in x-direction
Wedge within the given bounding box (m), oriented in x-direction.

default: not set (no object)

○ ○ ● ○

A 588 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing wedge object in y-direction
Wedge within the given bounding box (m), oriented in y-direction.

default: not set (no object)

○ ○ ● ○

A 589 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e Direct forcing wedge object in z-direction
Wedge within the given bounding box (m), oriented in z-direction.

default: not set (no object)

○ ○ ● ○

A 590 int Direct forcing solid from STL file
Reads the file 'solid.stl' and represents the geometry by direct forcing. It can be scaled, translated and rotated with A 591–A 593.

IMPORTANT: the STL file needs to be in ASCII format! Most CAD programs export to a binary STL file. It is possible to use e.g. Paraview for STL conversion from binary to ASCII.

0 OFF

1 ON

default: 0

○ ○ ● ○

A 591 double s_x, double s_y, double s_z Scale STL geometry
Scaling factors in x-, y- and z-direction for the A 590 geometry. They are applied before translation and rotation.

default: 1.0 ; 1.0 ; 1.0

○ ○ ● ○

A 592 double dx, double dy, double dz Translate STL geometry
Shift (m) of the A 590 geometry, applied after scaling.

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ○

A 593 double x_o, double y_o, double z_o, double ϕ, double θ, double ψ Rotate STL geometry
Rotates the A 590 geometry about the origin (x_o, y_o, z_o) (m) by the angles ϕ, θ, ψ (deg) about the x-, y- and z-axes. Applied after scaling and translation.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ○

A 599 int Use the distributed Lagrange multiplier method instead of direct forcing
With option 1, the direct forcing for solids and floating bodies is turned off and the DLM method is used for the A 584 cylinders.

0 OFF (direct forcing)

1 ON (DLM)

default: 0

6.2 B :: Boundary Conditions

Valid for:

S	F	N	C
○	○	○	●

B 10 int Wall functions for the velocities
Applies a logarithmic wall law to the tangential velocities at walls and bottom, using the roughness from B 50–B 56.

0 OFF

1 ON

default: 0

○ ○ ● ○

B 11 int Wall functions for the turbulence model
Wall-law boundary conditions for the turbulence quantities (k, ϵ/ω, eddy viscosity) at walls and bottom in NHFLOW. When OFF, the turbulence quantities are set to zero in the wall ghost cells.

0 OFF

1 ON

default: 1

○ ○ ○ ●

B 20 int Slip or no-slip boundary conditions for velocities
Ghost-cell treatment of the velocities at walls and at the bottom topography.

1 slip (zero gradient)

2 no-slip

3 no-slip via Dirichlet image-point extrapolation (see B 29)

4 no-slip at walls, slip at the bottom topography

default: 2

○ ○ ○ ●

B 21 int Preserve tangential velocities for forcing

Used by the direct forcing immersed boundary method together with B 20 1 or B 20 2. When ON, the tangential velocities at the solid surface are preserved in the forcing; when OFF, all velocity components inside the solid are simply forced to zero.

0 OFF

1 ON

default: 1

○ ○ ● ○

B 22 int Smooth or rough wall law for the bed shear stress

Wall law used to compute the bed shear stress for sediment transport in NHFLOW when S 16 1 is selected. The roughness is the effective bed roughness of the sediment model.

1 smooth wall law (iterative)

2 rough wall law

default: 2

○ ○ ○ ●

B 23 int Extrapolation or reflection for ghostcell updates

1 extrapolation

2 reflection

default: 1

○ ○ ○ ●

B 29 double Image point distance for ghost-cell Dirichlet extrapolation

Distance of the image point from the wall as a fraction γ of the cell size, used by the Dirichlet ghost-cell extrapolation (e.g. B 20 3). Range $0 < \gamma \leq 1$.

default: 0.5

○ ○ ○ ●

B 30 int Type of pressure reference point

Defines how the pressure level is fixed. The reference value is given in B 31; how it is applied is set by B 33.

0 OFF

1 fixed location (see B 32)

2 at the free surface above the inlet: the gage x- and y-location is set automatically to the inlet cell with the smallest x- and y-coordinates (outlet with the largest x-coordinate when B 99 $\geq$ 3), the z-location follows the free surface

3 atmospheric pressure B 31 imposed at the top boundary (lid)

4 normalize the pressure so that its mean value at the free surface (zero level set contour) is zero

default: 0

○ ○ ○ ●

B 31 double Pressure reference value
Pressure in Pa at the reference point (B 30 1, 2) or at the lid (B 30 3).

default: 0.0

○ ○ ○ ●

B 32 double x, double y, double z Location of the fixed pressure reference point
Coordinates in m, used with B 30 1. If B 32 is not given, the location is set automatically to the inlet cell with the smallest x- and y-coordinates and the highest z-coordinate (outlet with the largest x-coordinate when B 99 $\geq$ 3). For B 30 2 and 3 the location is always set automatically and B 32 is overwritten.

default: 0.0 ; 0.0 ; 0.0 (determined automatically, see above)

○ ○ ○ ●

B 33 int Type of pressure reference gage normalization
Virtual normalization only stores the gage offset and subtracts it in the output, inline normalization subtracts it from the pressure field in every time step. For certain configurations, inline pressure normalization causes instabilities.

1 virtual

2 inline

default: 1

● ○ ● ●

B 50 double Global wall roughness
Equivalent sand-grain roughness k_s in m, used for the bed and all walls unless overwritten by B 51–B 56 (CFD) or B 57 (NHFLOW side walls). In SFLOW it is the bed roughness used for the bottom friction.

default: 0.0001

○ ○ ○ ●

B 51 double Wall 1 roughness

Roughness k_s in m for the boundary side facing the negative x-direction (x_{min}). A value ≤ 0 means not set, B 50 is used.

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 52 double Wall 2 roughness

Roughness k_s in m for the boundary side facing the positive y-direction (y_{max}).

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 53 double Wall 3 roughness

Roughness k_s in m for the boundary side facing the negative y-direction (y_{min}).

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 54 double Wall 4 roughness

Roughness k_s in m for the boundary side facing the positive x-direction (x_{max}).

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 55 double Wall 5 roughness

Roughness k_s in m for the boundary side facing the negative z-direction (bottom).

default: -1.0 (not set, B 50 is used)

○ ○ ○ ●

B 56 double Wall 6 roughness

Roughness k_s in m for the boundary side facing the positive z-direction (top).

default: -1.0 (not set, B 50 is used)

○ ○ ● ○

B 57 double Side wall roughness for NHFLOW

Roughness k_s in m for the side walls, the top and solid bodies in the NHFLOW wall law (velocities with A 519 2, and the k-ω wall boundary conditions). The bottom uses B 50.

default: 0.0001

● ○ ● ●

B 60 int Enable ioFlow for open channel flow

This parameter is important for open channel flow calculations. The ioFlow module makes sure that the inflow discharge as given in W 10 is constant throughout the computation even though the inflow water level may change. This is the counterpart to B 90 for open channel flow. When selecting the hydrograph inflow option, make sure that a file with the name "hydrograph.dat" is present in the simulation folder. When selecting the hydrograph outflow option, make sure that a file with the name "hydrograph_out.dat" is present in the simulation folder. The data format of the hydrograph file consists of two columns, the first one gives the time in s and the second column the discharge in m^3/s.

0 OFF

1 constant inflow

2 hydrograph inflow

3 hydrograph outflow

4 hydrograph inflow and outflow

default: 0

○ ○ ● ●

B 61 int Inflow profile for ioFlow

Velocity profile at the inflow boundary for B 60. The logarithmic profiles use the roughness B 50. "Phase 1" refers to the water phase.

1 constant velocity

2 logarithmic profile bed

3 constant velocity only for phase 1

4 logarithmic profile only for phase 1, bed

5 logarithmic profile only for phase 1, sidewalls and bed

default: 2

○ ○ ○ ●

B 71 * double value, double distance, double line angle, double line x-origin, double line y-origin
Relaxation of the free surface towards a fixed water level for the initialization only (first time step), for open channel flow with B 60. value: water level z-coordinate in m; distance: length of the relaxation zone in m, measured from the line; line angle in deg; x- and y-origin of the line in m. Multiple zones are possible.

default: na

○ ○ ○ ●

B 75 int Outflow boundary condition
Boundary condition for the velocities at the outflow boundary.

1 zero gradient

2 convection condition

3 u-velocity scaled so that the outflow discharge matches the inflow discharge W 10 (scaling limited to 0.1–2.0), convection condition for v and w

default: 1

○ ○ ● ●

B 76 int Pressure inflow boundary conditions

0 hydrostatic pressure

1 zero gradient pressure

default: 1

● ○ ○ ●

B 77 int Pressure outflow boundary conditions
Option 2 controls the outflow water level with F 62. SFLOW only uses option 10.

0 zero gradient pressure

1 controlled outflow with pressure condition

2 controlled outflow with free surface condition (see F 62)

10 free stream outflow (zero pressure)

default: 1

● ● ● ●

B 81 double x_f, double y_f, double t_f Focus location and focus time for focused waves
x- and y-location in m and time in s at which the wave components of the focused wave theories (B 92 41–43) are in phase. For a 2D simulation set y_f to 0.0.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

B 82 int Type of focusing amplitude calculation
Amplitudes A_i of the wave components of focused waves. For NewWave the amplitudes are scaled with the spectrum so that their sum equals half of the wave height given in B 91 or B 93. Options 11–14 are identical to 1–4 but shift all phases by π, so that a wave trough instead of a crest is focused. The phase shift is only applied for unidirectional focused waves (B 130 0).

1 NewWave

2 Spectrum, $A_i = \sqrt{2S(\omega_i)\Delta\omega_i}$

3 Constant wave steepness spectrum, see B 83

4 Constant wave amplitude, $A_i = H/2$

11–14 as 1–4, trough focused

default: 1

● ● ● ●

B 83 double Gain factor per wave component for constant wave steepness spectrum focused waves
The amplitudes of the wave components are calculated using $A_i = \frac{B83}{k_i}$ (B 82 3). The frequency distribution is calculated as defined in B 84. For the Gaussian wavepackets (B 85 6) this value is used as the width of the Gaussian distribution in rad/s.

default: 0.0025

● ● ● ●

B 84 int Frequency spectrum discretisation method

1 Peak enhance method

2 Equal energy method

3 Uniform distribution

default: 1

● ● ● ●

B 85 int Wave spectrum for irregular waves

For the spectrum file, provide a "spectrum-file.dat" with the two columns "ω" and "S". For the 2D spectrum file, provide a "spectrum-file-2d.dat": the header line starts with a label followed by the directions, each following row gives the frequency and the spectral values for each direction; the number of frequencies and directions then replaces B 86 and B 133. The JONSWAP-type spectra use γ from B 88.

1 Pierson-Moskowitz

2 JONSWAP

3 Torsethaugen

4 Wavepackets

5 Wavepackets for steep waves

6 Gaussian Wavepackets

10 spectrum file

11 2D frequency-direction spectrum file

21 Goda JONSWAP

22 TMA

default: 1

● ● ● ●

B 86 int Number of regular waves for irregular wave generation

Number of frequency components used to discretise the spectrum. With directional spreading (B 130) the total number of components is B 86 $\times$ B 133.

default: 10

● ● ● ●

B 87 double ω_s, double ω_e Frequency range for irregular wave generation

Start and end angular frequency in rad/s of the discretised spectrum. When not given, the model will calculate the values automatically.

default: not set (calculated automatically)

● ● ● ●

B 88 double Peak enhancement factor γ

Used by the JONSWAP, Goda JONSWAP and TMA spectra (B 85 2, 21, 22).

default: 3.3

○ ● ● ●

B 89 int Wave generation optimization through space-time decomposition

Splits the input wave into precomputed spatial and temporal parts, which makes wave generation with many components considerably cheaper. Supported for the wave theories B 92 5, 31, 41 and 51. In CFD it applies to relaxation generation (B 98 2), in FNPF and NHFLOW to relaxation and Dirichlet generation. Not available in SFLOW.

0 OFF

1 ON

default: 0

● ● ● ●

B 90 int Enable ioWave for numerical wave tank

This parameter turns on the numerical wave tank. It is the counterpart to B 60 for wave simulations. A wave tank setup needs at least: the wave theory (B 92), the wave parameters (B 91 or B 93), the wave generation method (B 98), the numerical beach (B 99) and, for relaxation zones, their lengths (B 96).

0 OFF

1 ON

default: 0

● ● ● ●

B 91 double wave height, double wave length

Wave input with the wave length known: H in m and L in m. The wave period is calculated from the dispersion relation of the selected wave theory. For irregular, focused and reconstructed waves (B 92 31–53) the first value is the significant wave height H_s and the second the peak wave length, from which the peak period is calculated with linear dispersion. Use either B 91 or B 93; if both are given, B 93 is used. Exactly two values must be given, otherwise the program stops with an input error.

default: 0.0 ; 0.0

● ● ● ●

B 92 int Wave type

Different wave theories can be used as input for the numerical wave tank. It is important to check beforehand whether the selected theory fits the given wave conditions consisting of wave height, wave length/wave period (B 91 or B 93) and still water depth (B 94). The table below lists which generation methods (B 98) are available for each theory and model.

For the piston and flap wavemaker theories with η timeseries input (20, 24), provide a "wavemaker_eta.dat" input file with two columns "t" and "η".

For the piston and flap wavemaker theories (21, 22), provide a "wavemaker.dat" input file with two columns "t" and "X" or "t" and "β", depending on input from B 116. The vertical extent of the paddle is set with B 110 (piston), B 111 (flap) or B 112 (double-hinged flap). A second-order correction of the paddle signal can be added with B 113 and B 114, and in FNPF the moving-paddle boundary terms with B 119.

For the double-hinged flap wavemaker, two "X" or two "β" columns are expected. For giving two "X" columns, this will be the total excitation at the end of each flap. For giving two "β" columns, the second angle is the additional excitation of the second flap.

For the wave reconstruction, provide a "waverecon.dat" input file with three columns giving "A" and "ω" and phase "ϕ" for each of the harmonic wave components.

Wave theories which prescribe the potential directly are marked with a $*$ and are suitable for relaxation zone wave generation in FNPF. Those wave theories without $*$ require B 98 3 or B 98 4 for FNPF.

1 Shallow Water Waves

2 Linear Waves $*$

3 Deep Water Waves $*$

4 2*nd*-order Stokes Waves $*$

5 5*th*-order Stokes Waves (Fenton) $*$

6 Shallow Water Cnoidal Waves

7 1*st*-order Cnoidal Waves

8 5*th*-order Cnoidal Waves

9 1*st*-order Solitary Wave

10 3*rd*-order Solitary Wave

11 5*th*-order Stokes Waves (Skjelbreia)

20 Piston Wavemaker based on η timeseries

21 Piston Wavemaker

22 Flap Wavemaker

23 Double-hinged Flap Wavemaker

24 Flap Wavemaker based on η timeseries

31 1*st*-order Irregular Waves $*$

32 2*nd*-order Irregular Waves (Longuet-Higgins)

33 2*nd*-order Irregular Waves (Schäffer) $*$

41 1*st*-order Focused Waves $*$

42 2*nd*-order Focused Waves (Longuet-Higgins)

43 2*nd*-order Focused Waves (Schäffer) $*$

51 Wave Reconstruction with 1*st*-order Irregular Waves $*$

52 Wave Reconstruction with 2*nd*-order Irregular Waves (Longuet-Higgins)

53 Wave Reconstruction with 2*nd*-order Irregular Waves (Schäffer) $*$

61 Hydrodynamic Coupling HDC (FNPF to CFD or SFLOW to CFD)

Wave Generation Compatibility					
B 92	Wave Type	SFLOW	FNPF	NHFLOW	CFD
1	Shallow Water Waves	D + R	D	D + R	D + R
2	Linear Waves	D + R	D + R	D + R	D + R
3	Deep Water Waves	D + R	D + R	D + R	D + R
4	2*nd*-order Stokes Waves	D + R	D + R	D + R	D + R
5	5*th*-order Stokes Waves (Fenton)	D + R	D + R	D + R	D + R
6	Shallow Water Cnoidal Waves	D + R	D	D + R	D + R
7	1*st*-order Cnoidal Waves	D + R	D	D + R	D + R
8	5*th*-order Cnoidal Waves	D + R	D	D + R	D + R
9	1*st*-order Solitary Wave	D + R	D	D + R	D + R
10	3*rd*-order Solitary Wave	D + R	D	D + R	D + R
11	5*th*-order Stokes Waves (Skjelbreia)	D + R	D	D + R	D + R
20	Piston Wavemaker, η timeseries	D	D	D	D
21	Piston Wavemaker	D	D	D	D
22	Flap Wavemaker	D	D	D	D
23	Double-hinged Flap Wavemaker	D	D	D	D
24	Flap Wavemaker, η timeseries	D	D	D	D
31	1*st*-order Irregular Waves	D + R	D + R	D + R	D + R
32	2*nd*-order Irregular Waves (Longuet-Higgins)	D + R	D + R	D + R	D + R
33	2*nd*-order Irregular Waves (Schäffer)	D + R	D + R	D + R	D + R
41	1*st*-order Focused Waves	D + R	D + R	D + R	D + R
42	2*nd*-order Focused Waves (Longuet-Higgins)	D + R	D + R	D + R	D + R
43	2*nd*-order Focused Waves (Schäffer)	D + R	D + R	D + R	D + R
51	Wave Reconstruction with 1*st*-order Irregular Waves	D + R	D + R	D + R	D + R
52	Wave Reconstruction with 2*nd*-order Irregular Waves (Longuet-Higgins)	D + R	D + R	D + R	D + R
53	Wave Reconstruction with 2*nd*-order Irregular Waves (Schäffer)	D + R	D + R	D + R	D + R
61	Hydrodynamic Coupling HDC		D + R		D + R
70	Steady Surface Gravity Wave SSGW		R	R	

Table 6.1: Wave generation compatibility for B 92, D = Dirichlet/Neumann generation (B 98 3, 4, 5); R = Relaxation generation (B 98 2).

70 Steady Surface Gravity Wave SSGW $*$ (currently FNPF relaxation wavegen only, see B 170)

default: 0

● ● ● ●

B 93 double wave height, double wave period

Wave input with the wave period known: H in m and T in s. The wave length is calculated from the dispersion relation of the selected wave theory. For irregular, focused and reconstructed waves (B 92 31–53) the first value is the significant wave height H_s and the second the peak period T_p. Use either B 91 or B 93; if both are given, B 93 is used. Exactly two values must be given, otherwise the program stops with an input error.

default: 0.0 ; 0.0

● ● ● ●

B 94 double Water depth d for wave theory

Water depth in m used to evaluate the wave theory. If not given, d is set to the still water level F 60, which requires the bed at the wave generation to be located at $z = 0.0$ m.

default: not set (still water level F 60)

● ● ● ●

B 96 double dist1, double dist2 Lengths of the relaxation zones

Length in m of the wave generation relaxation zone (dist1), measured from the start of the domain (x_{min}), and length in m of the numerical beach relaxation zone (dist2), measured from the end of the domain (x_{max}). The relaxation zones are only active for relaxation generation (B 98 2) and relaxation beaches (B 99 1 or 2). If B 108 or B 107 are given, they replace the default zones and the respective value of B 96 is not used. A typical choice, used in the tutorials, is a generation zone of about one wavelength and a numerical beach of about two wavelengths, e.g. B 96 4.0 8.0 for L = 4.0 m. Exactly two values must be given, otherwise the program stops with an input error.

default: 0.0 ; 0.0

● ● ● ●

B 98 int Wave Generation Method

Relaxation generation ramps the solution towards the wave theory inside the generation zone (B 96, B 108) following the relaxation function by Jacobsen et al. [2012]; the zone also absorbs waves reflected back towards the inlet. For the Dirichlet type, waves are generated by assigning the values for the free surface and the velocity at the inflow

boundary only, which needs no extra domain length. Active absorption additionally corrects the inflow velocity with the difference between the target and the computed free surface elevation using shallow water theory, $U_c = (\eta_T - \eta_M)\sqrt{g/d}$, so that reflected waves leave the domain through the inlet.

Overrides (control_parse.cpp): B 98 1 (the former two-zone relaxation method by Afshar [2010]) is silently changed to B 98 2. For NHFLOW (A 10 5), B 98 3 is changed to B 98 4, i.e. Dirichlet generation in NHFLOW always includes active absorption, and B 98 5 is changed to B 98 3 to obtain plain Dirichlet generation without absorption. In SFLOW and FNPF, options 3 and 4 are identical (Dirichlet). In CFD with B 98 2 and W 10 $>$ 0, the discharge of W 10 is imposed at the inlet in addition to the waves. In NHFLOW, the mass-conservative Dirichlet BC A 515 4 imposes the prescribed mass flux exactly at the inflow face only for plain Dirichlet generation, i.e. with B 98 5 in the input (changed to 3 internally); with B 98 3 or 4 (active absorption) this flux correction is not applied.

0 OFF

1 changed to 2

2 Relaxation Method

3 Dirichlet (NHFLOW: with active absorption, see above)

4 Dirichlet with Active Absorption, based on shallow water theory (CFD, NHFLOW)

5 Dirichlet without absorption, NHFLOW only

default: 0

● ● ● ●

B 99 int Numerical Beach Method

Both relaxation methods dissipate the wave energy by ramping the free surface down to the still water level, the velocity to zero and the pressure to its hydrostatic distribution for still water conditions inside the beach zone (B 96, B 107). Typically the length of the dissipating relaxation zone should be two wavelengths. Option 1 uses a softer relaxation function than option 2 (the relaxation coordinate is scaled by 1/2). The Active Wave Absorption avoids reflections by generating a wave opposite of the reflected wave at the outflow boundary, thus canceling it out. The method is more efficient in computational terms, as it does not require additional space. Option 4 uses the wave frequency and wave number of the input wave (B 91 or B 93). SFLOW supports options 1–4.
In order to use the Active Wave Absorption beach, keep in mind to use the active beach boundary condition in the C data set in DIVEMesh.

0 OFF

1 Relaxation Method 1

2 Relaxation Method 2

3 Active Wave Absorption, based on shallow water theory

4 Active Wave Absorption, based on intermediate water theory

5 Active Wave Absorption, using a flap wavemaker (see B 123)

default: 0

● ● ● ●

B 101 int Ramp function for wave generation

Ramps the generated wave up smoothly from zero, $f = t/t_r - \frac{1}{\pi}\sin(\pi t/t_r)$ for $t < t_r$, with the ramp duration t_r from B 102.

0 OFF

1 ON, t_r = B 102 × wave period

2 ON, t_r = B 102 in s of simulation time

default: 0

● ● ● ●

B 102 double Ramp function duration

Duration of the ramp in wave periods (B 101 1) or in s (B 101 2).

default: 1.0

● ● ● ●

B 105 double angle β, double x_0, double y_0 Wave direction and origin for the generated wave

Propagation direction of the generated wave in deg (measured from the x-axis) and the x- and y-coordinates in m of the origin from which the wave phase distance is measured. If not given, the origin is the minimum x- and y-coordinate of the domain (0.0 ; 0.0 for HDC, B 92 61).

default: 0.0 ; domain x_{min} ; domain y_{min}

● ● ● ●

B 106 * double angle β, double x, double y Origin of the wave generation relaxation zone

Angle in deg and x-, y-coordinates in m. In the current code the input has no effect on the relaxation zones; use B 108 to place wave generation zones.

default: B 105 angle ; domain x_{min} ; domain y_{min}

● ● ● ●

B 107 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double d Numerical beach relaxation zones

The line from (x_{start}, y_{start}) to (x_{end}, y_{end}) in m defines where the waves are fully damped; the relaxation zone extends the acting distance d in m to both sides of this line, and the damping decreases with the distance from the line. As default, without any user input for B 107, a relaxation zone for the numerical beach is created at the end of the domain, i.e. at the maximum x-coordinate of the domain, with an orientation perpendicular to the x-axis over the full domain width, using the distance dist2 from B

96. If user input is provided for B 107, the default relaxation zone will not be created and the input from B 96 will not be used. The B 107 zones are active even without B 99 1 or 2. Multiple numerical beach zones are possible.

default: not set (zone from B 96, see above)

● ● ● ●

B 108 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double d Wave generation relaxation zones
The line from (x_{start}, y_{start}) to (x_{end}, y_{end}) in m defines where the wave theory is imposed fully; the relaxation zone extends the acting distance d in m to both sides of this line, and the influence of the wave theory decreases with the distance from the line. As default, without any user input for B 108, a relaxation zone for the wave generation is created at the beginning of the domain, i.e. at the minimum x-coordinate of the domain, with an orientation perpendicular to the x-axis over the full domain width, using the distance dist1 from B 96. If user input is provided for B 108, the default relaxation zone will not be created and the input from B 96 will not be used. Requires B 98 2. Multiple wave generation zones are possible.

default: not set (zone from B 96, see above)

● ● ● ●

B 110 double z_{start}, double z_{end} Vertical extent of the piston wavemaker
z-coordinates in m between which the piston velocity is applied (B 92 20, 21). Outside this range the velocity is zero.

default: not set (whole water column)

● ● ● ●

B 111 double z_{start}, double z_{end} Vertical extent of the flap wavemaker
z-coordinate of the hinge (z_{start}) and of the top of the flap (z_{end}) in m for B 92 22 and 24. Must be set when these theories are used.

default: 0.0 ; 0.0

● ● ● ●

B 112 double z_{start}, double z_{hinge_2}, double z_{end} Vertical extent of the double-hinged flap wavemaker
z-coordinates in m of the lower hinge, the second hinge and the top of the upper flap for B 92 23.

default: 0.0 ; 0.0 ; 1.0

● ● ● ●

B 113 int mode, int J Second-order wavemaker correction

Second-order wavemaker theory of Schäffer [1996] for the piston, flap and double-hinged flap wavemakers driven by "wavemaker.dat" (B 92 21, 22, 23). At start-up the first-order paddle signal is resampled, decomposed by FFT (at most 500 components, frequency band B 114), and the paddle displacement $X^{(2)}(t)$ that cancels the spurious free second-order waves is added to the input signal. For the double-hinged flap the correction is split over the two flaps in the proportion of the first-order motion. J is the number of evanescent modes per frequency (0: progressive modes only). The water depth is that of the wave theory (B 94). The correction assumes that the paddle reaches above the still water level; a warning is written if B 111 or B 112 z_{end} lies below it. The applied maximum correction is written to the screen. In FNPF, use together with B 119 1 to include the second-order paddle boundary terms consistently.

0 OFF

1 sub- and superharmonic correction

2 subharmonic correction only

3 superharmonic correction only

default: 0 ; 10

● ● ● ●

B 114 double f_{min}, double f_{max}, double $f_{2,min}$ Frequency band for the second-order wavemaker correction

Frequencies in Hz for B 113: first-order components between f_{min} and f_{max} are used ($f_{max} \leq 0$: chosen automatically; the same band is used for the paddle terms of B 119), and second-order components below $f_{2,min}$ are not corrected ($f_{2,min} \leq 0$: 0.1 times the peak frequency).

default: 0.0 ; 0.0 ; 0.0

○ ● ○ ○

B 119 int Moving-paddle boundary condition for FNPF wavemakers

For the piston and flap wavemakers (B 92 21, 22, 23) with Dirichlet generation in FNPF, the inflow velocity $\phi_x = X_t$ at $x = 0$ is extended by the second-order Taylor terms of the moving paddle, $X_z\phi_z - X\phi_{xx}$. These terms are evaluated from the linear paddle solution of the first-order signal (progressive and B 113 J evanescent modes, band B 114), tabulated at start-up, and not from the computed flow. With B 119 1 the numerical paddle behaves like a physical moving paddle, and the B 113 correction then also cancels these terms.

0 OFF

1 ON

default: 0

○ ○ ● ●

B 115 int Vertical velocity component for flap wavemaker kinematics
Includes the vertical velocity of the rotating flap (B 92 22, 23) in the generated velocities.

0 OFF

1 ON

default: 0

● ● ● ●

B 116 int Flap wavemaker kinematics input as paddle motion or angle
Meaning of the second (and third) column in "wavemaker.dat" for the flap wavemakers (B 92 22, 23).

1 paddle displacement $X(t)$ in m

2 paddle angle $\beta(t)$ in rad

default: 1

● ● ● ●

B 117 double Starting time shift for wavemaker kinematics input
Time in s: entries of "wavemaker.dat" or "wavemaker_eta.dat" before this time are skipped, and the remaining timeseries is shifted to start at $t = 0$.

default: 0.0

● ● ● ●

B 118 double Correction factor for the wavemaker velocities
Scaling factor applied to the paddle velocities of the wavemaker theories (B 92 21–24) and to the moving wavemaker objects in the 6DOF module. It was introduced to correct wavemaker generation in NHFLOW.

default: 1.0

● ● ● ●

B 120 double Phase angle shift for periodic regular waves
Phase shift in deg added to the phase of the regular wave theories.

default: -90.0°

○ ● ● ●

B 123 double z-coordinate of the hinge for flap-type active wave absorption

z-coordinate in m of the hinge for B 99 5. Below the hinge the absorbing velocity is zero, above it increases linearly towards the free surface.

default: 0.0

○ ● ○ ●

B 125 double y-coordinate for the hydrodynamic coupling slice

Uses the 2D slice of the HDC input data at this y-coordinate in m for the whole coupling boundary (B 92 61), e.g. to drive a 3D domain with results from a 2D simulation.

default: not set (full 3D input)

○ ● ○ ●

B 127 int Remove y-component from hydrodynamic coupling

Sets the y-velocity of the HDC input (B 92 61) to zero.

0 OFF

1 ON

default: 0

● ● ● ●

B 130 int Enable directional spreading for irregular waves

Generates short-crested waves by distributing each frequency component over B 133 directions between B 132 around the main direction B 131. The cosine-power function is $D = \frac{2}{\pi}\cos^s(\beta - \beta_0)$ with s = B 134; use B 134 2 for the cosine-squared function (the factor $\frac{2}{\pi}$ normalizes $s = 2$). The Mitsuyasu function uses s = B 134, or a frequency-dependent s from B 135 if B 134 ≤ 0.

0 OFF

1 Cosine power (cosine squared with B 134 2)

2 Mitsuyasu

default: 0

● ● ● ●

B 131 double Main direction for multidirectional irregular waves

Main wave direction β_0 in deg.

default: 0.0°

● ● ● ●

B 132 double β_s, double β_e Start and end angle for directional spreading
Angles in deg.

default: -90.0° ; 90.0°

● ● ● ●

B 133 int Number of angle intervals for directional spreading
If directional spreading is active (B 130 $>$ 0) and B 133 $\leq$ 1, the number is set to 3.

default: 1

● ● ● ●

B 134 double Spreading parameter s
Exponent s of the directional spreading function (B 130). Must be set to a positive value for B 130 1. For B 130 2, a positive value gives a constant s (recommended, $s \leq 85$); a value ≤ 0 switches to the frequency-dependent s from B 135.

default: -1.0 (not set)

● ● ● ●

B 135 double Peak spreading parameter s_{max} for Mitsuyasu spreading
Used for B 130 2 when B 134 $\leq$ 0: $s = s_{max}(\omega/\omega_p)^5$ for $\omega \leq \omega_p$ and $s = s_{max}(\omega/\omega_p)^{-2.5}$ above. A code comment marks this option as not correct; use B 134 instead.

default: 10.0

● ● ● ●

B 136 int Type of multi-directional irregular wave calculation

1 DSM - double summation method

2 PSSM - Pascal's Single Summation Method

3 YSSM - Yu's Single Summation Method

4 EEM - equal energy method, to be used together with B 84 2

default: 1

● ● ● ●

B 138 int seed 1, int seed 2 Seed numbers for the random phase generation of multi-directional irregular waves
If not given, the random numbers are based on the computer clock.

default: not set (clock based)

● ● ● ●

B 139 int Seed number for the random phase generation of irregular waves
IMPORTANT: seed numbers are not machine transferable. In order to reliably repeat a given wave spectrum, the wave component file (see REEF3D_Log-Wave folder) needs to be re-used and renamed to "waverecon.dat" (B 92 51–53).

0 time based seed number

>**0** input based seed number

default: 0

● ○ ○ ○

B 160 int Number of vertical layers for depth-averaged velocities in the wave generation for REEF3D::SFLOW
The wave theory velocity is evaluated at B 160 + 1 points over the water depth and averaged.

default: 5

○ ● ○ ○

B 170 int Number of Fourier modes for calculating the steady surface gravity wave (B 92 70)

default: 1024

○ ○ ○ ●

B 180 int Moving tank through modified gravity
Represents a translating tank (e.g. sloshing) by adding the tank acceleration from B 181–B 183 to the gravity vector W 20–W 22. Required for B 181–B 183; the rotations B 191 and B 192 switch this mode on by themselves. Cannot be combined with B 60 or B 90.

0 OFF

1 ON

default: 0

○ ○ ○ ●

B 181 double amplitude, double frequency, double phase Harmonic tank motion in x-direction
Displacement amplitude a in m, frequency f in Hz and phase φ in deg of a harmonic motion; the corresponding acceleration $a(2\pi f)^2 \sin(2\pi f t + \varphi)$ is added to the gravity in x-direction. Requires B 180 1.

default: 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

B 182 double amplitude, double frequency, double phase Harmonic tank motion in y-direction
As B 181, for the y-direction. Requires B 180 1.

default: 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

B 183 double amplitude, double frequency, double phase Harmonic tank motion in z-direction
As B 181, for the z-direction. Requires B 180 1.

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ●

B 191 double angle, double frequency, double y, double z Harmonic tank rotation around the x-axis
Rotation amplitude θ in deg and frequency f in Hz of the roll motion $\theta(t) = \theta \sin(2\pi f t)$, and the y- and z-coordinates in m of the rotation axis. Active between the times given in B 194.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ●

B 192 double angle, double frequency, double x, double z Harmonic tank rotation around the y-axis
Rotation amplitude in deg and frequency in Hz of the pitch motion, and the x- and z-coordinates in m of the rotation axis. Active between the times given in B 194.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ●

B 194 double t_s, double t_e Start and end time of the rotation
Time window in s in which the rotations B 191 and B 192 are applied.

default: $-1.0 \cdot 10^9$; $1.0 \cdot 10^9$ (always active)

○ ○ ● ○

B 201 double n, double d_{50}, double α, double β VRANS porous media parameters
Porosity n, mean grain diameter d_{50} in m and the linear and nonlinear resistance coefficients α and β for all NHFLOW VRANS objects (B 210–B 230).

default: 1.0 ; 0.01 ; 0.0 ; 0.0

○ ○ ● ○

B 208 int Print VRANS porous media forces
Calculates the total force of the flow on the NHFLOW VRANS porous media and writes it to the folder REEF3D_NHFLOW_VRANS_Force.

0 OFF

1 ON

default: 0

○ ○ ● ○

B 209 double Thickness of porous Heaviside layer
Half thickness of the smoothed transition at the surface of the NHFLOW VRANS objects, as a factor times the horizontal cell size.

default: 1.1

○ ○ ● ○

B 210 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS rectangular box
Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 212 * double x_{center}, double z_{center}, double y_{start}, double y_{end}, double $radius$ VRANS cylinder in y-direction
Coordinates and radius in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 213 * double x_{center}, double y_{center}, double z_{start}, double z_{end}, double $radius$ VRANS cylinder in z-direction
Coordinates and radius in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 214 * VRANS cylinder with flexible orientation and front face orthogonal to the cylinder axis

double x_{start}, double y_{start}, double z_{start}, double $radius_{start}$, double x_{end}, double y_{end}, double z_{end}, double $radius_{end}$
Coordinates and radii in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 215 * double x_{center}, double y_{center}, double z_{center}, double $radius$ VRANS sphere
Coordinates and radius in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 217 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS wedge object in x-direction
Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 218 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS wedge object in y-direction
Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 219 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} VRANS wedge object in z-direction
Coordinates in m. Porous parameters from B 201.

default: na

○ ○ ● ○

B 230 int VRANS, read STL file and generate porous solid

Reads the geometry from a file named "solid.stl". The geometry is scaled, translated and rotated with A 591–A 593. Porous parameters from B 201.

IMPORTANT: the STL file needs to be in ASCII format! Most CAD programs export to a binary STL file. It is possible to use e.g. Paraview for STL conversion from binary to ASCII.

0 OFF

1 ON

default: 0

○ ○ ○ ●

B 240 * double C, double D, double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Darcy porous media as rectangular box

Resistance coefficients C and D of the Darcy-Forchheimer source term and box coordinates in m. The active directions are set with B 241–B 243.

default: na

○ ○ ○ ●

B 241 int Porous media in x-direction for B 240

0 OFF

1 ON

default: 1

○ ○ ○ ●

B 242 int Porous media in y-direction for B 240

0 OFF

1 ON

default: 1

○ ○ ○ ●

B 243 int Porous media in z-direction for B 240

0 OFF

1 ON

default: 1

○ ○ ● ●

B 260 double Added mass coefficient c for VRANS porous media
Coefficient of the added mass (inertia) term in the VRANS equations.

default: 0.34

○ ○ ● ●

B 264 double KC number for the VRANS porous media resistance
Keulegan-Carpenter number used in the nonlinear resistance term, which is multiplied by $(1 + 7.5/KC)$. The default sentinel $1.0 \cdot 10^{20}$ means not set, i.e. no KC correction.

default: $1.0 \cdot 10^{20}$ (not set, no KC correction)

○ ○ ○ ●

B 270 * VRANS porous media as rectangular box

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double n, double d_{50}, double α, double β
Coordinates in m, porosity n, mean grain diameter d_{50} in m, resistance coefficients α and β.

default: na

○ ○ ○ ●

B 274 * VRANS porous media as vertical cylinder

double x_{center}, double y_{center}, double z_{start}, double z_{end}, double $radius$, double n, double d_{50}, double α, double β
Coordinates and radius in m, porous parameters as in B 270.

default: na

○ ○ ○ ●

B 281 * VRANS porous media as wedge in x-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double n, double d_{50}, double α, double β
Coordinates in m, porous parameters as in B 270.

default: na

○ ○ ○ ●

B 282 * VRANS porous media as wedge in y-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double n, double d_{50}, double α, double β
Coordinates in m, porous parameters as in B 270.

default: na

○ ○ ○ ●

B 291 * VRANS porous media as plate in x-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double $d_{thickness}$, double n, double d_{50}, double α, double β
Coordinates and plate thickness in m, porous parameters as in B 270.

default: na

○ ○ ● ●

B 295 int VRANS source treatment for turbulence model

0 OFF

1 VRANS source terms

2 turn off turbulence inside porous media

default: 1

○ ○ ○ ●

B 308 int Consider vegetation porosity effects on fluid acceleration
When ON, the porosity inside the vegetation zones is set to $1 - N\pi D^2/4$. Giving B 308 in the control file also switches on the vegetation model, which is required for B 310–B 322.

0 OFF

1 ON

default: 1

○ ○ ○ ●

B 309 double Inertia coefficient C_M for vegetation

default: 2.0

○ ○ ○ ●

B 310 * VRANS vegetation as rectangular box

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double N, double D, double C_d
Coordinates in m, N the number of cylinders per unit area in $1/m^2$, D the diameter of the cylinders in m and C_d the drag coefficient. Requires B 308 to activate the vegetation model.

default: na

○ ○ ○ ●

B 321 * VRANS vegetation as wedge in x-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double N, double D, double C_d
Parameters as in B 310. Requires B 308.

default: na

○ ○ ○ ●

B 322 * VRANS vegetation as wedge in y-direction

double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double N, double D, double C_d
Parameters as in B 310. Requires B 308.

default: na

● ○ ○ ●

B 411 * int ID, double Q patchBC discharge
Discharge in m^3/s for the patch with this ID (see B 440–B 442).

default: na

● ○ ○ ●

B 412 * int ID, double p patchBC pressure
Pressure in Pa for the patch with this ID.

default: na

● ○ ○ ●

B 413 * int *ID*, double *h* patchBC waterlevel

Water level (free surface z-coordinate) in m for the patch with this ID.

default: na

● ○ ○ ●

B 414 * int *ID*, double *U* patchBC velocity perpendicular to face

Velocity in m/s normal to the patch with this ID.

default: na

● ○ ○ ●

B 415 * int *ID*, double *U*, double *V*, double *W* patchBC velocity components

Velocity components in m/s for the patch with this ID.

default: na

● ○ ○ ●

B 416 * int *ID*, double α patchBC horizontal inflow angle

Horizontal inflow angle in deg for the patch with this ID.

default: na

● ○ ○ ○

B 418 * int *ID*, int flag patchBC free stream outflow

Free stream outflow pressure condition for the patch with this ID in SFLOW. The condition is activated for the ID whenever the command is given.

0 OFF

1 ON

default: 0

● ○ ○ ●

B 421 * int *ID*, int flag patchBC discharge hydrograph

Requires a file named e.g. "hydrograph_Q_1.dat", where the number is the patchBC ID, with the columns time in s and discharge in m^3/s. The hydrograph is activated for the ID whenever the command is given.

0 OFF

1 ON

default: 0

● ○ ○ ●

B 422 * int ID, int flag patchBC waterlevel hydrograph
Requires a file named e.g. "hydrograph_FSF_1.dat", where the number is the patchBC ID, with the columns time in s and water level in m. The hydrograph is activated for the ID whenever the command is given.

0 OFF

1 ON

default: 0

● ○ ○ ●

B 440 * int ID, int $face$, double x_{start}, double x_{end}, double y_{start}, double y_{end} patchBC as line
The patch boundary condition will convert solid boundaries to inflow or outflow conditions. Patches are addressed with the ID, with which it is possible to prescribe different flow conditions (see B 411–B 422). When flow values are not prescribed, the default are zero-gradient boundary conditions. The face selects the side of the boundary: 1 = x_{min}, 2 = y_{max}, 3 = y_{min}, 4 = x_{max}, 5 = bottom, 6 = top. Coordinates in m. The patch BC is available for SFLOW (A 10 2, only B 440) and CFD (A 10 6).

default: na

○ ○ ○ ●

B 441 * int ID, int $face$, double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} patchBC as rectangular box
As B 440, with the patch area given as a box in m.

default: na

○ ○ ○ ●

B 442 * int ID, int $face$, double x_M, double y_M, double z_M, double $radius$ patchBC as circle
As B 440, with the patch area given as a circle with center and radius in m.

default: na

6.3 C :: Concentration

Transport of a passive or density-active scalar concentration in the CFD solver. The concentration c modifies the fluid properties as $\rho = \rho_{1,2} + c \cdot C1/C3$ and $\nu = \nu_{1,2} + c \cdot C2/C4$ in phase 1 and phase 2, respectively. The concentration module is activated with C 10.
Valid for:

S	F	N	C
○	○	○	●

C 1 double Additional density from concentration in phase 1
Density increase per unit concentration in phase 1 (water), in kg/m^3: $\rho = W1 + c \cdot C1$.

default: 10.0 kg/m^3

○	○	○	●

C 2 double Additional viscosity from concentration in phase 1
Kinematic viscosity increase per unit concentration in phase 1, in m^2/s: $\nu = W2 + c \cdot C2$.

default: 0.0 m^2/s

○	○	○	●

C 3 double Additional density from concentration in phase 2
Density increase per unit concentration in phase 2 (air), in kg/m^3: $\rho = W3 + c \cdot C3$.

default: 10.0 kg/m^3

○	○	○	●

C 4 double Additional viscosity from concentration in phase 2
Kinematic viscosity increase per unit concentration in phase 2, in m^2/s: $\nu = W4 + c \cdot C4$.

default: 0.0 m^2/s

○	○	○	●

C 10 int Time scheme concentration function
Activates the concentration transport and selects its time integration. Concentration diffusion is controlled by D 20.

0 OFF
1 Adams-Bashforth
2 2nd-order TVD Runge-Kutta
3 3rd-order TVD Runge-Kutta

default: 0

○ ○ ○ ●

C 15 int Convection discretization for concentration function
Use together with C 10. When suspended sediment transport is active (S 12 $\geq$ 1), the concentration convection is replaced by an implicit WENO HJ scheme.

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

7 WENO3 FLUX

8 WENO3 HJ

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

42 High Resolution TVD scheme with Weller limiter

default: 0

○ ○ ○ ●

C 50 double c_1, double c_2 Initial concentration inside and outside area 1
The concentration is initialized to c_2 everywhere and set to c_1 inside the area defined by C 51 – C 58 and C 75.

default: 1.0 ; 0.0

○ ○ ○ ●

C 51 double Area 1 start, x-direction
Area 1 is the box C 51 $\leq x <$ C 54, C 52 $\leq y <$ C 55, C 53 $\leq z <$ C 56. The large default values mean that the box is not limited in this direction.

default: $-1.0 \cdot 10^7$ m

○ ○ ○ ●

C 52 double Area 1 start, y-direction

default: $-1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 53 double Area 1 start, z-direction

default: $-1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 54 double Area 1 end, x-direction

default: $1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 55 double Area 1 end, y-direction

default: $1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 56 double Area 1 end, z-direction

default: $1.0 \cdot 10^7$ *m*

○ ○ ○ ●

C 57 double *a*, double *b*, double *c*, double *d* Area 1 as half-space below a plane
Cells with $ax + by + cz < d$ are set to the concentration c_1 of C 50. Only active if at least one of the four values is positive.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

C 58 double x_0, double y_0, double z_0, double *r* Area 1 as sphere
Center coordinates and radius of a sphere in *m* in which the concentration is set to c_1 of C 50. Only active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

C 75 * double x_0, double z_0, double α, double s, double l, double v Area 1 as tilted box
Tilted rectangle in the x-z plane (extended over the whole y-direction) starting at the corner (x_0, z_0) in m, inclined by the angle α in deg, with length s in m along the inclination and thickness l in m normal to it. Inside, the concentration is set to c_1 of C 50 and the initial velocity is set to $u = v\cos\alpha$, $w = -v\sin\alpha$ with v in m/s. Can be used multiple times.

default: not set

6.4 D :: Discretization

Here the discretization of the convection and diffusion terms in the momentum equations of the CFD solver and the pressure algorithm are chosen. All convection methods are implemented in terms of conservative finite differences. The exception are the WENO HJ schemes, which are the non-conservative form of the WENO scheme.
Valid for:

S	F	N	C
○	○	○	●

D 10 int Convection discretization
Convection scheme for the momentum equations. WENO5 FLUX is the recommended default.

0 OFF
1 FOU
2 CDS2
3 QUICK
4 WENO5 FLUX
5 WENO5 HJ
6 CDS4
7 WENO3 FLUX
8 WENO3 HJ
10 High Resolution TVD scheme with Minmod limiter
11 High Resolution TVD scheme with van Leer limiter
12 High Resolution TVD scheme with Umist limiter
13 High Resolution TVD scheme with van Albada limiter
14 High Resolution TVD scheme with Superbee limiter
15 High Resolution scheme with Smart limiter
16 High Resolution TVD scheme with Limo3 limiter
60 CDS6, sixth-order central scheme

default: 4

○ ○ ○ ●

D 11 int Convection velocities for momentum equations

Interpolation of the convecting velocities to the cell faces. Applies to the flux-based convection schemes (FOU, CDS2, QUICK, WENO FLUX, CDS4, CDS6, High Resolution, HRIC, CICSAM) of D 10, F 35, F 85 and C 15.

1 FOU

2 CDS2

default: 2

○ ○ ○ ●

D 20 int Diffusion discretization

Selecting the implicit treatment of the diffusion term in the momentum equations for explicit time stepping has the advantage of removing diffusion from the CFL criterion, which determines the time step size. As the diffusion term enters the CFL criterion in the order of $1/dx^2$, effectively larger time steps for D 20 = 2 will be used on especially finer grids. When RANS turbulence models are used, the effective diffusion can be several magnitudes higher than the molecular diffusion. Then implicit diffusion treatment will increase the time step significantly. D 20 also sets the diffusion treatment of the concentration (C 10) and of the turbulence quantities (off for D 20 = 0).

0 OFF

1 explicit

2 implicit

3 implicit Crank-Nicolson (3D only)

default: 2

● ○ ● ●

D 21 int Print out implicit diffusion solver time and iterations

Prints the solver iterations and time of the implicit diffusion step to the screen every P 12 iterations.

0 OFF

1 ON

default: 0

○ ○ ○ ●

D 22 int Wall boundary condition for implicit diffusion

Treatment of solid neighbour cells in the implicit momentum diffusion (D 20 = 2). If X 10 > 0 (6DOF), D 22 is set to 2 automatically.

1 Coefficients towards solid cells and direct-forcing solid regions are replaced by the value of the cell itself (zero normal gradient); velocities inside direct-forcing solid regions are set to zero.

2 Coefficients towards solid cells use the ghost cell values of the wall boundary condition.

default: 1

○ ○ ○ ●

D 30 int Pressure algorithm

The projection method (PJM) solves a Poisson equation for the pressure correction every time step. Option 10 computes a hydrostatic pressure $p = \phi \rho |g|$ from the level set function without solving a Poisson equation.

0 OFF

1 PJM CORR, projection method with pressure correction

10 hydrostatic pressure

default: 1

6.5 F :: Free Surface

Free surface capturing (level set and VOF) and the initial free surface of all models. The level set function ϕ is positive in phase 1 (water) and negative in phase 2 (air). The initial free surface commands (F 51 – F 72, F 112 – F 115) are applied in the order in which they are listed here; later commands overwrite earlier ones.
Valid for:
S F N C

○ ○ ○ ●

F 30 int Time scheme level set

Activates the level set method for two-phase flow. The level set time scheme is used for the segregated momentum schemes N 40 = 12, 13, 14, 44. For the fully coupled schemes (N 40 = 2, 3, 4, 22, 23, 24, 33) the level set is advanced inside the momentum time scheme, but F 30 $>$ 0 is still required for the two-phase fluid properties.

0 OFF

2 2nd-order TVD Runge-Kutta

3 3rd-order TVD Runge-Kutta

default: 3

○ ○ ○ ●

F 35 int Convection discretization for level set
If F 80 $>$ 0 (VOF), F 35 is set to 0 automatically and the VOF convection scheme F 85 is used instead.

0 OFF
1 FOU
2 CDS2
3 QUICK
4 WENO5 FLUX
5 WENO5 HJ
6 CDS4
7 WENO3 FLUX
8 WENO3 HJ
10 High Resolution TVD scheme with Minmod limiter
11 High Resolution TVD scheme with van Leer limiter
12 High Resolution TVD scheme with Umist limiter
13 High Resolution TVD scheme with van Albada limiter
14 High Resolution TVD scheme with Superbee limiter
15 High Resolution scheme with Smart limiter
16 High Resolution TVD scheme with Limo3 limiter
42 High Resolution TVD scheme with Weller limiter
default: 5

○ ○ ○ ●

F 40 int Reinitialization time scheme
PDE-based reinitialization of the level set function to a signed distance function. F 40 $>$ 0 is also required for the level set initialization with F 51 – F 72 and F 112 – F 115.

0 OFF
3 3rd-order TVD Runge-Kutta
default: 3

○ ○ ○ ●

F 42 double Length for initial level set reinitialization
Length in m over which the level set is reinitialized at the start of the simulation; the number of initial reinitialization steps is $2\,L/(F\,43 \cdot dx)$. If F 42 $\geq$ 0 it overrides the default length L, which is the maximum domain length. Use it to reduce the initialization time in long domains.

default: -1.0 (use maximum domain length)

○ ● ○ ●

F 43 double Factor for reinitialization time step size
Pseudo time step of the reinitialization as a fraction of the smallest local cell size: $\Delta\tau = F\,43 \cdot dx$. Also used for the coastline reinitialization in FNPF and for the topography and solid reinitialization.

default: 0.55

○ ○ ○ ●

F 44 int Number of reinitialization time steps
Number of reinitialization steps per time step (after the initial reinitialization). For the fully coupled momentum schemes (N 40 = 2, 3, 4, 22, 23, 24, 33) and VOF-PLIC (F 80 = 4) this value is set internally for each Runge-Kutta stage.

default: 3

○ ○ ○ ●

F 45 double Factor for the calculation of the interface thickness ϵ
Half-thickness of the smoothed interface as a multiple of the cell size, $\epsilon = F\,45 \cdot dx$, over which density and viscosity change between the phases.

default: 2.1

○ ○ ○ ●

F 46 int Type of Picard iterations for volume correction
Mass conservation correction: the level set is shifted by a constant so that the volume of phase 1 matches the reference volume. The number of iterations is given by F 47.

0 OFF

2 Volume correction for level set and reinitialization step using the volume from the previous step

3 Volume correction for level set and reinitialization step using the total volume balance (inflow and outflow discharge)

default: 0

○ ○ ○ ●

F 47 int Number of Picard iterations for volume correction
Use together with F 46.

default: 10

● ○ ● ●

F 50 int Fix free surface at inflow or outflow

Selects at which open boundaries the level set function (CFD) or free surface elevation (SFLOW, NHFLOW) is prescribed; at the other open boundaries it is extrapolated.

1 Inflow fixed

2 Outflow fixed

3 Fix both

4 Fix none

default: 2

○ ○ ○ ●

F 51 double Phase 1 start, x-direction

Phase 1 (water) is initialized in the box F 51 $\leq x <$ F 54, F 52 $\leq y <$ F 55, F 53 $\leq z <$ F 56. The box is only used if at least one of F 51 – F 56 is given; the large default values mean that the box is not limited in this direction. Setting only F 56 therefore defines a horizontal free surface at z = F 56.

default: $-1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 52 double Phase 1 start, y-direction

default: $-1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 53 double Phase 1 start, z-direction

default: $-1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 54 double Phase 1 end, x-direction

default: $1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 55 double Phase 1 end, y-direction

default: $1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 56 double Phase 1 end, z-direction

default: $1.0 \cdot 10^{20}$ m

○ ○ ○ ●

F 57 double a, double b, double c, double d Phase 1 as half-space below a plane
Cells with $ax + by + cz < d$ are set to phase 1. Only active if at least one of the four values is positive.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

F 58 double x_m, double y_m, double z_m, double r Phase 1 as sphere
Center coordinates and radius in m of a sphere of phase 1, e.g. a droplet. Only active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

F 59 double x_m, double y_m, double z_{start}, double z_{end}, double r Phase 1 as vertical cylinder
Center coordinates, vertical extent and radius in m of a vertical cylinder of phase 1, e.g. a water column. Only active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

F 60 double Initial still water level for the whole domain
Still water level in m (z-coordinate). In CFD it initializes the level set as $\phi = F60 - z$ and overrides F 51 – F 59. In SFLOW, FNPF and NHFLOW it is the reference water level. It is also the starting value of the outflow water level ramp (F 64).

default: $-1.0 \cdot 10^{20}$ m (not set)

● ○ ○ ●

F 61 double Water level at the inflow boundary
Water level in m (z-coordinate) prescribed at the inflow boundary. In SFLOW it sets the water depth in the inflow ghost cells.

default: $-1.0 \cdot 10^{20}$ m (not set)

● ○ ● ●

F 62 double Water level at the outflow boundary
Water level in m (z-coordinate) prescribed at the outflow boundary (use F 50 = 2 or 3). The outflow condition is only applied for F 62 > 0. Use F 64 to ramp up from F 60 to F 62 and F 63 to initialize a sloping free surface.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 63 double x_{start} for linear free surface interpolation
From x = F 63 in m to the end of the domain, the initial free surface is interpolated linearly from the still water level (F 60 or F 56) to the outflow water level F 62. Requires F 62.

default: $-1.0 \cdot 10^{20}$ m (not set)

● ○ ● ●

F 64 int Number of iterations for outflow water level ramp up
The outflow water level is ramped from F 60 to F 62 over F 64 time steps with a cosine function. 0 applies F 62 directly.

default: 0

○ ○ ○ ●

F 70 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 box
Rectangular box in m that is set to phase 1 (water). Can be used multiple times.

default: not set

○ ○ ○ ●

F 71 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 2 box
Rectangular box in m that is set to phase 2 (air). Can be used multiple times.

default: not set

● ○ ● ●

F 72 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double h Region with prescribed initial water level
In the horizontal region given in m, the initial free surface is set to the water level h in m (z-coordinate). Can be used multiple times, e.g. for dam break cases.

default: not set

○ ○ ○ ●

F 80 int Time scheme VOF
Activates the volume of fluid (VOF) method instead of the level set method. F 35 is then set to 0 and F 85 selects the VOF convection scheme. The VOF implementation is in beta: the algorithms for open channel flow and wave generation are designed for the level set method. VOF-PLIC (option 4) works with N 40 = 3, 13, 23, 33.

0 OFF

1 Adams-Bashforth

3 3rd-order TVD Runge-Kutta

4 VOF-PLIC, geometric reconstruction of the interface (under development)

default: 0

○ ○ ○ ●

F 84 double c_α factor for VOF compression
Scales the interface compression velocity $u_c = c_\alpha |u| \, n$ for F 80 = 1 and 3. 0 switches the compression off.

default: 1.0

○ ○ ○ ●

F 85 int Convection discretization VOF
Recommended scheme for VOF convection is HRIC modified.

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

42 High Resolution TVD scheme with Weller limiter

51 HRIC

52 HRIC modified

53 CICSAM

default: 0

VOF-PLIC settings (F 88 – F 98). The following commands are only used with F 80 = 4 (VOF with piecewise linear interface construction). This method is under development; the defaults are recommended.

○ ○ ○ ●

F 88 int VOF-PLIC interface normal calculation
Method to compute the interface normal vector for the PLIC reconstruction.

2 Weymouth

3 Wang

4 first-order

5 least squares

6 WENO

7 from the level set function

8 mass centre

9 ELVIRA 2D

10 MYC 2D

11 MYC 2D, version 2

12 MYC 2D, version 3

13 MYC 3D

14 MYC 3D, version 2

default: 10

○ ○ ○ ●

F 90 int VOF-PLIC transport scheme
Geometric transport of the VOF field.

4 COSMIC 2D, advected distance $u\Delta t$

5 as 4, with a two-step velocity estimate for the advected distance

default: 4

○ ○ ○ ●

F 91 double Density threshold factor for N 40 = 33
For the momentum scheme N 40 = 33 (with and without VOF-PLIC), the velocity in cells with density below $\rho_{thres} = F91 \cdot W1 + W3$ is blended with the previous velocity to stabilize low-density cells near the interface.

default: 0.1

○ ○ ○ ●

F 92 int VOF-PLIC density method
Calculation of density and viscosity from the VOF field. Options 22 and 32 are identical to 2 and 3, except that in the implicit PLIC diffusion (F 80 4 with D 20 2) the eddy viscosity is weighted with the volume fraction, so that turbulent viscosity only acts in the water phase.

1 from a redistanced level set function (with reinitialization, F 40)

2 VOF-based, cell-weighted average at the cell faces

3 VOF-based with sub-cell fractions

22 as 2, eddy viscosity weighted with the volume fraction

32 as 3, eddy viscosity weighted with the volume fraction

default: 2

○ ○ ○ ●

F 93 double VOF-PLIC air phase threshold
Cells with a volume fraction below F 93 are treated as pure air.

default: 0.001

○ ○ ○ ●

F 94 double VOF-PLIC water phase threshold
Cells with a volume fraction above F 94 are treated as pure water.

default: 0.999

○ ○ ○ ●

F 95 double VOF-PLIC water threshold for the phase marker
Cells with a volume fraction above F 95 are marked as water in the phase marker functions used for the reconstruction and the spray deletion (F 98).

default: 0.999

○ ○ ○ ●

F 96 double x_{start}, double x_{end} VOF-PLIC density softening region
Between x_{start} and x_{end} in m, the density method F 92 = 3 is replaced by the method F 92 = 2.

default: not set

○ ○ ○ ●

F 98 int VOF-PLIC spray deletion
Removes small isolated fluid fractions (spray) after each time step based on the phase marker (F 93, F 95).

0 OFF

1 ON

default: 1

○ ○ ○ ●

F 112 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as wedge in x-direction
Inside the horizontal region given in m, phase 1 fills the space from $\min(z_{start}, z_{end})$ up to the line that rises linearly from z_{start} at x_{start} to z_{end} at x_{end}. Can be used multiple times.

default: not set

○ ○ ○ ●

F 113 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as wedge in y-direction
As F 112, with the sloping line from z_{start} at y_{start} to z_{end} at y_{end}. Can be used multiple times.

default: not set

○ ○ ○ ●

F 114 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as inverse wedge in x-direction
Inside the horizontal region given in m, phase 1 fills the space above the line from z_{start} at x_{start} to z_{end} at x_{end}, up to $\max(z_{start}, z_{end})$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 115 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Phase 1 as inverse wedge in y-direction
As F 114, with the sloping line from z_{start} at y_{start} to z_{end} at y_{end}. Can be used multiple times.

default: not set

○ ○ ○ ●

F 150 int Benchmarks
Initializes the level set function and prescribes the velocity field according to standard interface advection benchmark cases.

0 OFF

1 Vortex

2 Slotted disc

3 3D deformation

4 3D Taylor-Green vortex

11 1D convection of step, triangle and sine profiles

21 Slotted disc in the y-z plane

22 Slotted disc in the x-y plane

default: 0

○ ○ ○ ●

F 151 int Inverse sign of the benchmark level set
Inverts the sign of the initial level set function of the 3D deformation benchmark (F 150 = 3).

0 OFF

1 ON

default: 0

Three-phase flow (F 300 – F 395). Two level set functions ϕ_1 and ϕ_2 describe three fluids: phase 1 (water, W 1, W 2) where $\phi_1 > 0$; where $\phi_1 < 0$, phase 3 (e.g. oil, W 6, W 7) where $\phi_2 > 0$ and phase 2 (air, W 3, W 4) where $\phi_2 < 0$. Both functions are initialized to -1 and then set by the commands below; the interface thickness follows F 45.

○ ○ ○ ●

F 300 int Three-phase flow level set time scheme

0 OFF

3 3rd-order TVD Runge-Kutta

default: 0

○ ○ ○ ●

F 305 int Three-phase flow level set convection discretization

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

10 – 16, 42 High Resolution TVD schemes, limiters as for F 35

default: 5

○ ○ ○ ●

F 310 int Three-phase flow reinitialization

0 OFF

3 3rd-order TVD Runge-Kutta

default: 3

○ ○ ○ ●

F 360 double Initial interface of ϕ_1 in x-direction
$\phi_1 = F360 - x$, i.e. phase 1 for $x <$ F 360 in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 361 double Initial interface of ϕ_1 in y-direction
$\phi_1 = F\,361 - y$, i.e. phase 1 for $y <$ F 361 in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 362 double Initial interface of ϕ_1 in z-direction
$\phi_1 = F\,362 - z$, i.e. phase 1 below z = F 362 in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 369 * double x_0, double z_0, double α, double s, double l, double v Tilted box of phase 1 for ϕ_1
Tilted rectangle in the x-z plane as for C 75: corner (x_0, z_0) in m, angle α in deg, length s and thickness l in m. Inside, $\phi_1 > 0$ and the initial velocity is $u = v\cos\alpha$, $w = -v\sin\alpha$ with v in m/s. Can be used multiple times.

default: not set

○ ○ ○ ●

F 370 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Positive box for ϕ_1
Rectangular box in m where $\phi_1 > 0$ (phase 1). Can be used multiple times.

default: not set

○ ○ ○ ●

F 371 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Negative box for ϕ_1
Rectangular box in m where $\phi_1 < 0$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 374 * double x_c, double z_c, double r Positive cylinder in y-direction for ϕ_1
Horizontal cylinder along the y-axis with center (x_c, z_c) and radius r in m: $\phi_1 > 0$ inside and $\phi_1 < 0$ outside (the whole domain is overwritten). Can be used multiple times.

default: not set

○ ○ ○ ●

F 375 * double x_c, double z_c, double r Negative cylinder in y-direction for ϕ_1
As F 374 with inverted sign: $\phi_1 < 0$ inside and $\phi_1 > 0$ outside. Can be used multiple times.

default: not set

○ ○ ○ ●

F 380 double Initial interface of ϕ_2 in x-direction
$\phi_2 = F\,380 - x$ in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 381 double Initial interface of ϕ_2 in y-direction
$\phi_2 = F\,381 - y$ in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 382 double Initial interface of ϕ_2 in z-direction
$\phi_2 = F\,382 - z$ in m.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ○ ●

F 390 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Positive box for ϕ_2
Rectangular box in m where $\phi_2 > 0$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 391 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Negative box for ϕ_2
Rectangular box in m where $\phi_2 < 0$. Can be used multiple times.

default: not set

○ ○ ○ ●

F 394 * double x_c, double z_c, double r Positive cylinder in y-direction for ϕ_2
As F 374, for ϕ_2. Can be used multiple times.

default: not set

○ ○ ○ ●

F 395 * double x_c, double z_c, double r Negative cylinder in y-direction for ϕ_2
As F 375, for ϕ_2. Can be used multiple times.

default: not set

6.6 H :: Heat Transfer

Valid for:
S F N C
○ ○ ○ ●

H 1 double Thermal diffusivity of water
Thermal diffusivity in m^2/s of the water phase. Which level-set phase is water is set with H 9.

default: $1.4 \cdot 10^{-7}$ m^2/s

○ ○ ○ ●

H 2 double Thermal diffusivity of air
Thermal diffusivity in m^2/s of the air phase. Which level-set phase is air is set with H 9.

default: $2.216 \cdot 10^{-5}$ m^2/s

○ ○ ○ ●

H 3 int Fluid property calculation
How density and viscosity depend on the temperature when heat transfer is active (H 10 > 0, with a level-set free surface, F 30 > 0). With option 2 the density varies linearly with the temperature, $\rho = \rho_0 (1 - \beta(T - T_0))$, where ρ_0 is the reference density (W 1 or W 3), T_0 the reference temperature from H 50 and β the expansion coefficient from H 4; without H 4, $\beta = 1/T_0$ with T_0 in K. Viscosity stays constant (W 2, W 4).

1 density and viscosity interpolated from tabulated properties of water and air

2 Boussinesq approximation (density only)

default: 1

○ ○ ○ ●

H 4 double β_1, double β_2 Thermal expansion coefficients

Thermal expansion coefficients in $1/K$ of water (β_1) and air (β_2) for the Boussinesq approximation (H 3 = 2). Setting H 4 activates the use of these values; if H 4 is not given, $\beta = 1/T_0$ is used.

default: not set ($\beta = 1/T_0$)

○ ○ ○ ●

H 9 int Assignment of water and air to the two phases

Selects which level-set phase (phase 1: $\phi > 0$, phase 2: $\phi < 0$) is water and which is air. This determines which of H 1/H 2, H 4 and H 50 apply to which phase.

1 phase 1: water, phase 2: air

2 phase 1: air, phase 2: water

default: 1

○ ○ ○ ●

H 10 int Time scheme for heat transfer

Turns on the solution of the temperature transport equation and selects its time scheme. Requires a level-set free surface (F 30 $>$ 0); the fluid properties then follow H 3.

0 OFF

1 2nd-order Adams-Bashforth

2 2nd-order TVD Runge-Kutta

3 3rd-order TVD Runge-Kutta

4 3rd-order Runge-Kutta with Crank-Nicolson diffusion

default: 0

○ ○ ○ ●

H 15 int Convection discretization for heat transfer

Spatial discretization of the convection term of the temperature equation (H 10 $>$ 0).

0 OFF

1 FOU

2 CDS2

3 QUICK

4 WENO5 FLUX

5 WENO5 HJ

6 CDS4

7 WENO3 FLUX

8 WENO3 HJ

9 WENO5 FLUX (uniform-grid formulation)

10 High Resolution TVD scheme with Minmod limiter

11 High Resolution TVD scheme with van Leer limiter

12 High Resolution TVD scheme with Umist limiter

13 High Resolution TVD scheme with van Albada limiter

14 High Resolution TVD scheme with Superbee limiter

15 High Resolution scheme with Smart limiter

16 High Resolution TVD scheme with Limo3 limiter

60 CDS6

default: 5

○ ○ ○ ●

H 50 double T_1, double T_2 Temperature 1 and temperature 2

Initial temperatures in $°C$. The whole domain is first set to T_2; cells inside the region defined by H 51–H 56, H 57 or H 58 are set to T_1. Since the default box of H 51–H 56 covers the whole domain, the domain starts at T_1 unless the box is narrowed. With H 3 = 2 the two values are also the reference temperatures T_0 of water (T_1) and air (T_2) for H 9 = 1, swapped for H 9 = 2.

default: 20.0 $°C$; 20.0 $°C$

○ ○ ○ ●

H 51 double Temperature 1 region start, x-direction

Lower x-bound in m of the box initialized with T_1 (H 50). Use together with H 52–H 56.

default: $-1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 52 double Temperature 1 region start, y-direction

Lower y-bound in m of the box initialized with T_1 (H 50).

default: $-1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 53 double Temperature 1 region start, z-direction
Lower z-bound in m of the box initialized with T_1 (H 50).

default: $-1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 54 double Temperature 1 region end, x-direction
Upper x-bound in m of the box initialized with T_1 (H 50).

default: $1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 55 double Temperature 1 region end, y-direction
Upper y-bound in m of the box initialized with T_1 (H 50).

default: $1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 56 double Temperature 1 region end, z-direction
Upper z-bound in m of the box initialized with T_1 (H 50).

default: $1.0 \cdot 10^7$ m (no bound)

○ ○ ○ ●

H 57 double a, double b, double c, double d Temperature 1 region bounded by a plane
Cells with $ax + by + cz < d$ (x, y, z in m) are set to T_1 (H 50), in addition to the H 51–H 56 box. Active only if at least one of the four values is positive.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 (inactive)

○ ○ ○ ●

H 58 double x_0, double y_0, double z_0, double r Temperature 1 region as sphere
Cells inside the sphere with center (x_0, y_0, z_0) and radius r (all in m) are set to T_1 (H 50). Active for $r > 0$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 (inactive)

○ ○ ○ ●

H 61 double Fixed temperature on surface 1
Sets a Dirichlet temperature in $°C$ on the domain boundary surface 1 (x_{min}). Surfaces are numbered 1: x_{min}, 2: y_{max}, 3: y_{min}, 4: x_{max}, 5: bottom (z_{min}), 6: top (z_{max}). Without the command the default boundary treatment for the temperature applies on this surface.

default: not set

○ ○ ○ ●

H 62 double Fixed temperature on surface 2
Dirichlet temperature in $°C$ on surface 2 (y_{max}), see H 61.

default: not set

○ ○ ○ ●

H 63 double Fixed temperature on surface 3
Dirichlet temperature in $°C$ on surface 3 (y_{min}), see H 61.

default: not set

○ ○ ○ ●

H 64 double Fixed temperature on surface 4
Dirichlet temperature in $°C$ on surface 4 (x_{max}), see H 61.

default: not set

○ ○ ○ ●

H 65 double Fixed temperature on surface 5
Dirichlet temperature in $°C$ on surface 5 (bottom, z_{min}), see H 61.

default: not set

○ ○ ○ ●

H 66 double Fixed temperature on surface 6
Dirichlet temperature in $°C$ on surface 6 (top, z_{max}), see H 61.

default: not set

6.7 I :: Initialization

Valid for:

S	F	N	C
●	○	●	●

I 10 int Initialize everything

Shortcut that overrides I 11, I 12 and I 13: option 1 sets I 11 = 1, I 12 = 2, I 13 = 1; option 2 sets I 11 = 2, I 12 = 2, I 13 = 1. Option 1 also initializes the inflow velocities in the ioflow/iowave boundary routines at start-up. Option 2 (potential flow including the free surface) is available in CFD only. I 10 is reset to 0 when a restart is requested (I 40 $>$ 0), and I 13 is reset to 0 without turbulence model (T 10 = 0).

0 OFF

1 ON

2 ON, potential flow initialization including the free surface (CFD)

default: 0

●	○	●	●

I 11 int Initialize velocities with potential flow solver

Solves a potential flow problem once before the first time step and uses the resulting velocities as initial condition. The linear solver is the one selected with N 10; for NHFLOW use N 10 = 1 or 11–19. Set automatically by I 10; reset to 0 if I 40 $>$ 0.

0 OFF

1 ON

2 ON, potential flow in the water phase only, including the free surface (CFD)

default: 0

○	○	○	●

I 12 int Initialize pressure

Initializes the pressure field hydrostatically, $p = \rho g h$, plus the reference pressure I 55. Skipped for options 1–3 if the numerical wave tank is fully initialized (I 30 = 1 together with B 90 $>$ 0). Set automatically to 2 by I 10; reset to 0 if I 40 $>$ 0.

0 OFF

1 hydrostatic, based on the vertical distance to the still water level

2 hydrostatic, based on the level-set function

3 hydrostatic, based on the vertical distance to the top of the domain (to the still water level with a free surface, F 30 $>$ 0)

default: 0

○ ○ ○ ●

I 13 int Initialize turbulence model

Calls the initialization of the selected turbulence model (T 10) before the first time step. Set automatically by I 10; reset to 0 if I 40 $>$ 0 or T 10 = 0.

0 OFF

1 ON

default: 0

○ ○ ○ ●

I 21 double Phase 2 velocities after potential flow initialization

Controls the velocities in the air phase (level set $\phi < 0$) after the potential flow initialization (I 11 $>$ 0).

0 keep the potential flow solution

1 set to zero

default: 0

● ● ● ●

I 30 int Full initialization of the numerical wave tank

Initializes the free surface, the velocities (and in FNPF the velocity potential) in the whole domain with the wave theory selected in B 92 at $t = 0$, instead of only in the wave generation zone. Requires wave generation (B 90 $>$ 0).

0 OFF

1 ON

default: 0

○ ○ ○ ●

I 40 int Initialization from state file (restart)

Restarts a simulation from a CFD state file written with P 40, selected with I 41. The state files are written per MPI process, so the restart must use the same decomposition (M 10). Simulation time and print counters continue from the state file. Any I 40 $>$ 0 resets I 10–I 13 to 0.

0 OFF

1 ON, from state file (requires I 41)

default: 0

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I 41 int State file number for restart
Number of the state file to restart from (I 40 = 1), i.e. the first number in the state file names `REEF3D-CFD-State-<number>-<rank>.r3d` in the folder `REEF3D_CFD_STATE`.

default: 0

○ ● ○ ○

I 44 int Read FNPF state file with 3D velocity potential
When restarting FNPF from a state file, also read the 3D velocity potential (state file written with the potential included, see P 44).

0 OFF

1 ON

default: 0

● ● ● ●

I 50 double Start time of the simulation
Initial value of the simulation time in s. With a restart (I 40 = 1) the time from the state file is used.

default: 0.0 s

○ ○ ○ ●

I 55 double Reference pressure
Constant pressure in Pa used as the initial pressure field and added to the hydrostatic pressure initialization (I 12).

default: 0.0 Pa

○ ○ ○ ●

I 56 int Zero pressure in the air phase
If ON, the hydrostatic pressure is set to zero above the free surface (level set $\phi < 0$), both in the pressure initialization (I 12) and in the hydrostatic pressure boundary values of ioflow/iowave.

0 OFF

1 ON

default: 0

○ ○ ○ ●

I 58 double w_0, double r_2 Initial vertical velocity around the F 58 sphere
Sets the vertical velocity to w_0 in m/s inside the sphere defined by F 58 and decreases it linearly to zero between the F 58 radius and the radius r_2 in m (r_2 larger than the F 58 radius). Active for $r_2 > 0$, e.g. for droplet impact cases.

default: 0.0 ; 0.0 (inactive)

○ ○ ○ ●

I 230 int Read FlowFile for inflow boundary condition
Uses a FlowFile (velocities, pressure, level set over time) from a previous 2D CFD simulation, printed with P 230, as inflow boundary condition. The number is the ID of the FlowFile gage. When active, the iowave wave generation at the inflow (B 98) is bypassed, so B 98 should be set to 0. See also I 231.

default: 0 (OFF)

● ● ● ●

I 231 double x-offset for hydrodynamic coupling input; vertical offset of FlowFile inflow
For hydrodynamic coupling (B 92 = 61), shift in m added to the x-coordinate when the input data is interpolated to the current grid. For FlowFile inflow (I 230, CFD), vertical offset in m of the FlowFile profile, rounded to whole cells.

default: 0.0 m

● ● ● ●

I 232 double y-offset for hydrodynamic coupling input
For hydrodynamic coupling (B 92 = 61), shift in m added to the y-coordinate when the input data is interpolated to the current grid.

default: 0.0 m

● ● ● ●

I 233 double z-offset for hydrodynamic coupling input
For hydrodynamic coupling (B 92 = 61), shift in m added to the z-coordinate (relative to the still water depth) when 3D input data (from FNPF or NHFLOW) is interpolated to the current grid.

default: 0.0 m

● ● ● ●

I 241 double Time offset for hydrodynamic coupling input

For hydrodynamic coupling (B 92 = 61), time in *s* added to the current wave time when the input data is read and interpolated in time, i.e. the coupling starts at time I 241 of the source simulation.

default: 0.0 *s*

6.8 L :: Lagrangian Particles (NHFLOW)

The L commands control the Lagrangian particle tracking of REEF3D::NHFLOW, which can be used for general particle tracing and for the transport of floating or buoyant marine (macro)plastics, in the spirit of OpenDrift. The particles are passive: they are moved by the flow but do not act back on it. The tracking is activated with L 10 1 and the particles are defined by one or more releases (L 21 – L 23). Each release carries its own rise or settling velocity, windage coefficient and particle mode, so that different particle classes can be simulated together. The particle tracking is not available in the other modules.

The deterministic particle velocity is the sum of the interpolated fluid velocity, the rise velocity w_s in the vertical and, for particles at the free surface, the windage $c_w U_{10}$ in the wind direction. It is integrated with Heun's method: the velocity at the old position and old time level is evaluated before the momentum step, and the corrector uses the new flow field after the momentum step, with the same time step as the flow solver. The fluid velocity is interpolated bilinearly in the horizontal over the surrounding cell-centre columns and linearly in the vertical within each column, using that column's σ-levels. Above the top cell centre the velocity is extrapolated linearly to the free surface, below the lowest cell centre it is held constant, and dry columns are excluded from the interpolation. Since NHFLOW resolves the individual waves, the Stokes drift follows from the tracking of the orbital velocities and no Stokes drift parameterization is added. The bed is the NHFLOW bed level (including the topography); particles above the free surface are moved onto it. For regular waves with $kh = 1.9$, the drift of surface particles agrees with linear Stokes theory within about 1%.

Turbulent dispersion is modelled as a random walk that is added after the Heun step: $\Delta x = \sqrt{2K_h \Delta t}\, R$ in each horizontal direction and, for a vertical diffusivity that varies over the depth, $\Delta z = K_v'(z)\Delta t + \sqrt{2K_v(z + K_v'\Delta t/2)\Delta t}\, R$ [Visser, 1997], where R is a standard normal random number. Particles that reach the free surface are placed on it (state 1) and are then subject to windage; surface-trapped particles (mode 1) always follow the free surface. At the bed, particles are reflected or deposited (L 52). Particles in dry cells are stranded (state 2) and either refloat when the cell becomes wet again or stay beached permanently (L 51). Particles that leave the computational domain through its lateral sides are removed or reflected (L 53).

In parallel runs, the particles are exchanged with the neighbouring processes after every time step. The release schedule, the particle ids and the positions of the box releases do not depend on the domain decomposition, so without random walk the results are identical for

any number of processes. The random numbers are drawn per process (L 11), so trajectories with random walk depend on the decomposition, while their statistics do not.

Output is written into the folder `REEF3D_NHFLOW_Particles` with the interval L 61: `REEF3D-NHFLOW-Particles-<n>.vtp` for ParaView (point data: velocity, id, release, state, age; the simulation time is stored as TimeValue), `REEF3D-NHFLOW-Particles-Tracks.csv` with the columns t, id, release, x, y, z, u, v, w, state, and `REEF3D-NHFLOW-Particles-Log.dat` with the number of released particles, the number of particles in the domain, in the water column, at the surface, stranded and on the bed, and the cumulative number of particles that left the domain. Positions and velocities are written in world coordinates. The particle states are 0: water column, 1: free surface, 2: stranded in a dry cell, 3: deposited on the bed.

Example: floating macroplastics with 3% windage and a sinking fraction, released continuously over 600 s:

```
L 10 1
L 41 10.0 0.0
L 23 100.0 200.0 50.0 80.0 0.0 0.0 5000 0.0 600.0 0.0 0.03 1
L 23 100.0 200.0 50.0 80.0 -1.0 -1.0 2000 0.0 600.0 -0.01 0.0 0
L 31 0.05
L 61 5.0
L 62 3
```

Valid for:

S	F	N	C
○	○	●	○

L 10 int Lagrangian particle tracking

Activates the particle tracking for NHFLOW (A 10 5). The particles are defined with L 21 – L 23.

0 OFF

1 ON

default: 0

S	F	N	C
○	○	●	○

L 11 int Random seed

Seed of the random numbers for the random walk (L 31, L 32). Each process uses the seed plus a process-dependent offset. Change the seed to obtain an independent realization of the random walk.

default: 1

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L 21 * double x, double y, double z, int n, double t_s, double t_e, double w_s, double c_w, int mode Point release
Releases n particles at the point (x, y, z) in m. The release times are distributed evenly over $[t_s, t_e)$ in s; with $t_e \leq t_s$ all particles are released at t_s. w_s is the rise velocity in m/s ($w_s > 0$: buoyant, $w_s < 0$: settling), c_w the windage coefficient as fraction of U_{10} (e.g. $0.01 - 0.05$ for floating plastics with emerged area). With mode 0 the particles move in three dimensions, with mode 1 they are trapped at the free surface and z is ignored. z is given in the absolute vertical coordinate of the grid (the same as for the probes, P 65); a particle released above the free surface is placed on it, one released below the bed is reflected into the water column (or deposited, L 52). x and y are given in world coordinates.

default: not set

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L 22 * double x_s, double y_s, double z_s, double x_e, double y_e, double z_e, int n, double t_s, double t_e, double w_s, double c_w, int mode Line release
Releases n particles along the line from (x_s, y_s, z_s) to (x_e, y_e, z_e) in m, evenly spaced at the positions $s = (q + 0.5)/n$, $q = 0 \ldots n - 1$, along the line. For a release over a time window, the particles are released in this order, i.e. the line is seeded from start to end over the window. The remaining values are the same as for L 21.

default: not set

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L 23 * double x_s, double x_e, double y_s, double y_e, double z_s, double z_e, int n, double t_s, double t_e, double w_s, double c_w, int mode Box release
Releases n particles at uniformly distributed random positions inside the box in m. The positions are reproducible and do not depend on L 11 or on the domain decomposition. The remaining values are the same as for L 21.

default: not set

○ ○ ● ○

L 31 double Horizontal diffusivity K_h
Constant horizontal diffusivity in m^2/s for the random walk. It represents the dispersion by processes that are not resolved by the grid. It does not act on stranded or deposited particles.

default: 0.0

○ ○ ● ○

L 32 int Vertical diffusion
Vertical random walk of the three-dimensional particles (mode 0). With option 2 the diffusivity is $K_v(z) = K_{v,0} + \nu_t / Sc_t$ with the background value L 33, the eddy viscosity ν_t of the NHFLOW turbulence model (A 560) and the turbulent Schmidt number L 34; the gradient of K_v is taken into account with the drift correction of Visser [1997].

0 OFF

1 constant diffusivity L 33

2 L 33 plus eddy viscosity / L 34

default: 0

○ ○ ● ○

L 33 double Vertical diffusivity $K_{v,0}$
Constant (L 32 1) or background (L 32 2) vertical diffusivity in m^2/s.

default: 0.0

○ ○ ● ○

L 34 double Turbulent Schmidt number Sc_t
Used with L 32 2.

default: 1.0

○ ○ ● ○

L 41 double U_{10}, double direction Wind for windage
Wind speed at 10 m height (m/s) and the direction it blows towards (deg), measured counter-clockwise from the positive x-axis, as for A 571. The windage $c_w U_{10}$ of each release acts on the particles at the free surface. If L 41 is not given and the NHFLOW wind forcing is active (A 570 $>$ 0), the values of A 571 are used. Note that the wind stress of A 570 – A 578 already drives a surface current in NHFLOW; c_w should therefore only represent the direct wind drag on the emerged part of the particles.

default: 0.0 ; 0.0 (or A 571)

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L 51 int Beaching
Treatment of stranded particles, i.e. particles in dry cells (wetting and drying, A 540 – A 544). Stranded particles are placed on the bed and do not move.

0 stranded, refloat at the free surface when the cell becomes wet again

1 permanent beaching

default: 0

○ ○ ● ○

L 52 int Bed contact
Treatment of three-dimensional particles that reach the bed. Deposited particles stay on the bed; resuspension is not included.

0 reflection

1 deposition of settling particles ($w_s < 0$), reflection of all others

default: 1

○ ○ ● ○

L 53 int Lateral domain boundaries
Treatment of particles that leave the horizontal extent of the computational domain. The boundary type from DIVEMesh is not taken into account. Removed particles are counted in the log file.

0 remove at all boundaries

1 reflect at all boundaries

default: 0

○ ○ ● ○

L 61 double Print interval
Output interval of the particle files in s of simulated time. With a value ≤ 0 the particles are written every time step. The first output is written at the start of the simulation.

default: 1.0

○ ○ ● ○

L 62 int Output format
The log file is always written.

1 VTP files

2 CSV track file

3 VTP files and CSV track file

default: 1

6.9 M :: MPI

Valid for:

S	F	N	C
●	●	●	●

M 10 int Number of processes for parallel computations

Must be set, also for serial runs (M 10 1). The value must equal the M 10 used in DIVEMesh to decompose the grid and the number of MPI processes REEF3D is started with (e.g. `mpirun -n 4` for M 10 4). REEF3D compares all three at start-up and stops with the message "Inconsistent M 10 parameter" if they differ.

default: 0 (not set)

6.10 N :: Numerical Options

Valid for:

S	F	N	C
●	●	●	●

N 10 int Linear solver for the Poisson/Laplace equation

Selects the solver for the Laplace equation of the velocity potential (FNPF), the pressure equation (NHFLOW, CFD) and the potential flow initialization (I 11). Use with a matching preconditioner in N 11. The default is the HYPRE Struct BiCGStab solver with PFMG preconditioning (N 10 14, N 11 11); it is available in all four modules and is the fastest HYPRE combination. From the HYPRE AIJ solvers, PCG with BoomerAMG (N 10 21, N 11 21) is fastest. For more information on the HYPRE solvers and preconditioners, see the HYPRE user manual Cen [2015].

Module restrictions: SFLOW always uses HYPRE Struct and accepts only 11–19. The internal BiCGStab solver (3) with Jacobi preconditioning works for the CFD pressure only. HYPRE AIJ (21–25) and SStruct (31–38) work for FNPF, NHFLOW and CFD, but not for the NHFLOW potential flow initialization (I 11 1).

REEFMG (N 10 1) is REEF3D's own multigrid solver: a semicoarsening multigrid in x and y with exact vertical line relaxation, used as preconditioner for a BiCGStab iteration (N 11 is ignored). It is made for the FNPF Laplace equation and the NHFLOW pressure, and the code also supports the CFD pressure (D 30 1–3) and the potential flow initialization. It is controlled by N 12–N 15. REEFMG requires that the grid is not decomposed in the vertical direction, that all ranks at the same x (y) position have the same number of cells in x (y), and that there are no periodic boundaries; otherwise it stops with a message.

1 REEFMG, BiCGStab with semicoarsening multigrid (FNPF, NHFLOW, CFD)

3 BiCGStab with Jacobi preconditioning (internal, CFD)

11 PCG (HYPRE Struct)
12 GMRES (HYPRE Struct)
13 LGMRES (HYPRE Struct)
14 BiCGStab (HYPRE Struct)
15 Hybrid-PCG (HYPRE Struct)
16 Hybrid-GMRES (HYPRE Struct)
17 Hybrid-BiCGStab (HYPRE Struct)
18 PFMG (HYPRE Struct, geometric multigrid)
19 SMG (HYPRE Struct, geometric multigrid)

21 PCG (HYPRE AIJ)
22 GMRES (HYPRE AIJ)
23 LGMRES (HYPRE AIJ)
24 BiCGStab (HYPRE AIJ)
25 AMG (HYPRE AIJ, BoomerAMG)

31 PCG (HYPRE SStruct)
32 GMRES (HYPRE SStruct)
33 LGMRES (HYPRE SStruct)
34 BiCGStab (HYPRE SStruct)
38 SysPFMG (HYPRE SStruct, geometric multigrid)
default: 14

● ● ● ●

N 11 int Preconditioner for the Poisson/Laplace equation
Preconditioner for the Krylov solvers selected in N 10; use only options from the same solver family. The multigrid solvers used on their own (N 10 18, 19, 25, 38) and REEFMG (N 10 1) ignore N 11. With HYPRE Struct and PFMG (11), N 20 and N 23–N 26 can switch the preconditioner to SMG during the run.

0 OFF
10 Jacobi (HYPRE Struct, N 10 11–14)
11 PFMG (HYPRE Struct, N 10 11–17)
12 SMG (HYPRE Struct, N 10 11–17)

21 BoomerAMG (HYPRE AIJ, N 10 21–24)

31 SysPFMG (HYPRE SStruct, N 10 31–34)
32 Split with PFMG (HYPRE SStruct, N 10 31–34)
33 Split with SMG (HYPRE SStruct, N 10 31–34)
default: 11

○ ● ● ●

N 12 int REEFMG line smoother
Ordering of the vertical line relaxation in REEFMG (N 10 1).

0 one forward sweep before and one backward sweep after the coarse-grid correction

1 symmetric forward and backward sweeps

2 red-black line smoother (experimental, slower on CPUs)

default: 0

○ ● ● ●

N 13 int REEFMG maximum number of coarse levels
Limits the number of coarsening steps per horizontal direction in REEFMG (N 10 1). With 0 the grid is coarsened until a local extent falls below 8 cells.

default: 0 (automatic)

○ ● ● ●

N 14 int REEFMG multigrid precision
Floating-point precision of the REEFMG multigrid hierarchy (N 10 1). With 32 the hierarchy is stored in single precision (about 25% less solver memory and a faster V-cycle); the BiCGStab iteration and the convergence test remain in double precision, so the converged result is unchanged. Any other value uses double precision.

32 single precision

64 double precision

default: 32

○ ● ● ●

N 15 int REEFMG coarse-grid agglomeration
If ON, the coarsest REEFMG level (N 10 1) is gathered onto every rank and solved there, which replaces many halo exchanges by one all-gather per V-cycle and may pay off at high numbers of processes. It is refused if the gathered problem is too large.

0 OFF

1 ON

default: 0

○ ● ● ○

N 18 int Index ordering of the HYPRE Struct grid

Ordering of the grid axes passed to HYPRE Struct (N 10 11–19). Option 2 makes the vertical index the fastest-running one; it is implemented for the FNPF Laplace equation and the NHFLOW pressure only.

1 standard (x, y, z)

2 permuted (z, y, x)

default: 1

○ ● ● ●

N 20 int Iteration threshold for preconditioner switch

If a HYPRE Struct solve with PFMG preconditioning (N 11 11) needs more than N 20 iterations, the preconditioner is switched to SMG for the rest of the simulation. The switch is enabled per module with N 23 (FNPF), N 25 (NHFLOW) and N 26 (CFD).

default: 20

○ ● ○ ○

N 23 int Preconditioner switch for FNPF

Enables the PFMG-to-SMG preconditioner switch (N 20) for FNPF.

0 OFF

1 ON

default: 0

○ ○ ● ○

N 25 int Preconditioner switch for NHFLOW

Enables the PFMG-to-SMG preconditioner switch (N 20) for NHFLOW.

0 OFF

1 ON

default: 0

○ ○ ○ ●

N 26 int Preconditioner switch for CFD

Enables the PFMG-to-SMG preconditioner switch (N 20) for CFD.

0 OFF

1 ON

default: 1

○ ○ ○ ●

N 40 int Time scheme for the momentum equations

In the coupled schemes the free surface (level set or VOF) is advanced within the Runge-Kutta stages of the momentum equations; in the separate schemes the level set is advanced with its own time scheme (F 30 = 2 or 3). Options 22, 23 and 24 are identical to 2, 3 and 4. With PLIC (F 80 = 4), options 3, 13 and 33 use their PLIC variants. Override: with 6DOF (X 10 > 0), N 40 = 3 or 23 is changed to 4 and N 40 = 13 to 14.

0 OFF

2 2nd-order TVD Runge-Kutta (momentum-free surface coupled)

3 3rd-order TVD Runge-Kutta (momentum-free surface coupled)

4 3rd-order low-storage Runge-Kutta (momentum-free surface coupled)

5 3rd-order Runge-Kutta with Crank-Nicolson diffusion

12 2nd-order TVD Runge-Kutta (separate free surface)

13 3rd-order TVD Runge-Kutta (separate free surface)

14 3rd-order low-storage Runge-Kutta (separate free surface), with its own time loop; direct-forcing variant for 6DOF (X 10 = 1) or FSI (Z 10 > 0)

33 3rd-order TVD Runge-Kutta (momentum-free surface coupled), conservative formulation with face densities

44 3rd-order low-storage Runge-Kutta (separate free surface)

default: 3

● ● ● ●

N 41 double Maximum simulated time

End time of the simulation in s. Of N 41 and N 45, whichever criterion is reached first ends the simulation.

default: $1.0 \cdot 10^{19}$ s (no limit)

○ ○ ● ●

N 43 double Stopping criterion for the convection-diffusion solver

Convergence tolerance of the iterative solver for the implicit transport equations (implicit diffusion, turbulence, scalars), i.e. the internal BiCGStab solver and the HYPRE Struct/SStruct BiCGStab used for these equations.

default: $1.0 \cdot 10^{-5}$

● ● ● ●

N 44 double Stopping criterion for the Poisson/Laplace solver

Convergence tolerance of the solver selected in N 10 (pressure, Laplace and potential flow equations) and of the SFLOW solvers.

default: $1.0 \cdot 10^{-8}$

● ● ● ●

N 45 int Maximum number of time steps

Maximum number of outer iterations (time steps). Of N 41 and N 45, whichever criterion is reached first ends the simulation.

default: $1.0 \cdot 10^{8}$ (effectively no limit)

● ● ● ●

N 46 int Maximum number of solver iterations

Maximum number of iterations of the linear solvers per solve. REEFMG prints a warning when the limit is reached. The SFLOW potential flow initialization uses a fixed limit of 2500.

default: 250

● ● ● ●

N 47 double Relaxation factor for time step size

Safety factor (Courant number) that multiplies the maximum stable time step from the CFL criterion for adaptive time stepping (N 48 1). In SFLOW with the Boussinesq option A 220 = 4 values above 0.25 are reduced to 0.25.

default: 0.3

● ● ● ●

N 48 int Adaptive time stepping

0 OFF, fixed time step N 49

1 ON, time step from the CFL criterion with N 47

default: 1

● ● ● ●

N 49 double Time step size for fixed time stepping
Time step in s for fixed time stepping (N 48 0). In FNPF and NHFLOW it is also the upper limit of the adaptive time step.

default: 1.0 s

○ ○ ○ ●

N 50 double Adaptive time stepping method
Formulation of the CFL criterion for adaptive time stepping in CFD (N 48 1).

1 standard, based on the velocity magnitude and the smallest cell size

2 component-wise, based on each velocity component and the cell size in that direction (less conservative)

default: 1

● ● ● ●

N 61 double Critical velocity for emergency stop
If the maximum velocity in any direction exceeds this value in m/s, the simulation stops and writes a final output file. In FNPF, velocities in the first five cells at x_{min} are additionally limited to $0.95 \cdot$ N 61.

default: 500.0 m/s

6.11 P :: Printer

The P commands control the output of REEF3D. All output is written into folders inside the directory from which the simulation is started. The folder names start with `REEF3D_` followed by the name of the hydrodynamic model, which corresponds to the validity columns: SFLOW (S), FNPF (F), NHFLOW (N) and CFD (C), e.g. `REEF3D_CFD_VTU` or `REEF3D_FNPF_WSF`. In the descriptions below, `<M>` stands for this model name. Model-independent output goes to `REEF3D_Log` (log files, discharge gauges), `REEF3D_Log-Wave` (wave time series), `REEF3D_SOLID` and `REEF3D_FlowFile`.

Many outputs can be triggered either by the iteration count or by the simulation time. An iteration-based interval (e.g. P 20, P 41, P 54, P 181, P 191) writes a file every n^{th} time step. A time-based interval in seconds (e.g. P 30, P 42, P 55, P 182, P 192) writes a file at the first time step after each multiple of the interval, and once at the start of the simulation. The time-based setting takes precedence: the iteration-based output of the same kind is only active while the corresponding time-based value is negative. Negative defaults (e.g. -10 or -1.0) mean inactive. In addition, P 35, P 184, P 185, P 194 and P 195 define print-out windows with their own interval; they can be given several times and act on top of the regular

intervals. P 34 and P 183 use the sediment (morphological) time instead of the hydrodynamic time. With the defaults, no periodic 3D field output is written: set P 20 or P 30 (and P 181 or P 182 for free-surface files and for SFLOW). Point gauges and probes are written every time step into one continuous file, line output (e.g. P 52, P 62, P 67, P 144) writes one new file per print-out.

Valid for:
S F N C

● ● ● ●

P 10 int Format of the 3D field output (ParaView binary)
The 3D files are written into `REEF3D_<M>_VTU`, `REEF3D_<M>_VTS` or `REEF3D_<M>_VTR` for FNPF, NHFLOW and CFD. VTR is only available for CFD; in FNPF and NHFLOW option 3 switches the 3D output off. SFLOW writes 2D VTP files instead (`REEF3D_SFLOW_VTP_FSF` and `REEF3D_SFLOW_VTP_BED`, see P 181), which are only written for P 10 = 1.

0 OFF

1 VTU (unstructured grid)

2 VTS (structured grid)

3 VTR (rectilinear grid, CFD only)

default: 1

● ● ● ●

P 12 int Terminal and log print frequency
Writes the iteration summary to the terminal and to the log files in `REEF3D_Log` (main, max, solver and volume log) every n^{th} iteration. Use larger values to reduce terminal output for long simulations.

default: 1

● ● ● ●

P 15 int Numbering of the output files
Applies to the 3D and VTP field files and to most other numbered output files.

1 consecutive print-out counter

2 iteration number

default: 1

● ● ● ●

P 16 int Add time stamp to ParaView files
Writes the simulation time as `TimeValue` field data into the VTU/VTS/VTR and VTP files, so that ParaView shows the physical time instead of the file number. For the 3D files the sediment time is used when P 34 > 0.

0 OFF

1 ON

default: 0

○ ○ ● ○

P 19 int Print interval for NHFLOW floating body files
Selects which print-out interval is used for the floating body VTP/STL files in `REEF3D_NHFLOW_6DOF_VTP` and `REEF3D_NHFLOW_6DOF_STL`. In CFD the floating body files always follow the 3D interval (P 20, P 30, P 35), in SFLOW always the VTP interval (P 181, P 182, P 185).

1 3D (VTU) interval: P 20, P 30 or P 35

2 VTP interval: P 181, P 182 or P 185

default: 2

○ ● ● ●

P 20 int Print 3D results every n^{th} iteration
Writes the 3D field files (see P 10) into `REEF3D_<M>_VTU`. Only active if P 30 and P 34 are negative. Choose one of P 20, P 30, P 34 or P 35. For SFLOW use P 181. Also sets the interval of the floating body files in CFD (and NHFLOW with P 19 = 1).

default: -10 (inactive)

○ ○ ○ ●

P 21 int Print time-averaged velocity, pressure and temperature
Adds `velocity_mean`, `pressure_mean` and, with heat transfer (H 10), `T_mean` to the CFD VTU files. Averaging starts at the time given with P 22.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 22 double Start time for time averaging
Simulation time in s after which the averaging for P 21 starts. Set it after the initial transients have left the domain.

default: 0.0

● ● ● ●

P 23 int Print test array
Adds the internal `test` array to the 3D files and the VTP files. Used for debugging and development.

0 OFF
1 ON
default: 0

○ ○ ○ ●

P 24 int Print density to the VTU file
Adds the density `rho` in kg/m^3. Ignored for multiphase flow (F 300 > 0), which writes its own density field.

0 OFF
1 ON
default: 0

○ ● ● ●

P 25 int Print solid to the VTU file
Adds the solid level set function `solid`. In NHFLOW, the `Heaviside` function of the solid is written as well.

0 OFF
1 ON
default: 0

○ ○ ● ●

P 26 int Print suspended sediment concentration to the VTU file
Adds `ST_conc`. Use together with sediment transport (S 10).

0 OFF
1 ON
default: 0

○ ○ ○ ●

P 27 int Print topo to the VTU file
Adds the bed level set function `topo`. Set to 1 automatically when sediment transport is active (S 10 $\geq$ 1) or when a topography is read from the DIVEMesh grid file.

0 OFF

1 ON

default: 0

● ○ ● ●

P 28 int Print floating body level set function
Adds the floating body level set function (`floating` in the 3D files, `fb` in the SFLOW and NHFLOW free surface VTP files).

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 29 int Print wall distance to the VTU file
Adds `walldist`, the distance to the nearest wall in m.

0 OFF

1 ON

default: 0

○ ● ● ●

P 30 double Print 3D results every *t* seconds
Time-based alternative to P 20, writes into `REEF3D_<M>_VTU`. Only active if P 34 is negative. Choose one of P 20, P 30, P 34 or P 35. For SFLOW use P 182.

default: -1.0 (inactive)

○ ○ ● ●

P 34 double Print 3D results every *t* seconds of sediment time
Uses the morphological (sediment) time instead of the hydrodynamic time. Only active if P 30 is negative. Choose one of P 20, P 30, P 34 or P 35.

default: -1.0 (inactive)

○ ● ● ●

P 35 * double t_{start}, double t_{end}, double Δt Print 3D results in a time window
Writes the 3D files every Δt seconds between t_{start} and t_{end} (all in s). Several windows can be defined; they act in addition to P 20 or P 30. In CFD the floating body files follow these windows as well.

default: na ; na ; na

● ● ● ●

P 40 int Print state files
Writes binary state files into `REEF3D_<M>_STATE` with the interval of P 41 or P 42. They are used for restart (see I 40 for full initialization from a state file), for hydrodynamic coupling (HDC, e.g. SFLOW or FNPF to CFD; see the H options in the DIVEMesh manual for building the HDC input from state files) and for grid nesting. The CFD state file contains the velocities, pressure, level set function, k, eddy viscosity, ϵ or ω, topo, cbed and conc. Related commands: P 43 to P 47.

0 OFF

1 ON

default: 0

● ● ● ●

P 41 int Print state files every n^{th} iteration
Only used if P 42 is negative. With the default, a state file is written every iteration when P 40 = 1.

default: 1

● ● ● ●

P 42 double Print state files every t seconds
Time-based alternative to P 41. For hydrodynamic coupling, match the interval to the time step expected by the receiving simulation (see I 241).

default: -1.0 (inactive)

● ● ● ●

P 43 double x_{start}, double x_{end}, double y_{start}, double y_{end} Print area for state files
Restricts the state files of P 40 to the given horizontal area in m. Reduces the file size when only a part of the domain is needed, e.g. for HDC or nesting. Without P 43 the whole domain is written.

default: na ; na ; na ; na

○ ● ● ○

P 44 int Components of the FNPF and NHFLOW state files
For FNPF the options below apply. For NHFLOW, η and the velocities are always written; P 44 = 1 adds the pressure.

1 η, free surface potential, velocities (standard for HDC)

2 η, free surface potential, velocities, 3D potential

3 η, free surface potential (more efficient for FNPF nesting)

default: 1

● ● ● ●

P 45 int State file type

1 single file per print-out (numbered according to P 15)

2 continuous file (one file per processor)

default: 2

● ● ● ●

P 46 int i_{start}, int i_{end} Iteration window for state files
State files (P 40, P 41) are only written for iterations $i_{start} \leq n < i_{end}$.

default: na ; na (inactive)

● ● ● ●

P 47 int t_{start}, int t_{end} Time window for state files
Intended to restrict the time-based state files (P 40, P 42) to $t_{start} \leq t < t_{end}$. Note: the code currently reads both values as integers and compares them with the iteration number, not the simulation time.

default: na ; na (inactive)

● ● ● ●

P 50 * double x, double y Wave gauge for the theoretical wave
Location in m. Records the free surface elevation of the wave theory selected for wave generation every time step into `REEF3D_<M>_WSF/REEF3D-<M>-WSF-HG-THEORY.dat`. Requires wave generation (B 90). Useful for comparing with P 51 at the same locations.

default: na ; na

● ● ● ●

P 51 * double x, double y Wave gauge

Location in m. Records the simulated free surface elevation every time step into `REEF3D_<M>_WSF/REEF3D-<M>-WSF-HG.dat`, one column per gauge. In CFD the surface is found from the level set function (from the VOF function for F 80 = 4).

default: na ; na

● ● ● ●

P 52 * double y Free surface line in x-direction

y-location in m of a line along the x-direction. The free surface along the line is written into `REEF3D_<M>_WSFLINE`, one file per print-out with the interval of P 54 or P 55.

default: na

● ● ● ●

P 53 int Add the theoretical wave to the free surface line files

Adds the free surface of the wave theory to the files of P 52 and P 56. Requires wave generation (B 90).

0 OFF

1 ON

default: 0

● ● ● ●

P 54 int Print line output every n^{th} iteration

Interval for the free surface lines (P 52, P 56), the CFD line probes (P 62), the SFLOW depth-averaged probes (P 63) and the NHFLOW profiles and line probes (P 67, P 144). Only used if P 55 is negative (P 63 always uses P 54).

default: 10

● ● ● ●

P 55 double Print line output every t seconds

Time-based alternative to P 54 for P 52, P 56, P 62, P 67 and P 144.

default: -1.0 (inactive)

● ● ● ●

P 56 * double x Free surface line in y-direction
x-location in m of a line along the y-direction. Written into `REEF3D_<M>_WSFLINE_Y` with the interval of P 54 or P 55.

default: na

● ● ● ●

P 58 * double x, double y, double T Theoretical wave time series
Writes the free surface elevation of the wave theory at (x, y) in m from $t = 0$ to T in s, sampled every 0.1 s, into `REEF3D_Log-Wave/REEF3D-Wave-Timeseries-<n>.dat`. The series is computed once at the start of the simulation. Requires wave generation (B 90); not written for B 92 = 20, 21, 22, 23 and 61. If no P 58 is given, one series at $(0, 0)$ with $T = 12800$ s is written.

default: 0.0 ; 0.0 ; 12800.0

○ ● ○ ○

P 59 int Print breaking wave log
Writes the time and the (x, y) location of every cell flagged as breaking, every time step, into `REEF3D_FNPF_Breaking_Log`.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 61 * double x, double y, double z Point probe
Location in m. Records u, v, w, p, k, ϵ or ω and the eddy viscosity every time step into `REEF3D_CFD_ProbePoint/REEF3D-CFD-Probe-Point-<n>.dat`.

default: na ; na ; na

○ ○ ○ ●

P 62 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Line probe
Start and end point of the line in m. Writes x, y, z, u, v, w, p, k, ω, eddy viscosity and level set function along the line into `REEF3D_CFD_ProbeLine`, one file per print-out with the interval of P 54 or P 55.

default: na ; na ; na ; na ; na ; na

● ○ ○ ○

P 63 * double x, double y Depth-averaged point probe
Location in m. Records the depth-averaged velocities U, V, W, the pressure, η and the bed level every P 54 iterations into `REEF3D_SFLOW_ProbePoint/REEF3D-SFLOW-Probe-Point-<n>.dat`.

default: na ; na

○ ○ ● ●

P 64 * double x, double y, double z Pressure probe
Location in m. Records the pressure every time step into `REEF3D_CFD_PressureProbe/REEF3D-CFD-Probe-Pressure-<n>.dat` (CFD) or `REEF3D_NHFLOW_ProbePoint/REEF3D-NHFLOW-Probe_Press-<n>.dat` (NHFLOW).

default: na ; na ; na

○ ● ● ●

P 65 * double x, double y, double z Velocity probe
Location in m. Records u, v, w (FNPF also the potential) every time step into `REEF3D_<M>_ProbePoint/REEF3D-<M>-Vel-Probe-<n>.dat`.

default: na ; na ; na

○ ● ● ●

P 66 * double x, double y, double z Velocity probe for the theoretical wave
Location in m. Records the velocities of the wave theory every time step into `REEF3D_<M>_ProbePoint/REEF3D-<M>-Vel-Probe-Theory-<n>.dat`. Requires wave generation (B 90).

default: na ; na ; na

○ ○ ● ○

P 67 * double x, double y Vertical velocity profile
Location in m. Writes the vertical profile of the horizontal velocity u over z into `REEF3D_NHFLOW_U-Profile/REEF3D-NHFLOW-U-Profile-<n>-<i>.dat` (with i the iteration number), one file per print-out with the interval of P 54 or P 55.

default: na ; na

○ ○ ○ ●

P 71 int Print viscosity to the VTU file
Adds the molecular `viscosity` field in m^2/s.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 72 int Print VOF function to the VTU file
Adds the volume fraction as point data (`VOF`) and as cell data (`VOF_C`).

0 OFF

1 ON

default: 0

○ ○ ● ○

P 74 int Print sigma-coordinate vertical velocity to the VTU file
Adds `omega_sig`, the vertical velocity relative to the sigma levels in m/s.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 75 int Print vorticity to the VTU file

0 OFF

1 ON

default: 0

● ○ ● ●

P 76 int Print sediment bedload: q_{be}, q_b
Written to the CFD VTU and topo VTP files (P 190) and to the bed VTP files of SFLOW and NHFLOW. Use together with sediment transport (S 10).

0 OFF

1 ON

default: 0

● ○ ● ●

P 77 int Print bed slope sediment parameters: $\alpha, \beta, \theta, \gamma, \phi$
Written to the same files as P 76.

0 OFF

1 ON

default: 0

● ○ ● ●

P 78 int Print sediment parameters: dh, bedchange, reduce, threshold, slideflag
Written to the same files as P 76.

0 OFF

1 ON

default: 0

● ○ ● ●

P 79 int Print bed shear stress
Writes the effective and the critical value of the selected quantity. Written to the same files as P 76.

0 OFF

1 bed shear stress

2 shear velocity

3 Shields parameter

default: 0

○ ● ● ○

P 80 int Print P 85 and P 88 results every n^{th} iteration

default: 1

○ ○ ● ●

P 81 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Force calculation box
Box in m. Integrates pressure and viscous forces on the solid surfaces (solids and topography from DIVEMesh) inside the box and writes F_x, F_y, F_z in N every time step into `REEF3D_CFD_Force/REEF3D_CFD_Force-<n>.dat` or `REEF3D_NHFLOW_Force/REEF3D_NHFLOW_Force-<n>.dat`. The triangulated surface is written once at the start into `REEF3D_SOLID` (CFD) or `REEF3D_NHFLOW_SOLID`. See also P 82, P 91 and P 92.

default: na ; na ; na ; na ; na ; na

○ ○ ● ●

P 82 int Add eddy viscosity to the viscous forces
Includes the turbulent eddy viscosity in the viscous part of the force calculation of P 81 and of the floating body forces (6DOF).

0 OFF

1 ON

default: 0

○ ● ● ○

P 85 * double x_c, double y_c, double r, double C_d, double C_m Morison force on a vertical cylinder
Centre and radius in m, drag and inertia coefficients. Computes the horizontal forces F_x, F_y in N from the Morison equation using the simulated kinematics (ALE approach) every P 80 iterations. Output: `REEF3D_FNPF_Force_ALE/REEF3D_FNPF_ALE_Force-<n>.dat` or `REEF3D_NHFLOW_Force_ALE/REEF3D_NHFLOW_ALE_Force-<n>.dat`.

default: na ; na ; na ; na ; na

○ ● ○ ○

P 88 * double x, double y Print wave kinematics
Location in m. Writes the time and, for each vertical level, z, u, $\partial u/\partial t$, v, $\partial v/\partial t$ (binary, double precision, after a one-line text header) every P 80 iterations into `REEF3D_FNPF_Kinematics/REEF3D_FNPF-Kinematics-<n>.dat`.

default: na ; na

○ ○ ● ●

P 91 double Pressure sampling distance for the force calculation
The pressure is interpolated at a distance of P 91 times the local cell size from the solid surface, along the surface normal. Used by P 81 and by the CFD floating body forces.

default: 0.25

○ ○ ○ ●

P 92 int Force from water or water and air combined

For P 81. With 0, only surface elements in water or close to the free surface (level set $> -1.6\Delta x$) contribute.

0 force only from water

1 force from water and air combined

default: 0

○ ○ ○ ●

P 101 double x_m, double y_m, double z_s, double z_e, double r_1, double r_2 Print sloshing force and moment

Activates the output of the pressure forces on the left and right tank walls and on the bottom and the moment, every time step, into `REEF3D_CFD_Force/REEF3D-CFD-Sloshing-Force.dat`. The moment is taken about the rotation point of B 192. The six values must be given but are currently not used in the calculation.

default: na ; na ; na ; na ; na ; na (inactive)

● ● ● ○

P 110 int Print significant wave height

Computes $H_s = 4\sqrt{\mathrm{var}(\eta)}$ in m at every surface point from the start time of P 111 and adds the field `Hs` to the VTU and free surface VTP files.

0 OFF

1 ON

default: 0

● ● ● ○

P 111 double Start time for the significant wave height calculation

Simulation time in s after which the statistics for P 110 are collected. Set it after the initial transients.

default: 0.0

● ○ ● ●

P 121 * double x, double y Bed level gauge

Location in m. Records the bed level at every sediment time step into `REEF3D_<M>_Sediment/REEF3D-<M>-Sediment-Point.dat`. Use together with sediment transport (S 10).

default: na ; na

● ○ ● ●

P 122 int Maximum bed change gauge

Writes the maximum bed change into `REEF3D_<M>_Sediment/REEF3D-<M>-Sediment-Max.dat`.

0 OFF

1 ON

default: 0

● ○ ● ●

P 123 * double *y* Bed profile in x-direction

y-location in m of a line along the x-direction. The bed level along the line is written into `REEF3D_<M>_Sediment/Line`, one file per sediment time step.

default: na

● ○ ● ●

P 124 * double *x* Bed profile in y-direction

x-location in m of a line along the y-direction. Written into `REEF3D_<M>_Sediment/Line`, one file per sediment time step.

default: na

● ○ ● ●

P 125 * double *x*, double *y* Bed shear stress gauge

Location in m. Records the bed shear stress in N/m^2 into `REEF3D_<M>_Sediment/REEF3D-<M>-Sediment-Bedshear.dat`.

default: na ; na

● ○ ● ●

P 126 int Maximum bed shear stress print-out

Writes the maximum bed shear stress over time into `REEF3D_NHFLOW_Sediment/REEF3D-NHFLOW-Sediment-Bedshear-Max.dat` for NHFLOW and into `REEF3D_CFD_Sediment/REEF3D-CFD-Sediment-Bedshear-Max.dat` for the other models.

0 OFF

1 ON

default: 0

○ ○ ● ○

P 131 int Print maximum wet-dry state (inundation) into the free surface VTP file

Adds `wetdry_max`, which marks every cell that has been wet at least once. Requires P 180 = 1.

0 OFF

1 ON

default: 0

○ ○ ● ○

P 133 * double y Run-up gauge in x-direction

y-location in m of a cross section along the x-direction. Records the x-position and elevation of the wet-dry front (the wet cell furthest in positive x-direction) every time step into `REEF3D_NHFLOW_RUNUP/REEF3D-NHFLOW-runup-x.dat`.

default: na

○ ○ ● ○

P 134 * double y Maximum run-up gauge in x-direction

As P 133, but records the maximum run-up reached so far and the time it occurred into `REEF3D_NHFLOW_RUNUP/REEF3D-NHFLOW-runup-max-x.dat`. The file is overwritten every time step.

default: na

○ ● ○ ○

P 140 * double x_c, double y_c Run-up on a vertical cylinder

Centre of the cylinder in m, radius from P 141. Evaluates six empirical run-up formulas (R1 to R6, e.g. MacCamy and Fuchs, Hallermeier, Peng et al.) from the local wave kinematics every time step and writes them into `REEF3D_FNPF_Runup/REEF3D_Runup-<n>.dat`.

default: na ; na

○ ● ○ ○

P 141 double Cylinder radius for the run-up gauges

Radius in m, used for all P 140 gauges.

default: 0.0

○ ○ ● ○

P 144 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, int n Depth-averaged velocity line probe
Start and end point of the line in m and number of points along the line. Writes the depth-averaged velocities along the line into `REEF3D_NHFLOW_DEP-AVG-VEL-LINE`, one file per print-out with the interval of P 54 or P 55.

default: na ; na ; na ; na ; na

○ ○ ● ○

P 145 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, int n, double t_{start}, double t_{end} Depth- and time-averaged velocity line probe
As P 144, but the depth-averaged velocities are also averaged in time between t_{start} and t_{end} in s. One file is written at t_{end} into `REEF3D_NHFLOW_DEP_TIME_AVG_VEL_LINE`.

default: na ; na ; na ; na ; na ; na ; na

○ ○ ● ○

P 146 * double y, double t_{start}, double t_{end} Time-averaged free surface line in x-direction
y-location of the line in m and averaging window in s. One file is written at t_{end} into `REEF3D_NHFLOW_TIME_AVG_WSFLINE`.

default: na ; na ; na

○ ○ ● ○

P 147 * double x, double t_{start}, double t_{end} Time-averaged free surface line in y-direction
x-location of the line in m and averaging window in s. One file is written at t_{end} into `REEF3D_NHFLOW_TIME_AVG_WSFLINE_Y`.

default: na ; na ; na

○ ○ ● ○

P 148 * double x, double y, double t_{start}, double t_{end} Time-averaged horizontal velocity profile
Location in m and averaging window in s. Writes the time-averaged vertical profiles of u, v and of the horizontal velocity magnitude once at t_{end} into `REEF3D_NHFLOW_TIME_AVG_VEL_PROFILE`.

default: na ; na ; na ; na

○ ○ ○ ●

P 151 int Type of the DIVEMesh data array

When the DIVEMesh grid file contains an additional data array, it is read automatically and written to the CFD VTU files as `data`. P 151 selects how the values are interpreted.

1 scalar

2 signed distance function (converted from a surface elevation)

default: 1

○ ○ ○ ●

P 152 int Type of boundary condition for the DIVEMesh data array

Ghost cell treatment of the data array of P 151.

1 x-velocity

2 y-velocity

3 z-velocity

4 scalar with Neumann boundary conditions

default: 4

○ ○ ○ ●

P 166 int Print discharge of the P 167 cross sections to the terminal

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 167 * double x Discharge gauge

x-location in m of a cross section normal to the x-direction. The water discharge in m^3/s through the cross section is written every time step into `REEF3D_Log/REEF3D-discharge_x.dat`.

default: na

○ ○ ○ ●

P 168 * double x, double z_{start}, double z_{end} Discharge window gauge

As P 167, but only the part of the cross section between z_{start} and z_{end} in m is evaluated. Output: `REEF3D_Log/REEF3D-discharge_window_x.dat`.

default: na ; na ; na

○ ● ● ●

P 180 int Print free surface VTP files

Writes the free surface as 2D VTP files into `REEF3D_CFD_FSF`, `REEF3D_FNPF_VTP_FSF` or `REEF3D_NHFLOW_VTP_FSF`, with the interval of P 181, P 182, P 183, P 184 or P 185, and once at the end of the simulation. NHFLOW with sediment transport also writes the bed into `REEF3D_NHFLOW_VTP_BED`. SFLOW always writes its VTP files and does not need P 180.

0 OFF

1 ON

default: 0

● ● ● ●

P 181 int Print VTP files every n^{th} iteration

Interval for the free surface files of P 180. For SFLOW this is the main output: the free surface and bed VTP files in `REEF3D_SFLOW_VTP_FSF` and `REEF3D_SFLOW_VTP_BED`. Also used for the floating body files in SFLOW and NHFLOW (see P 19). Only active if P 182 is negative (and P 183 for NHFLOW).

default: -10 (inactive)

● ● ● ●

P 182 double Print VTP files every t seconds

Time-based alternative to P 181, for the same files (free surface for CFD and FNPF, free surface and bed for SFLOW and NHFLOW).

default: -1.0 (inactive)

○ ○ ● ○

P 183 double Print VTP files every t seconds of sediment time

As P 182, but based on the sediment (morphological) time. Only active if P 182 is negative.

default: -1.0 (inactive)

○ ● ● ●

P 184 * int i_{start}, int i_{end}, int Δi Print VTP files in an iteration window

Writes the free surface files of P 180 every Δi iterations between iteration i_{start} and i_{end}. Several windows can be defined; they act in addition to P 181 or P 182.

default: na ; na ; na

● ● ● ●

P 185 * double t_{start}, double t_{end}, double Δt Print VTP files in a time window
Writes the free surface files of P 180 every Δt seconds between t_{start} and t_{end} (in s). Several windows can be defined. In SFLOW only the floating body files use P 185.

default: na ; na ; na

○ ○ ● ○

P 186 * double t_{start}, double t_{end} Time-averaged free surface VTP file
Averages velocity, η and water level between t_{start} and t_{end} in s and writes one VTP file at t_{end} into `REEF3D_NHFLOW_VTP_TIME_AVG_FSF`. Several windows can be defined. Works independently of P 180.

default: na ; na

○ ○ ○ ●

P 190 int Print topo VTP files
Writes the bed surface as VTP files into `REEF3D_CFD_Topo`, with the interval of P 191, P 192, P 194 or P 195. Sediment fields are added with P 76 to P 79.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 191 int Print topo VTP files every n^{th} iteration
Only active if P 192 is negative. Note: the code currently also requires P 182 to be negative.

default: -10 (inactive)

○ ○ ○ ●

P 192 double Print topo VTP files every t seconds
Time-based alternative to P 191. Avoid combining it with P 182, as both currently share one print timer.

default: -1.0 (inactive)

○ ○ ○ ●

P 194 * int i_{start}, int i_{end}, int Δi Print topo VTP files in an iteration window
Writes the topo files every Δi iterations between iteration i_{start} and i_{end}. Several windows can be defined.

default: na ; na ; na

○ ○ ○ ●

P 195 * double t_{start}, double t_{end}, double Δt Print topo VTP files in a time window
Writes the topo files every Δt seconds between t_{start} and t_{end} (in s). Several windows can be defined.

default: na ; na ; na

○ ○ ○ ●

P 230 * double x Print FlowFile
x-location in m of a vertical cross section in a 2D CFD simulation. The flow in the cross section is written every time step into `REEF3D_FlowFile` (one header file and one file per time step for each location). FlowFiles can be used as inflow boundary condition of another simulation, see I 230.

default: na

○ ● ○ ○

P 240 * double x Print PotentialFile
x-location in m for PotentialFile output from 2D FNPF simulations. Note: this output is currently not functional in the code.

default: na

○ ○ ○ ●

P 250 int Print CPM sediment variables to the VTU file
Adds `CPM_Ts`, `CPM_Tau`, `CPM_press` and `CPM_test`. Only for the particle-based sediment model (S 10 $\geq$ 1 with Q 10 = 1).

0 OFF

1 ON

default: 0

○ ● ○ ○

P 310 int Print breaking flag to the VTU file
Adds `breaking`, which marks the cells where wave breaking is detected.

0 OFF

1 ON

default: 0

○ ● ○ ○

P 311 int Print breaking viscosity to the VTU file
Adds `vb`, the artificial viscosity used by the breaking model, in m^2/s.

0 OFF

1 ON

default: 0

○ ○ ○ ●

P 351 * double *x*, double *y* Wave gauge for multiphase level set 1
Location in m. For multiphase flow (F 300 > 0), records the interface of the first level set function every time step into `REEF3D_CFD_WSF/REEF3D-CFD-WSF-HG-1.dat`.

default: na ; na

○ ○ ○ ●

P 352 * double *x*, double *y* Wave gauge for multiphase level set 2
As P 351 for the second level set function. Output: `REEF3D_CFD_WSF/REEF3D-CFD-WSF-HG-2.dat`.

default: na ; na

6.12 Q :: Particles (MP-PIC sediment)

The Q commands control the particle-based sediment transport model of REEF3D::CFD. It is activated with S 10 1 (or S 10 2 for a porous VRANS bed) together with Q 10 1; with Q 10 0 the Exner-based sediment model is used. Instead of solving the Exner equation, the sediment below the initial bed is represented by computational parcels, which are seeded in all cells below the bed (Q 24, Q 29). Each parcel carries the grain size d_{50} (S 20) and density (S 22) and represents $(1-n)\Delta x^3/(N_{ppc}\,\pi d_{50}^3/6)$ grains, with the porosity n (S 24) and the number of particles per cell N_{ppc} (Q 24). The parcels are moved by the drag of the interpolated fluid velocity; with Q 11 2 the multiphase particle-in-cell (MP-PIC) formulation

adds the gradient of a lithostatic pressure and of a continuum inter-particle stress following Snider (2001), $\tau = P_s\theta^\beta / \max(\theta_{cp}-\theta, \epsilon(1-\theta))$, where θ is the particle volume fraction. The bed level is updated from the net number of parcels entering or leaving each water column. The start of the particle transport follows S 41/S 43/S 45/S 47, the sediment time step S 13 – S 15; the bed shear stress (S 16, S 30), the slope reduction (S 80) and the sandslide (S 90) options of section S also apply.

Note: the constants of the Snider stress model are currently fixed in the code ($P_s = 100\,Pa$, $\beta = 5$, $\epsilon = 10^{-7}$, $\theta_{cp} = 1 - n + 0.05$).

Valid for:

S	F	N	C
○	○	○	●

Q 10 int Sediment model type

Requires S 10 > 0. With Q 10 1 the fluid density used in the pressure solver is additionally blended to the sediment density S 22 below the bed level (standard level set setup without 6DOF, heat, concentration or rheology).

0 Exner-based sediment model (section S)

1 particle-based sediment model

default: 0

○	○	○	●

Q 11 int Particle transport model

1 plain particle transport: drag against the frictional resistance Q 30, two-stage Runge-Kutta time stepping

2 MP-PIC with Snider (2001) inter-particle stress and lithostatic pressure, explicit Euler time stepping

default: 1

○	○	○	●

Q 24 int Number of particles per cell

Number of parcels seeded per cell below the initial bed. It must be set to a positive value, because the default 0 is invalid. For the regular seeding types (Q 29 1 and 2) the parcels are placed on a regular sub-grid with $\mathrm{round}(Q24^{1/3})$ parcels per direction, e.g. Q 24 8 gives $2 \times 2 \times 2$ parcels.

default: 0 (must be set)

○ ○ ○ ●

Q 29 int Particle seeding type

Parcels above the bed or inside solid bodies are removed after seeding.

1 regular spacing

2 regular spacing with random jitter of up to $0.1\Delta x$

3 fully random

default: 1

○ ○ ○ ●

Q 30 double Friction factor for the particle resistance force

Dimensionless friction coefficient of the resistance force $F_s = Q30\,(\rho_s - \rho_w)\,|g|\,\pi d_{50}^3/6$ of a grain. The driving drag force is reduced by F_s and set to zero where it is smaller than F_s. Used with Q 11 1.

default: 0.58

○ ○ ○ ●

Q 73 * double value, double distance, double β, double x_0, double y_0 Relaxation zone for the particle forces along a line

The forces on the parcels are zero on the straight line through (x_0, y_0) in m and increase smoothly to their full value at the given distance in m from the line, so that the particles are held in place near e.g. the inflow boundary. The line angle β is given in degrees; $\beta = 0$ gives a line parallel to the y-axis. The first value is currently not used, but must be given. Several lines can be combined.

default: not set

○ ○ ○ ●

Q 180 int Print particles

Writes the parcels as ParaView .vtp/.pvtp files to the folder REEF3D_CFD_CPM_Particle. The output interval is given with either Q 181 or Q 182.

0 OFF

1 ON

default: 0

○ ○ ○ ●

Q 181 int Particle print interval in iterations

Prints the particles every Q 181 iterations. Only active if Q 181 > 0 and Q 182 < 0.

default: -10 (inactive)

○ ○ ○ ●

Q 182 double Particle print interval in time

Prints the particles every Q 182 seconds of simulation time. Only active if Q 182 > 0 and Q 181 < 0.

default: -1.0 *s* (inactive)

6.13 S :: Sediment

Valid for:

S	F	N	C
●	○	●	●

S 10 int Sediment transport module

Switches on morphodynamics. Any value > 0 also activates the topography handling (G 1 is set to 1) and the printing of the bed (P 27 is set to 1). In REEF3D::CFD the particle-based sediment model is selected with Q 10 1 instead of the default Exner-based model (see section Q).

0 OFF

1 ON, impermeable sediment bed (SFLOW, NHFLOW, CFD)

2 ON, porous sediment bed with VRANS (CFD only, requires A 10 6; porosity S 24, VRANS coefficients S 26)

12 ON, fully coupled RK2 sediment time stepping (NHFLOW only)

default: 0

● ○ ● ●

S 11 int Bedload transport formula

Equilibrium bedload transport capacity $q_{b,e}$ used in the Exner equation. All formulas use d_{50} (S 20), the sediment density (S 22) and the critical Shields parameter (S 30).

0 OFF

1 van Rijn

2 Meyer-Peter Müller

3 Engelund and Fredsøe (dynamic friction coefficient S 18)

4 Engelund and Hansen

5 Einstein-Brown

default: 0

○ ○ ● ●

S 12 int Suspended sediment transport
Solves an implicit advection-diffusion equation for the suspended sediment concentration with the settling velocity from S 23/S 25. The near-bed reference concentration follows van Rijn; its reference height is set with S 61, the coupling with the Exner equation with S 62.

0 OFF

1 ON, van Rijn reference concentration

default: 0

● ○ ● ●

S 13 double Maximum sediment time step
Upper limit of the morphodynamic time step in *s* for S 15 0, and the fixed time step for S 15 2.

default: 10.0 *s*

● ○ ● ●

S 14 double Courant number for the sediment time step
The morphodynamic time step is decoupled from the hydrodynamic time step (see S 15). It is determined from a Courant criterion on the rate of bed elevation change, $\Delta t_{sed} = S14 \cdot \Delta x / \max|\partial z_b / \partial t|$. For the particle model (Q 10 1) the maximum horizontal particle velocity is used instead of the bed change rate. Recommended range: 0.1 – 0.5.

default: 0.3

● ○ ● ●

S 15 int Sediment time step selection
All options are multiplied by the factor S 17 (Exner model only).

0 adaptive: minimum of S 13 and the Courant criterion S 14

1 adaptive: minimum of the flow time step and the Courant criterion S 14

2 fixed time step S 13

3 flow solver time step (Exner model only)

default: 0

● ○ ● ●

S 16 int Bed shear stress formulation
The available options depend on the flow model. Log-law based options use the effective roughness $k_s = S21 \cdot d_{50}$ (see also S 36).

1 wall function / log-law, velocity based (SFLOW: Manning formulation with k_s)

2 log-law, velocity based, limited from below by the turbulent kinetic energy (SFLOW: depth-averaged log-law)

3 viscosity based: $\tau = \rho(\nu + \nu_t)\, u/\Delta z$ (SFLOW: depth-averaged log-law)

4 turbulent kinetic energy based: $\tau = 0.3\rho k$

5 geometric mean of options 1 and 4 (CFD only)

6 depth-averaged velocity, Chezy (CFD) or Manning (NHFLOW) formula

7 log-law with the grain roughness, horizontal velocity only (CFD only)

8 log-law with iterative roughness correction for transitional flow (CFD only)

default: 1

● ○ ● ●

S 17 double Factor for the sediment time step
Dimensionless factor applied to the sediment time step selected with S 15 (for S 15 1 and 3 it multiplies the flow time step). Values > 1 accelerate the morphodynamics relative to the hydrodynamics (morphological acceleration). Not used by the particle model and not applied when S 29 is used.

default: 1.0

● ○ ● ●

S 18 double Dynamic friction coefficient μ_d for the Engelund and Fredsøe bedload formula
Only used with S 11 3.

default: 0.823

● ○ ● ●

S 19 double Maximum simulated sediment time
The simulation stops when the accumulated morphodynamic time reaches S 19 in s (in addition to the limits N 41 and N 45). The default is a large sentinel value, i.e. inactive.

default: $1.0 \cdot 10^{19}$ s

● ○ ● ●

S 20 double Sediment grain size d_{50}

Median grain diameter in m. It enters the transport formulas, the critical bed shear stress, the bed roughness (S 21) and, for the particle model, the parcel size.

default: 0.001 m

● ○ ● ●

S 21 double Roughness factor

The equivalent sand roughness of the sediment bed is $k_s = S21 \cdot d_{50}$. It is used in the bed shear stress calculation (S 16) and, with S 28 1, in the wall boundary condition of the flow over the bed.

default: 3.0

● ○ ● ●

S 22 double Sediment density

default: 2650.0 kg/m^3

● ○ ● ●

S 23 double Sediment settling velocity

Settling velocity w_s in m/s used for suspended sediment transport. Only used if the automatic calculation is switched off (S 25 0).

default: 0.001 m/s

● ○ ● ●

S 24 double Porosity of the sediment bed

Volume fraction of pores n in the bed. It enters the Exner equation as $1/(1-n)$, the VRANS porosity for S 10 2, and the parcel size and packing limit of the particle model.

default: 0.5

● ○ ● ●

S 25 int Automatic settling velocity calculation

When ON, w_s is calculated from d_{50}, the sediment and water densities and the viscosity W 2: Stokes law for $d_{50} < 0.1\,mm$, the van Rijn formula for $0.1 \leq d_{50} < 1\,mm$ and $1.1\sqrt{g d_{50}(\rho_s/\rho_w - 1)}$ for $d_{50} \geq 1\,mm$.

0 OFF, use S 23

1 ON

default: 1

○ ○ ○ ●

S 26 double α, double β VRANS resistance coefficients for the porous sediment bed
Dimensionless linear (α) and non-linear (β) resistance coefficients of the VRANS porous media model inside the sediment bed. Only used with S 10 2 and A 10 6.

default: 650.0 ; 2.2

○ ○ ○ ●

S 27 int Number of Exner sub-iterations per sediment step
Number of times the Exner equation (and the sandslide algorithm) is advanced with Δt_{sed} during one call of the sediment module, without updating the bed shear stress and transport rates in between.

default: 1

○ ○ ● ●

S 28 int Roughness of the sediment bed in the flow boundary condition
Selects the roughness used in the wall boundary condition of the hydrodynamics (velocities and turbulence) at the sediment bed when S 10 > 0.

0 $k_s = d_{50}$

1 $k_s = S21 \cdot d_{50}$

default: 1

● ○ ● ●

S 29 double t_s, double t_e, double Δt_s, double Δt_e Sediment time step ramp
Linear ramp of the maximum sediment time step from Δt_s to Δt_e (in s) between the sediment times t_s and t_e (in s). Before t_s the value Δt_s, after t_e the value Δt_e is used. The ramped value limits the Courant-based time step (S 14); S 15 and S 17 are then ignored. Exner model only.

default: not set (inactive)

● ○ ● ●

S 30 double Critical Shields parameter
θ_{cr} for incipient motion on a flat bed. The critical bed shear stress is $\tau_{cr} = r\,\theta_{cr}(\rho_s - \rho_w)g d_{50}$ with the slope reduction factor r (S 80).

default: 0.047

● ○ ● ●

S 31 int Type of Exner formulation
Determines how the transport direction enters the divergence of the bedload flux.

1 scalar q_b, upwinding with the flow direction at the cell faces

2 vector flux $(q_b s_x, q_b s_y)$ with the cell-centred flow direction

3 directional derivative of q_b along the cell-centred flow direction

default: 2

● ○ ● ●

S 32 int Exner equation discretization

1 FOU

2 CDS2

3 CDS2 HJ

4 WENO5 FLUX

5 WENO5 HJ

default: 1

● ○ ● ●

S 33 int Non-equilibrium bedload transport
With non-equilibrium transport the bedload flux q_b relaxes towards the transport capacity $q_{b,e}$ over the adaptation length L_s along the transport path, $q_b + L_s \hat{s} \cdot \nabla q_b = q_{b,e}$. The equation is solved with a Gauss-Seidel method in every sediment step (maximum number of sweeps S 49). For options 2 and 3, L_s is typically only a few grain diameters, so the result is practically identical to equilibrium transport on usual grids.

0 equilibrium transport, $q_b = q_{b,e}$

1 constant adaptation length L_s = S 40 in m

2 van Rijn (1984) saltation length $L_s = 3 d_{50} D_*^{0.6} T^{0.9}$

3 Phillips and Sutherland (1989) $L_s = 4000(\theta - \theta_{cr}) d_{50}$

4 depth scaled L_s = S 40 $\cdot h$

default: 0

○ ○ ● ●

S 34 int Type of suspended load bed change calculation
Currently without effect: the implicit suspended sediment solver used for S 12 1 does not evaluate this option.

1 bedload-suspended load interface

2 flow depth integration

default: 1

● ○ ● ●

S 35 double Exner magnification factor
Dimensionless factor multiplying the bed change rate in the Exner equation (morphological acceleration). It is also included in the bed celerity limit S 103.

default: 1.0

● ○ ● ●

S 36 int Effective bed roughness calculation

0 grain roughness only, $k_s = S21 \cdot d_{50}$

1 grain roughness plus van Rijn's bedform roughness, based on the transport stage and the water depth

default: 0

○ ○ ● ○

S 37 int Inertia contribution to the bed shear stress
Adds a bed shear stress contribution from the horizontal flow acceleration, $\tau_{acc} = \rho_w c_m c_v c_n S21\, d_{50} |\partial \mathbf{u}/\partial t|$, to the effective bed shear stress (NHFLOW only).

0 OFF

1 ON

default: 0

○ ○ ● ○

S 38 int Blend bed shear stress with shallow water formulation
In very shallow water the bed shear stress is blended with a depth-averaged Manning-type formulation (NHFLOW only). The blending starts below the water depth $h = S39 \cdot A544$; below $0.5\,S39 \cdot A544$ only the shallow water formulation is used.

0 OFF

1 ON

default: 0

○ ○ ● ○

S 39 double Blending depth factor for S 38
Dimensionless factor on the wetting and drying threshold A 544 that sets the water depth where the shallow water blending of S 38 starts. With the defaults this is $100 \cdot 0.001\,m = 0.1\,m$.

default: 100.0

● ○ ● ●

S 40 double Adaptation length for non-equilibrium bedload transport
Adaptation length L_s in m for S 33 1, or dimensionless factor on the water depth for S 33 4.

default: 0.1

● ○ ● ●

S 41 int Type of sediment start criterion
Sediment transport starts once the criterion is reached; before that only the bed shear stress is calculated.

1 number of flow iterations (S 43)

2 flow simulation time (S 45)

3 flow simulation time divided by the wave period, t/T (S 47)

default: 1

● ○ ● ●

S 42 int Type of sediment interval criterion
Interval between two calls of the sediment module (Exner model only; the particle model is called every flow time step).

1 number of flow iterations (S 44)

2 flow simulation time (S 46)

3 t/T (S 48)

default: 1

● ○ ● ●

S 43 int Number of flow iterations before sediment transport starts
Used with S 41 1. Also the start of the erosion-deposition coupling of suspended sediments (S 62 1) and of the NHFLOW RK2 sediment model (S 10 12).

default: 1000

● ○ ● ●

S 44 int Number of flow iterations between bed updates
Used with S 42 1.

default: 10

● ○ ● ●

S 45 double Flow simulation time before sediment transport starts
Used with S 41 2.

default: 1.0 s

● ○ ● ●

S 46 double Flow simulation time between bed updates
Used with S 42 2.

default: 1.0 s

● ○ ● ●

S 47 double t/T before sediment transport starts
Number of wave periods before sediment transport starts; used with S 41 3. Requires a wave generation with a defined wave period.

default: 1.0

● ○ ● ●

S 48 double t/T between bed updates
Number of wave periods between two bed updates; used with S 42 3.

default: 0.1

● ○ ● ●

S 49 int Maximum number of Gauss-Seidel sweeps for non-equilibrium bedload
Upper limit of sweeps per sediment step for S 33 > 0; the iteration stops earlier at convergence (relative tolerance 10^{-6}).

default: 200

● ○ ● ●

S 50 int Fixed bed level at inflow or outflow boundary
Boundary condition for the bed level at the inflow and outflow boundaries: a fixed bed does not change there, otherwise the bed level is extrapolated.

1 inflow fixed

2 outflow fixed

3 inflow and outflow fixed

4 none fixed

default: 4

● ○ ● ●

S 51 * double d_k, double $F_{a,k}$, double $F_{s,k}$ Sediment fraction for a graded bed
Each S 51 line defines one grain size class of a multi-fraction (graded) sediment bed: the diameter d_k in m, its initial volume fraction in the active layer $F_{a,k}$ and in the substrate $F_{s,k}$. Between 1 and 20 fractions are allowed; without S 51 the bed has a single fraction with d_{50} = S 20. The fractions of each layer are normalized to a sum of 1; if all $F_{s,k}$ are zero, the substrate gets the composition of the active layer. The bed consists of an active (mixing) layer of thickness S 52 on top of one well-mixed substrate layer of thickness S 53 (Hirano active-layer model). The bedload formula S 11 is evaluated once per fraction with d_k, i.e. with the Shields parameter $\theta_k = \tau_b/((\rho_s - \rho) g d_k)$ and the critical Shields parameter S 30 (including the slope reduction S 80) multiplied by the hiding/exposure factor (S 54), and the capacity is weighted with the local fraction in the active layer. The Exner equation, the sandslide and the bed relaxation zones act on each fraction, so that the volume of each fraction is conserved. Not available with the RK2 sediment solver S 10 12. The fractions and the local d_{50}, d_m, d_{90} and substrate thickness are added to the sediment output (ST_Fk, ST_d50, ST_dm, ST_d90, ST_Hsub); the bed volume per fraction is logged in `REEF3D_Log/REEF3D_sediment_mixture.dat`.

default: not set (single fraction, S 20)

● ○ ● ●

S 52 double Active layer thickness
Thickness L_a in m of the active layer of the graded bed (S 51). With a value ≤ 0, $L_a = 2\,d_{90}$ of the initial active-layer composition.

default: 0.0 ($2\,d_{90}$)

● ○ ● ●

S 53 double Initial substrate layer thickness
Initial thickness in m of the substrate layer below the active layer of the graded bed (S 51). The substrate grows with deposition and shrinks with erosion.

default: 10.0

● ○ ● ●

S 54 int Hiding/exposure function
Correction of the critical Shields parameter of each fraction of the graded bed (S 51) for hiding of fine grains between coarse grains and exposure of coarse grains. With option 1, $\theta_{cr,k}$ is multiplied by $\xi_k = (p_{e,k}/p_{h,k})^{-m}$ with $p_{h,k} = \sum_j F_j d_j/(d_k + d_j)$, $p_{e,k} = 1 - p_{h,k}$ and the exponent m from S 55 [Wu et al., 2000].

0 OFF

1 Wu, Wang and Jia (2000)

default: 1

● ○ ● ●

S 55 double Hiding/exposure exponent m
Exponent of the hiding/exposure function S 54 1.

default: 0.6

● ○ ● ●

S 56 int Grain size for the bed roughness of a graded bed
Diameter d in the bed roughness k_s = S 21 $\cdot d$ when a graded bed (S 51) is used. Options 1–3 use the local grain size of the active layer, so that the roughness follows the bed sorting.

0 S 20

1 local d_{50}

2 local mean diameter d_m

3 local d_{90}

default: 0

○ ○ ○ ●

S 57 double Initial flat sediment bed level
Sets a horizontal sediment bed at the elevation z = S 57 in m over the whole domain, overriding the topography from the grid. The default sentinel value means not set.

default: $-1.0 \cdot 10^{20}$ m (not set)

○ ○ ● ●

S 61 int Near-bed concentration for suspended sediment
Concentration c_b at the reference level used for the erosion-deposition balance with S 12 1.

1 concentration of the first cell above the bed

2 first cell concentration extrapolated to the reference height $2d_{50}$ with the Rouse profile

default: 1

○ ○ ● ●

S 62 int Coupling of suspended sediment with the Exner equation

1 erosion-deposition term $w_s(c_b - c_{b,e})$ is added to the Exner equation (after S 43 iterations)

2 depth-integrated suspended load flux is added to the bedload flux (computed by NHFLOW)

default: 1

● ○ ● ●

S 71 double Start of the erodible region in x-direction
The Exner equation is only solved for $S71 \leq x \leq S72$, and the slope reduction of the critical shear stress (S 80) is only applied there. The default sentinel means no limit. For smooth transitions S 73 or S 75 are preferred.

default: $-1.0 \cdot 10^{20}$ m

● ○ ● ●

S 72 double End of the erodible region in x-direction
See S 71.

default: $1.0 \cdot 10^{20}$ m

● ○ ● ●

S 73 * double z_b, double distance, double β, double x_0, double y_0 Relaxation zone for the sediment bed along a line
Inhibits sediment transport near a straight line through (x_0, y_0) in m and makes it smoothly transition to the active bed over the given distance in m. On the line the bed is held at the elevation z_b in m and the bed change, transport rates and bed shear stress are set to zero; they are blended with a smooth function to their computed values at the given distance. The line angle β is given in degrees; $\beta = 0$ gives a line parallel to the y-axis, i.e. the distance is measured in x-direction. Several lines can be combined.

default: not set

● ○ ● ●

S 74 * double x_s, double x_e, double y_s, double y_e Non-erodible rectangular zone
The bedload transport is set to zero inside the rectangle $x_s < x < x_e$, $y_s < y < y_e$ (in m). Several zones can be defined.

default: not set

● ○ ● ●

S 75 * double x_0, double distance Relaxation zone for the sediment bed in x-direction
Within the given distance in m from $x = x_0$ in m, the bed is smoothly relaxed towards its initial level and the transport rates and bed shear stress are reduced, reaching zero at x_0. This method is preferred over S 71 and S 72 because of the smooth transition of the sediment bed. Several locations can be defined.

default: not set

● ○ ● ●

S 77 double x_s, double x_e Active region of the bed update algorithms in x-direction
Restricts the bed filter (S 100), the sandslide algorithm (S 90), the relaxation S 73 and the transfer of the bed level to the CFD topography to $x_s < x < x_e$ (in m).

default: $-1.0 \cdot 10^{20}$; $1.0 \cdot 10^{20}$ m (whole domain)

● ○ ● ●

S 78 int Inflow guard
When ON, the bed is kept fixed in the cells at the inflow boundary and at inflow patches of the patchBC (B 400s).

0 OFF

1 ON

default: 1

● ○ ● ●

S 80 int Critical shear stress reduction for a sloping bed
Reduces or increases the critical bed shear stress on sloping beds with the factor r, limited to $0.01 \leq r \leq 1.5$. Uses the angle of repose S 81/S 82.

0 OFF

1 Parker and Kovacs

2 Dey empirical

3 Dey analytical

4 Fredsøe

5 Fredsøe, with longitudinal and transverse parts weighted by the flow direction

default: 0

● ○ ● ●

S 81 double Angle of repose
Mean angle of repose ϕ in degrees. The effective angle varies between $\phi \pm$ S 82 depending on the direction of the bed slope relative to the flow. Used by S 80 and the sandslide algorithm S 90.

default: 35.0 °

● ○ ● ●

S 82 double Variation of the angle of repose $\Delta\phi$
See S 81.

default: 5.0 °

● ○ ● ●

S 84 int Limiter for the critical bed shear stress reduction
Treatment of the reduction factor of S 80 where the bed is steeper than the angle of repose.

1 Fredsøe: reduction factor clipped to $0.01 \leq r \leq 1.5$

2 function based: $r = 0.1/|\gamma| + 0.1$ for slopes above the angle of repose

default: 1

● ○ ● ●

S 85 int Bedload direction correction
Corrects the bedload direction for the bed slope using the method proposed by Koch and Flokstra (1980).

0 OFF

1 ON

default: 1

● ○ ● ●

S 90 int Sandslide algorithm
Corrects bed slopes that exceed the angle of repose (S 81, S 82) after each bed update. Iterations S 91, correction factor S 92, extra angle S 93, calling interval S 94.

0 OFF

1 explicit full correction

2 explicit partial correction

3 steepest descent

4 weighted multi-direction

5 diffusion

default: 0

● ○ ● ●

S 91 int Maximum number of sandslide iterations
The iteration stops earlier when no cell needs correction.

default: 1

● ○ ● ●

S 92 double Sandslide correction factor
Dimensionless factor on the bed level correction per sandslide iteration (S 90 1 – 4).

default: 1.0

● ○ ● ●

S 93 double Delta angle for the sandslide correction
Additional angle in degrees by which an over-steep slope is flattened below the angle of repose (S 90 1 and 2).

default: 0.0 °

● ○ ● ●

S 94 int Sandslide interval

The sandslide algorithm is called every S 94 sediment iterations.

default: 1

● ○ ● ●

S 100 int Number of outer bed filter iterations

Spatial smoothing of the bed change rate before the bed update, restricted to sediment cells within S 77. Larger numbers increase the filter effect; 0 switches the filter off. The filter weight is S 102.

default: 3

● ○ ● ●

S 101 int Number of inner bed filter iterations

Corrector iterations of each outer filter iteration (Van Cittert sharpening). Larger numbers move the bed change closer to the unfiltered values; recommended range 1 – 5.

default: 5

● ○ ● ●

S 102 double Bed filter weight

Weight of the centre cell in the bed filter, $f = S102\,h + (1 - S102)\,\overline{h}_{neighbours}$. A value of 1.0 means no smoothing, smaller values smooth more.

default: 0.75

● ○ ● ●

S 103 double Bed celerity Courant number for the sediment time step

Additionally limits the sediment time step with the celerity of bed forms, $\Delta t_{sed} \leq S103\,\Delta x/c$ with $c = S35/(1-n)\,\partial q_b/\partial z_b$. Only bed level differences larger than d_{50} are considered. 0.0 switches the limit off.

default: 0.0

6.14 T :: Turbulence

Valid for:

S	F	N	C
○	○	○	●

T 10 int Turbulence model
Turbulence model of REEF3D::CFD. The turbulence models of SFLOW and NHFLOW are selected with A 260 and A 560.

0 OFF

1 k-ϵ model

2 k-ω model

12 EARSM based on k-ω model

21 URANS with k-ϵ model

22 URANS with k-ω model

31 LES with Smagorinsky SGS model

33 LES with WALE model

default: 0

S	F	N	C
○	○	○	●

T 12 int Convection discretization for the turbulence model

0 OFF

1 FOU

5 WENO HJ for non-uniform grids

55 WENO HJ

default: 5

S	F	N	C
○	○	○	●

T 21 int Type of LES filter
Used with T 10 31 and 33.

0 box

1 f1

2 f2

default: 0

○ ○ ○ ●

T 23 double C_3 coefficient for the URANS filter
Dimensionless coefficient of the URANS eddy viscosity filter of the k-ω model (T 10 22).

default: 1.0

● ○ ● ●

T 31 double Factor for the eddy viscosity limiter
Coefficient of the strain-based limiter $\nu_t \leq T31 \cdot k/|S|$ (activated with T 34 in CFD, A 564 in NHFLOW).

default: $1/\sqrt{3} \approx 0.577$

○ ○ ● ●

T 33 int Turbulent kinetic energy source term in the momentum equations
Adds the isotropic part of the Reynolds stresses, $\frac{2}{3}\nabla k$, as source term to the momentum equations (two-equation models).

0 OFF
1 ON
default: 0

○ ○ ○ ●

T 34 int Eddy viscosity limiter
Strain-based eddy viscosity limiter of the k-ω model, with the coefficient T 31.

0 OFF
1 ON
default: 1

○ ○ ○ ●

T 36 int Free surface boundary condition for the turbulent dissipation
Sets the specific dissipation near the free surface to damp the turbulence there, using the virtual origin T 37 and the interface width T 38.

0 OFF
1 ON
2 ON, including the distance from the nearest wall
3 ON, including the water depth (k-ω only)
default: 0

○ ○ ● ●

T 37 double Virtual origin for the distance y' above the free surface
Used by the free surface dissipation boundary condition (T 36 in CFD, A 567 in NHFLOW); for T 36 3 or A 567 3 it is a dimensionless factor on the water depth.

default: 0.07

○ ○ ○ ●

T 38 double Width factor for the free surface turbulence damping
Half-width ϵ of the free surface zone for T 36 and T 39, as a multiple of the average cell size.

default: 1.6

○ ○ ○ ●

T 39 int Minimum eddy viscosity at the free surface
Within the free surface zone (width T 38) the eddy viscosity of the k-ω model is limited from below by a Smagorinsky subgrid-scale eddy viscosity ($C_s = 0.2$).

0 OFF

1 ON

default: 0

○ ○ ○ ●

T 41 int k-ω stabilization
Stabilization of the eddy viscosity in nearly potential flow regions after Larsen and Fuhrman (2018), with the coefficient T 42. For NHFLOW use A 565.

0 OFF

1 ON

default: 0

○ ○ ● ●

T 42 double Coefficient λ_2 for the k-ω stabilization
Coefficient of the stabilization limiter (T 41 in CFD, A 565 in NHFLOW); smaller values give a weaker limitation.

default: 0.05

○ ○ ○ ●

T 44 double Height factor for the wall velocity at the sediment bed
In the k-ω wall boundary condition at a sediment bed (S 10 > 0) the velocity is interpolated at the height T 44 $\cdot \Delta z$ above the bed.

default: 1.0

○ ○ ○ ●

T 45 int Buoyancy term in the k-equation
Adds turbulence production or damping due to density stratification (k-ω model).

0 OFF

1 ON

default: 0

6.15 W :: Water and Material Properties

Valid for:

S	F	N	C
●	●	●	●

W 1 double Density of phase 1 (water)

default: 998.2 kg/m^3

● ● ● ●

W 2 double Kinematic viscosity of phase 1 (water)

default: $1.004 \cdot 10^{-6}$ m^2/s

○ ○ ○ ●

W 3 double Density of phase 2 (air)

default: 1.205 kg/m^3

○ ○ ○ ●

W 4 double Kinematic viscosity of phase 2 (air)

default: $1.41 \cdot 10^{-5}$ m^2/s

○ ○ ○ ●

W 5 double Surface tension coefficient
Surface tension between phase 1 and phase 2; 0.0 switches surface tension off.

default: 0.0 N/m

○ ○ ○ ●

W 6 double Density of phase 3
Density of the third fluid (e.g. oil) for the multiphase level set model (F 300).

default: 840.0 kg/m^3

○ ○ ○ ●

W 7 double Kinematic viscosity of phase 3
Viscosity of the third fluid for the multiphase level set model (F 300).

default: $3.0 \cdot 10^{-4}$ m^2/s

● ○ ● ●

W 10 double Discharge
Inflow discharge used by ioFlow (B 60) and by the wave generation with current.

default: 0.0 m^3/s

○ ○ ○ ●

W 11 double u, double v, double w Inflow velocities on surface 1
Fixed velocities in m/s on surface 1 (requires an inflow boundary condition on surface 1 in DIVEMesh). Options W 11 – W 16 are made for benchmark and other simple test and validation cases. For more complex and real world cases, either B 60 1 (ioFlow) or the B 400s (patchBC) options are recommended.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 12 double u, double v, double w Inflow velocities on surface 2
Requires an inflow boundary condition on surface 2 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 13 double u, double v, double w Inflow velocities on surface 3
Requires an inflow boundary condition on surface 3 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 14 double u, double v, double w Inflow velocities on surface 4
Requires an inflow boundary condition on surface 4 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 15 double u, double v, double w Inflow velocities on surface 5
Requires an inflow boundary condition on surface 5 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ○ ●

W 16 double u, double v, double w Inflow velocities on surface 6
Requires an inflow boundary condition on surface 6 in DIVEMesh; see W 11.

default: 0.0 ; 0.0 ; 0.0 m/s

○ ○ ● ●

W 20 double Acceleration due to gravity, x-component
In NHFLOW only used in the force calculation of floating bodies; SFLOW and FNPF only use W 22.

default: 0.0 m/s^2

○ ○ ● ●

W 21 double Acceleration due to gravity, y-component
See W 20.

default: 0.0 m/s^2

● ● ● ●

W 22 double Acceleration due to gravity, z-component
Must be set for all free surface simulations, typically -9.81.

default: 0.0 m/s^2

○ ○ ○ ●

W 29 double x, double y, double z Additional constant pressure gradient
Constant source term added to the momentum equations like a body force, i.e. the pressure gradient divided by the density in m/s^2, e.g. to drive the flow with periodic boundary conditions.

default: 0.0 ; 0.0 ; 0.0 m/s^2

○ ○ ○ ●

W 30 int Compressibility of air
Calculates the density of phase 2 from the ideal gas law using the local pressure and the temperature W 31.

0 OFF

1 ON

default: 0

○ ○ ○ ●

W 31 double Air temperature for the compressibility model
Temperature in $°C$ used with W 30 1.

default: 20.0 $°C$

○ ○ ○ ●

W 41 * double x_c, double y_c, double z_s, double z_e, double U, double β Velocity vertical line source for phase 1
Vertical line source at (x_c, y_c) in m between z_s and z_e in m with the velocity U in m/s. Here, β is the direction of the flow in degrees.

default: not set

○ ○ ○ ●

W 50 double Air inflow velocity

Velocity in m/s in x-direction imposed in the air phase at the inflow boundary, e.g. to model wind. Only active if the command is given.

default: 0.0 m/s (not set)

● ○ ○ ●

W 90 int Non-Newtonian rheology

Rheology of phase 1. In SFLOW only option 1 is available, as a Bingham-type bed friction.

0 OFF

1 Herschel-Bulkley (yield stress W 96 or W 101, consistency W 97, exponent W 98)

2 three-phase mixture: Herschel-Bulkley slurry (W 98, W 105, W 106) and Coulomb viscoplastic gravel (W 106, W 107), weighted with W 108 (CFD only)

default: 0

○ ○ ○ ●

W 95 double Maximum kinematic viscosity ν_0

Upper limit of the rheological viscosity for W 90 > 0; it regularizes the viscosity at vanishing shear rates.

default: 0.001 m^2/s

● ○ ○ ●

W 96 double Yield stress τ_0

Prescribed yield stress for W 101 0. It is also used in the first time step for all W 101 options.

default: 1.0 Pa

● ○ ○ ●

W 97 double Consistency index K

Consistency index of the Herschel-Bulkley model, $\tau = \tau_0 + K\dot{\gamma}^n$, in Pas^n.

default: $1.0 \cdot 10^{-5}$ Pas^n

○ ○ ○ ●

W 98 double Power law index n

$n < 1$: shear thinning, $n > 1$: shear thickening, $n = 1$: Bingham fluid.

default: 1.0

● ○ ○ ●

W 101 int Yield stress model

Pressure-dependent yield stress models use the angle of repose and cohesion W 102, the pressure selected with W 111 and the regularization W 103. Options $3-5$ use the buoyant part of the pressure (interstitial fluid density $1000\,kg/m^3$). SFLOW supports options 0 and 1.

0 prescribed yield stress W 96

1 Mohr-Coulomb, $\tau_0 = p\tan\phi + c$

2 Drucker-Prager type, $\tau_0 = (p\tan\phi + c)(1 - e^{-W103\dot{\gamma}})$

3 Herschel-Bulkley-Coulomb with buoyant pressure

4 as 3, with shear rate generated excess pore pressure, $p\,e^{-W104\dot{\gamma}}$

5 as 3, with linear shear rate coupling, $p - W104\dot{\gamma}$

default: 0

● ○ ○ ●

W 102 double ϕ, double c Angle of repose and cohesion for the Mohr-Coulomb yield stress

Angle of repose ϕ in degrees and cohesion c in Pa, used with W 101 > 0.

default: 30.0 ; 0.0

○ ○ ○ ●

W 103 double Transition factor for the Mohr-Coulomb yield stress

Regularization time scale in s in the factor $(1 - e^{-W103\dot{\gamma}})$ of W 101 $2-5$; larger values give a sharper transition.

default: 1.0 s

○ ○ ○ ●

W 104 double Shear rate coupling factor for the pore pressure

Coefficient of the shear rate dependent reduction of the effective pressure: time scale in s for W 101 4, and coefficient in Pas for W 101 5.

default: 1.0

○ ○ ○ ●

W 105 double τ_{00}, double C, double C_{kc}, double C_i, double C_m Slurry yield stress parameters for W 90 2

Calibration yield stress τ_{00} in Pa (accounts for the grid dependency of the shear rate), total volumetric solid concentration C, and the volumetric concentrations of kaolinite and chlorite C_{kc}, illite C_i and montmorillonite C_m (all in m^3/m^3). The slurry yield stress is $\tau_y = \tau_0 C^2 e^{22CP_1}$, with $\tau_0 = \tau_{00}$ for $C < 0.47$ and $\tau_0 = \tau_{00} e^{5(C-0.47)}$ otherwise, and P_1 derived from $C_{kc} + 1.3C_i + 1.7C_m$.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

W 106 double b, double m_y Slurry consistency factor and gravel regularization for W 90 2

The slurry consistency is $k = b\tau_y$. m_y in s is the regularization parameter of the Coulomb viscoplastic gravel viscosity, $\mu = \mu_{min} + p\sin\delta\,(1 - e^{-m_y\dot{\gamma}})/\dot{\gamma}$.

default: 0.3 ; 0.2

○ ○ ○ ●

W 107 double μ_{min}, double δ, double μ_{max} Gravel viscosity parameters for W 90 2

Minimum dynamic viscosity μ_{min} in Pas, internal friction angle δ in degrees and maximum dynamic viscosity μ_{max} in Pas of the Coulomb viscoplastic gravel phase. Note that the default $\mu_{max} = 0$ switches the gravel viscosity off, so it must be set.

default: 0.0 ; 0.0 ; 0.0

○ ○ ○ ●

W 108 double Slurry volume fraction a_2 for W 90 2

The mixture viscosity is $(a_2\mu_{slurry} + (1 - a_2)\mu_{gravel})/\rho$, with ρ = W 1 as mean density of the mixture.

default: 0.3

○ ○ ○ ●

W 110 int Numerical treatment of the rheology

Determines whether the viscous (rheology) part and the yield stress part of the Herschel-Bulkley model enter as viscosity or as momentum source term.

1 viscosity for rheology and yield stress

2 viscosity for rheology, momentum source term for yield stress

3 momentum source term for rheology and yield stress

4 viscosity for rheology, yield stress gradient as momentum source term

5 rheology and yield stress gradient as momentum source term

6 viscosity for rheology, frictional source term from the pressure gradient, $\tan\phi\nabla p$

7 as 6

default: 1

○ ○ ○ ●

W 111 int Pressure for the Mohr-Coulomb yield stress

1 hydrostatic

2 total pressure (hydrostatic and dynamic)

3 hydrostatic pressure away from the interface and total pressure close to the interface (see W 112)

4 average of hydrostatic and total pressure

5 hydrostatic for the first 10 iterations, total pressure afterwards

default: 1

○ ○ ○ ●

W 112 double Threshold factor for W 111 3

The total pressure is used where the level set distance from the interface is smaller than W 112 $\cdot\Delta x$.

default: 2.1

6.16 X :: 6DOF

The X commands control the rigid-body (6DOF) module for floating and moving bodies, including moorings, nets and prescribed-motion wavemakers. The module is available for CFD and NHFLOW (two-way and one-way coupling), for FNPF (two-way coupling of triangulated bodies, X 110, X 163, X 164, X 180; moorings, no nets) and, in a simplified form, for SFLOW (one-way coupling and ship-induced pressure fields). A minimal free-floating body needs:

- X 10 to activate the module (X 10 1 for two-way coupling),
- X 11 to select the degrees of freedom (default: all six are solved),
- X 21 (density) or X 22 (mass) of the body; the centre of gravity (X 23) and moments of inertia (X 24) are computed from the geometry if they are not given,
- the body geometry: either primitives (X 110, X 131–X 133, X 153, X 163, X 164) or an ASCII STL file `floating.stl` (X 180, optionally scaled, moved and rotated with X 181–X 183).

The initial position is the position of the geometry as defined; an initial rotation around the centre of gravity can be added with X 101 and initial velocities with X 102 and X 103.

Valid for:

S	F	N	C
●	●	●	●

X 10 int Turn on 6DOF algorithm for floating bodies

Activates the rigid-body module. In FNPF (A 10 3) the bodies are resolved in the Laplace problem (Section 5.2); moorings are supported there, nets are ignored. Whenever X 10 > 0, D 22 is set to 2, and the NHFLOW/CFD time scheme N 40 is changed from 3 or 23 to 4 and from 13 to 14.

0 OFF

1 Two-way coupling: body motion from the hydrodynamic forces (CFD, NHFLOW and FNPF)

2 One-way coupling: the body moves according to X 11 and the prescribed or initial velocities without hydrodynamic feedback (SFLOW and NHFLOW)

3 Ship-wave mode: the body acts as an external moving pressure field on the free surface (SFLOW and NHFLOW, requires X 400)

4 Prescribed wavemaker geometry from X 170 or X 172 driven by `wavemaker.dat` (NHFLOW)

default: 0

● ● ● ●

X 11 int u, int v, int w, int p, int q, int r Degrees of freedom of the floating body

Sets the treatment of the three linear (u, v, w) and three angular (p, q, r) velocities individually. Prescribed degrees of freedom take their velocity from X 210/X 211 or from a motion file (X 240). In 2D simulations v is always switched off.

0 fixed (OFF)

1 solve the 6DOF equation of motion

2 prescribed

default: 1 ; 1 ; 1 ; 1 ; 1 ; 1

○ ○ ○ ●

X 14 int Preserve tangential velocities for forcing

Controls how the body velocity is imposed by the direct forcing in CFD. With 1, cells inside the body receive the full rigid-body velocity, while in the smoothing layer outside the body (X 41) each velocity component is forced only with the weight of the corresponding component of the surface normal, which preserves tangential velocities.

0 OFF (full forcing everywhere)

1 ON

default: 1

○ ○ ● ○

X 15 int Type of direct forcing near the body surface for NHFLOW
Controls how the body velocity is imposed by the direct forcing in NHFLOW (also for the X 10 4 wavemakers). The forcing layer thickness is set with A 526.

0 full forcing of all velocity components (no-slip)

1 full forcing inside the body; outside, each component weighted with the magnitude of the corresponding normal component

2 full forcing inside the body; outside, only the normal velocity component is forced (free slip, tangential velocity preserved)

default: 1

○ ○ ● ○

X 16 double n, double d_{50}, double α, double β Porous floating body
Treats the NHFLOW floating body (X 10 1) as a moving porous medium instead of a rigid body imposed by direct forcing. Given are the porosity n of the body, the characteristic grain or element size d_{50} in m and the linear (Darcy) and quadratic (Forchheimer) resistance coefficients α and β, with the same closure as the NHFLOW VRANS objects (B 201; the KC correction B 264 is applied as well). The porosity field $1 - H(1 - n)$ follows the body, and the Darcy-Forchheimer resistance acts on the fluid velocity relative to the body velocity; it is integrated implicitly over each Runge-Kutta stage, so that large coefficients remain stable. Its reaction is added to the body forces (written to the viscous force columns of the force file), the pressure load on the body surface is scaled with $(1 - n)$, and no viscous wall forces are computed on the envelope. The forces are always integrated over the triangulated surface (as X 60 1). Can be combined with static VRANS structures (B 200 1). Giving X 16 activates the porous body.

default: not set (rigid body); 0.5 ; 0.01 ; 1000.0 ; 1.1

● ● ● ●

X 19 int Print out interval for 6DOF log files
The 6DOF log files (position, velocity and forces in the folder `REEF3D_*_6DOF`) are written every X 19 time steps. Also used for the FSI log files (Z 10).

default: 1

○ ○ ○ ●

X 20 int Number of floating bodies
Number of 6DOF bodies in CFD. Body 0 reads its geometry from `floating.stl`, body $n \geq 1$ from `floating-n.stl` (X 180 1). All other X settings are shared by all bodies. NHFLOW and SFLOW always use one body.

default: 1

○ ● ● ●

X 21 double Floating body homogeneous density ρ in [kg/m^3]
The mass is computed from this density and the body volume. The density is used by default; if X 22 is given, X 22 takes precedence. For a porous floating body (X 16), X 21 and X 22 define the mass of the solid skeleton, i.e. X 21 is the bulk density $\rho_s(1-n)$.

default: 900.0

○ ● ● ●

X 22 double Floating body mass m in [kg]
If given, the mass is prescribed and the density is computed from mass and body volume; X 21 is then ignored.

default: 0.0 (not set, mass from X 21)

● ● ● ●

X 23 double x_G, double y_G, double z_G Floating body centre of gravity in [m]
If not given, it is calculated from the geometry. The centre of gravity is also the origin of the body-fixed (non-inertial) coordinate system and the centre of rotation.

default: calculated from the geometry

○ ● ● ●

X 24 double I_x, double I_y, double I_z Floating body moments of inertia in [kg m^2]
Principal moments of inertia about the centre of gravity. If given, the products of inertia are set to zero. If not given, the full inertia tensor is calculated from the geometry and the density.

default: calculated from the geometry

○ ● ● ●

X 25 double C_p, double C_q, double C_r Damping coefficients for rotational motion around the three axes
Linear damping moment $-C\,\omega$ added to the moment balance of the solved rotational degrees of freedom, in [N m s].

default: 0.0 ; 0.0 ; 0.0

○ ● ● ●

X 26 double C_u, double C_v, double C_w Damping coefficients for translational motion along the three axes
Linear damping force $-C\,u$ added to the force balance of the solved translational degrees of freedom, in [N s/m].

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ●

X 38 int Type of viscous force calculation on the body surface
Used with the STL force calculation (X 60 1). The velocity is sampled at the distance X 43 from the surface.

0 CFD: viscous stresses from the local velocity gradients with the molecular viscosity; NHFLOW: no viscous force

1 CFD: wall-gradient approximation with molecular and eddy viscosity; NHFLOW: skin friction with $C_f = 10^{-4}$

2 CFD: logarithmic wall function (roughness $k_s = 10^{-5}$ m)

default: 0

● ○ ● ●

X 41 double Thickness of the continuous direct forcing layer
Half-width of the smoothed Heaviside function around the body surface, in multiples of the mean cell size. In CFD it is used for floating bodies and for the direct forcing of solids and topography. In SFLOW and in the NHFLOW ship-wave mode (X 10 3) it sets the width of the pressure field transition; otherwise NHFLOW uses A 526.

default: 0.6

○ ○ ● ●

X 42 double Distance for pressure force evaluation
Distance from the surface triangles along the outward normal at which the pressure is interpolated for the force integration (X 60 1), in multiples of the local cell size.

default: 0.0

○ ○ ● ●

X 43 double Distance for shear stress evaluation
Distance from the surface triangles along the outward normal at which the velocities are interpolated for the viscous forces (X 38), in multiples of the local cell size.

default: 1.0

○ ○ ○ ●

X 45 int Type of free surface level set convection discretization at the floating body
Modifies the stencil of the fifth-order WENO Hamilton-Jacobi scheme near the floating body. Only effective with F 35 5.

0 standard stencil, convection also inside the floating body

1 gradients set to zero for stencil cells closer than $0.6\,dx$ to the body surface

2 gradients weighted with the averaged body indicator function

3 gradients set to zero inside the body

4 gradients weighted with the upwind body indicator function

5 gradients set to zero inside solids and topography

6 gradients inside the body set to a large value (10^{20})

default: 0

○ ○ ○ ●

X 48 int Level set extrapolation into direct forcing solids
If ON, the level set and the k-ω turbulence quantities are copied from the first fluid cell into the two adjacent cells inside direct-forcing bodies (floating bodies, solids, topography).

0 OFF

1 ON

default: 0

● ● ● ●

X 50 int Type of print out format for 6DOF structure
Since the VTP print out is binary, it is significantly faster than the STL print out. This becomes runtime relevant for STL files with a large number of triangles.

1 VTP

2 STL

default: 1

○ ○ ● ●

X 60 int Type of hydrodynamic force calculation
With 1 the pressure (X 42) and viscous stresses (X 38, X 43) are integrated over the surface triangles. With 2 the forces are computed from a triangulation reconstructed from the level set function; the surface refinement X 185 is then skipped.

1 STL

2 LSM

default: 1

● ● ● ●

X 101 double ϕ_0, double θ_0, double ψ_0 Initial Euler angles in [deg]
Rotates the body around its centre of gravity at the start. Positive angles rotate the body clockwise. Mooring end points on the body are rotated with it only if X 313 1.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 102 * double u_0, double v_0, double w_0 Initial linear floating body velocity in [m/s]
If given several times, the values are added.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 103 double p_0, double q_0, double r_0 Initial angular floating body velocity in [rad/s]

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 110 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Rectangular box floating body in [m]
Several boxes can be combined into one body. The first box also defines length and width of the pressure field for X 400 2.

default: na

● ○ ● ●

X 131 double r, double h, double x_c, double y_c, double z_c Cylinder in x-direction floating body
Radius r, length h and centre point (x_c, y_c, z_c) in [m]. Not available in FNPF, where the program stops with an error message.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ○ ● ●

X 132 double r, double h, double x_c, double y_c, double z_c Cylinder in y-direction floating body

Radius r, length h and centre point (x_c, y_c, z_c) in [m]. Not available in FNPF, where the program stops with an error message.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ○ ● ●

X 133 double r, double h, double x_c, double y_c, double z_c Cylinder in z-direction floating body

Radius r, height h and centre point (x_c, y_c, z_c) in [m]. Not available in FNPF, where the program stops with an error message.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ○ ● ●

X 153 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end} Two-sided wedge in z-direction floating body in [m]

Not available in FNPF, where the program stops with an error message.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 163 * double x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6 Wedge floating body

Each of the 6 corner points is given by its coordinates in [m], see the figure below.

default: na

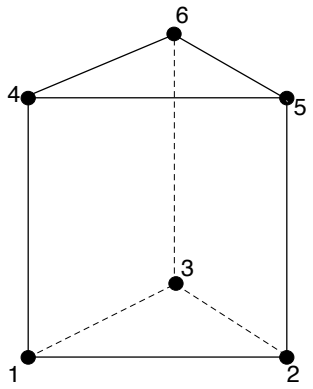

Figure 6.1: Definition of the wedge points for X 163

● ● ● ●

X 164 * double x_1, y_1, z_1, x_2, y_2, z_2, x_3, y_3, z_3, x_4, y_4, z_4, x_5, y_5, z_5, x_6, y_6, z_6, x_7, y_7, z_7, x_8, y_8, z_8
Hexahedron floating body
Each of the 8 corner points is given by its coordinates in [m], see the figure below.

default: na

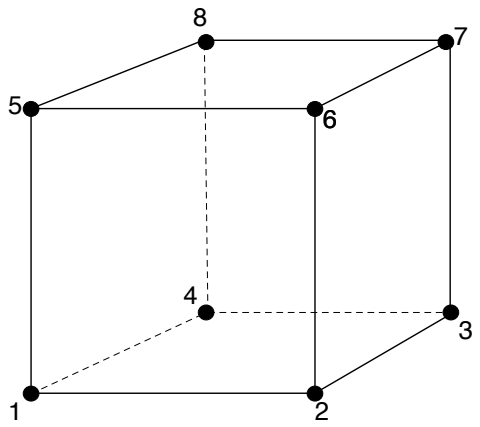

Figure 6.2: Definition of the hexahedron points for X 164

○ ○ ● ○

X 170 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}
Piston wavemaker in [m]
Box-shaped piston moved with the displacement time series in `wavemaker.dat` (columns t [s], x [m]; time shifted by X 241). Requires X 10 4.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ○

X 171 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{hinge}, double z_{end} Flap wavemaker in [m]
Intended for X 10 4; currently not implemented.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ○

X 172 double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{hinge1}, double z_{hinge2}, double z_{end} Double-hinged flap wavemaker in [m]
Requires X 10 4. The motion is read from `wavemaker.dat` (columns t [s], horizontal displacement of the lower and of the upper flap segment [m], or angles in [rad] with B 116 2); lines with t smaller than B 117 are skipped.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 180 int Read STL file `floating.stl` for floating body geometry
The file must be in ASCII STL format. With X 20 > 1, the further bodies are read from `floating-1.stl`, `floating-2.stl`, ...

0 OFF

1 ON

default: 0

● ● ● ●

X 181 double s_x, double s_y, double s_z Scale STL geometry in x-, y-, and z-direction
Scaling factors applied to the STL coordinates, before X 182 and X 183.

default: 1.0 ; 1.0 ; 1.0

● ● ● ●

X 182 double Δx, double Δy, double Δz 3D linear translation of STL object in [m]
Applied after the scaling X 181 and before the rotation X 183.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 183 double x_{origin}, double y_{origin}, double z_{origin}, double ϕ, double θ, double ψ 3D rotation of STL object
Rotation around the point $(x_{origin}, y_{origin}, z_{origin})$ in [m] by the angles in [deg], applied after X 181 and X 182. Positive angles rotate the body clockwise.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

● ● ● ●

X 185 int Type of geometry refinement for triangulated surface mesh
Surface triangles larger than the critical length (reference cell size times X 186) are split. The refinement is skipped for X 60 2 and for the wavemakers X 170–X 172.

0 OFF

1 based on averaged dx

2 based on global minimum dx

3 based on local minimum dx

default: 1

● ● ● ●

X 186 double Geometry refinement factor for triangulated surface mesh
Critical triangle size for X 185 relative to the reference cell size.

default: 0.7

○ ○ ○ ●

X 188 int Ray casting algorithm for the floating body level set
Method used to compute the distance function of the body from the surface triangles in CFD.

1 directional ray casting in x-, y- and z-direction

2 direct distance calculation

default: 1

● ● ● ●

X 205 int Ramp up function type
Shape of the ramp functions X 206 and X 207.

1 linear

2 sine

default: 1

● ● ● ●

X 206 double t_s, double t_e External motion ramp up start time and end time in [s]
Ramps the prescribed velocities (X 210, X 211, X 240) from zero at t_s to full value at t_e; before t_s the prescribed velocities are zero.

default: not set (no ramp)

● ○ ● ○

X 207 double t_s, double t_e Draft ramp up start time and end time in [s]
Ramps the external pressure field of the ship-wave mode (X 10 3, X 400). Without X 207 the free surface is initialised with the depression of the full draft.

default: not set (no ramp)

● ● ● ●

X 210 double u, double v, double w Prescribed constant linear velocities of the floating body in [m/s]
Applied to the degrees of freedom set to 2 in X 11, ramped with X 206.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 211 double p, double q, double r Prescribed constant angular velocities of the floating body in [rad/s]
Applied to the degrees of freedom set to 2 in X 11, ramped with X 206.

default: 0.0 ; 0.0 ; 0.0

● ● ● ●

X 240 int Read motion file for 6DOF motions
The motion is read from `6DOF_motion.dat`; the velocities are computed from the differences between consecutive lines and applied to the degrees of freedom set to 2 in X 11 (ramped with X 206). CoG = course over ground (yaw angle) in [deg].

0 OFF

1 Translation format: t, x, y, z

11 Course over Ground format: t, x, y, CoG (acts on u, v and r)

21 Piston wavemaker format: t, x (acts on u)

default: 0

● ● ● ●

X 241 double Time shift for motion files in [s]
Added to the time column of `6DOF_motion.dat` (X 240) and of the piston `wavemaker.dat` (X 170).

default: 0.0

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X 310 int Turn on mooring modelling for floating bodies
The lines are defined with X 311 (options 1–3) or X 312 (option 4). Results are written to `REEF3D_*_6DOF_Mooring`.

0 OFF

1 Static catenary

2 Quasi-static bar model

3 Dynamic model

4 Spring (using X 312)

default: 0

○ ● ● ●

X 311 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double w, double ρ_c, double EA, double d, double l, double H, double P, double Q Mooring line configuration

Each mooring line has to be specified separately with this command. $\vec{x}_{start}$ is the anchor point at the bottom and $\vec{x}_{end}$ the fairlead on the body, in [m]. w is the mass per unit length in air [kg/m], ρ_c the density of the line material [kg/m^3] (used for the submerged weight), EA the axial stiffness [N], d the line diameter [m] and l the unstretched length [m]. H is the number of elements for the numerical approaches. P and Q are read but currently not used. Do not combine with X 312.

default: na

○ ● ● ●

X 312 * double x_{start}, double x_{end}, double y_{start}, double y_{end}, double z_{start}, double z_{end}, double k, double T_0 Spring configuration

Each spring has to be specified separately with this command (X 310 4). $\vec{x}_{start}$ is the fixed point at the bottom and $\vec{x}_{end}$ the attachment point on the body, in [m]. k is the stiffness [N/m] and T_0 the pre-tension [N]. Do not combine with X 311.

default: na

○ ● ● ●

X 313 int Initial rotation of mooring end points with the floating body

If ON, the fairlead points $\vec{x}_{end}$ of X 311/X 312 are rotated with the initial body rotation X 101.

0 OFF

1 ON

default: 0

○ ● ● ●

X 314 double T_1, T_2, ... Breaking tension of the mooring lines in [N]

One value for each mooring line (in the order of X 311 or X 312), all on one line. A line breaks when its tension reaches the value. Set the value to zero to disable breaking for an individual line.

default: not set (no breaking)

○ ● ● ●

X 315 double t_1, t_2, ... Breaking time of the mooring lines in [s]
One value for each mooring line (in the order of X 311 or X 312), all on one line. A line breaks at the given simulation time. Set the value to zero to disable breaking for an individual line.

default: not set (no breaking)

○ ○ ● ●

X 320 * int Turn on net modelling
Each net has to be specified separately with this command, together with one X 321 and one X 322 per net. Nets are attached to the floating body, or fixed if X 10 is 0. Results are written to `REEF3D_*_6DOF_Net`.

0 OFF

2 Static net cylinder

3 Static net wall

4 Static sheets following floating body

12 Dynamic net cylinder

13 Dynamic net wall

14 Dynamic net cone

default: 0

○ ○ ● ●

X 321 * double Sn, double d_t, double l_t, double d_k, double ρ, double N_d, double N_l Net material

Each net has to be specified separately with this command. Sn is the solidity, d_t the twine diameter [m], l_t the twine length [m], d_k the knot diameter [m] (0.0 if knotless) and ρ the density of the net material [kg/m^3]. The net is discretised with N_d meshes along D and N_l meshes along L (X 322).

default: na

○ ○ ● ●

X 322 * double D, double L, double x_0, double y_0, double z_0, double ϕ, double θ, double ψ
Net dimensions

Each net has to be specified separately with this command. D is the diameter or width of the net and L its height, in [m]. The structure is first rotated around the origin by the angles around the x-, y- and z-axis in [deg]. Then the centre point of the bottom of the structure is moved to the position (x_0, y_0, z_0) in [m].

default: na

○ ○ ● ●

X 323 double m_s, double d_s, double l_s Sinker for dynamic net models
Mass in air [kg], diameter [m] and length [m] of the sinkers, used for all dynamic nets (X 320 12–14).

default: 0.0 ; 0.0 ; 0.0

○ ○ ● ●

X 324 * double x, double y, double z Knot probes for dynamic net models
Location in [m]; the dynamic net model tracks the knot nearest to this point.

default: na

○ ○ ● ●

X 325 double Δt, double r_x, double r_y, double r_z Time step and relaxation for dynamic net models
Time step of the net solver in [s] and relaxation factors in x-, y- and z-direction. Set the time step to zero to use the time step of the fluid solver.

default: 0.001 ; 0.01 ; 0.01 ; 0.01

● ○ ● ○

X 400 int External moving pressure field for the ship-wave mode
Type of pressure field representing the body in X 10 3. The field follows the body motion (X 11, X 210, X 211, X 240) and is ramped with X 207.

0 OFF

2 Box: $p = -p_0\left(1 - c_l(x/L)^4\right)\left(1 - c_b(y/B)^2\right)\exp\left(-a(y/B)^2\right)$ inside the box, with length L and width B from the first X 110 and x, y relative to the centre of gravity

3 1D Gaussian: $p = p_0 \exp\left(-(x/a)^2\right)$, with x relative to the centre of gravity and a in [m]

10 STL: $p = -\rho g \cdot draft$ from the body geometry (see X 410)

default: 0

● ○ ● ○

X 401 double p_0, double c_l, double c_b, double a Coefficients for the external moving pressure field

Pressure amplitude p_0 in [Pa] and shape coefficients for X 400 2 and 3 (see there).

default: 0.0 ; 2.0 ; 16.0 ; 16.0

● ○ ● ○

X 410 int Add local free surface for draft calculation

For X 400 10: adds the free surface elevation sampled just outside the body edge (offset X 41 cells) to the draft.

0 OFF

1 ON

default: 0

6.17 Z :: FSI

Valid for:

S	F	N	C
○	○	○	●

Z 10 int Turn on FSI algorithm for flexible bodies

Log files are written with the interval X 19.

0 OFF

1 Strips

default: 0

○ ○ ○ ●

Z 11 * double x, double y, double z, double L, double W, double T, double ρ, double E, double I_x, double I_y, double I_z, double ν, double N Flexible strip definition

Each strip has to be specified separately with this command. (x, y, z) is the initial position of the central bottom point of the strip in [m]. L is the length, W the width and T the thickness of the strip in [m] ($L \gg W \gg T$ must hold). ρ is the density of the material [kg/m^3], E the Young modulus [N/m^2], (I_x, I_y, I_z) are the second moments of area [m^4] and ν is the Poisson ratio. N is the number of elements per strip.

default: na

○ ○ ○ ●

Z 12 double $c_{d,X}$, double $c_{d,Y}$, double $c_{d,Z}$, double $c_{k,X}$, double $c_{k,Y}$, double $c_{k,Z}$ Structural damping of the strips

Velocity-related viscous damping in the local X,Y,Z directions of the beam, applied to all strips. $\vec{c}_d$ acts on the translational motions, $\vec{c}_k$ on the rotational motions.

default: 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0 ; 0.0

○ ○ ● ●

Z 20 int Flexible rod trees

Branched flexible rods (soft corals, gorgonians, flexible vegetation) that are thinner than the grid cells, coupled to the flow in an unresolved way (actuator-line type): the fluid velocity is sampled at Lagrangian points along each rod element, Morison-type loads (normal and tangential drag, Froude-Krylov and added-mass inertia, buoyancy and gravity, scaled with the submerged fraction) act on the elements, and the reaction is spread back to the fluid with a 4-point Peskin kernel as a body force. The structure is a discrete Cosserat rod of rigid cylinder elements connected by elastic joints, with the roots clamped to the bed; it is advanced once per time step in the final Runge-Kutta stage. Independent of Z 10. The colonies are read from the file `rodtree.dat` in the case directory (the program stops if it is missing or invalid; # starts a comment). Global keywords: `integrator` implicit|explicit, `substeps` n, `mode` dynamic|quasistatic|rigid, `solver` tree|sparse, `reaction` full|drag|none. Each colony is a block `colony` [name] ... `end` with `node` id x y z r (m), `edge` parent child, `clamp` node, `material` E (Pa) ν ρ (kg/m^3), `hydro` C_{dn} C_{dt} C_a [C_m, default $1+C_a$], `damping` β (s), `refine` n, and optionally `instance` Δx Δy Δz (copies of the colony), `represent` weight spacing (representative colonies for canopies), `polyps` h_p ϕ_p $C_{d,p}$ [k_t] and `polyp_response` U_r ΔU_r τ_p. Per colony, the tip displacement, the base force, the hydrodynamic force, the number of submerged elements and the polyp extension are written every time step to `REEF3D_RodTree/REEF3D_RodTree_<name>.dat`.

0 OFF

1 rod trees from `rodtree.dat`, unresolved coupling

default: 0

○ ○ ● ●

Z 21 double Print interval for the rod trees

Interval in s at which the rod-tree geometry is written as VTP files (`REEF3D_RodTree/REEF3D-RodTree-<n>.vtp`, Z 20 1). A value ≤ 0 writes no VTP files.

default: 0.0 (off)

7. Compiling the Code

REEF3D is written in C++20 and has the following dependencies:

- a C++ compiler with C++20 support (e.g. GCC or Clang), used through the MPI compiler wrapper `mpicxx`,
- MPI for the parallelization (OpenMPI is recommended),
- HYPRE for the solution of the pressure and Poisson equations,
- Eigen 5.0.0 (linear algebra, used by the 6DOF, mooring, net and beam modules). Eigen is a header-only library that is shipped with the REEF3D source code (folder `ThirdParty/eigen-5.0.0`), so it does not need to be installed separately.

DIVEMesh only requires a C++20 compiler (g++); it does not use MPI or HYPRE. Neither REEF3D nor DIVEMesh contain Fortran code, so a Fortran compiler (gfortran) is not needed for them. It is only used by HYPRE for its optional Fortran interface; HYPRE can be configured without it (Section 7.1.4).

The recommended route is:

1. install the dependencies, preferably with the package manager of the system (Section 7.1) or from source (Sections 7.1.3 and 7.1.4),
2. download the source code (Section 7.2),
3. compile DIVEMesh and REEF3D with the Makefiles (Section 7.3) or with CMake (Section 7.4).

Windows users can work in the Windows Subsystem for Linux (Section 7.6). Alternatively, Docker can be used on all operating systems (Section 7.7). Developers can set up the CodeLite IDE as described in Section 7.5.

7.1 Install Required Packages

7.1.1 Ubuntu and Debian

Compilers, build tools and OpenMPI are installed with:
sudo apt update
sudo apt install build-essential git cmake openmpi-bin libopenmpi-dev

HYPRE is available as a distribution package as well:
sudo apt install libhypre-dev

Both the Makefile and CMake expect the HYPRE header files in `$HYPRE_DIR/include` and the library in `$HYPRE_DIR/lib` (default `HYPRE_DIR=/usr/local/hypre`). If the distribution package installs HYPRE in a different layout, either adapt the include and library paths in the Makefile, or build HYPRE from source into `/usr/local/hypre` as described in Section 7.1.4. Building HYPRE from source is also what the Docker image does and is the most robust option.

7.1.2 macOS

- Install the Apple command line tools, which include the C and C++ compilers:
  xcode-select --install

  If problems occur in the compilation although the command line tools are installed, a possible solution is to remove and reinstall them. The command to remove the command line tools is:
  sudo rm -rf /Library/Developer/CommandLineTools

- With Homebrew (https://brew.sh), OpenMPI, HYPRE and CMake are installed with:
  brew install open-mpi hypre cmake

  The HYPRE installation directory is then given by **brew --prefix hypre**; pass it to the build as shown in Sections 7.3 and 7.4.

- Confirm that the compiler wrapper works (in a new terminal):
  mpicxx --version

7.1.3 Installing OpenMPI from source

REEF3D depends on MPI for the parallelization of the code. MPI is very portable and the de-facto standard for the parallelization of high-performance computations; see https://www.mpi-forum.org/. We recommend OpenMPI for macOS and Linux. On HPC clusters, use the MPI installation provided by the cluster (usually loaded with `module load`). If no package is available, OpenMPI can be compiled from source:

- Download the source code from https://www.open-mpi.org/software/ and unpack the tar.gz archive.

- Open a terminal, change into the unpacked folder and type:
  ./configure --prefix=/usr/local/openmpi

- Compile:
  make -j n all (with n being the number of available cores on your computer)

- Install:
  sudo make install

- Add the OpenMPI executables to the PATH and the libraries to the library path, e.g. by adding the following lines to the file ∼/.bashrc (or ∼/.zshrc on macOS):
  export PATH=/usr/local/openmpi/bin:$PATH
  export LD_LIBRARY_PATH=/usr/local/openmpi/lib:$LD_LIBRARY_PATH

  The file `.bashrc` is a hidden file in your home folder; **ls -a** shows it.

- Open a new terminal and check the installation with **which mpicxx** and **mpirun --version**.

7.1.4 Installing HYPRE from source

HYPRE Cen [2015], Falgout et al. [2006] is a library for high-performance linear solvers and preconditioners. It can be downloaded from the GitHub repository: https://github.com/hypre-space/hypre/releases. HYPRE has to be compiled with the same MPI installation that is used for REEF3D.

- Download the source code and unpack the archive.
- Open a terminal, change into the `src` folder of the HYPRE archive and type:
  ./configure --prefix=/usr/local/hypre
  If no Fortran compiler is installed, add --**disable-fortran**.
- Compile:
  make -j n all (with n the number of available cores)
- Install:
  sudo make install

The installation creates the folders `/usr/local/hypre/include` and `/usr/local/hypre/lib`, which is the default location used by the REEF3D Makefile and CMake. If HYPRE is built as a shared library, add `/usr/local/hypre/lib` to `LD_LIBRARY_PATH` in the same way as for OpenMPI.

7.2 Getting the Source Code

The source code of REEF3D and DIVEMesh is downloaded from GitHub with:
git clone https://github.com/REEF3D/REEF3D.git
git clone https://github.com/REEF3D/DIVEMesh.git

Alternatively, the source code can be downloaded as archives from www.reef3d.com or the GitHub release pages. To update an existing clone, type **git pull** inside the folder and recompile.

7.3 User Compilation & Installation: Makefile

For users who are interested in running the code only, the easiest way is the compilation of REEF3D and DIVEMesh with the Makefiles given in the code repositories. Open the terminal and go to the main directory of each code.

- For DIVEMesh, type:
  make -j n (with n the number of available cores)
  The executable is `bin/DiveMESH`. With **make -j n OPENMP=1**, DIVEMesh is compiled with OpenMP.

- For REEF3D, type:
  make -j n
  The executable is `bin/REEF3D`; the object files are placed in the `build` folder.

If HYPRE is not installed in `/usr/local/hypre`, the path is given on the command line, e.g. with Homebrew:
make -j n HYPRE_DIR=$(brew --prefix hypre)

The REEF3D Makefile offers the following targets:

release (default) optimized build with `-O3 -march=native -flto`. Because of `-march=native`, the executable is optimized for the processor of the computer on which it is compiled and may not run on older or different processor types. On clusters, compile on the type of node the simulations run on.

all optimized build (`-O3`) without processor-specific optimization.

dev optimized build with extensive compiler warnings.

debug build without optimization and with debug symbols (`-O0 -g`).

clean removes the `build` and `bin` folders.

For example, **make -j 8 debug**. Run **make clean** before switching between targets. In case of errors during the compilation, check that `mpicxx` is found and that `HYPRE_DIR` is correct.

7.4 User Compilation & Installation: CMake

REEF3D can also be compiled with CMake (version 3.21 or newer). In the main directory of REEF3D, type:
**cmake -S . -B build-cmake -DCMAKE_BUILD_TYPE=Release **
** -DHYPRE_DIR=/usr/local/hypre**
cmake --build build-cmake -j n

The executable is written to `bin/REEF3D`, as with the Makefile. A separate build folder such as `build-cmake` avoids mixing the CMake files with the object files of the Makefile in `build`. The following options are available:

CMAKE_BUILD_TYPE `Release` (default), `Debug`, `Dev` or `All`; these correspond to the Makefile targets release, debug, dev and all.

HYPRE_DIR HYPRE installation prefix (default `/usr/local/hypre`). If no HYPRE library is found in `$HYPRE_DIR/lib`, CMake downloads HYPRE 3.2.0 from GitHub and builds it as part of REEF3D. This requires git and internet access.

MPI is located automatically by CMake and has to be installed beforehand. Eigen is taken from `ThirdParty/eigen-5.0.0`; if that folder is missing, an installed Eigen 5.0.0 is used or it is downloaded.

7.5 Developer Compilation & Installation: CodeLite

REEF3D is written in highly modular C++ in order to provide efficient code development and maintenance. REEF3D is designed with cross-platform usage in mind. The source code can be compiled under macOS, Windows (WSL) and Linux. Since the user experience should be the same on all platforms when developing the code, CodeLite was chosen for the Integrated Development Environment (IDE). The reasons for this are:

- it is free under the GPL license
- it is cross-platform and works under macOS, Windows and Linux
- it works with a variety of compilers
- it works with the MPI library
- support for parallel builds

CodeLite can be downloaded from https://codelite.org. The REEF3D source contains a CodeLite project file (`src/REEF3D.project`), so all header and source files are sorted in a tree structure. This should make the orientation and navigation in the source code easier.

The following settings are needed in CodeLite to compile REEF3D. First, go to Settings > Build Settings and add a new compiler named mpicxx. The remaining settings are shown in the following screenshots. Check the location of your compilers with **which gcc**, **which g++** and **which mpicxx** to enter the appropriate paths. The compiler options have to include `-std=c++20`; compare with the flags in the Makefile if in doubt.

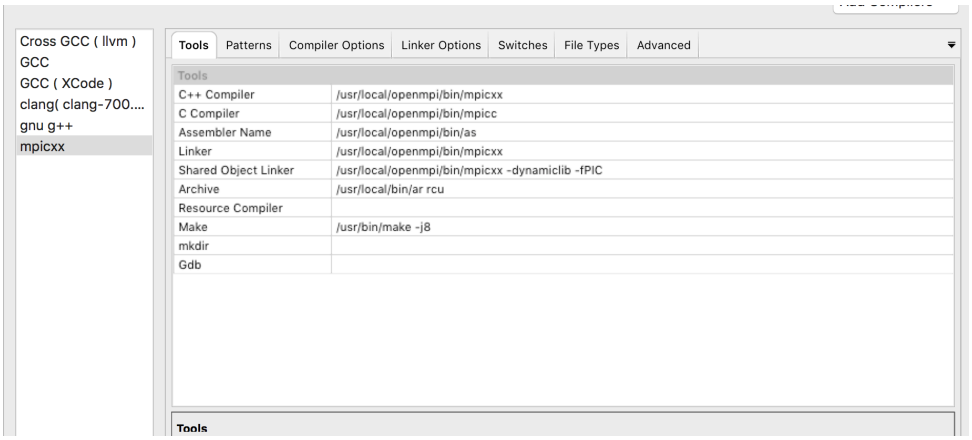


Figure 7.1: Setup of the mpicxx compiler in CodeLite.

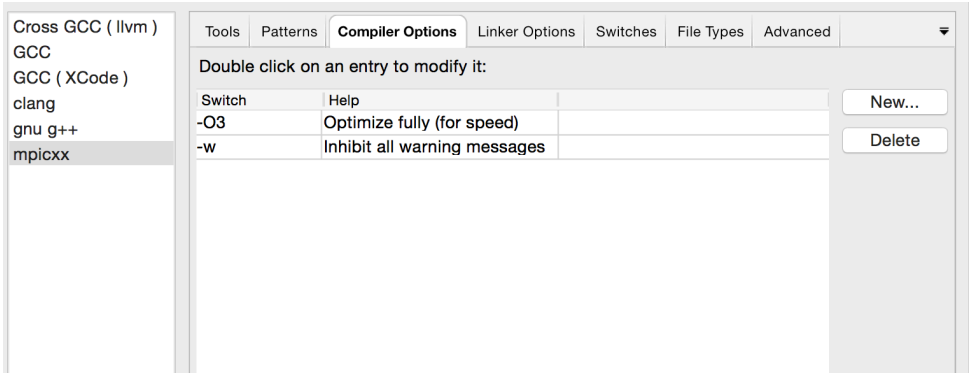


Figure 7.2: Compiler options for mpicxx.

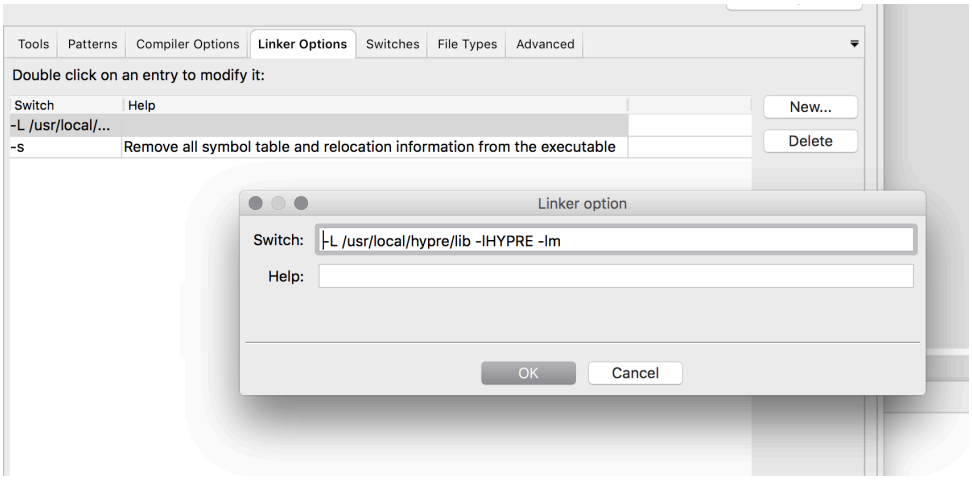


Figure 7.3: Linker options for mpicxx, link to HYPRE.

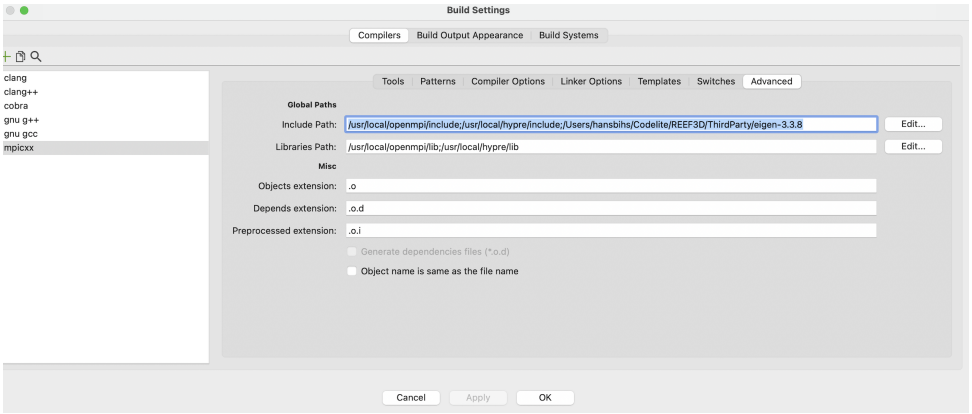


Figure 7.4: Add the include and library paths in the advanced tab of the mpicxx compiler: the HYPRE include and lib folders, and the Eigen headers shipped with REEF3D in `ThirdParty/eigen-5.0.0`. Adapt the paths to the file structure of the computer.

Afterwards, the code can be compiled by clicking Build > Build Project or pressing F7.

7.6 Installing REEF3D on Windows (WSL)

On Windows 10 (version 2004 or newer) and Windows 11, REEF3D is compiled and run in the Windows Subsystem for Linux (WSL 2), which provides a full Ubuntu installation.

- Open PowerShell as administrator and type:
  wsl --install
  This installs WSL 2 with Ubuntu as the default distribution. Restart the computer when prompted.

- Start Ubuntu from the Start menu. At the first start, you are asked for a user name and password for the Linux system; these are independent of your Windows credentials.

- Install the dependencies and compile DIVEMesh and REEF3D exactly as described for Ubuntu in Sections 7.1, 7.2 and 7.3.

- The Windows drives are available inside Ubuntu under `/mnt/c/`, `/mnt/d/`, ... For better performance, keep the source code and the simulation folders in the Linux file system (e.g. in your Ubuntu home folder) rather than under `/mnt/c/`.

- The Linux file system is accessible from the Windows Explorer under
  `\\wsl.localhost\Ubuntu\home\<user name>`
  (also shown as "Linux" in the navigation pane). ParaView installed on Windows can open the result files directly from there (Section 9.2).

7.7 Installing & Compiling REEF3D using Docker

Docker provides an alternative route for the compilation and installation of REEF3D and DIVEMesh that is independent of the operating system. The `Dockerfile` in the REEF3D repository creates an Ubuntu image in which OpenMPI and HYPRE are compiled from source, and the current versions of DIVEMesh and REEF3D are cloned from GitHub and compiled. The installation steps are as follows:

- Download and install Docker Desktop from:
  https://www.docker.com/products/docker-desktop/

  Adjust the number of CPU cores and the RAM available to Docker in the settings (optional), see Fig. 7.5.

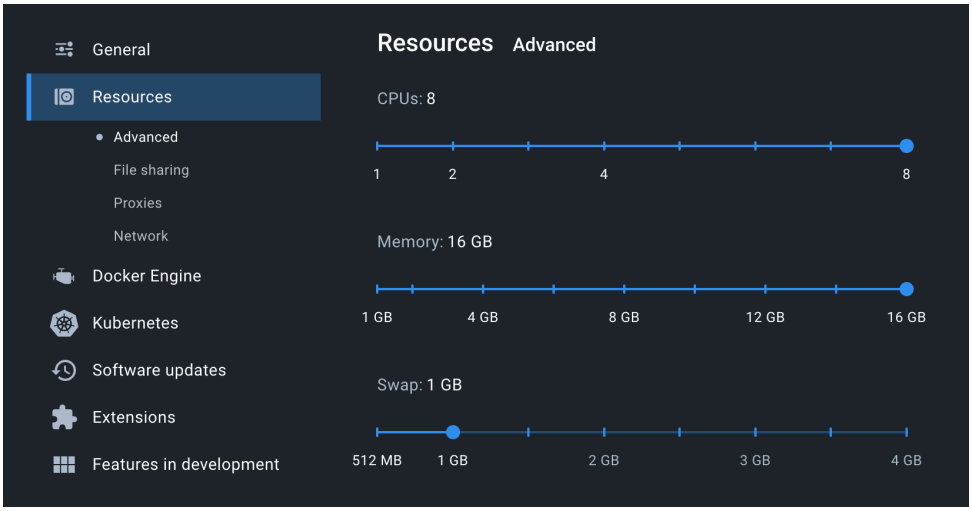


Figure 7.5: Docker resources

- Copy the file `Dockerfile` from the REEF3D repository into an empty folder and open a terminal in this folder. Only the Dockerfile is needed; the source code is downloaded during the build.

- Build the image, called reef3d, with:
  docker build -t reef3d .

  The process takes several minutes. Afterwards, all dependencies are installed and DIVEMesh and REEF3D are compiled. Because the Dockerfile always clones the latest source code, an update requires rebuilding the image without cache: **docker build --no-cache -t reef3d .**

- Create a folder called *simulations* on your computer. It is mounted into the container as `/simulations` and is the interface between the container and the local machine: everything written there is visible on both sides, everything else stays inside the container.

- Create and start the container. Replace `</local/path/simulations>` with the full path of your *simulations* folder.

  attached (the terminal is connected to the container directly):
  docker run -it --name reef3d -v </local/path/simulations>:/simulations reef3d

  detached (the container runs in the background):
  docker run -dit --name reef3d -v </local/path/simulations>:/simulations reef3d
  and connect to it with:
  docker attach reef3d

  In both cases, a bash shell opens in the folder `/simulations`. When the shell starts, the executables `DiveMESH` and `REEF3D` are copied into `/simulations`, which is on the PATH inside the container. Leaving the shell with **exit** stops the container; pressing Ctrl+P followed by Ctrl+Q detaches from it and leaves it running.

- Place each case in a subfolder of *simulations*, change into it inside the container and run DIVEMesh and REEF3D as described in Chapter 8, e.g. **DiveMESH** followed by **mpirun -n 4 REEF3D**.

- Other useful commands:
  start the stopped container: **docker start -ai reef3d**
  stop the container: **docker stop reef3d**
  restart the container: **docker restart reef3d**
  open an additional shell in a running container: **docker exec -it reef3d bash**

As in the Makefile build, the executables in the image are compiled with `-march=native` for the processor on which the image was built.

8. Running the Code

Every simulation is run in its own folder, the simulation folder. A simulation consists of two steps: DIVEMesh generates the grid and splits it into subdomains, and REEF3D then runs the simulation in parallel on these subdomains.

8.1 The Simulation Folder

Before the simulation, the folder contains the two input files and any further input files that the commands refer to:

```
my_case/
    control.txt        input for DIVEMesh (grid, geometry, M 10)
    ctrl.txt           input for REEF3D (see Chapter Functions)
    ...                optional input, e.g. STL files or bathymetry data
```

DIVEMesh adds the folder `DIVEMesh_Grid` with one grid file per subdomain (`grid-000001.dat`, `grid-000002.dat`, ...) and further folders with log and visualization files (e.g. `DIVEMesh_Log`, `DIVEMesh_Paraview`). REEF3D reads the grid files and writes all results into folders starting with `REEF3D_` (see Section 8.4 and Chapter 9). Result folders from a previous run in the same folder are not deleted automatically: files with the same name are overwritten, others remain. Delete or move old results before a new run.

The executables `DiveMESH` and `REEF3D` (in the `bin` folders of the source code after compilation) are either copied into the simulation folder or, more conveniently, made available through the PATH, e.g. by adding the following line to ∼/`.bashrc`:
export PATH=/path/to/REEF3D/bin:/path/to/DIVEMesh/bin:$PATH
In the following, the executables are assumed to be on the PATH; if they have been copied into the simulation folder instead, use **./DiveMESH** and **./REEF3D**.

8.2 DIVEMesh

Open a terminal in the simulation folder and type:
DiveMESH
DIVEMesh reads `control.txt` (see the DIVEMesh User Guide), runs serially and generates the grid files. The number of subdomains is set with **M 10** in `control.txt`. If DIVEMesh stops with a message about a missing input parameter, check the commands in `control.txt` against the DIVEMesh User Guide. The DIVEMesh output in `DIVEMesh_Paraview` can be opened in ParaView to check the grid, the geometry and the domain decomposition before the simulation is started.

8.3 REEF3D

REEF3D is started with MPI from the simulation folder:
mpirun -n *number of processes* **REEF3D**
For example, for a grid decomposed into four subdomains:
mpirun -n 4 REEF3D
Alternatively, **mpiexec -n 4 REEF3D** can be used.

The number of MPI processes has to be the same as **M 10** in `control.txt` (the number of subdomains generated by DIVEMesh) and **M 10** in `ctrl.txt`. REEF3D checks this when reading the grid and stops with the message `Inconsistent M 10 parameter` if the three numbers differ. If DIVEMesh has not been run, REEF3D stops with `Could not open DIVEMesh grid file`. As a guideline, M 10 should not exceed the number of physical cores available.

At the start, REEF3D prints `read ctrl` while reading `ctrl.txt`. If a command has too few values, REEF3D stops with `missing input parameter in ctrl.txt`; unknown commands are ignored without a warning (see Chapter 6). During the simulation, the iteration number, the simulated time, the time step and solver information are printed to the terminal every P 12 iterations. The simulation ends when the maximum simulated time N 41 or the maximum number of iterations N 45 is reached. To keep the terminal output, redirect it to a file, e.g.:
mpirun -n 4 REEF3D > reef3d_output.txt 2>&1

8.4 Log Files

Besides the result files, REEF3D writes log files that are useful to monitor a simulation and to check it afterwards:

REEF3D_Log (CFD and NHFLOW) contains `REEF3D_Version.dat` (version and git branch of the executable), `REEF3D_mainlog.dat` (per iteration: time step, simulated time, computing time, pressure solver iterations and time, volumes, in- and outflow), `REEF3D_maxlog.dat` (maximum velocities, viscosity, turbulence quantities and pressure), `REEF3D_volumelog.dat` and `REEF3D_solverlog.dat` (iterations and time of the individual solvers). If activated, the sediment log and discharge gauges are written here as well.

REEF3D_SFLOW_Log (SFLOW) contains `REEF3D_SFLOW_mainlog.dat` with the same type of information.

REEF3D_Log-Probes (CFD and NHFLOW) contains the locations of the wave gauges as a `.vtu` file, which can be loaded in ParaView together with the results.

REEF3D_Log-Wave contains the theoretical time series of the generated waves at the locations given with P 58 (by default at $x = y = 0$).

FNPF does not write a main log file; its progress is followed through the terminal output. The version log is also helpful when reporting a problem, as it identifies the exact code version used.

8.5 Running on HPC Clusters

On HPC clusters, simulations are submitted to a queueing system instead of being started directly. The details (module names, partitions, accounts, whether `srun` or `mpirun` is used) differ between clusters; consult the documentation of your cluster. REEF3D should be compiled on the cluster with the MPI library provided there (Chapter 7). DIVEMesh runs serially and is usually fast, so it can be run on the login node or as the first step of the job. The following generic job script for the Slurm queueing system runs a case with 128 subdomains on two nodes with 64 cores each (`M 10 128` in both input files):

```
#!/bin/bash
#SBATCH --job-name=my_case
#SBATCH --nodes=2
#SBATCH --ntasks-per-node=64
#SBATCH --time=24:00:00
#SBATCH --account=<project>        # cluster specific

module load OpenMPI HYPRE          # module names are cluster specific

cd $SLURM_SUBMIT_DIR
DiveMESH                           # optional, if not run before
srun REEF3D > reef3d_output.txt 2>&1
                                   # or: mpirun -n $SLURM_NTASKS REEF3D
```

The total number of tasks (here $2 \times 64 = 128$) has to be equal to M 10. The job is submitted with **sbatch job.sh**, its status is checked with **squeue -u $USER**, and it is cancelled with **scancel** *job-id*. Large simulations write large amounts of output; use the work or scratch file system of the cluster for the simulation folder and choose the print intervals (P 20, P 30) accordingly.

9. Post-Processing

9.1 Output Files

REEF3D writes its results into folders in the simulation folder whose names start with `REEF3D_` followed by the module name. Field data are written in the VTK XML formats: every MPI process writes one file per output time (`.vtu` or `.vtp`), and a small parallel master file (`.pvtu` or `.pvtp`) that references the files of all processes. **Always open the .pvtu or .pvtp files in ParaView**, not the individual per-process files. File names contain an eight-digit output counter, e.g. `REEF3D-CFD-00000012.pvtu`, and the simulated time is stored in each file, so that ParaView loads the files of one folder as a time series.

The most important folders are:

- `REEF3D_CFD_VTU`, `REEF3D_NHFLOW_VTU`, `REEF3D_FNPF_VTU`: three-dimensional flow field (`.pvtu`), written with the print intervals P 20 (iterations) or P 30 (simulated time). The format is selected with P 10; with the non-default formats the folders end in `_VTS` (`.pvts`) or, for CFD, `_VTR` (`.pvtr`).
- `REEF3D_CFD_FSF`, `REEF3D_NHFLOW_VTP_FSF`, `REEF3D_FNPF_VTP_FSF`: free surface as a surface (`.pvtp`), activated with P 180.
- `REEF3D_SFLOW_VTP_FSF`, `REEF3D_SFLOW_VTP_BED`: SFLOW results. SFLOW is depth-averaged and writes its results as surfaces (`.pvtp`); the print interval is set with P 181 or P 182.
- `REEF3D_NHFLOW_VTP_BED`, `REEF3D_FNPF_VTP_BED`, `REEF3D_CFD_Topo`: bed surface; the CFD sediment bed is activated with P 190.
- `REEF3D_*_6DOF_VTP`: surface of floating bodies (6DOF).
- `REEF3D_*_WSF`: time series of the wave gauges (P 51, P 50) as text files, see Section 9.5.
- `REEF3D_NHFLOW_Particles`, `REEF3D_CFD_Particles`: Lagrangian particles as point clouds (`.vtp`), CSV track file and log file, activated with L 10 (Section 6.8).
- `REEF3D_*_ProbePoint`, `REEF3D_CFD_Force`, `REEF3D_*_Sediment`, ...: further time series from probes, force calculations, sediment transport and other options, as text files (`.dat`).

The log folders are described in Section 8.4. Solid geometries and the bathymetry prepared by DIVEMesh can be visualized from the `.vtp` files that DIVEMesh writes.

9.2 Paraview

ParaView (www.paraview.org) is an open-source visualization tool. It is cross-platform and works on macOS, Windows and Linux. ParaView supports parallel output files as well as parallel rendering through distributed computing. Because it is very flexible, efficient and free, it is the preferred program for visualizing REEF3D's results.

9.3 Windows Users (WSL)

When REEF3D runs in WSL (Section 7.6), ParaView can be installed on Windows as usual. The result files in the Linux file system are opened directly in the Windows version of ParaView under `\\wsl.localhost\Ubuntu\home\<user name>\...`. For very large data sets, the ParaView server `pvserver` of a Linux ParaView installation can alternatively be started in Ubuntu and connected to from the Windows ParaView client with File > Connect; client and server must have the same ParaView version.

9.4 Visualising Two-Phase Flow Results in Paraview

REEF3D can write out a range of different result files, depending on the hydrodynamic module used. In this section, the procedure is presented how to visualize the free surface obtained from two-phase flow simulations with the CFD module. For this type of calculation, the level set method is employed, which defines the free surface implicitly: the free surface is the zero contour of the level set function (the variable phi in the ParaView drop-down menu). In the following, the procedure is presented on how to extract the zero level set contour from the results of the CFD module.

Open the .pvtu file series in the folder `REEF3D_CFD_VTU` in ParaView through File > Open. Click on Outline in the ParaView toolbar as shown in Fig. 9.1 to obtain the outline of the wave tank. Click on the Slice filter on the toolbar (circled in Fig. 9.1) and select "y-normal" in the properties sidebar to generate a 2D slice along the length of the numerical wave tank as shown in Fig. 9.2. The slice can be colored according to the various variables available in the drop-down menu. To obtain the free surface, select the slice in the pipeline browser, click on the Contour filter on the toolbar and select Contour by: phi in the properties sidebar with the value 0, as shown in Fig. 9.3. The geometry of the structure used in the wave tank is included by adding the .vtp file generated by DIVEMesh to obtain the result shown in Fig. 9.4.

To obtain a 3D visualization of the free surface, open the .pvtu file in ParaView and directly apply the Contour filter as shown in Fig. 9.5. Alternatively, the free surface can be written directly as a surface with P 180 (folder `REEF3D_CFD_FSF`).

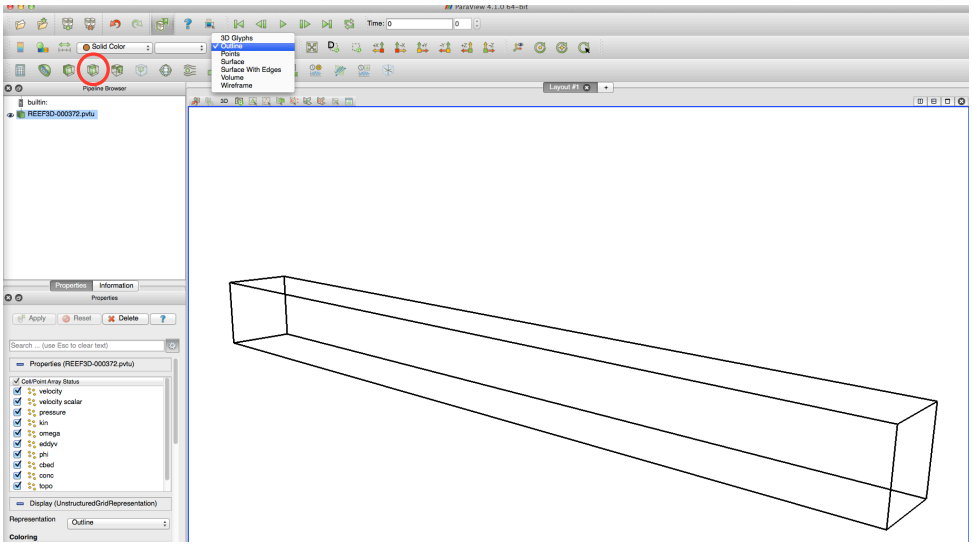


Figure 9.1: Outline of the numerical wave tank

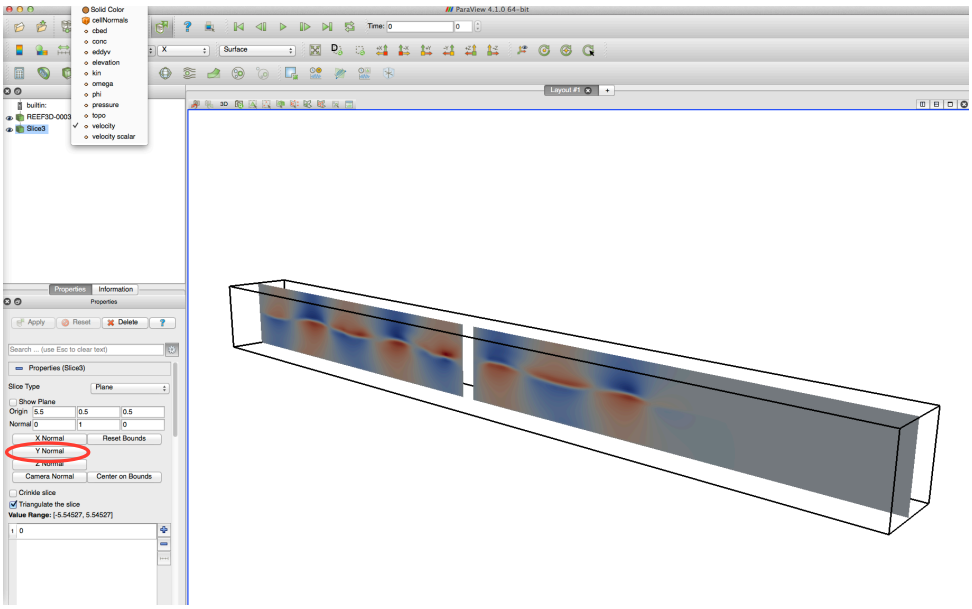


Figure 9.2: Creating a 2D slice

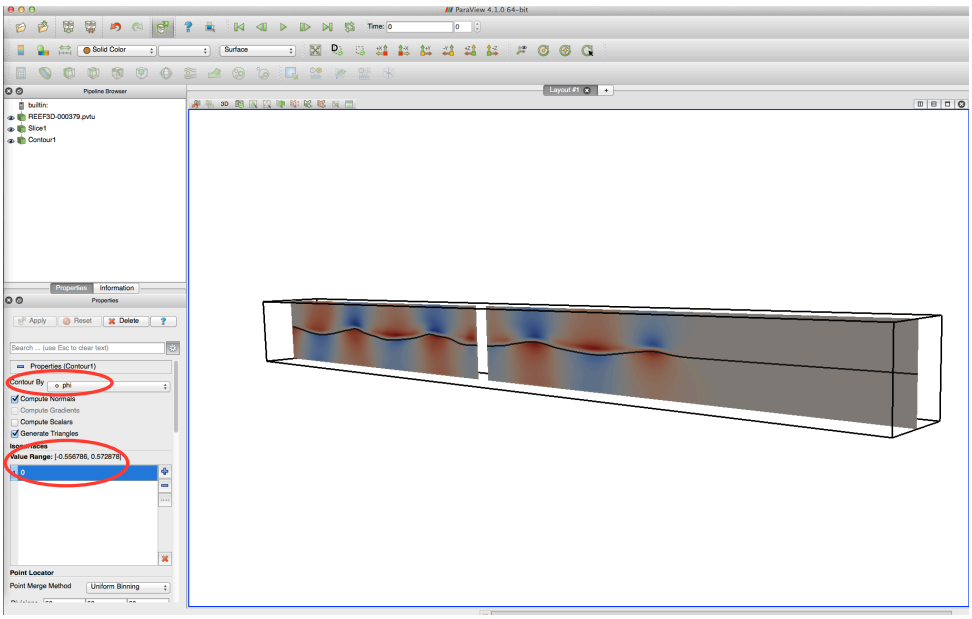


Figure 9.3: Generating the free surface

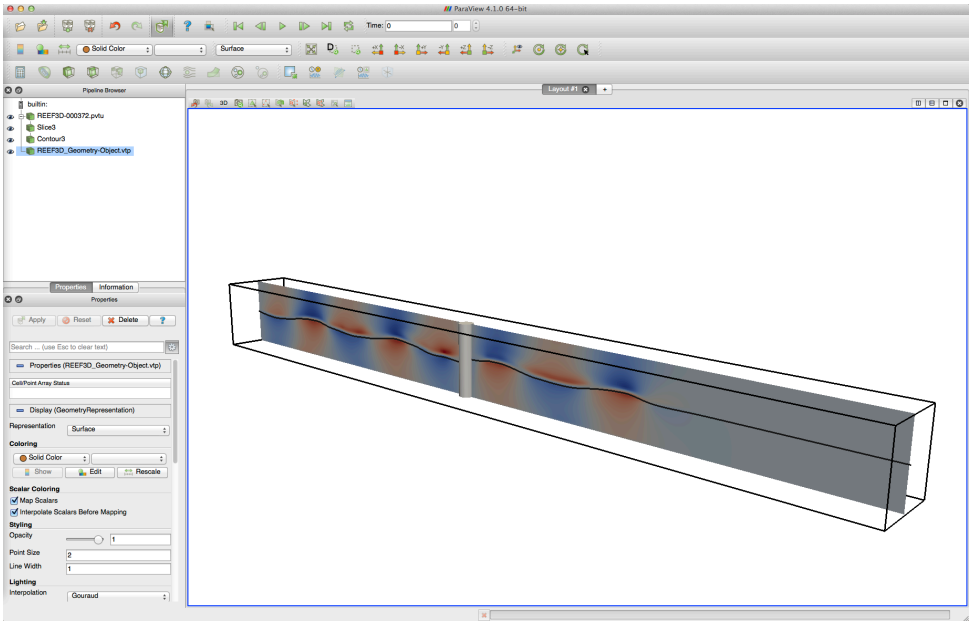


Figure 9.4: Including the geometry of the structure

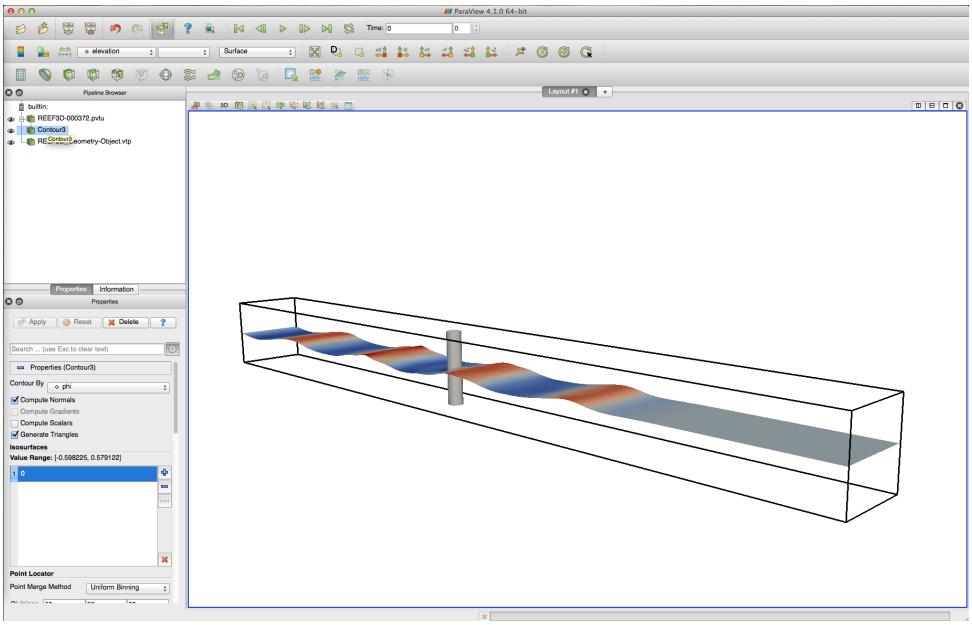


Figure 9.5: Visualizing the free surface in 3D

Generating screenshots and animations from Paraview

Use File > Save Screenshot from the drop-down menu, input the desired resolution, click OK and select the file type in the next window to save a screenshot from ParaView.

To generate an animation, use File > Save Animation from the drop-down menu. Enter the desired frame rate and the range of time steps to be included in the animation in Frame range. For an animation in real time, the frame rate is $1/\Delta t_{P30}$ frames per second, where Δt_{P30} is the print interval set with P 30 (e.g. 10 frames per second for P 30 0.1). Select the desired file type in the next window.

9.5 Time Series from Gauges and Probes

Wave gauges (P 51), probes and force calculations write plain text files that can be plotted with any tool, e.g. Python, Matlab or a spreadsheet program. As an example, the wave gauge file of the FNPF module `REEF3D_FNPF_WSF/REEF3D-FNPF-WSF-HG.dat` (and the corresponding files `REEF3D-CFD-WSF-HG.dat`, `REEF3D-NHFLOW-WSF-HG.dat` and `REEF3D-SFLOW-WSF-HG.dat` of the other modules) has the following structure:

```
number of gauges:  2

x_coord     y_coord
1    5.0    0.5
2    10.0   0.5


time    P1    P2

0.0      0.0      0.0
0.0125   0.0003  -0.0001
...
```

The header lists the number and the x- and y-coordinates of the gauges. It is followed by one line per time step with the simulated time in s and the free surface value of each gauge in m. Because of the adaptive time stepping, the time interval between the lines is in general not constant; interpolate the data to a constant time step before a spectral analysis. FNPF, NHFLOW and SFLOW write the free surface elevation η relative to the still water level. CFD writes the vertical coordinate z of the free surface (the zero level set), so the still water depth has to be subtracted to obtain η. A value of -10^{20} means that no free surface was found at the gauge location, e.g. because it lies outside the domain.

The file can be read and plotted with Python as follows:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
fname = "REEF3D_FNPF_WSF/REEF3D-FNPF-WSF-HG.dat"
with open(fname) as f:
    lines = f.readlines()

# data start after the header line beginning with "time"
start = next(i for i, l in enumerate(lines) if l.startswith("time")) + 1
data = np.loadtxt(lines[start:], ndmin=2)
t, eta = data[:, 0], data[:, 1:]      # eta[:, n] is gauge n+1

plt.plot(t, eta[:, 0], label="gauge 1")
plt.xlabel("t [s]")
plt.ylabel("eta [m]")
plt.legend()
plt.show()
```

10. Tutorial | REEF3D::SFLOW

REEF3D::SFLOW solves the non-hydrostatic shallow water equations: all flow quantities are depth-averaged, so the model runs on a horizontal 2D grid and is very efficient for wave propagation in shallow and intermediate water depth and for open channel flow. The tutorials below are 2D wave flumes with a single cell in y-direction. For visualization, open the REEF3D_SFLOW_VTP_FSF folder (free surface: eta, velocity, pressure, ...) and the REEF3D_SFLOW_VTP_BED folder (bed) in ParaView; wave gauge time series are written to REEF3D_SFLOW_WSF. Each tutorial folder contains the DIVEMesh input 'control.txt' and the REEF3D input 'ctrl.txt'. Run DIVEMesh (**./DiveMESH**) in the folder first, then REEF3D with **mpirun -n** *M 10* **reef3d**.

Note that in SFLOW the ParaView output interval is set with P 182 (time) or P 181 (iterations), not with P 30. The ctrl.txt files in the repository contain P 30, which SFLOW does not read, so without further changes only the initial state is written to REEF3D_SFLOW_VTP_FSF and REEF3D_SFLOW_VTP_BED. Add e.g. P 182 0.1 to obtain a time series of .vtp files.

10.1 2nd-order Stokes waves

Repository folder: `Tutorials/REEF3D_SFLOW/8_1 Stokes Wave Propagation`

Regular 2nd-order Stokes waves propagate over a flat bed in a 28 m long 2D flume. The waves are generated in a relaxation zone at the left end and absorbed by a relaxation beach at the right end. The case demonstrates regular wave propagation with SFLOW and the comparison of the computed surface elevation with the wave theory at three gauges (P 50 and P 51 at the same locations).

- Wave: H = 0.05 m, L = 4.0 m (B 91), water depth d = 0.5 m (F 60)
- Domain: 28 m × 0.02 m, cell size dx = 0.02 m (1400 cells in x)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

10.1.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 28.0 0.0 0.02 0.0 1.0 // rectangular domain x_s, x_e, y_s, y_e, z_s, z_e [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)
```

10.1.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 91 0.05 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.00 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.5 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtp)
P 30 0.1 // no effect in SFLOW (the .vtp interval is set with P 182)

P 50 4.0 0.01 // theoretical wave gauge 1 at x, y [m]
P 51 4.0 0.01 // wave gauge 1 at x, y [m]

P 50 14.5 0.01 // theoretical wave gauge 2 at x, y [m]
P 51 14.5 0.01 // wave gauge 2 at x, y [m]

P 50 19.0 0.01 // theoretical wave gauge 3 at x, y [m]
P 51 19.0 0.01 // wave gauge 3 at x, y [m]

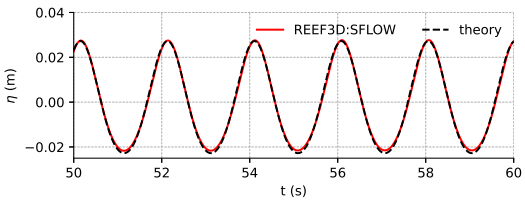
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

W 10 0.0 // discharge [m^3/s], not used (ioFlow is off)
W 22 -9.81 // gravity in z-direction [m/s^2]

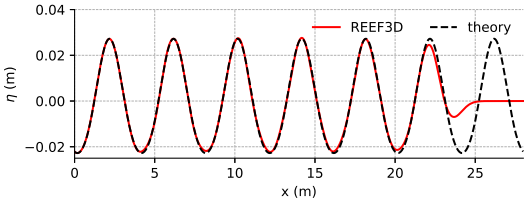
10.1.3 What to look at

Plot the time series in REEF3D_SFLOW_WSF: 'REEF3D-SFLOW-WSF-HG.dat' contains the computed surface elevation at the three P 51 gauges (x = 4.0, 14.5 and 19.0 m) and 'REEF3D-SFLOW-WSF-HG-THEORY.dat' the theoretical elevation at the same P 50 locations. For ParaView, open REEF3D_SFLOW_VTP_FSF and color by eta or velocity (add P 182 for more than the initial state, see the chapter introduction). The spatial profile shown below requires a free surface line: P 54 alone has no effect, add e.g. P 52 0.01 to write REEF3D_SFLOW_WSFLINE files.

10.1.4 Results



(a) Free surface elevation evolution over time at wave gauge 2 (x =14.5 m)



(b) Free surface elevation in space at t =60 s

Figure 10.1: Free surface elevations in the simulation of 2nd-order Stokes waves in intermediate water depth.

10.2 Cnoidal waves over constant water depth

Repository folder: `Tutorials/REEF3D_SFLOW/8_2 Cnoidal Wave Propagation`

Cnoidal waves propagate over a flat bed in the same 28 m flume as the Stokes wave case. In addition to the wave input, this ctrl.txt sets the SFLOW numerics explicitly with the A 2xx commands: 3rd-order Runge-Kutta time stepping, WENO flux reconstruction and the quadratic non-hydrostatic pressure. Note that A 243 0 and A 246 0 switch off wetting-drying and the breaking wave algorithm, which are both on by default.

- Wave: H = 0.21 m, L = 4.0 m (B 91), water depth d = 0.5 m (F 60)
- Domain: 28 m × 0.02 m, cell size dx = 0.02 m (1400 cells in x)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

10.2.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 28.0 0.0 0.02 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

10.2.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

A 210 3 // SFLOW time scheme: 3rd-order Runge-Kutta
A 211 4 // SFLOW convection: WENO flux reconstruction

A 220 2 // SFLOW quadratic non-hydrostatic pressure
A 223 0.5 // blending factor for current/previous η in the hydrostatic pressure gradient

A 240 1 // SFLOW free surface algorithm on
A 243 0 // wetting and drying off
A 246 0 // SFLOW breaking wave algorithm off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 91 0.21 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.00 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.5 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtp)
P 30 0.5 // no effect in SFLOW (the .vtp interval is set with P 182)

P 50 4.0 0.01 // theoretical wave gauge 1 at x, y [m]
P 51 4.0 0.01 // wave gauge 1 at x, y [m]

P 50 14.5 0.01 // theoretical wave gauge 2 at x, y [m]
P 51 14.5 0.01 // wave gauge 2 at x, y [m]

P 50 19.0 0.01 // theoretical wave gauge 3 at x, y [m]
P 51 19.0 0.01 // wave gauge 3 at x, y [m]

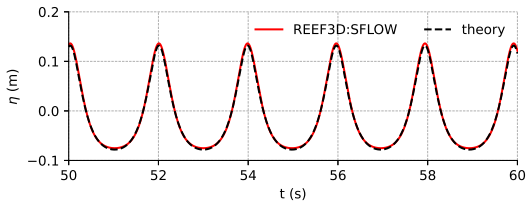
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

W 22 -9.81 // gravity in z-direction [m/s^2]

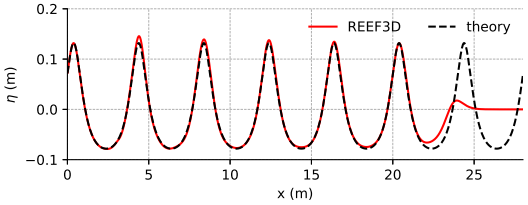
10.2.3 What to look at

Compare the columns of 'REEF3D-SFLOW-WSF-HG.dat' (P 51) and 'REEF3D-SFLOW-WSF-HG-THEORY.dat' (P 50) in REEF3D_SFLOW_WSF at x = 4.0, 14.5 and 19.0 m; the cnoidal wave shape with its sharp crests and flat troughs should be preserved along the flume. As in the previous case, add P 182 for a time series of REEF3D_SFLOW_VTP_FSF files and P 52 for the spatial profile.

10.2.4 Results



(a) Free surface elevation evolution over time at wave gauge 2 (x =14.5 m)



(b) Free surface elevation in space at t =60 s

Figure 10.2: Free surface elevations in the simulation of cnoidal waves in intermediate water depth.

10.3 Solitary wave over constant water depth

Repository folder: `Tutorials/REEF3D_SFLOW/8_3 Solitary Wave Propagation`

A single solitary wave is generated at the left end of the flume and propagates over a flat bed. The case shows that SFLOW keeps the shape of the solitary wave during propagation; the surface elevation at a gauge in the middle of the flume is compared with the theoretical solution. The numerical settings (A 2xx) are the same as in the cnoidal wave case.

- Wave: H = 0.05 m (B 91), water depth d = 0.5 m (F 60)
- Domain: 28 m × 0.02 m, cell size dx = 0.02 m (1400 cells in x)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 20 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

10.3.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 28.0 0.0 0.02 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

10.3.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

A 210 3 // SFLOW time scheme: 3rd-order Runge-Kutta
A 211 4 // SFLOW convection: WENO flux reconstruction

A 220 2 // SFLOW quadratic non-hydrostatic pressure
A 223 0.5 // blending factor for current/previous η in the hydrostatic pressure gradient

A 240 1 // SFLOW free surface algorithm on
A 243 0 // wetting and drying off
A 246 0 // SFLOW breaking wave algorithm off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 9 // wave type: 1st-order solitary wave
B 91 0.05 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.00 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.5 // still water level [m]

N 41 20.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 30 0.5 // no effect in SFLOW (the .vtp interval is set with P 182)

P 50 14.5 0.01 // theoretical wave gauge 1 at x, y [m]
P 51 14.5 0.01 // wave gauge 1 at x, y [m]

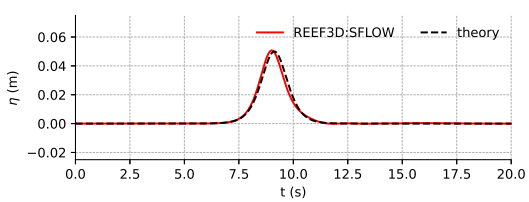
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

W 22 -9.81 // gravity in z-direction [m/s^2]

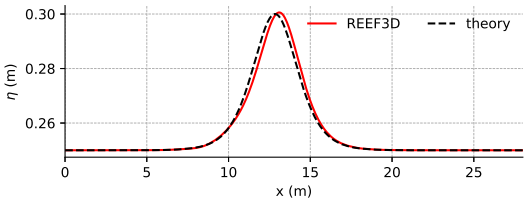
10.3.3 What to look at

Plot the single gauge at x = 14.5 m from 'REEF3D-SFLOW-WSF-HG.dat' (P 51) together with 'REEF3D-SFLOW-WSF-HG-THEORY.dat' (P 50) in REEF3D_SFLOW_WSF. Note that this ctrl.txt has no P 10 line; P 10 is 1 by default, so the initial .vtp files are still written. Add P 182 for a time series and P 52 for the spatial profile.

10.3.4 Results



(a) Free surface elevation evolution over time at the wave gauge (x =14.5 m)



(b) Free surface elevation in space at t =8.4155 s

Figure 10.3: Free surface elevations in the simulation of a solitary wave in intermediate water depth.

10.4 Wave propagation over a submerged bar

Repository folder: `Tutorials/REEF3D_SFLOW/8_4 Wave Propagation over a Submerged Bar`

Regular waves propagate over a submerged trapezoidal bar following the laboratory experiments of Beji and Battjes [1993]. The bar is built from solid objects in DIVEMesh: a front slope from x = 11 m to 17 m, a 2 m long crest and a back slope to x = 22 m, 0.3 m high. On the front slope the waves steepen and higher harmonics are generated, which are released behind the bar. The case tests the dispersion properties of SFLOW.

- Wave: 2nd-order Stokes, H = 0.021 m, T = 2.525 s (B 93), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 38 m × 0.02 m, cell size dx = 0.02 m (1900 cells in x)
- Wave generation zone 5 m, numerical beach 9.5 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

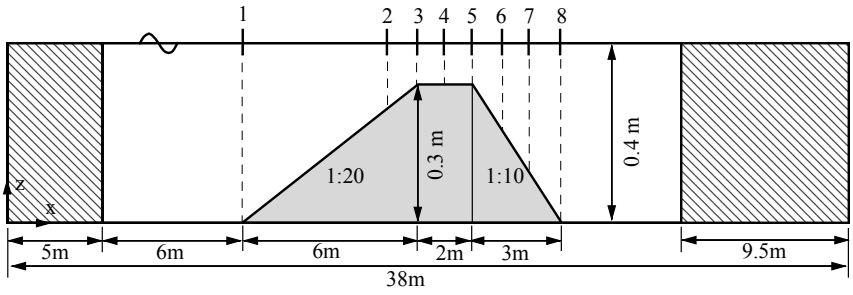


Figure 10.4: Experimental setup submerged bar

10.4.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane

C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 0.0 38.0 0.0 0.02 0.0 0.8 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 11.0 17.0 0.0 0.02 0.0 0.3 // front slope of the bar, rising from z = 0 to 0.3 m
S 10 17.0 19.0 0.0 0.02 0.0 0.3 // bar crest, 0.3 m high
S 61 19.0 22.0 0.0 0.02 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

10.4.2 REEF3D: ctrl.txt

A 10 2 // hydrodynamic model: REEF3D::SFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 0.021 2.525 // wave height H [m], wave period T [s]
B 96 5.0 9.50 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1
b 120 0.0 // lower-case letter: this line is ignored (deactivated)

F 60 0.40 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 51 11.0 0.01 // wave gauge 1 at x, y [m]
P 51 16.0 0.01 // wave gauge 2 at x, y [m]
P 51 17.0 0.01 // wave gauge 3 at x, y [m]
P 51 18.0 0.01 // wave gauge 4 at x, y [m]
P 51 19.0 0.01 // wave gauge 5 at x, y [m]
P 51 20.0 0.01 // wave gauge 6 at x, y [m]
P 51 21.0 0.01 // wave gauge 7 at x, y [m]
P 51 22.0 0.01 // wave gauge 8 at x, y [m]
P 51 24.0 0.01 // wave gauge 9 at x, y [m]

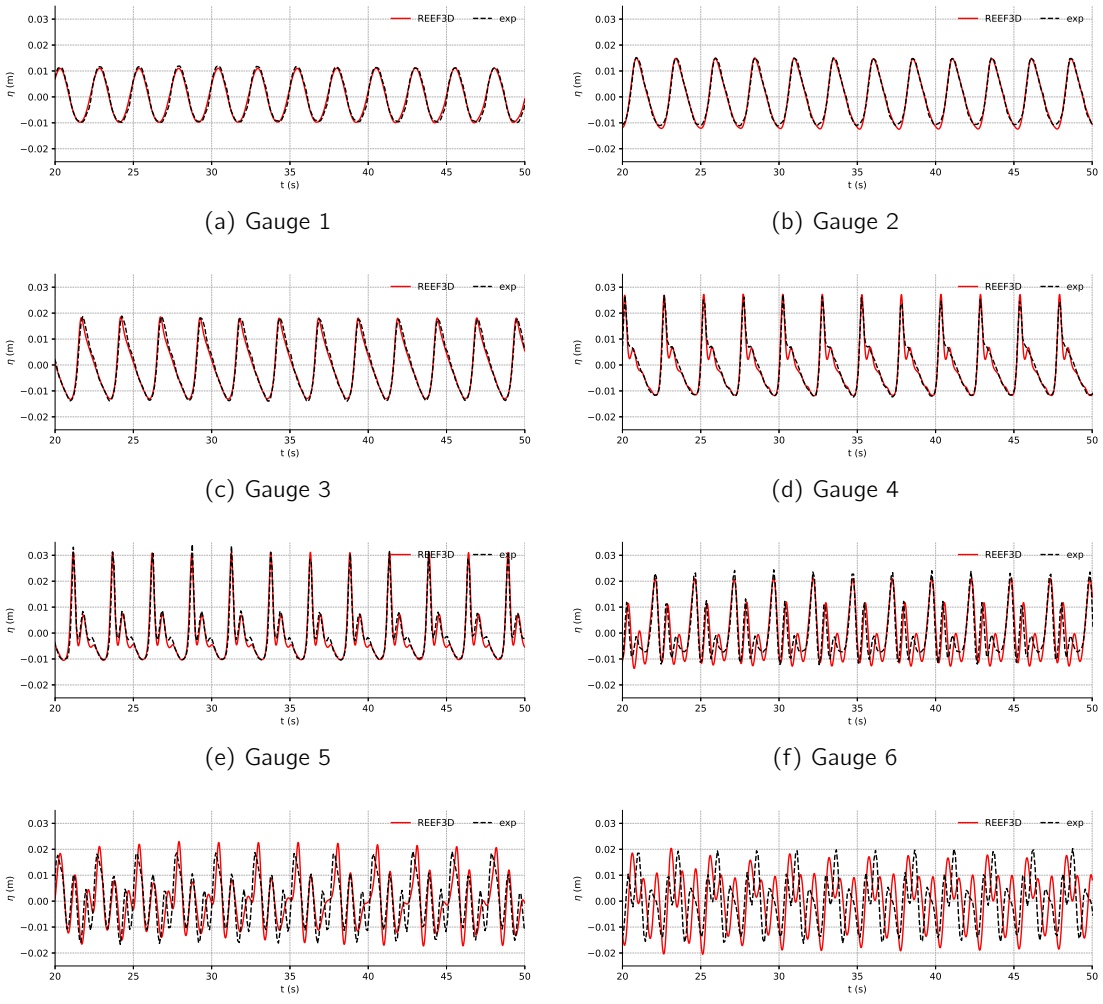
W 10 0.0 // discharge [m^3/s], not used (ioFlow is off)
W 22 -9.81 // gravity in z-direction [m/s^2]

10.4.3 What to look at

The nine P 51 gauges at x = 11, 16, 17, 18, 19, 20, 21, 22 and 24 m are written to 'REEF3D-SFLOW-WSF-HG.dat' in REEF3D_SFLOW_WSF, one column per gauge in the order given in ctrl.txt. Plot them over the last wave periods and compare with the experimental data of Beji and Battjes [1993]; the figures below show the first eight gauges. The bar geometry is visible in the bed elevation of the REEF3D_SFLOW_VTP_BED file.

10.4.4 Results

Note that the high-frequency bound waves released behind the top of the submerged bar are in relatively deep water, which explains the discrepancies with the experiment at wave gauges 6 and onwards. Be aware of the water depth condition when applying depth-averaged models.



(a) Gauge 1

(b) Gauge 2

(c) Gauge 3

(d) Gauge 4

(e) Gauge 5

(f) Gauge 6

(g) Gauge 7

(h) Gauge 8

Figure 10.5: Free surface elevations in the simulation of wave propagation over a submerged bar.

11. Tutorial | REEF3D::FNPF

REEF3D::FNPF is a fully nonlinear potential flow model: the Laplace equation for the velocity potential is solved on a sigma grid that follows the bed and the free surface. It is used for wave propagation from deep to shallow water over large domains and for long irregular sea states. For visualization, open the REEF3D_FNPF_VTU folder (.pvtu files with the velocity potential Fi, velocity and elevation) and, where P 180 is set, the REEF3D_FNPF_VTP_FSF folder (free surface: eta, Fifsf, breaking, ...); the bed is written to REEF3D_FNPF_VTP_BED. Wave gauge time series are written to REEF3D_FNPF_WSF. Each tutorial folder contains 'control.txt' for DIVEMesh and 'ctrl.txt' for REEF3D; run DIVEMesh (**./DiveMESH**) first and then **mpirun -n** *M 10* **reef3d**.

In FNPF the vertical extent given in B 10 of control.txt does not set the water depth: the vertical grid is normalized to a sigma coordinate between the bed and the free surface, whose still water level is given by F 60. Only the number of vertical cells (B 2) and their distribution (B 103, B 113, B 116) matter.

11.1 2nd-order Stokes wave in intermediate water depth

Repository folder: `Tutorials/REEF3D_FNPF/9_1 Regular Wave Propagation`

Regular 2nd-order Stokes waves propagate over a flat bed in a 40 m long 2D wave tank. The waves are generated in a relaxation zone at the left end and absorbed in a relaxation beach at the right end. The vertical grid is clustered towards the free surface. The computed surface elevation is compared with the wave theory both in time (gauges) and in space (free surface line).

- Wave: H = 0.04 m, L = 4.0 m (B 91), water depth d = 2.0 m (F 60)
- Domain: 40 m long, 400 × 1 × 10 cells (B 2), dx = 0.1 m
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 40 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.1.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.1 // uniform cell size dx [m]
B 2 400 1 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 40.0 0.0 0.1 0.0 2.0 // rectangular domain x_s,x_e,y_s,y_e,z_s,z_e [m]
```

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 2.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.1.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 91 0.04 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 2.0 // still water level [m]

N 41 40.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

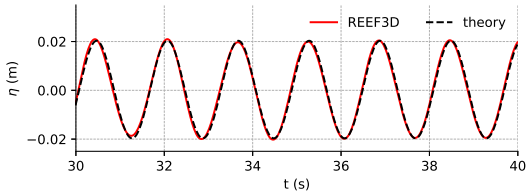
P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]
P 50 15.0 0.005 // theoretical wave gauge 1 at x, y [m]
P 50 20.0 0.005 // theoretical wave gauge 2 at x, y [m]
P 50 25.0 0.005 // theoretical wave gauge 3 at x, y [m]
P 51 15.0 0.005 // wave gauge 1 at x, y [m]
P 51 20.0 0.005 // wave gauge 2 at x, y [m]
P 51 25.0 0.005 // wave gauge 3 at x, y [m]
P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 54 100 // wsfline print interval in iterations (needs P 52)

W 22 -9.81 // gravity in z-direction [m/s^2]

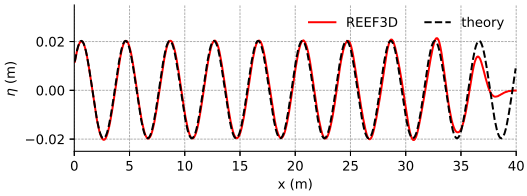
11.1.3 What to look at

Plot 'REEF3D-FNPF-WSF-HG.dat' (P 51) against 'REEF3D-FNPF-WSF-HG-THEORY.dat' (P 50) in REEF3D_FNPF_WSF for the gauges at x = 15, 20 and 25 m. The files in REEF3D_FNPF_WSFLINE contain the free surface along the tank at y = 0.005 m every 100 iterations, together with the theoretical profile (P 53 1). In ParaView, open REEF3D_FNPF_VTU (every 0.5 s) and color by Fi or velocity.

11.1.4 Results



(a) Free surface elevation evolution over time at wave gauge 2 (x =20 m)



(b) Free surface elevation in space at t =40 s

Figure 11.1: Free surface elevations in the simulation of 2nd-order Stokes waves in intermediate water depth.

11.2 Wave propagation over a submerged bar

Repository folder: `Tutorials/REEF3D_FNPF/9_2 Wave Propagation Over a Submerged Bar`

Regular waves propagate over a submerged trapezoidal bar following the laboratory experiments of Beji and Battjes [1993]. The bar is built from solid objects in DIVEMesh (front slope x = 11–17 m, crest x = 17–19 m, back slope x = 19–22 m, 0.3 m high), and the sigma grid follows the resulting bed. The generation of higher harmonics on the bar and their release behind it test the nonlinear and dispersive properties of FNPF.

- Wave: 2nd-order Stokes, H = 0.021 m, T = 2.525 s (B 93), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 40 m long, 1000 × 1 × 10 cells (B 2), dx = 0.04 m
- Wave generation zone 5 m, numerical beach 10 m (B 96), simulated time 100 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

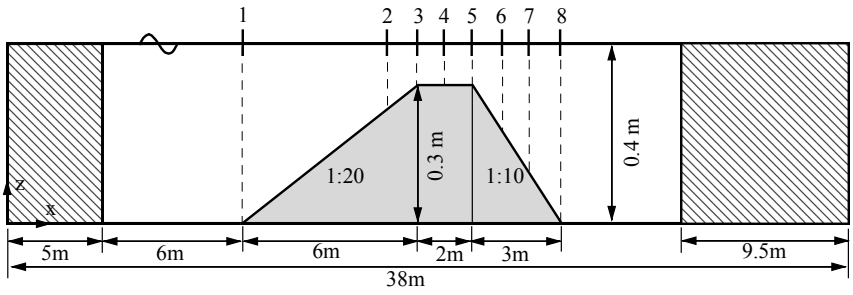


Figure 11.2: Experimental setup submerged bar

11.2.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane

C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.04 // uniform cell size dx [m]
B 2 1000 1 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 40.0 0.0 0.04 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 11.0 17.0 0.0 0.04 0.0 0.3 // front slope of the bar, rising from $z = 0$ to 0.3 m
S 10 17.0 19.0 0.0 0.04 0.0 0.3 // bar crest, 0.3 m high
S 61 19.0 22.0 0.0 0.04 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 3.0 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.2.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 0.021 2.525 // wave height H [m], wave period T [s]
B 96 5.0 10.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 0.4 // still water level [m]

N 41 100.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]

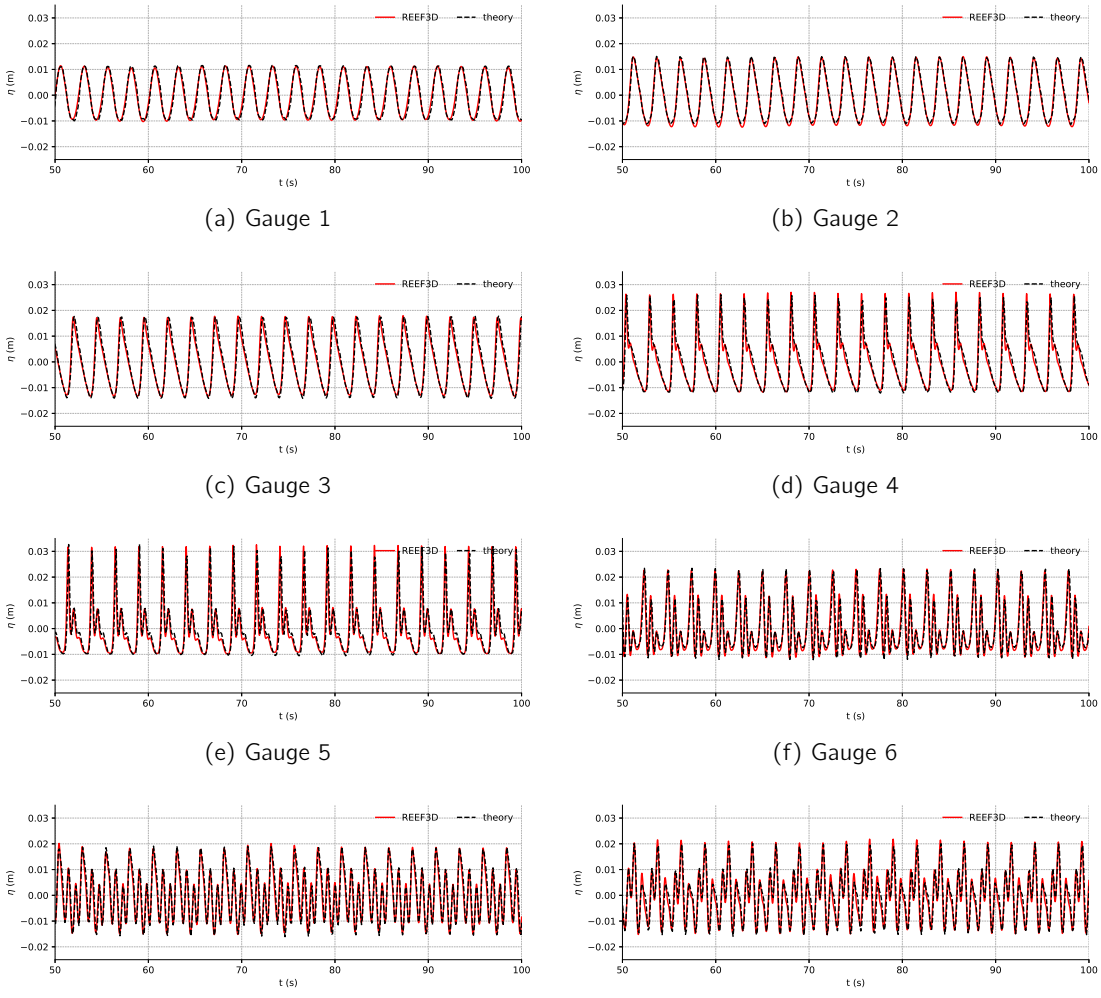
P 51 11.0 0.005 // wave gauge 1 at x, y [m]
P 51 16.0 0.005 // wave gauge 2 at x, y [m]
P 51 17.0 0.005 // wave gauge 3 at x, y [m]
P 51 18.0 0.005 // wave gauge 4 at x, y [m]
P 51 19.0 0.005 // wave gauge 5 at x, y [m]
P 51 20.0 0.005 // wave gauge 6 at x, y [m]
P 51 21.0 0.005 // wave gauge 7 at x, y [m]
P 51 22.0 0.005 // wave gauge 8 at x, y [m]
P 51 24.0 0.005 // wave gauge 9 at x, y [m]

W 22 -9.81 // gravity in z-direction [m/s^2]

11.2.3 What to look at

'REEF3D-FNPF-WSF-HG.dat' in REEF3D_FNPF_WSF contains one column per P 51 gauge (x = 11, 16, 17, 18, 19, 20, 21, 22 and 24 m) in the order given in ctrl.txt; compare them with the measurements of Beji and Battjes [1993]. The figures below show the first eight gauges. In ParaView, REEF3D_FNPF_VTU (every 0.5 s) shows the wave field over the bar.

11.2.4 Results



(a) Gauge 1

(b) Gauge 2

(c) Gauge 3

(d) Gauge 4

(e) Gauge 5

(f) Gauge 6

(g) Gauge 7

(h) Gauge 8

Figure 11.3: Free surface elevations in the simulation of wave propagation over a submerged bar.

11.3 Wave breaking over a mild slope

Repository folder: `Tutorials/REEF3D_FNPF/9_3 Wave Breaking Over a Mild Slope`

Cnoidal waves shoal and break on a mild slope of about 1:35, similar to the plunging breaker experiment of Ting and Kirby [1995] that is also used in the CFD tutorials (shifted by 8 m in x). The waves are generated with the Dirichlet method (B 98 3), which is required in FNPF for wave theories that do not prescribe the velocity potential directly, such as cnoidal waves (see B 92). Breaking is modelled with the viscous damping algorithm (A 350 1) with breaking detection and filtering for shallow and deep water; wetting and drying (A 343 1) handles the shoreline on the slope.

- Wave: 5th-order cnoidal, H = 0.128 m, T = 5.0 s (B 93), water depth d = 0.4 m
- Domain: 30 m long, 600 × 1 × 10 cells (B 2), dx = 0.05 m, slope from x = 5.8 m
- Numerical beach 9 m (B 96), simulated time 160 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.3.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]
B 2 600 1 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 30.0 0.0 0.05 0.0 0.748 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 1.0 // stretching factor in z-direction
B 116 0.4 // focus point of the z-stretching [m]

S 61 5.8 32.0 0.0 0.05 0.0 0.748 // bed slope (about 1:35) starting at x = 5.8 m

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.3.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF

A 341 2.0 // coastline relaxation zone width, factor times dx
A 343 1 // wetting and drying on
A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 3 // breaking detection: shallow and deep water
A 352 3 // additional filtering for breaking in shallow and deep water
A 365 0.0025 // viscosity for breaking wave energy dissipation

B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 93 0.128 5.0 // wave height H [m], wave period T [s]
B 96 9.5 9.0 // lengths of wave generation zone and numerical beach [m]
B 98 3 // wave generation: Dirichlet (DWG)
B 99 1 // numerical beach: relaxation method 1

F 60 0.4 // still water level [m]

N 41 160.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 23 1 // add the test array to the ParaView output
P 30 0.1 // print interval in simulated time [s]

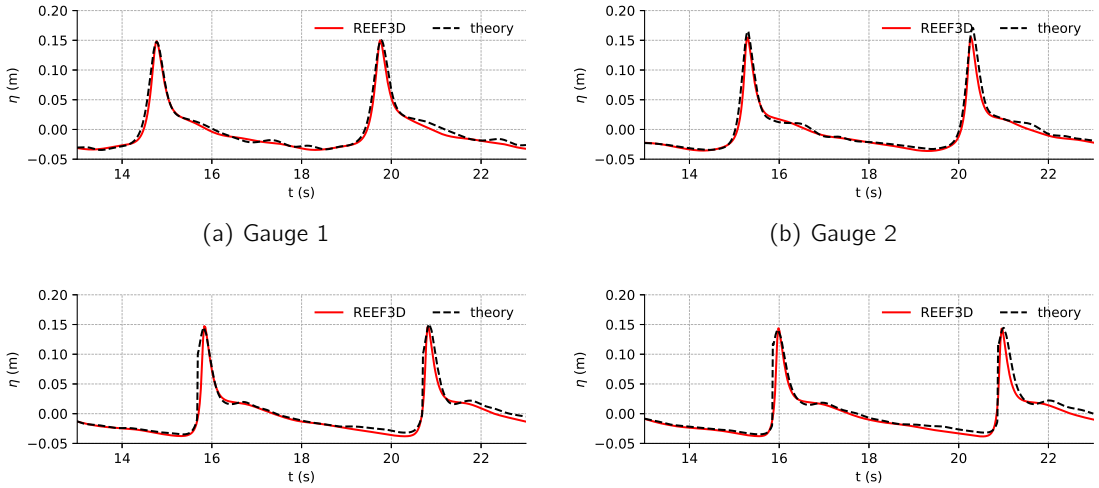
P 51 11.8 0.0025 // wave gauge 1 at x, y [m]
P 51 12.8 0.0025 // wave gauge 2 at x, y [m]
P 51 13.8 0.0025 // wave gauge 3 at x, y [m]
P 51 14.1 0.0025 // wave gauge 4 at x, y [m]

W 22 -9.81 // gravity in z-direction [m/s^2]

11.3.3 What to look at

Plot the four P 51 gauges at x = 11.8, 12.8, 13.8 and 14.1 m from 'REEF3D-FNPF-WSF-HG.dat' in REEF3D_FNPF_WSF; they lie around the breaking point on the slope. In ParaView, open REEF3D_FNPF_VTU (every 0.1 s) and color the free surface by the breaking flag to see where the breaking algorithm is active; P 23 1 adds the test array to these files.

11.3.4 Results



(a) Gauge 1

(b) Gauge 2

(c) Gauge 3

(d) Gauge 4

Figure 11.4: Free surface elevations in the simulation of breaking waves over a mild slope.

11.4 Irregular wave sea state in 2D

Repository folder: `Tutorials/REEF3D_FNPF/9_4 Irregular Wave Propagation`

A long-crested irregular sea state based on a JONSWAP spectrum is simulated in a 6 km long 2D wave tank in deep water for 12800 s, i.e. more than the typical 3-hour design sea state. The spectrum is discretized into 2048 components with the peak enhance method (B 84 1). A fixed time step is used, and the breaking algorithm with deep water detection keeps steep waves stable. To keep the output small, the .vtu files are only written in the last 50 s (P 35).

- Wave: JONSWAP, H_s = 5.0 m, T_p = 12.0 s, γ = 3.0, ω = 0.32–1.4 rad/s, water depth d = 600 m
- Domain: 6000 m long, 6000 × 1 × 20 cells (B 2), dx = 1.0 m
- Wave generation zone 600 m, numerical beach 1200 m (B 96), fixed time step 0.03 s, simulated time 12800 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.4.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane
```

B 1 1.0 // uniform cell size dx [m]
B 2 6000 1 20 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 6000.0 0.0 1.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 3.0 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.4.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off
A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 2 // breaking detection: deep water
A 352 0 // no additional filtering in breaking regions

B 84 1 // spectrum discretisation: peak enhance method
B 85 2 // JONSWAP spectrum
B 86 2048 // number of wave components
B 87 0.32 1.4 // lower and upper angular frequency ω_s, ω_e [rad/s]
B 88 3.0 // JONSWAP peak enhancement factor γ
B 90 1 // turn on ioWave (numerical wave tank)
B 92 31 // wave type: 1st-order irregular waves
B 93 5.00 12.00 // significant wave height H_s [m], peak period T_p [s]
B 96 600.0 1200.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1
b 139 16000 // lower-case letter: ignored (B 139 would set the random seed of the spectrum)

F 60 600.0 // still water level [m]

N 41 12800.0 // maximum simulated time [s]
n 47 1.0 // lower-case letter: ignored (not needed with the fixed time step)
N 48 0 // adaptive time stepping off
N 49 0.03 // fixed time step [s]

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 12750.0 12800.0 1.0 // print window: start [s], end [s], interval [s]

P 51 300.00 0.5 // wave gauge 1 at x, y [m]
P 51 1500.0 0.5 // wave gauge 2 at x, y [m]
P 51 3500.0 0.5 // wave gauge 3 at x, y [m]

W 22 -9.81 // gravity in z-direction [m/s^2]

11.4.3 What to look at

The main result is the surface elevation time series in 'REEF3D-FNPF-WSF-HG.dat' (REEF3D_FNPF_WSF) at x = 300, 1500 and 3500 m. Compute the spectrum of a gauge time series (discarding the initial ramp-up) and compare it with the input JONSWAP spectrum, see the figure below for gauge 2. REEF3D_FNPF_VTU contains snapshots every 1 s between 12750 s and 12800 s.

11.4.4 Results

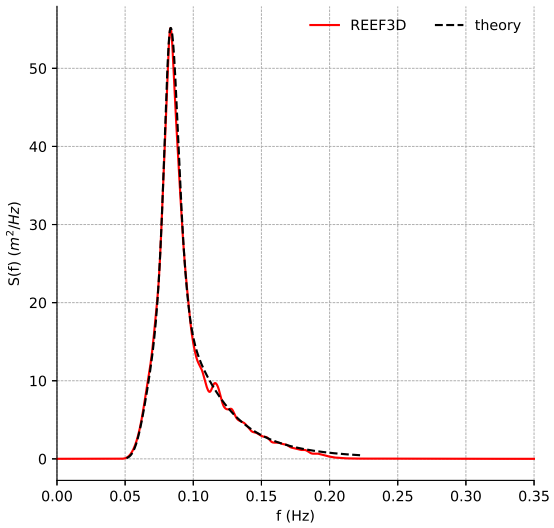


Figure 11.5: Wave spectrum in the simulation of a 3-hour irregular sea state at wave gauge 2.

11.5 Wave propagation over natural bathymetry

Repository folder: `Tutorials/REEF3D_FNPF/9_5 Wave Propagation Over Natural Bathymetry`

Regular waves propagate over a measured natural bathymetry in a 4 km × 4 km domain. DIVEMesh interpolates the bed from the point cloud in 'geo.dat' (about 10^6 x, y, z points, required in the case folder) with local inverse distance interpolation. Waves are generated along part of the left boundary (B 108), and relaxation beaches are placed along the remaining open boundaries (B 107). Coastlines are treated with wetting-drying and a coastline relaxation zone, and breaking with the viscous damping algorithm including filtering.

- Wave: 2nd-order Stokes, H = 2.0 m, T = 10.0 s (B 93), still water level z = 300 m (F 60), bed from geo.dat
- Domain: 4000 m × 4000 m, 800 × 800 × 10 cells (B 2), dx = 5.0 m
- Relaxation zones 200 m wide (B 107, B 108), simulated time 600 s
- Extra input file: geo.dat (read by DIVEMesh)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.5.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 5.0 // uniform cell size dx [m]
B 2 800 800 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 4000.0 0.0 4000.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

G 10 1 // turn on geodat interpolation of the bed from geo.dat
G 15 2 // local inverse distance interpolation
G 20 0 // automatic grid size from geo.dat extent: off
G 31 14 // number of outer smoothing filter iterations
G 41 1 // print SWAN bottom file from the interpolated bed

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.5.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 5 // FNPF free surface BCs: 5th-order WENO

A 341 2.0 // coastline relaxation zone width, factor times dx
A 343 1 // wetting and drying on
A 344 0.1 // wetting-drying threshold water depth [m]
A 346 2.1 // damping viscosity in the coastline relaxation zone

A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 3 // breaking detection: shallow and deep water
A 352 3 // additional filtering for breaking in shallow and deep water
A 361 5 // breaking filter: outer iterations
A 362 2 // breaking filter: inner iterations
A 365 1.86 // viscosity for breaking wave energy dissipation

B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 2.0 10.0 // wave height H [m], wave period T [s]
B 96 200.0 400.0 // lengths of wave generation zone and numerical beach [m]
B 107 0.0 4000.0 0.0 0.0 200.0 // beach zone along the side $y = 0$, 200 m wide

B 107 0.0 4000.0 4000.0 4000.0 200.0 // beach zone along the side y = 4000 m
B 107 4000.0 4000.0 0.0 4000.0 200.0 // beach zone along the end x = 4000 m
B 107 0.0 0.0 2900.0 3500.0 200.0 // beach zone on the inflow side x = 0 for y = 2900–3500 m
B 108 0.0 0.0 200.0 2900.0 200.0 // wave generation zone on x = 0 for y = 200–2900 m
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

F 60 300.0 // still water level [m]

I 30 0 // full wave tank initialization off

N 41 600.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 10.0 // print interval in simulated time [s]

P 51 2000.0 2000.0 // wave gauge at the domain centre x = y = 2000 m

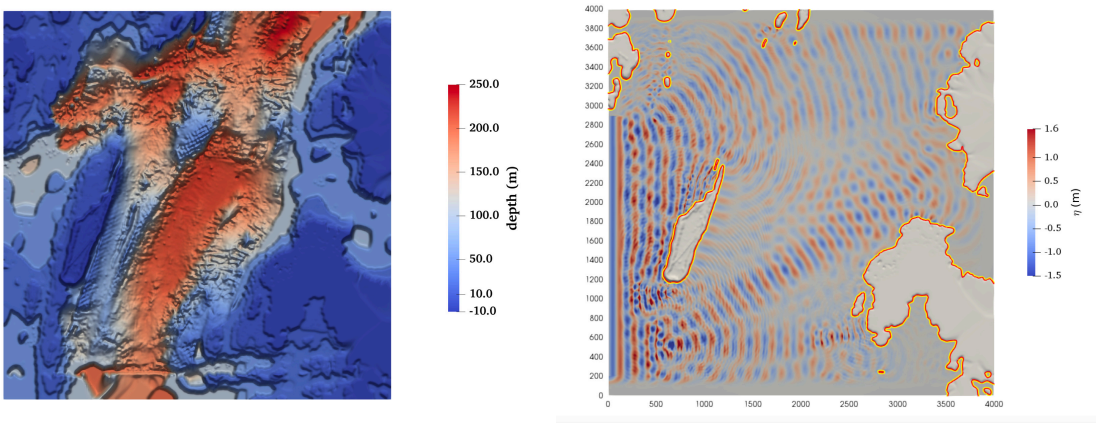
P 180 1 // print free surface .vtp files
P 185 0.0 120.0 0.1 // free surface .vtp print window: start [s], end [s], interval [s]
P 185 480.0 600.0 0.1 // free surface .vtp print window: start [s], end [s], interval [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

11.5.3 What to look at

Open REEF3D_FNPF_VTP_FSF in ParaView: free surface files are written every 0.1 s in the windows 0–120 s and 480–600 s (P 185); color by eta to see refraction and shoaling over the bathymetry, and by breaking and coastline to see where breaking and wetting-drying are active. The bed is in REEF3D_FNPF_VTP_BED, and REEF3D_FNPF_VTU contains the full field every 10 s. The single P 51 gauge at the domain centre is written to REEF3D_FNPF_WSF.

11.5.4 Results



(a) Bathymetry

(b) Free surface elevation

Figure 11.6: Free surface elevations in the simulation of wave propagation over natural bathymetry

11.6 3D short-crested multi-directional irregular wave

Repository folder: `Tutorials/REEF3D_FNPF/9_6 Short Crested Multi-directional Irregular Wave`

A short-crested irregular sea state is generated in a 2 km × 2 km basin in deep water. The JONSWAP spectrum is discretized with the equal energy method (B 84 2) and spread over 16 directions with a Mitsuyasu spreading function around the principal direction 0° (B 130–B 136). An array of six wave gauges (one centre gauge and five gauges on a circle of about 28 m radius) is used to estimate the directional spectrum. I 30 1 initializes the whole tank with the wave field, which is useful for a quick check of the setup; for production runs I 30 0 is recommended.

- Wave: JONSWAP, H_s = 4.5 m, T_p = 12.0 s, γ = 3.3, ω = 0.39–1.01 rad/s, spreading parameter 10, water depth d = 600 m
- Domain: 2000 m × 2000 m, 1000 × 1000 × 10 cells (B 2), dx = 2.0 m
- Wave generation zone 400 m, numerical beach 800 m (B 96), fixed time step 0.08 s, simulated time 12800 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

11.6.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane
```

B 1 2.0 // uniform cell size dx [m]
B 2 1000 1000 10 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 2000.0 0.0 2000.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.875 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

11.6.2 REEF3D: ctrl.txt

A 10 3 // hydrodynamic model: REEF3D::FNPF
A 310 3 // FNPF time scheme: 3rd-order TVD Runge-Kutta
A 311 4 // FNPF free surface BCs: 5th-order WENO
A 343 0 // wetting and drying off
A 350 1 // breaking wave algorithm: viscous damping (Baquet)
A 351 2 // breaking detection: deep water
A 352 0 // no additional filtering in breaking regions
A 365 1.86 // viscosity for breaking wave energy dissipation

B 84 2 // spectrum discretisation: equal energy method
B 85 2 // JONSWAP spectrum
B 86 1024 // number of wave components
B 87 0.39 1.01 // lower and upper angular frequency ω_s, ω_e [rad/s]
B 88 3.3 // JONSWAP peak enhancement factor γ
B 90 1 // turn on ioWave (numerical wave tank)
B 92 31 // wave type: 1st-order irregular waves
B 93 4.5 12.0 // significant wave height H_s [m], peak period T_p [s]
B 96 400.0 800.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1
B 130 2 // directional spreading: Mitsuyasu
B 131 0.0 // principal wave direction [deg]
B 132 -90.0 90.0 // start and end angle of the spreading [deg]
B 133 16 // number of directions
B 134 10.0 // spreading parameter of the spreading function
B 136 4 // multi-directional method: equal energy method (with B 84 2)

F 60 600.0 // still water level [m]

I 30 1 // full wave tank initialization on (quick setup check)

N 41 12800.0 // maximum simulated time [s]
N 48 0 // adaptive time stepping off
N 49 0.08 // fixed time step [s]

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 12700.0 12800.0 0.5 // print window: start [s], end [s], interval [s]

P 51 800.0 1000.0 // wave gauge array for the directional analysis: gauge 1
P 51 777.2464 983.4685 // gauge 2
P 51 808.6911 973.2515 // gauge 3
P 51 828.1250 1000.0000 // gauge 4
P 51 808.6911 1026.7485 // gauge 5
P 51 777.2464 1016.5315 // gauge 6

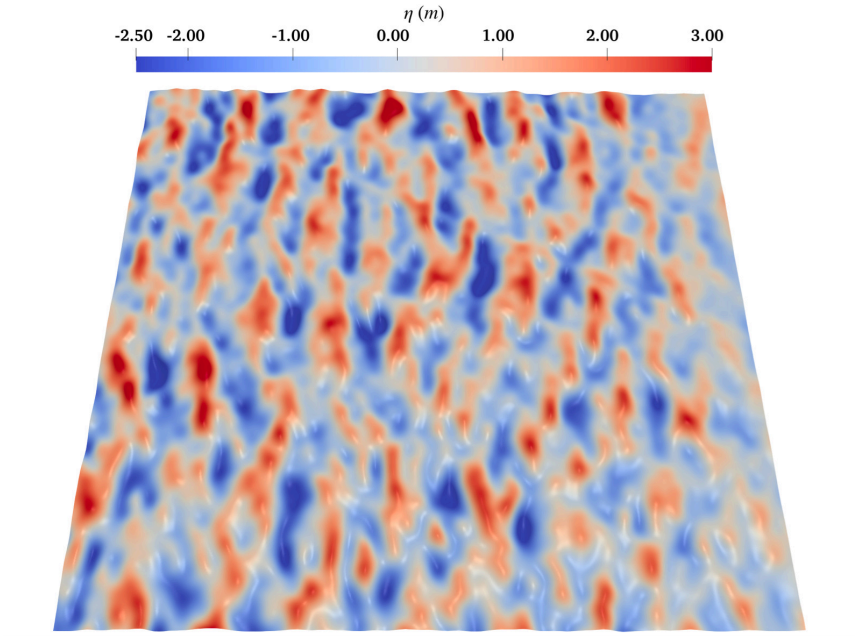
P 180 1 // print free surface .vtp files
P 185 0.0 300.0 0.5 // free surface .vtp print window: start [s], end [s], interval [s]
P 185 12500.0 12800.0 0.5 // free surface .vtp print window: start [s], end [s], interval [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

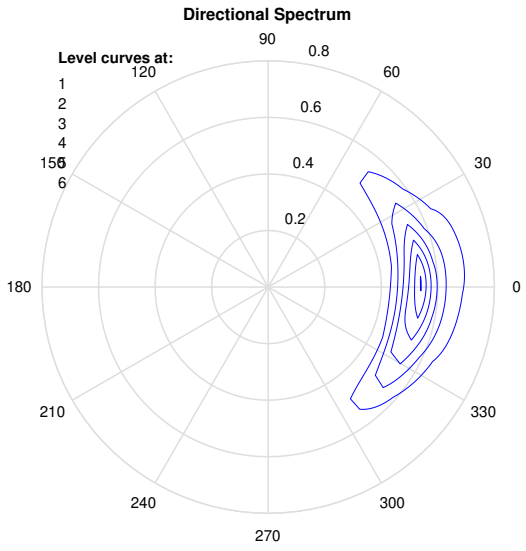
11.6.3 What to look at

Use the six columns of 'REEF3D-FNPF-WSF-HG.dat' in REEF3D_FNPF_WSF to estimate the directional spectrum and compare it with the input spreading. REEF3D_FNPF_VTP_FSF contains the free surface every 0.5 s for 0–300 s and 12500–12800 s (P 185), which shows the short-crested wave field; REEF3D_FNPF_VTU contains snapshots every 0.5 s between 12700 s and 12800 s (P 35).

11.6.4 Results



(a) Free surface elevation



(b) Directional spectrum

Figure 11.7: Free surface elevations and directional spectrum in the simulation of a short-crested irregular sea.

12. Tutorial | REEF3D::NHFLOW

REEF3D::NHFLOW is a three-dimensional non-hydrostatic wave model on a sigma grid that follows the bed and the free surface; the free surface is computed with a shock-capturing Godunov-type scheme and the dynamic pressure with a pressure correction. In contrast to the depth-averaged SFLOW, NHFLOW resolves the vertical flow structure, typically with only a few vertical cells. For visualization, open the REEF3D_NHFLOW_VTU folder (.pvtu files with velocity, pressure and elevation) and, where P 180 is set, the REEF3D_NHFLOW_VTP_FSF folder (free surface: eta, velocity, breaking, ...); wave gauge time series are written to REEF3D_NHFLOW_WSF. As in FNPF, the vertical extent of the DIVEMesh domain is normalized to the sigma coordinate: the water depth follows from F 60 and the bed, and B 2 sets the number of vertical cells. Run DIVEMesh (**./DiveMESH**) in the case folder first and then **mpirun -n** *M 10* **reef3d**.

12.1 5th-order Stokes waves

Repository folder: `Tutorials/REEF3D_NHFLOW/1_Stokes5`

Regular 5th-order Stokes waves propagate over a flat bed in a 200 m long 2D wave tank; the setup is the same as the CFD tutorial for 5th-order Stokes waves (`11_9`), which allows a direct comparison of the two models. The waves are generated in a relaxation zone of about one wavelength and absorbed in a relaxation beach of about two wavelengths. The vertical grid has only five cells, clustered towards the free surface (B 103, B 113, B 116). Computed and theoretical surface elevations are written both at gauges and along the tank.

- Wave: H = 1.0 m, T = 4.5 s (B 93), water depth d = 4.01 m (F 60), wavelength about 25 m
- Domain: 200 m long, 350 × 1 × 5 cells (B 2 overrides B 1), $dx \approx 0.57$ m
- Wave generation zone 25 m, numerical beach 50 m (B 96), simulated time 120 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

12.1.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]
B 2 350 1 5 // number of cells N_x, N_y, N_z (overrides B 1)
```

B 10 0.0 200.0 0.0 0.05 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]
B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

12.1.2 REEF3D: ctrl.txt

A 10 5 // hydrodynamic model: REEF3D::NHFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 93 1.0 4.5 // wave height H [m], wave period T [s]
B 96 25.0 50.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 1 // numerical beach: relaxation method 1

F 60 4.01 // still water level [m]

I 12 1 // hydrostatic pressure initialization (CFD only, no effect in NHFLOW)

N 41 120.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.25 // print interval in simulated time [s]
P 50 15.0 0.005 // theoretical wave gauge 1 at x, y [m]
P 50 50.0 0.005 // theoretical wave gauge 2 at x, y [m]
P 50 75.0 0.005 // theoretical wave gauge 3 at x, y [m]
P 50 125.0 0.005 // theoretical wave gauge 4 at x, y [m]

P 51 15.0 0.005 // wave gauge 1 at x, y [m]
P 51 50.0 0.005 // wave gauge 2 at x, y [m]
P 51 75.0 0.005 // wave gauge 3 at x, y [m]
P 51 125.0 0.005 // wave gauge 4 at x, y [m]

P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 55 0.05 // wsfline and line probe print interval [s]

P 180 1 // print free surface .vtp files
P 182 0.05 // free surface .vtp print interval [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

12.1.3 What to look at

Plot 'REEF3D-NHFLOW-WSF-HG.dat' (P 51) against 'REEF3D-NHFLOW-WSF-HG-THEORY.dat' (P 50) in REEF3D_NHFLOW_WSF at x = 15, 50, 75 and 125 m; compare crest height and phase with the theory along the tank. REEF3D_NHFLOW_WSFLINE contains the free surface along the tank with the theoretical profile every 0.05 s (P 52, P 53, P 55). In ParaView, open REEF3D_NHFLOW_VTP_FSF (every 0.05 s) for the free surface and REEF3D_NHFLOW_VTU (every 0.25 s) for the velocity field under the waves.

12.2 Wave propagation over a submerged bar

Repository folder: `Tutorials/REEF3D_NHFLOW/2_Submerged Bar`

Regular waves propagate over a submerged trapezoidal bar following the laboratory experiments of Beji and Battjes [1993], with the same geometry as the CFD tutorial (11_10): front slope x = 6–12 m, crest x = 12–14 m, back slope x = 14–17 m, 0.3 m high. The case tests how well NHFLOW with three vertical cells represents the generation of higher harmonics on the bar and their release behind it. Nine gauges are placed along the tank.

- Wave: linear, H = 0.02 m, T = 2.02 s (B 93), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 35 m long, 1200 × 1 × 3 cells (B 2), $dx \approx 0.029$ m
- Wave generation zone 3.73 m, numerical beach 8.73 m (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

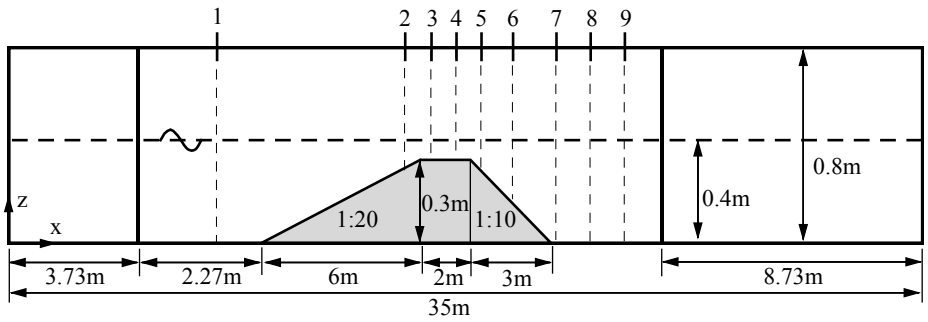


Figure 12.1: Experimental setup submerged bar

12.2.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach

C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 2 1200 1 3 // number of cells N_x, N_y, N_z (overrides B 1)
B 10 0.0 35.0 0.0 0.01 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 6.0 12.0 0.0 0.01 0.0 0.3 // front slope of the bar, rising from $z = 0$ to 0.3 m
S 10 12.0 14.0 0.0 0.01 0.0 0.3 // bar crest, 0.3 m high
S 61 14.0 17.0 0.0 0.01 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 1.0 // focus point of the z-stretching [m]

M 10 6 // number of processes (partitions)

12.2.2 REEF3D: ctrl.txt

A 10 5 // hydrodynamic model: REEF3D::NHFLOW

B 90 1 // turn on ioWave (numerical wave tank)
B 92 2 // wave type: linear waves
B 93 0.02 2.02 // wave height H [m], wave period T [s]
B 96 3.73 8.73 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

F 60 0.4 // still water level [m]

N 41 60.0 // maximum simulated time [s]
N 47 1.0 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 4.0 0.005 // wave gauge 1 at x, y [m]
P 51 10.5 0.005 // wave gauge 2 at x, y [m]
P 51 12.5 0.005 // wave gauge 3 at x, y [m]
P 51 13.5 0.005 // wave gauge 4 at x, y [m]
P 51 14.5 0.005 // wave gauge 5 at x, y [m]
P 51 15.7 0.005 // wave gauge 6 at x, y [m]
P 51 17.3 0.005 // wave gauge 7 at x, y [m]
P 51 19.0 0.005 // wave gauge 8 at x, y [m]
P 51 21.0 0.005 // wave gauge 9 at x, y [m]

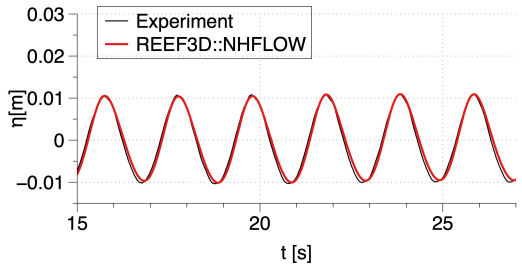
P 54 100 // wsfline interval in iterations; no effect here, no P 52 is given

12.2.3 What to look at

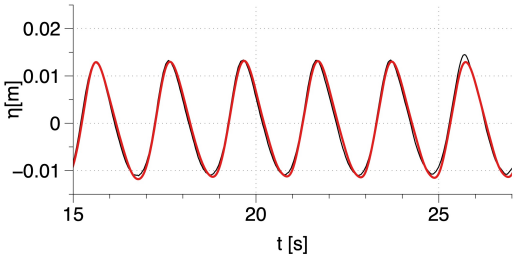
'REEF3D-NHFLOW-WSF-HG.dat' in REEF3D_NHFLOW_WSF contains one column per P 51 gauge (x = 4.0, 10.5, 12.5, 13.5, 14.5, 15.7, 17.3, 19.0 and 21.0 m) in the order given in ctrl.txt; compare them with the measurements of Beji and Battjes [1993], see the figures below. REEF3D_NHFLOW_VTU (every 0.5 s) shows the flow field over the bar. The P 54 line has no effect because no free surface line (P 52) is defined.

12.2.4 Results

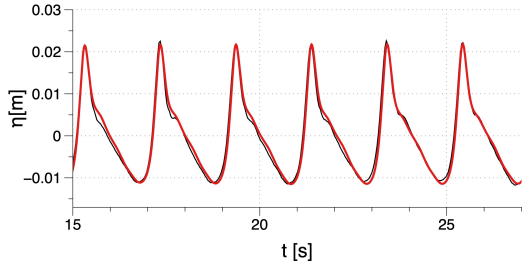
Behind the bar, the bound higher harmonics are released as free waves that are in relatively deep water. NHFLOW is a three-dimensional sigma-grid model, not a depth-averaged one, but with only three vertical cells (B 2) its dispersion accuracy for these short waves is limited, which explains the discrepancies with the experiment at wave gauges 6 and onwards. Be aware of the relative water depth of all relevant wave components when choosing the number of vertical cells.



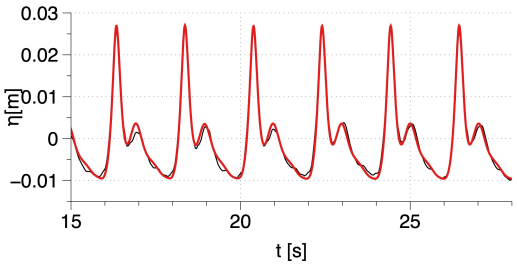
(a) Gauge 1



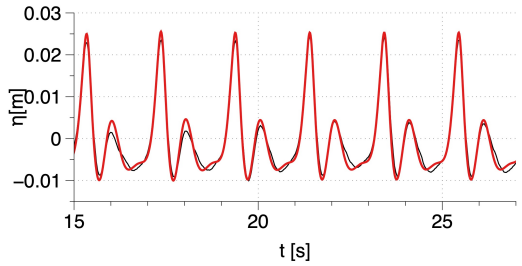
(b) Gauge 2



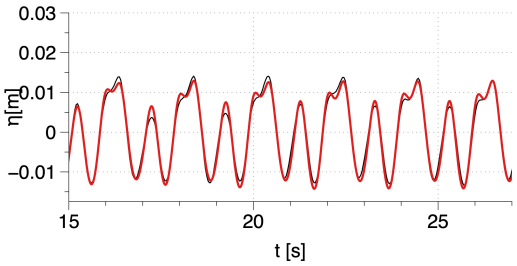
(c) Gauge 3



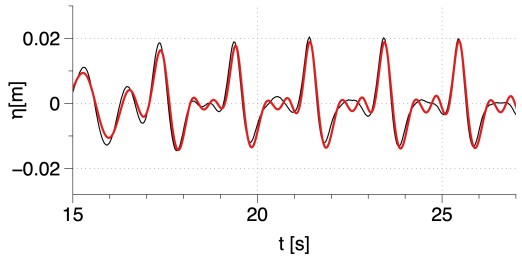
(d) Gauge 4



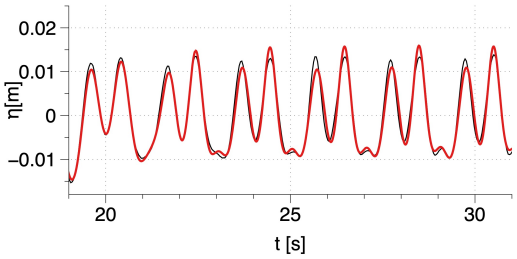
(e) Gauge 5



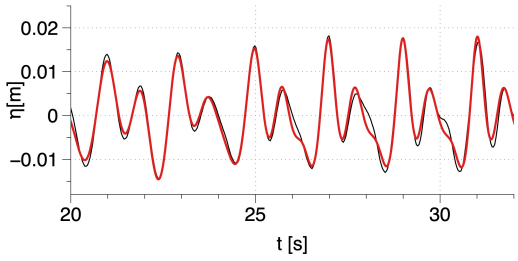
(f) Gauge 6



(g) Gauge 7



(h) Gauge 8



(i) Gauge 9

Figure 12.2: Free surface elevations in the simulation of wave propagation over a submerged bar.

13. Tutorial | REEF3D::CFD

REEF3D::CFD solves the three-dimensional incompressible Navier-Stokes equations on a staggered rectilinear grid, with the level set method for the free surface in two-phase flow. It is used for violent free surface flows, breaking waves, wave forces on structures, floating bodies and sediment transport. For visualization, open the REEF3D_CFD_VTU folder (.pvtu files with velocity, pressure, eddyv, phi, ...) in ParaView; the free surface is the zero contour of the level set function phi, see Chapter 9. Depending on the P commands, further results are written to REEF3D_CFD_WSF (wave gauges), REEF3D_CFD_WSFLINE (free surface lines), REEF3D_CFD_ProbePoint and REEF3D_CFD_ProbeLine (probes), REEF3D_CFD_Force (forces), REEF3D_CFD_FSF (free surface .vtp) and REEF3D_CFD_6DOF (floating bodies). Each tutorial folder contains 'control.txt' for DIVEMesh and 'ctrl.txt' for REEF3D; run DIVEMesh (**./DiveMESH**) first and then **mpirun -n** *M 10* **reef3d**. Commands that are not given in ctrl.txt take their default values (e.g. D 10 4, D 20 2, D 30 1, F 30 3, F 40 3, N 40 3, P 10 1).

13.1 2D Dam Break

Repository folder: `Tutorials/REEF3D_CFD/11_1 2D Dam Break`

A water column of 0.15 m width and 0.3 m height (F 54, F 56) collapses in a closed 2D box and hits the opposite wall. The case demonstrates the two-phase flow capabilities of REEF3D::CFD with the level set method for a strongly deforming free surface. All boundaries are walls, and no turbulence model is used. Comments to the control files are marked by blue fonts.

- Domain: 0.6 m × 0.005 m × 0.6 m, cell size dx = 0.005 m (120 × 1 × 120 cells)
- Water column: $x \leq 0.15$ m, $z \leq 0.3$ m; the run stops after 25000 iterations (N 45)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.1.1 DIVEMesh: control.txt

C 11 21 // side 1 (x_{min}): wall
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 21 // side 4 (x_{max}): wall
C 15 21 // side 5 (bottom): wall
C 16 21 // side 6 (top): wall

B 1 0.005 // uniform cell size dx [m]
B 10 0.0 0.6 0.0 0.005 0.0 0.6 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)

13.1.2 REEF3D: ctrl.txt

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 50 4 // level set not fixed at inflow or outflow
F 54 0.15 // water column ends at this x-coordinate [m]
F 56 0.3 // water column ends at this z-coordinate [m]

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 25000 // maximum number of iterations
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

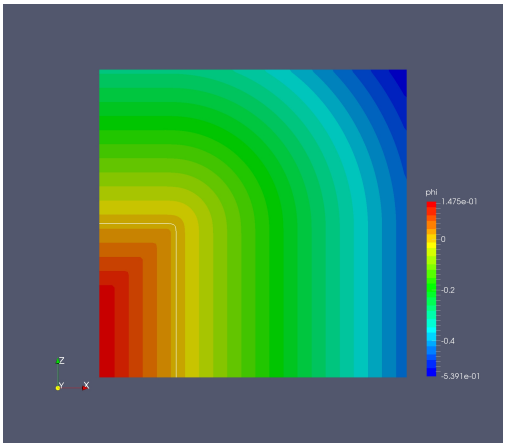
P 10 1 // print ParaView files (.vtu)
P 30 0.01 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

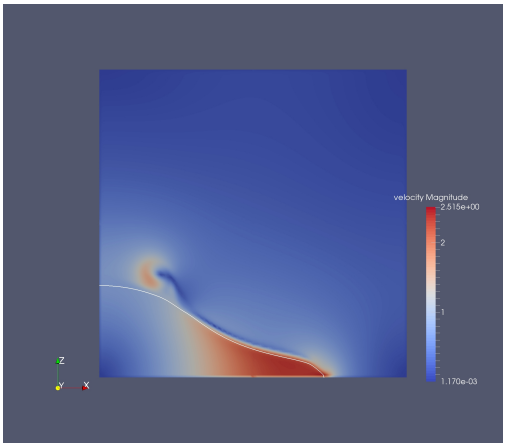
13.1.3 What to look at

Open REEF3D_CFD_VTU in ParaView (one file every 0.01 s), extract the free surface as the contour phi = 0 and color it or a slice by velocity, as in the figure below.

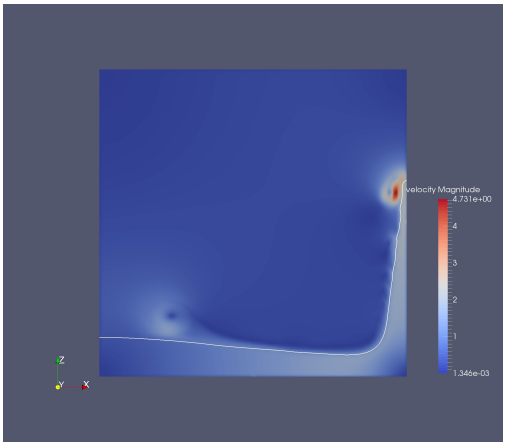
13.1.4 Results



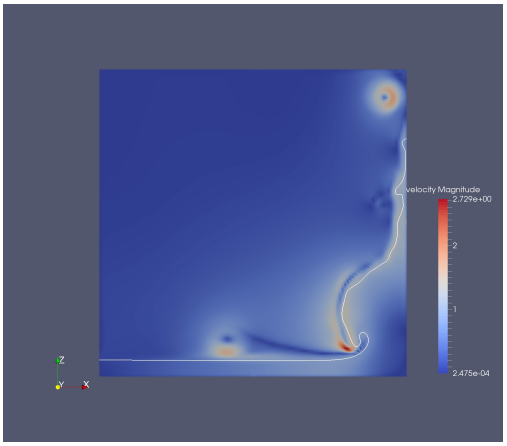
(a) Initial Condition with level set function



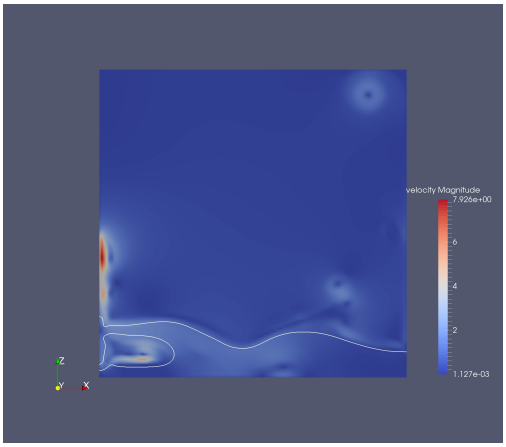
(b) after t = 0.2 sec with velocity magnitude



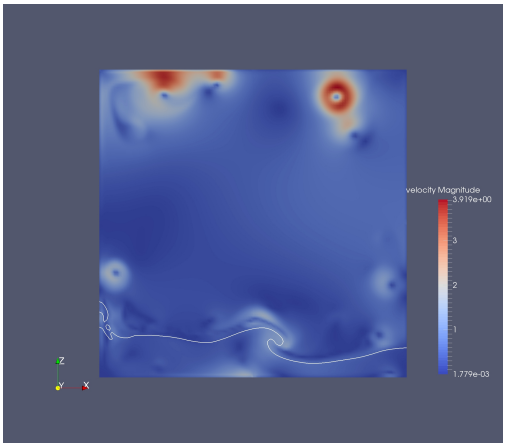
(c) after t = 0.4 sec with velocity magnitude



(d) after t = 0.7 sec with velocity magnitude



(e) after t = 1.0 sec with velocity magnitude



(f) after t = 1.5 sec with velocity magnitude

Figure 13.1: Dam Break

13.2 3D Dam Break with Obstacle

Repository folder: `Tutorials/REEF3D_CFD/11_2 3D Dam Break with Obstacle`

A water column over the full width of a closed 3D tank collapses and hits a rectangular obstacle. The flow splits around the obstacle and runs up on it, which gives a fully three-dimensional free surface. This case is optimized for fast processing. In order to get a more detailed free surface result, change B 1 to e.g. 0.0125.

- Domain: 2.0 m × 1.0 m × 1.0 m, cell size dx = 0.025 m (80 × 40 × 40 cells)
- Water column: $x \leq 0.5$ m, $z \leq 0.7$ m; obstacle x = 1.2–1.4 m, y = 0.4–0.6 m
- Simulated time 25 s (N 41), at most 50000 iterations (N 45)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.2.1 DIVEMesh: control.txt

C 11 21 // side 1 (x_{min}): wall
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 21 // side 4 (x_{max}): wall
C 15 21 // side 5 (bottom): wall
C 16 21 // side 6 (top): wall

B 1 0.025 // uniform cell size dx [m]
B 10 0.0 2.0 0.0 1.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 10 1.2 1.4 0.4 0.6 0.0 1.0 // rectangular obstacle (0.2 m × 0.2 m, full height)

M 10 6 // number of processes (partitions)

13.2.2 REEF3D: ctrl.txt

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 54 0.5 // water column ends at this x-coordinate [m]
F 56 0.7 // water column ends at this z-coordinate [m]

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 25.0 // maximum simulated time [s]
N 45 50000 // maximum number of iterations
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.01 // print interval in simulated time [s]

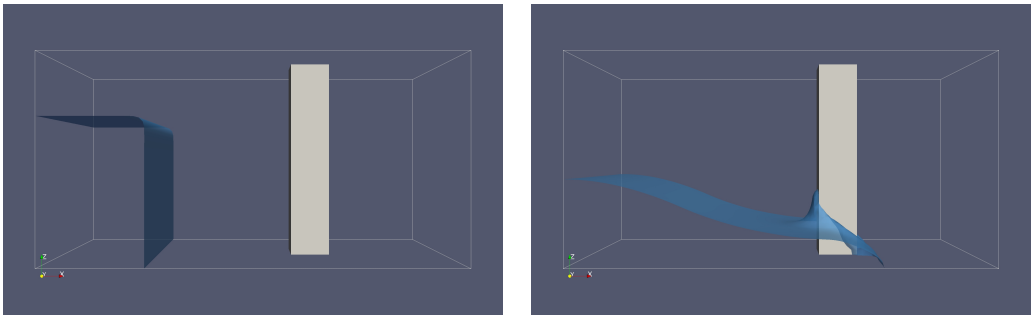
T 10 0 // no turbulence model

W 22 -9.81 // gravity in z-direction [m/s^2]

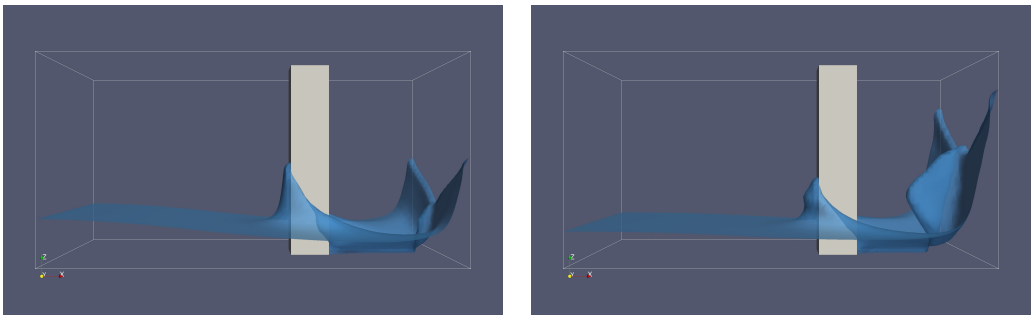
13.2.3 What to look at

Open REEF3D_CFD_VTU (one file every 0.01 s) and apply a contour filter for phi = 0 directly on the 3D data to see the free surface around the obstacle.

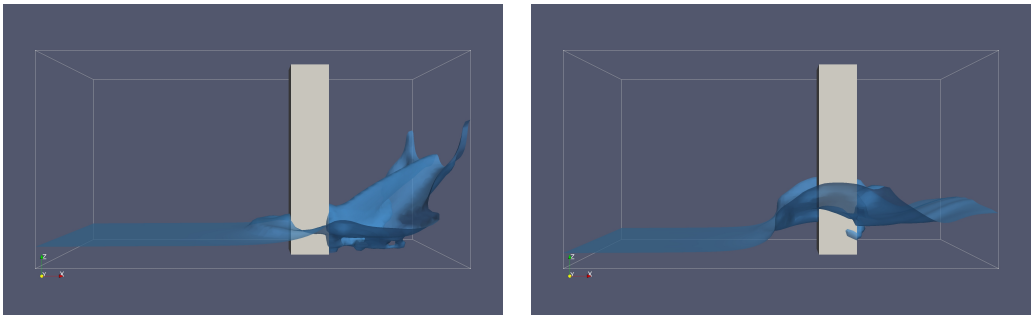
13.2.4 Results



(a) Initial Condition (b)



(c) (d)



(e) (f)

Figure 13.2: 3D Dam Break with Obstacle

13.3 2D Vortex Shedding

Repository folder: `Tutorials/REEF3D_CFD/11_3 2D Vortex Shedding`

Single-phase flow (F 30 0, F 40 0) passes a circular cylinder in a 2D channel with walls at the bottom and top. The inflow discharge is prescribed with ioFlow (B 60 1, W 10), and the velocities are initialized with the potential flow solver (I 11 1). With the mean velocity of $6 \cdot 10^{-4}$ m/s, the cylinder diameter of 0.2 m and the default water viscosity, the Reynolds number is about 120, so a laminar vortex street develops behind the cylinder; no turbulence model is used.

- Domain: 4.0 m × 0.01 m × 1.0 m, cell size dx = 0.01 m (400 × 1 × 100 cells)
- Cylinder: x = 1.0 m, z = 0.5 m, radius 0.1 m; discharge $6 \cdot 10^{-6}$ m^3/s
- 10000 iterations (N 45), output every 10 iterations (P 20)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.3.1 DIVEMesh: control.txt

```
C 11 1 // side 1 (x_min): inflow
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 2 // side 4 (x_max): outflow
C 15 21 // side 5 (bottom): wall
C 16 21 // side 6 (top): wall

B 1 0.01 // uniform cell size dx [m]
B 10 0.0 4.0 0.0 0.01 0.0 1.0 // rectangular domain x_s, x_e, y_s, y_e, z_s, z_e [m]

S 32 1.0 0.5 0.1 // cylinder in y-direction at x = 1.0 m, z = 0.5 m, radius 0.1 m

M 10 6 // number of processes (partitions)
```

13.3.2 REEF3D: ctrl.txt

```
B 10 0 // wall functions for the velocities off
B 50 0.0001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 1 // diffusion: explicit
D 30 1 // pressure: projection method

F 30 0 // level set transport off (single-phase flow)
F 40 0 // reinitialization off

I 11 1 // initialize velocities with the potential flow solver
```

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 10000 // maximum number of iterations
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations
P 62 0.0 0.0 0.005 0.005 0.0 1.0 // vertical line probe at the inlet x = 0 m

T 10 0 // no turbulence model

W 10 0.000006 // inflow discharge [m^3/s]

13.3.3 What to look at

Open REEF3D_CFD_VTU and color by velocity or pressure; streamlines or line integral convolution of the velocity show the vortex street as in the figure below. The vertical line probe at the inlet (P 62) is written to REEF3D_CFD_ProbeLine every 10 iterations (default P 54) and shows the inflow profile.

13.3.4 Results

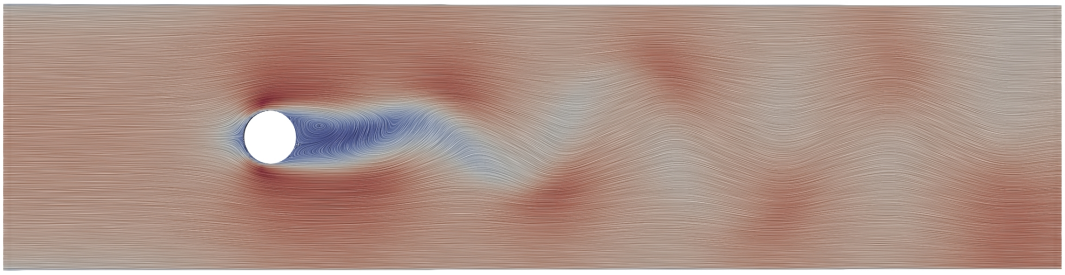


Figure 13.3: 2D vortex shedding with stream lines

13.4 Flow through a Narrow Contraction

Repository folder: `Tutorials/REEF3D_CFD/11_4 Narrow Contraction`

For the flow through a narrow contraction the open channel flow capabilities in ioFlow are used: a constant discharge enters at the inflow boundary and the water level is fixed at the outflow (F 50 2). The channel narrows from 0.6 m to 0.2 m width between x = 0.8 m and 1.1 m, stays narrow up to x = 1.4 m and widens again up to x = 1.7 m; the geometry is built from wedges (S 83) and boxes (S 10). The flow is initialized with the potential flow solver (I 10 1) and the k-ω model is used. Just for information: the grid is rather coarse for this flow situation. The grid can be easily refined by changing the B 1 parameter in the DIVEMesh 'control.txt' file, depending on the performance of your computer.

- Domain: 2.5 m × 0.6 m × 0.5 m, cell size dx = 0.025 m (100 × 24 × 20 cells)
- Water depth 0.15 m (F 60), discharge 0.05 m^3/s (W 10), i.e. about 0.56 m/s upstream
- 12000 iterations (N 45), output every 10 iterations (P 20)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.4.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 2.5 0.0 0.6 0.0 0.5 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 83 0.8 0.0 0.0 1.1 0.0 0.0 1.1 0.2 0.0 0.8 0.0 0.5 1.1 0.0 0.5 1.1 0.2 0.5 // wedge (six corner points): converging part of the contraction
S 10 1.1 1.4 0.0 0.2 0.0 0.5 // box: narrow section, side y = 0–0.2 m
S 83 1.4 0.0 0.0 1.7 0.0 0.0 1.4 0.2 0.0 1.4 0.0 0.5 1.7 0.0 0.5 1.4 0.2 0.5 // wedge (six corner points): diverging part of the contraction

S 83 0.8 0.6 0.0 1.1 0.6 0.0 1.1 0.4 0.0 0.8 0.6 0.5 1.1 0.6 0.5 1.1 0.4 0.5 // wedge (six corner points): converging part of the contraction
S 10 1.1 1.4 0.4 0.6 0.0 0.5 // box: narrow section, side y = 0.4–0.6 m
S 83 1.4 0.6 0.0 1.7 0.6 0.0 1.4 0.4 0.0 1.4 0.6 0.5 1.7 0.6 0.5 1.4 0.4 0.5 // wedge (six corner points): diverging part of the contraction

M 10 6 // number of processes (partitions)

13.4.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 0.5 // length for the initial reinitialization instead of the domain length [m]

F 50 2 // level set fixed at the outflow
F 60 0.15 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 12000 // maximum number of iterations
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations

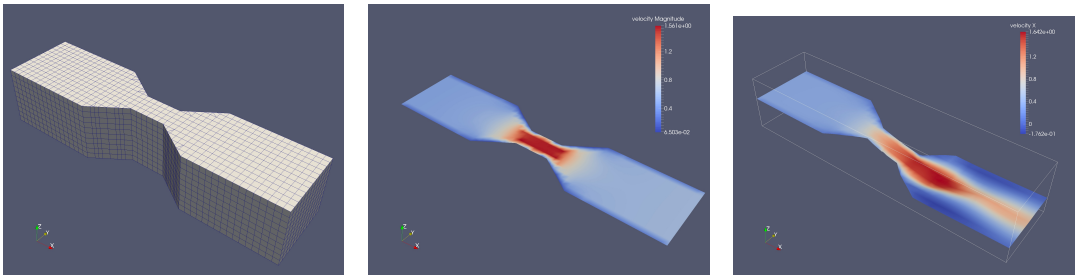
T 10 2 // k-ω turbulence model

W 10 0.05 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

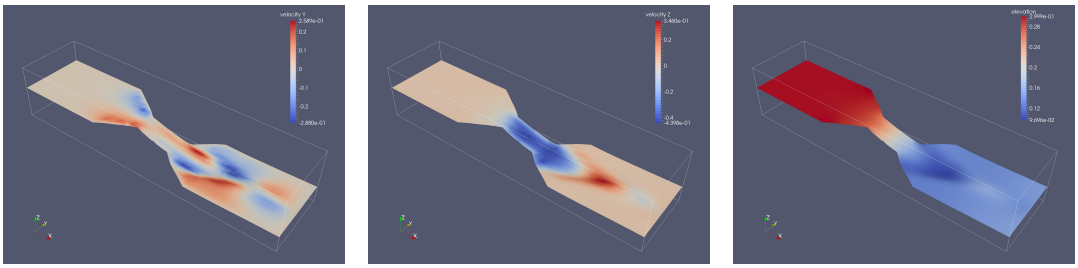
13.4.3 What to look at

Open REEF3D_CFD_VTU: the first file shows the initial potential flow velocities; later files show the acceleration through the contraction (velocity components U, V, W) and the drop of the free surface (contour phi = 0), as in the figure below.

13.4.4 Results



(a) Mesh (b) Initial velocity with potential flow solver (c) U



(d) V (e) W (f) Free surface elevation

Figure 13.4: Narrow Contraction

13.5 Flow around a Circular Pier

Repository folder: `Tutorials/REEF3D_CFD/11_5 Flow around a Circular Pier`

Open channel flow around a vertical circular cylinder. As in the previous case, ioFlow prescribes the discharge at the inflow and keeps the water level fixed at the outflow; the flow is initialized with the potential flow solver and the k-ω model is used. The case shows the free surface deformation and the three-dimensional flow field around the pier.

- Domain: 1.5 m × 1.0 m × 0.4 m, cell size dx = 0.025 m (60 × 40 × 16 cells)
- Pier at $x = y$ = 0.5 m, diameter 0.15 m; water depth 0.2 m, discharge 0.1 m^3/s (mean velocity 0.5 m/s)
- 100000 iterations (N 45), output every 10 iterations (P 20)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.5.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 1.5 0.0 1.0 0.0 0.4 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 33 0.5 0.5 0.075 // vertical cylinder at $x = y = 0.5$ m, radius 0.075 m

M 10 6 // number of processes (partitions)

13.5.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 50 2 // level set fixed at the outflow
F 60 0.2 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 45 100000 // maximum number of iterations
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations

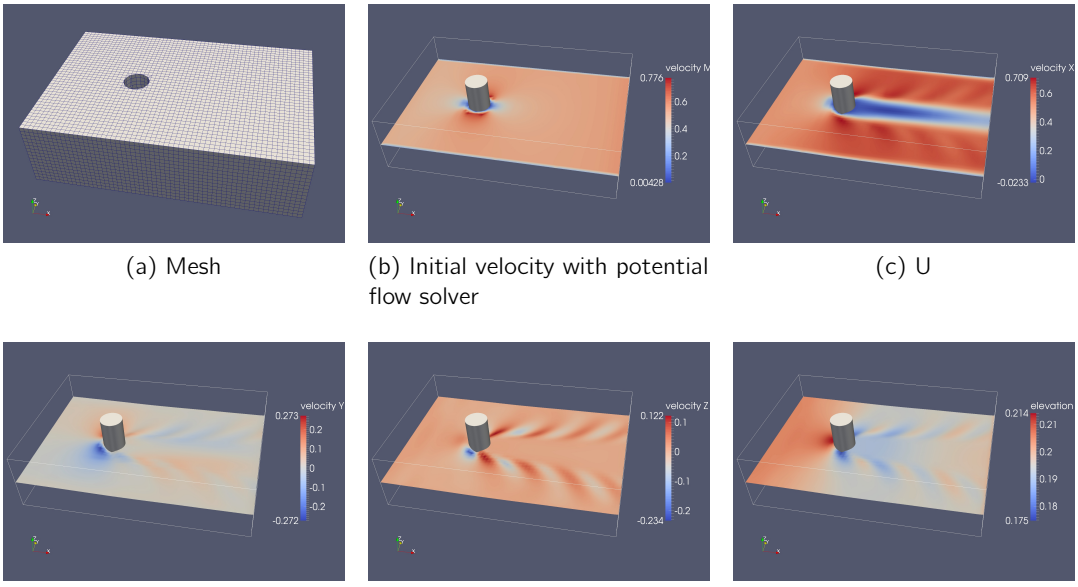
T 10 2 // k-ω turbulence model

W 10 0.1 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

13.5.3 What to look at

Open REEF3D_CFD_VTU and use a horizontal slice below the free surface to see the velocity field around the pier, and the contour phi = 0 for the free surface with the bow wave in front of the pier.

13.5.4 Results



(a) Mesh (b) Initial velocity with potential flow solver (c) U

(d) V (e) W (f) Free surface elevation

Figure 13.5: Flow around Circular Cylinder

13.6 Rectangular Wave Tank

Repository folder: `Tutorials/REEF3D_CFD/11_6 Rectangular Wave Tank`

For the case of a simple rectangular wave tank without obstacles, the simulation can be run in 2D. Linear waves are generated with the relaxation method, and the propagating waves are compared with the theoretical solution throughout the wave tank. The idea is that if the wave propagation is represented well by the numerical model, the numerical results will match the theoretical free surface profile in the tank. In this example, the wave has a relatively low steepness, and good convergence can be reached for dx = 0.05 m already, which improves further on finer grids. For steeper waves, finer grids may be required.

- Wave: linear, H = 0.02 m, L = 4.0 m (B 91), water depth d = 0.5 m
- Domain: 30 m × 0.05 m × 1.0 m, cell size dx = 0.05 m (600 × 1 × 20 cells)
- Wave generation zone 4 m, numerical beach 8 m (B 96), simulated time 90 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.6.1 DIVEMesh: control.txt

```
C 11 6 // side 1 (x_min): wave generation
C 12 3 // side 2 (y_max): symmetry plane
C 13 3 // side 3 (y_min): symmetry plane
C 14 7 // side 4 (x_max): numerical beach
```

C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]
B 10 0.0 30.0 0.0 0.05 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)

13.6.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 2 // wave type: linear waves
B 91 0.02 4.0 // wave height H [m], wavelength L [m]
B 96 4.0 8.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.5 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 90.0 // maximum simulated time [s]
N 47 0.25 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.05 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

P 52 0.025 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 54 500 // wsfline print interval in iterations (needs P 52)

13.6.3 What to look at

The files in REEF3D_CFD_WSFLINE contain the numerical free surface along the tank at y = 0.025 m every 500 iterations together with the theoretical profile (P 53 1); plot both to obtain the comparison in the last figure. In ParaView, REEF3D_CFD_VTU (every 0.05 s) shows the velocities U and W under the waves.

13.6.4 Results

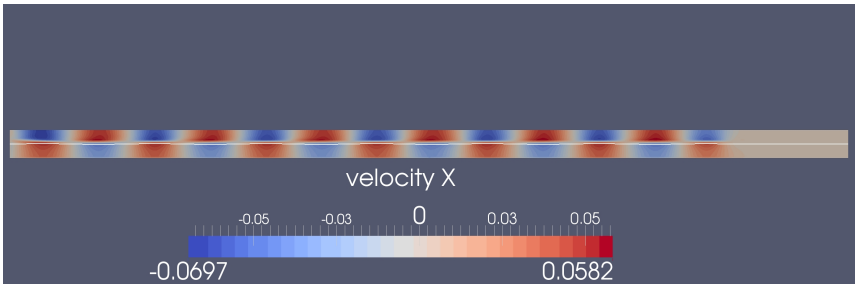


Figure 13.6: Wave tank with free surface and U

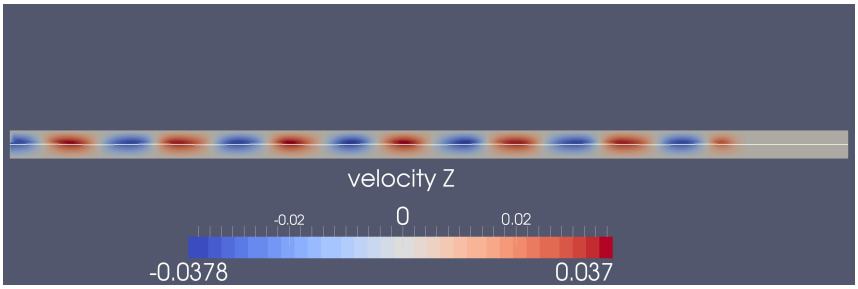


Figure 13.7: Wave tank with free surface and W

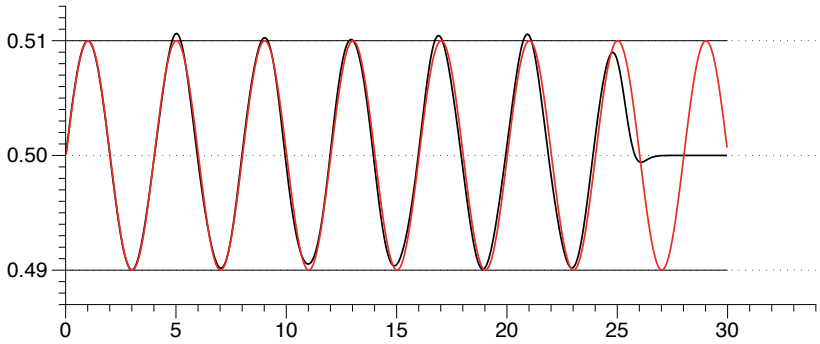


Figure 13.8: Free surface along the tank, numerical result (black line) vs theory (red line)

13.7 Rectangular Wave Tank with DWG and AWA

Repository folder: `Tutorials/REEF3D_CFD/11_7 Rectangular Wave Tank DWG-AWA`

Similar to the previous case, a two-dimensional rectangular wave tank is used. Now, different wave generation and absorption methods are used, namely Dirichlet wave generation (DWG) and active wave absorption (AWA). With DWG, the values for the horizontal and vertical velocities are prescribed at the inlet boundary without further modulation. The level set function is allowed to move freely, but due to the correct flow momentum it will follow the prescribed wave theory. AWA generates a wave opposite to the reflected one at the outlet boundary, thus cancelling out the unwanted reflections. AWA is based on shallow water theory, which also has consequences for the type of applications it should be used for, i.e. shallow to intermediate water with preferably longer waves. Overall, the DWG-AWA combination requires a smaller tank or basin compared to the relaxation methods (RM). AWA, DWG and RM can be used for generation and absorption in any combination.

- Wave: 5th-order Stokes, H = 0.1 m, L = 2.0 m (B 91), water depth d = 0.5 m
- Domain: 15 m × 0.025 m × 1.0 m, cell size dx = 0.025 m (600 × 1 × 40 cells)
- Simulated time 60 s; state files for a hot-start every 1 s (P 40, P 42)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.7.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 15.0 0.0 0.025 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 20 2 // decomposition method 2 (improved rectangular)

M 10 6 // number of processes (partitions)

13.7.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 91 0.1 2.0 // wave height H [m], wavelength L [m]
B 96 2.0 4.0 // lengths of wave generation zone and numerical beach [m]
B 98 3 // wave generation: Dirichlet (DWG)
B 99 3 // numerical beach: active wave absorption (AWA)

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.50 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.25 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.1 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 1.0 // state file interval [s]

T 10 0 // no turbulence model

W 22 -9.81 // gravity in z-direction [m/s^2]

P 52 0.0125 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 55 0.50 // wsfline and line probe print interval [s]

P 51 2.50 0.0125 // wave gauge 1 at x, y [m]
P 51 5.00 0.0125 // wave gauge 2 at x, y [m]
P 51 15.0 0.0125 // wave gauge on the outlet boundary (x_{max} = 15 m)
P 51 20.0 0.0125 // wave gauge outside the domain (x_{max} = 15 m)

13.7.3 What to look at

Compare the numerical and theoretical free surface profiles in REEF3D_CFD_WSFLINE (every 0.5 s, P 55) to check that no reflections build up in the short tank, and plot the gauges in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF). Note that the last two P 51 gauges are located at x = 15 m and 20 m, i.e. on the outlet boundary and outside the 15 m long domain. REEF3D_CFD_VTU (every 0.1 s) shows the velocities U and W.

13.7.4 Results

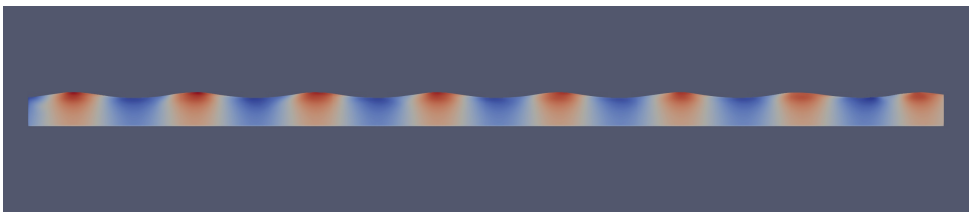


Figure 13.9: DWG-AWA Wave tank with free surface and U

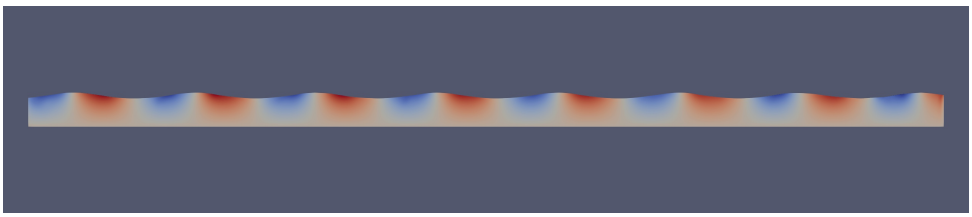


Figure 13.10: DWG-AWA Wave tank with free surface and W

13.8 Rectangular Wave Tank with Wavemaker

Repository folder: `Tutorials/REEF3D_CFD/11_8 Rectangular Wave Tank Wavemaker`

Following the previous NWT examples, now the waves are generated with wavemaker kinematics input, see e.g. Aggarwal et al. [2018a] for more information. The motion of a flap wavemaker is given as a time series of the flap angle in the file 'wavemaker.dat' (two columns: time t [s] and angle β [rad], B 116 2), which must be in the case folder. The flap kinematics are applied at the inlet boundary (B 98 3), and the waves are absorbed with active wave absorption at the outlet (B 99 3).

- Flap from z = -0.25 m to 1.46 m (B 111), input time shifted by 4.0 s (B 117), water depth d = 0.75 m
- Domain: 15 m × 0.025 m × 1.5 m, cell size dx = 0.025 m (600 × 1 × 60 cells)
- Simulated time 60 s; output every 10 iterations (P 20), state files every 1 s
- Extra input file: wavemaker.dat (read by REEF3D)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.8.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 8 // side 4 (x_{max}): numerical beach, type 8 (handled like type 7 in REEF3D)
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]
B 10 0.0 15.0 0.0 0.025 0.0 1.5 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.8.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 22 // wave type: flap wavemaker, kinematics from wavemaker.dat
B 98 3 // wave generation: Dirichlet (DWG)
B 99 3 // numerical beach: active wave absorption (AWA)
B 111 -0.25 1.46 // flap wavemaker from z_{start} = -0.25 m (below the bed) to z_{end} = 1.46 m
B 116 2 // flap wavemaker input as angle $\beta(t)$ [rad]

B 117 4.0 // time shift [s]: rows before t = 4.0 s in wavemaker.dat are skipped, t = 4.0 s becomes t = 0

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.5 // length for the initial reinitialization instead of the domain length [m]
F 60 0.75 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

T 10 0 // no turbulence model

W 22 -9.81 // gravity in z-direction [m/s^2]

P 10 1 // print ParaView files (.vtu)
P 20 10 // print interval in iterations
P 40 1 // print state files for hot-start
P 42 1.0 // state file interval [s]
P 52 0.015 // free surface line (wsfline) along x at this y [m]
P 55 0.01 // wsfline print interval [s]

P 51 5.0 0.015 // wave gauge 1 at x, y [m]
P 51 10.0 0.015 // wave gauge 2 at x, y [m]

13.8.3 REEF3D: wavemaker.dat

The 'wavemaker.dat' file contains the time dependent kinematics of the wavemaker.

13.8.4 What to look at

Plot the two gauges at x = 5 m and 10 m from 'REEF3D-CFD-WSF-HG.dat' in REEF3D_CFD_WSF, and the free surface along the tank in REEF3D_CFD_WSFLINE (every 0.01 s). REEF3D_CFD_VTU shows the generated waves with the velocities U and W, as in the figures below.

13.8.5 Results

Figure 13.11: Wavemaker generated waves with free surface and U

Figure 13.12: Wavemaker generated waves with free surface and W

13.9 5th-order Stokes Waves

Repository folder: `Tutorials/REEF3D_CFD/11_9 5th-order Stokes Waves`

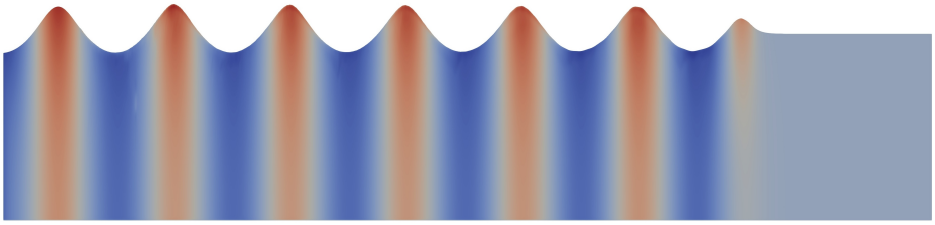


Figure 13.13: 5th-order Stokes Waves setup

This verification case shows the propagation of 5th-order Stokes waves in a 200 m long tank and highlights the use of velocity probes both from the model and the wave theory. The waves are generated in a relaxation zone of about one wavelength and absorbed in a relaxation beach of about two wavelengths. The same wave is used in the NHFLOW tutorial `1_Stokes5`. The numerical schemes not given in ctrl.txt take their default values.

- Wave: H = 1.0 m, T = 4.5 s (B 93), water depth d = 4.01 m (F 60), wavelength about 25 m
- Domain: 200 m × 0.025 m × 8.0 m, cell size dx = 0.025 m (8000 × 1 × 320 cells)
- Wave generation zone 25 m, numerical beach 50 m (B 96), simulated time 120 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.9.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane

C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 200.0 0.0 0.025 0.0 8.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.9.2 REEF3D: ctrl.txt

B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 93 1.0 4.5 // wave height H [m], wave period T [s]
B 96 25.0 50.0 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 30 2 // pressure: projection method (same solver as D 30 1 in the current code)

F 60 4.01 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 120.0 // maximum simulated time [s]
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.25 // print interval in simulated time [s]

P 50 15.0 0.005 // theoretical wave gauge 1 at x, y [m]
P 50 50.0 0.005 // theoretical wave gauge 2 at x, y [m]
P 50 75.0 0.005 // theoretical wave gauge 3 at x, y [m]
P 50 100.0 0.005 // theoretical wave gauge 4 at x, y [m]
P 50 125.0 0.005 // theoretical wave gauge 5 at x, y [m]

P 51 15.0 0.005 // wave gauge 1 at x, y [m]
P 51 50.0 0.005 // wave gauge 2 at x, y [m]
P 51 75.0 0.005 // wave gauge 3 at x, y [m]
P 51 100.0 0.005 // wave gauge 4 at x, y [m]
P 51 125.0 0.005 // wave gauge 5 at x, y [m]

P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 55 0.05 // wsfline and line probe print interval [s]

P 65 25.0 0.025 3.0 // velocity probe 1 at x, y, z [m]
P 66 25.0 0.025 3.0 // theoretical velocity probe 1 at x, y, z [m]

P 65 75.0 0.025 3.0 // velocity probe 2 at x, y, z [m]
P 66 75.0 0.025 3.0 // theoretical velocity probe 2 at x, y, z [m]

P 65 125.0 0.025 3.0 // velocity probe 3 at x, y, z [m]
P 66 125.0 0.025 3.0 // theoretical velocity probe 3 at x, y, z [m]

W 22 -9.81 // gravity in z-direction [m/s^2]

13.9.3 What to look at

Plot 'REEF3D-CFD-WSF-HG.dat' (P 51) against 'REEF3D-CFD-WSF-HG-THEORY.dat' (P 50) in REEF3D_CFD_WSF for the gauges at x = 15, 50, 75, 100 and 125 m. The velocity probes at z = 3.0 m write 'REEF3D-CFD-Vel-Probe-n.dat' (P 65) and the corresponding theoretical velocities 'REEF3D-CFD-Vel-Probe-Theory-n.dat' (P 66) to REEF3D_CFD_ProbePoint; compare the horizontal and vertical velocity at x = 25, 75 and 125 m. REEF3D_CFD_WSFLINE contains the numerical and theoretical free surface along the tank every 0.05 s. Note that the first gauge in the current ctrl.txt is at x = 15 m, while the first free surface figure below is labelled x = 25 m.

13.9.4 Free Surface Results

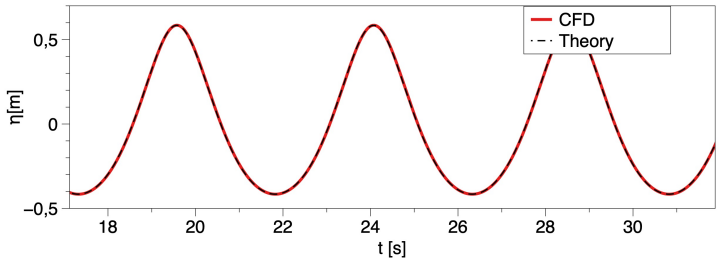


Figure 13.14: Wave gage 1, x = 25m

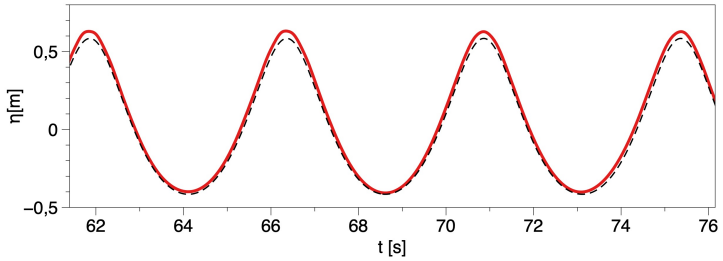


Figure 13.15: Wave gage 2, x = 50m

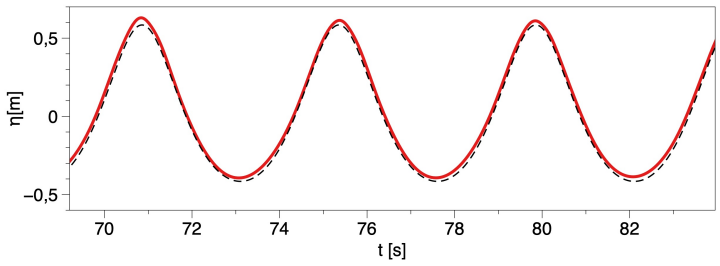


Figure 13.16: Wave gage 3, x = 75m

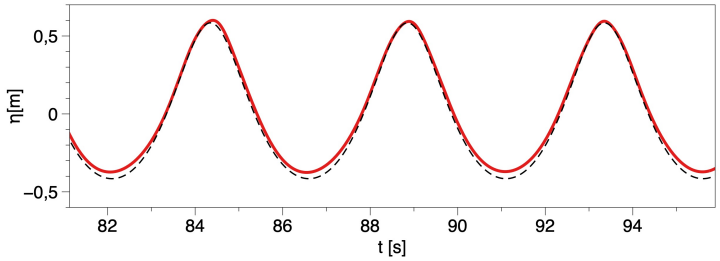


Figure 13.17: Wave gage 4, x = 100m

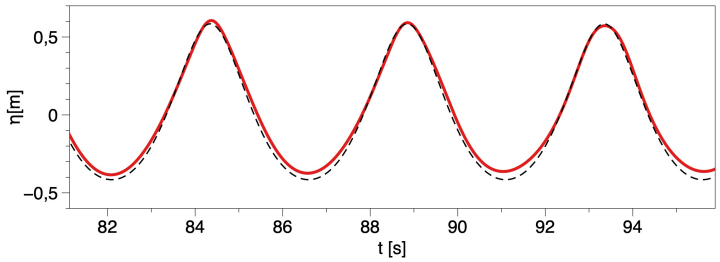


Figure 13.18: Wave gage 5, x = 125m

13.9.5 Velocity Results

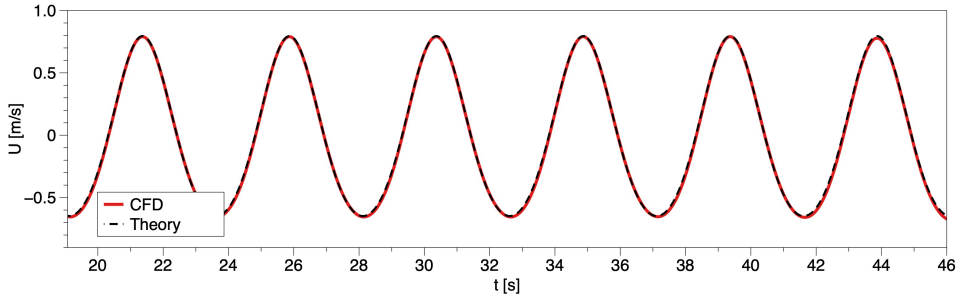


Figure 13.19: Horizontal Velocity, x = 25m, z=3m

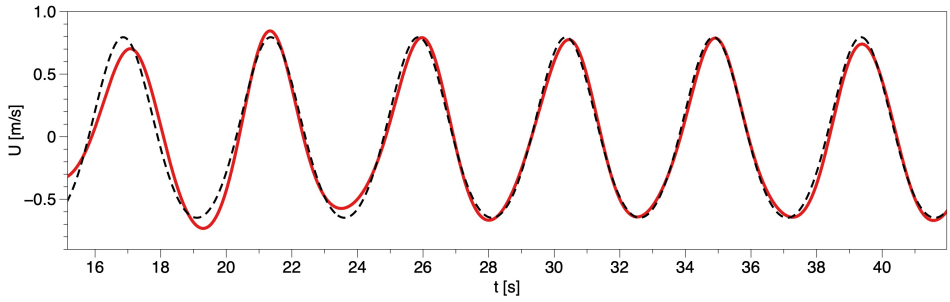


Figure 13.20: Horizontal Velocity, x = 75m, z=3m

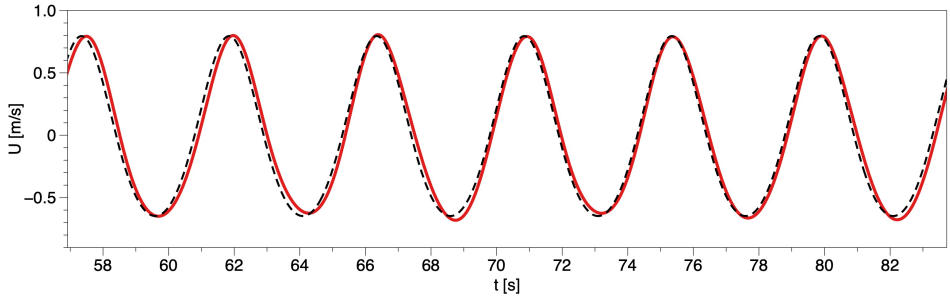


Figure 13.21: Horizontal Velocity, x = 125m, z=3m

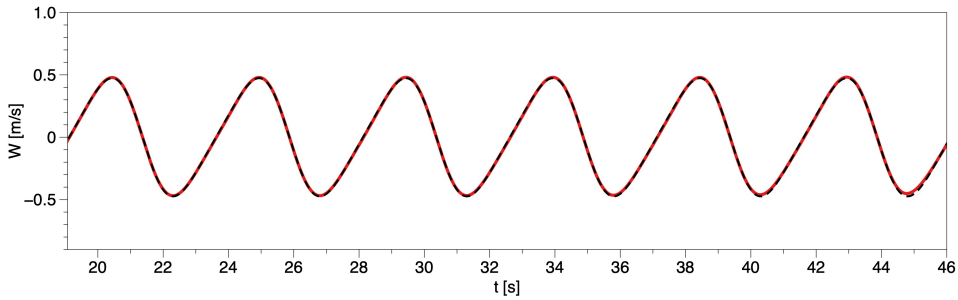


Figure 13.22: Vertical Velocity, x = 25m, z=3m

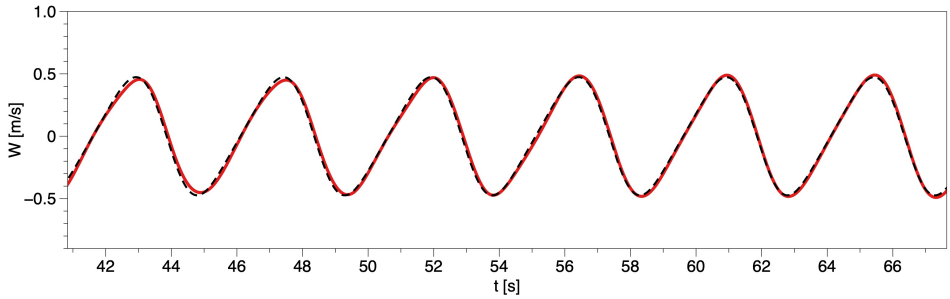


Figure 13.23: Vertical Velocity, x = 75m, z=3m

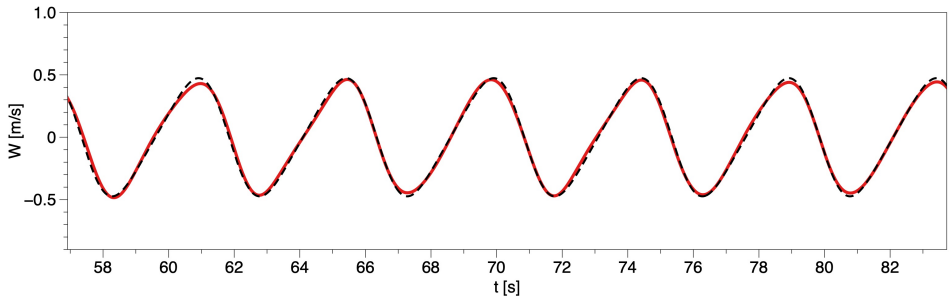


Figure 13.24: Vertical Velocity, x = 125m, z=3m

13.10 Wave Propagation over a Submerged Bar

Repository folder: `Tutorials/REEF3D_CFD/11_10 Submerged Bar Wave Propagation`

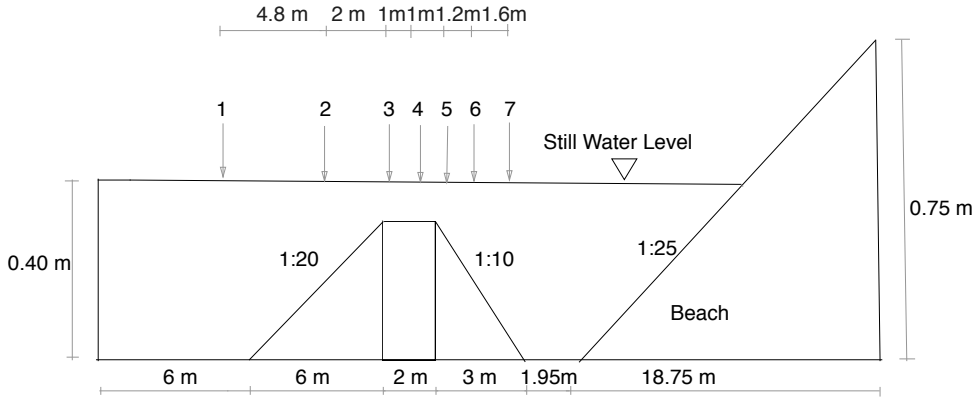


Figure 13.25: Experimental setup submerged bar

The well-known benchmark case for wave propagation over a submerged bar is presented here. The description of the experimental setup can be found in the original paper by Beji and Battjes [1993]. The bar consists of a front slope from x = 6 m to 12 m, a crest from 12 m to 14 m and a back slope to 17 m, 0.3 m high. The simulations are run in 2D with wave gauges at several locations along the wave tank.

- Wave: linear, H = 0.02 m, L = 3.73 m (B 91), water depth d = 0.4 m, 0.1 m on the bar crest
- Domain: 24 m × 0.01 m × 0.8 m, cell size dx = 0.01 m (2400 × 1 × 80 cells)
- Wave generation zone and numerical beach 3.73 m each (B 96), simulated time 60 s
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.10.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.01 // uniform cell size dx [m]
B 10 0.0 24.0 0.0 0.01 0.0 0.8 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 6.0 12.0 0.0 0.01 0.0 0.3 // front slope of the bar, rising from z = 0 to 0.3 m
S 10 12.0 14.0 0.0 0.01 0.0 0.3 // bar crest, 0.3 m high
S 61 14.0 17.0 0.0 0.01 0.3 0.0 // back slope of the bar, falling from 0.3 m to 0

M 10 6 // number of processes (partitions)

13.10.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 2 // wave type: linear waves
B 91 0.02 3.73 // wave height H [m], wavelength L [m]
B 96 3.73 3.73 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 0.8 // length for the initial reinitialization instead of the domain length [m]
F 60 0.4 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.5 // print interval in simulated time [s]

W 22 -9.81 // gravity in z-direction [m/s^2]

P 52 0.005 // free surface line (wsfline) along x at this y [m]
P 53 1 // add the theoretical wave to the wsfline files
P 54 10 // wsfline print interval in iterations (needs P 52)

P 51 2.0 0.005 // wave gauge 1 at x, y [m]
P 51 4.0 0.005 // wave gauge 2 at x, y [m]
P 51 5.2 0.005 // wave gauge 3 at x, y [m]
P 51 10.5 0.005 // wave gauge 4 at x, y [m]
P 51 12.5 0.005 // wave gauge 5 at x, y [m]
P 51 13.5 0.005 // wave gauge 6 at x, y [m]
P 51 14.5 0.005 // wave gauge 7 at x, y [m]

13.10.3 What to look at

'REEF3D-CFD-WSF-HG.dat' in REEF3D_CFD_WSF contains one column per P 51 gauge (x = 2.0, 4.0, 5.2, 10.5, 12.5, 13.5 and 14.5 m), in the order given in ctrl.txt; compare them with the measurements of Beji and Battjes [1993] as in the figures below. REEF3D_CFD_WSFLINE contains the free surface along the tank every 10 iterations together with the theoretical wave (P 53 1).

13.10.4 Results

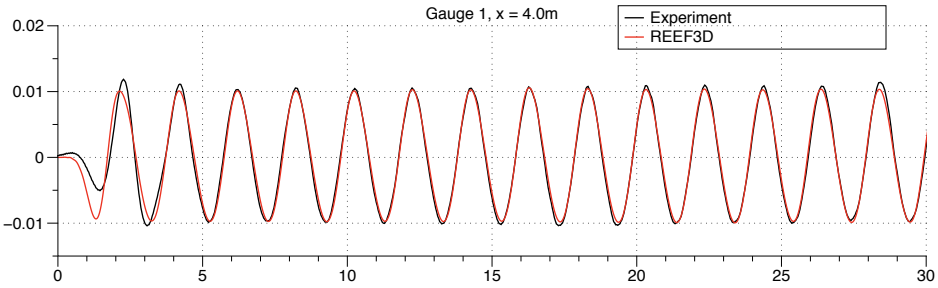


Figure 13.26: Wave gage 1

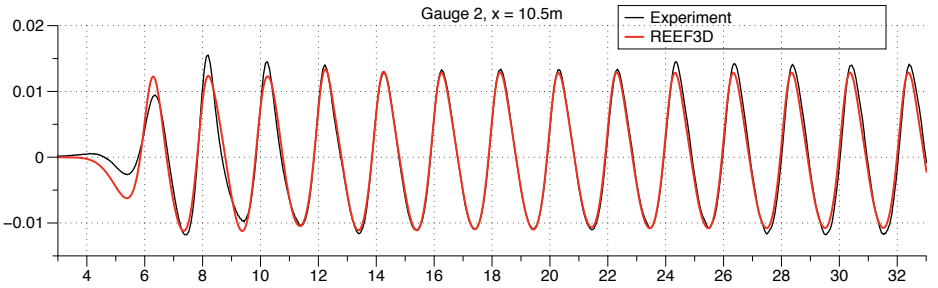


Figure 13.27: Wave gage 2

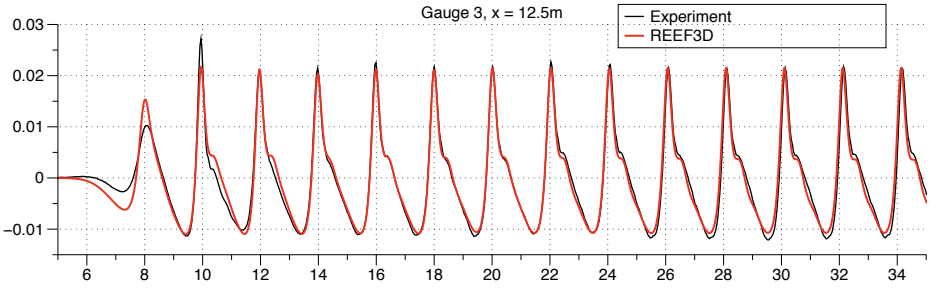


Figure 13.28: Wave gage 3

Figure 13.29: Wave gage 4

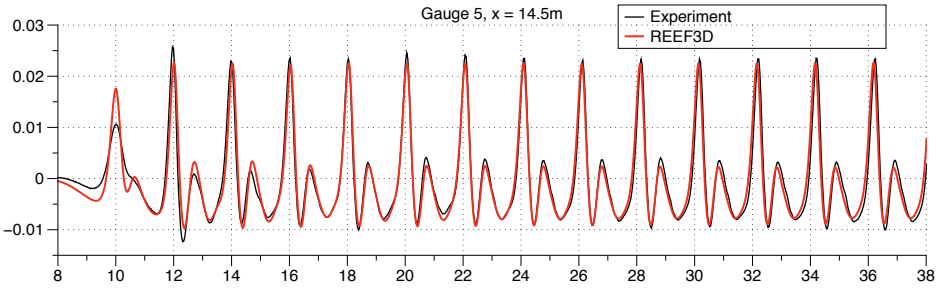


Figure 13.30: Wave gage 5

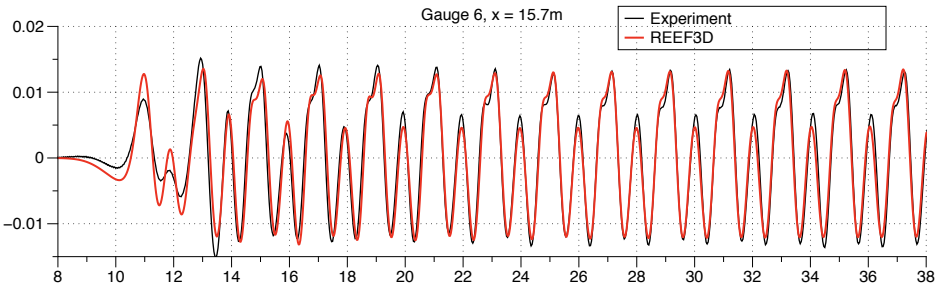


Figure 13.31: Wave gage 6

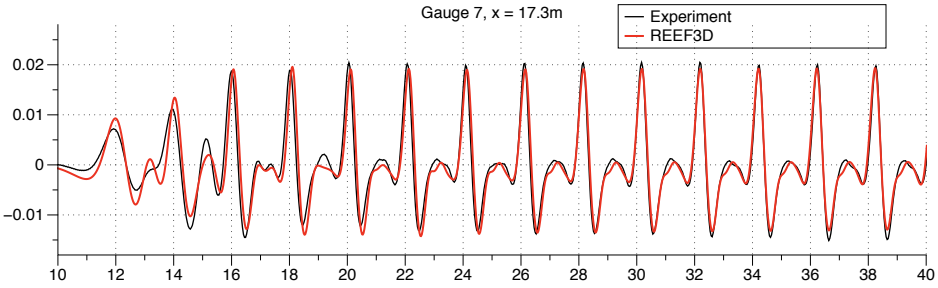


Figure 13.32: Wave gage 7

13.11 Plunging Breaking Waves over Slope

Repository folder: `Tutorials/REEF3D_CFD/11_11 Plunging Breaking Waves Slope`

The well-known benchmark case for plunging breaking waves over a slope by Ting and Kirby [1995] is shown here. Cnoidal waves are generated with the relaxation method and break on a 1:35 slope; there is no numerical beach (zero length in B 96), the wave energy is dissipated by breaking on the slope. The breaking waves require a relatively fine grid, thus 128 cores on NOTUR's supercomputer facilities were used to simulate the case. For more information on this case and breaking waves over slopes in general, please have a look at these in-depth studies: Alagan Chella et al. [2015b], Alagan Chella et al. [2015c] and Alagan Chella et al. [2015a].

- Wave: 5th-order cnoidal, H = 0.128 m, T = 5.0 s (B 93), water depth d = 0.4 m
- Domain: 40 m × 0.005 m × 1.0 m, cell size dx = 0.005 m (8000 × 1 × 200 cells), slope from x = 13.8 m

- Wave generation zone 9.8 m (B 96), simulated time 60 s, k-ω model with free surface damping (T 36 2)
- Run: **./DiveMESH**, then **mpirun -n 128 reef3d**

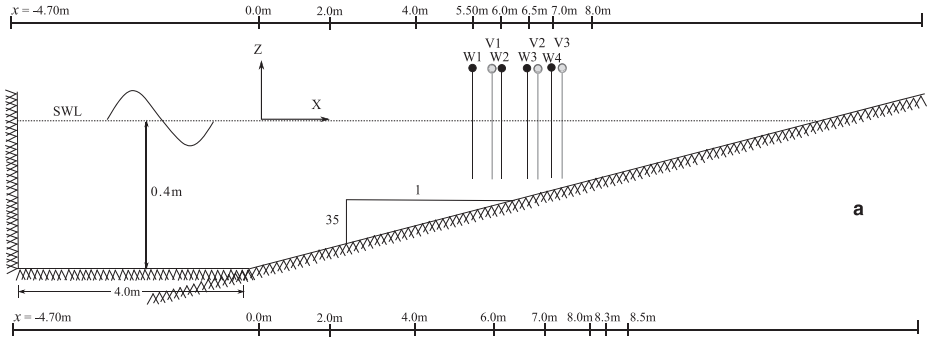


Figure 13.33: Experimental setup plunging breaking waves over slope

13.11.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.005 // uniform cell size dx [m]

B 10 0.0 40.0 0.0 0.005 0.0 1.00 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 61 13.8 40.0 0.0 0.005 0.0 0.748 // bed slope (about 1:35) starting at x = 13.8 m

M 10 128 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.11.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 93 0.128 5.0 // wave height H [m], wave period T [s]
B 96 9.8 0.0 // wave generation zone 9.8 m, beach length 0 m
B 99 2 // numerical beach: relaxation method 2 (zero length in B 96)
B 98 2 // wave generation: relaxation method 2

```
D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.4 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 128 // number of processes (same as in control.txt)

P 30 0.05 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 0.50 // state file interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 10 0.0 // discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 19.8 0.0025 // wave gauge 1 at x, y [m]
P 51 20.8 0.0025 // wave gauge 2 at x, y [m]
P 51 21.8 0.0025 // wave gauge 3 at x, y [m]
P 51 22.1 0.0025 // wave gauge 4 at x, y [m]

P 52 0.0025 // free surface line (wsfline) along x at this y [m]

P 61 21.095 0.0025 0.35 // point probe 1 at x, y, z [m]
P 61 21.095 0.0025 0.25 // point probe 2 at x, y, z [m]
P 61 21.095 0.0025 0.4 // point probe 3 at x, y, z [m]

P 61 22.025 0.0025 0.35 // point probe 4 at x, y, z [m]
P 61 22.025 0.0025 0.26 // point probe 5 at x, y, z [m]
P 61 22.025 0.0025 0.4 // point probe 6 at x, y, z [m]
P 61 21.595 0.0025 0.4 // point probe 7 at x, y, z [m]

P 61 22.145 0.0025 0.35 // point probe 8 at x, y, z [m]
P 61 22.145 0.0025 0.28 // point probe 9 at x, y, z [m]
P 61 22.145 0.0025 0.4 // point probe 10 at x, y, z [m]
```

13.11.3 What to look at

The four P 51 gauges at x = 19.8, 20.8, 21.8 and 22.1 m are written to 'REEF3D-CFD-WSF-HG.dat' in REEF3D_CFD_WSF; compare them with the measurements of Ting and Kirby [1995]. The P 61 point probes in the breaking region write velocities and pressure to 'REEF3D-CFD-Probe-Point-n.dat' in REEF3D_CFD_ProbePoint. REEF3D_CFD_VTU (every 0.05 s; P 10 is 1 by default) shows the plunging jet; color by velocity as in the first figure below. State files are written every 0.5 s (P 40, P 42).

13.11.4 Results

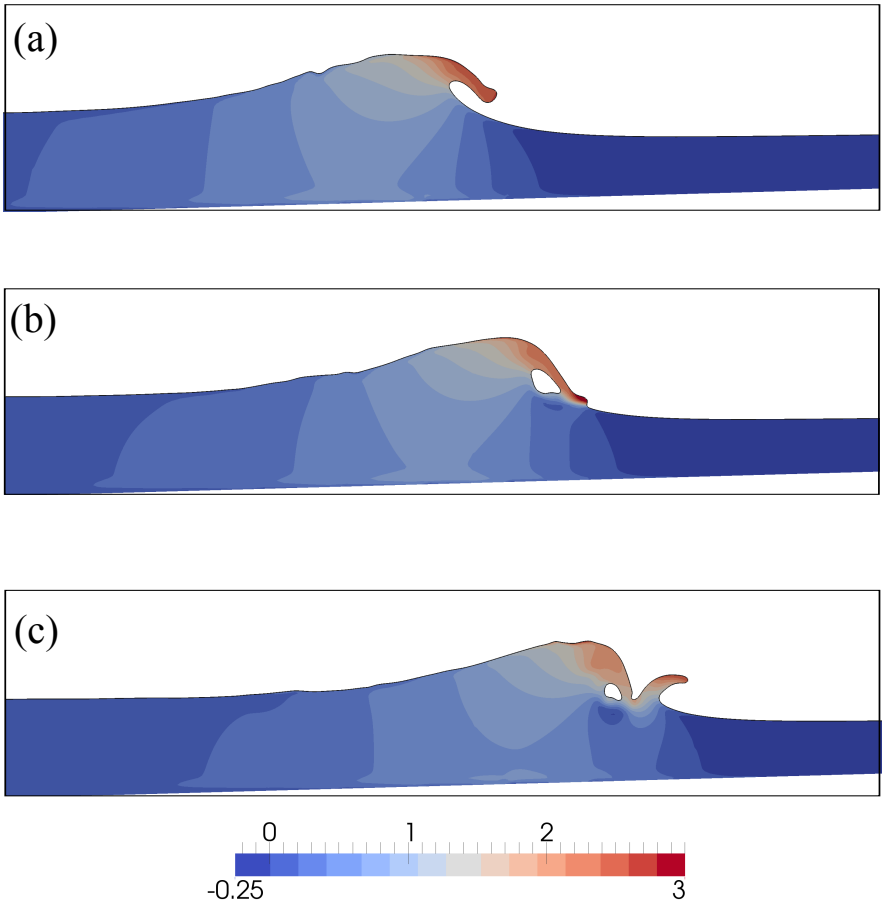


Figure 13.34: Variation of horizontal velocity, U (m/s) under the plunging breaker at t= 10.85s (a), 10.95s (b) and 11.05s (c)

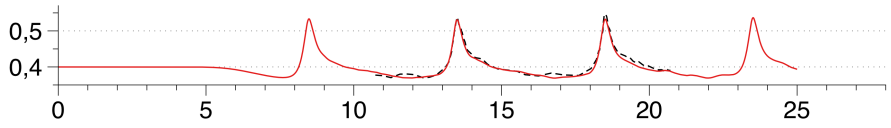


Figure 13.35: Wave gage 1, black experiment Ting and Kirby [1995], red REEF3D

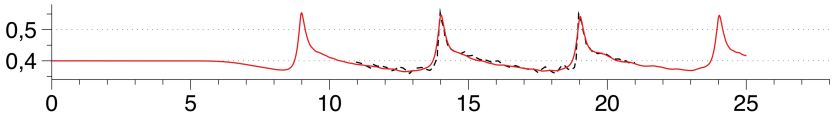


Figure 13.36: Wave gage 2, black experiment Ting and Kirby [1995], red REEF3D

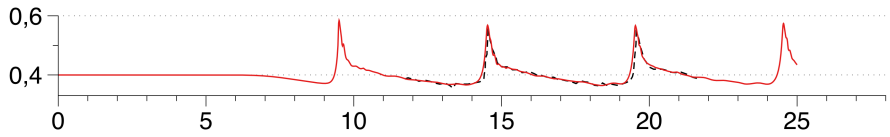


Figure 13.37: Wave gage 3, black experiment Ting and Kirby [1995], red REEF3D

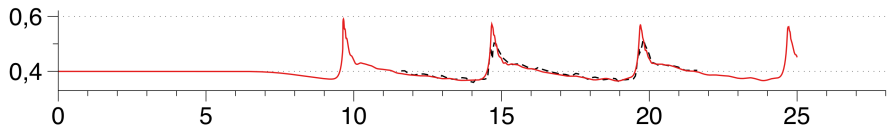


Figure 13.38: Wave gage 4, black experiment Ting and Kirby [1995], red REEF3D

13.12 Plunging Breaking Waves over Slope with Non-uniform Grid

Repository folder: `Tutorials/REEF3D_CFD/11_12 Plunging Breaking Waves Slope NUG`

The simulation of plunging breaking waves over a slope from the previous section by Ting and Kirby [1995] is shown here with a non-uniform grid (NUG) setup. The grid is refined in x-direction around the breaking region (x = 20 m, B 101 5) and in z-direction around the still water level (z = 0.4 m, B 103 5), and the tank is shortened to 30 m. This reduces the number of cells compared to the uniform grid, so the case runs with 6 processes. The ctrl.txt is identical to the one above, with the exception of the M 10 parameter.

- Domain: 30 m × 0.005 m × 1.0 m, 3000 × 1 × 80 cells (B 2) with sinh stretching (stretching factor 2.5) in x and z
- Wave and numerics as in the previous section
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.12.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 2 3000 1 80 // number of cells N_x, N_y, N_z (overrides B 1)

B 10 0.0 30.0 0.0 0.005 0.0 1.00 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

```
B 101 5 // x-direction stretching: point focus (sinh), uses B 111 and B 114
B 111 2.5 // stretching factor in x-direction
B 114 20.0 // focus point of the x-stretching [m]

B 103 5 // vertical stretching: point focus (sinh), uses B 113 and B 116
B 113 2.5 // stretching factor in z-direction
B 116 0.40 // focus point of the z-stretching [m]

S 61 13.8 30.0 0.0 0.005 0.0 0.463 // bed slope (about 1:35) starting at x = 13.8 m

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)
```

13.12.2 REEF3D: ctrl.txt

```
B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 8 // wave type: 5th-order cnoidal waves
B 93 0.128 5.0 // wave height H [m], wave period T [s]
B 96 9.8 0.0 // wave generation zone 9.8 m, beach length 0 m
B 99 2 // numerical beach: relaxation method 2 (zero length in B 96)
B 98 2 // wave generation: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.4 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 60.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 30 0.05 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 0.50 // state file interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 10 0.0 // discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]
```

P 51 19.8 0.0025 // wave gauge 1 at x, y [m]
P 51 20.8 0.0025 // wave gauge 2 at x, y [m]
P 51 21.8 0.0025 // wave gauge 3 at x, y [m]
P 51 22.1 0.0025 // wave gauge 4 at x, y [m]

P 52 0.0025 // free surface line (wsfline) along x at this y [m]

P 61 21.095 0.0025 0.35 // point probe 1 at x, y, z [m]
P 61 21.095 0.0025 0.25 // point probe 2 at x, y, z [m]
P 61 21.095 0.0025 0.4 // point probe 3 at x, y, z [m]

P 61 22.025 0.0025 0.35 // point probe 4 at x, y, z [m]
P 61 22.025 0.0025 0.26 // point probe 5 at x, y, z [m]
P 61 22.025 0.0025 0.4 // point probe 6 at x, y, z [m]
P 61 21.595 0.0025 0.4 // point probe 7 at x, y, z [m]

P 61 22.145 0.0025 0.35 // point probe 8 at x, y, z [m]
P 61 22.145 0.0025 0.28 // point probe 9 at x, y, z [m]
P 61 22.145 0.0025 0.4 // point probe 10 at x, y, z [m]

13.12.3 What to look at

The same output as in the previous section is written. Compare the gauges in REEF3D_CFD_WSF with those of the uniform grid case, and inspect the stretched grid in ParaView by showing the REEF3D_CFD_VTU data with the surface-with-edges representation.

13.13 Shoaling over Irregular Topography

Repository folder: `Tutorials/REEF3D_CFD/11_13 Shoaling over Irregular Bathymetry`

In this case, wave propagation over irregular topography is calculated using phase-accurate irregular waves based on Boers' experiment Boers [1996]. Here, the bed topography is generated using a 'geo.dat' file with x, y, z point data. These coordinates are interpolated using a local inverse distance algorithm in DIVEMesh. The waves are generated using wave reconstruction (B 92 51), a very convenient way to create phase-accurate irregular waves based on a single wave gauge Aggarwal et al. [2018b]; the wave components are read from 'waverecon.dat' and applied with Dirichlet wave generation. No numerical beach is used (B 99 0); the waves break on the bar and on the foreshore.

- Water depth d = 0.75 m at the wave maker, irregular waves from waverecon.dat
- Domain: 36.3 m × 0.01 m × 1.0 m, cell size dx = 0.01 m (3630 × 1 × 100 cells)
- Simulated time 200 s, k-ω model with free surface damping (T 36 2)
- Extra input files: geo.dat (read by DIVEMesh), waverecon.dat (read by REEF3D)
- Run: **./DiveMESH**, then **mpirun -n 128 reef3d**

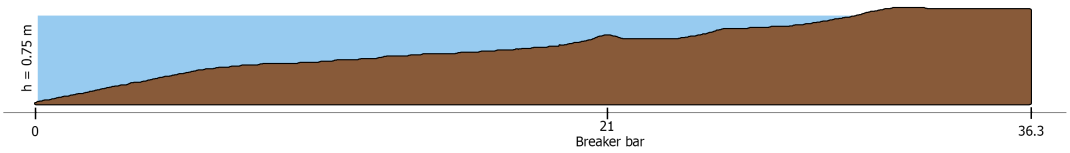


Figure 13.39: Experimental setup for shoaling over irregular bathymetry Elakel [2018]

13.13.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 3 // side 2 (y_{max}): symmetry plane
C 13 3 // side 3 (y_{min}): symmetry plane
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.01 // uniform cell size dx [m]

B 10 0.0 36.3 0.0 0.01 0.0 1.00 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

G 10 1 // turn on geodat interpolation of the bed from geo.dat

G 15 2 // local inverse distance interpolation
G 31 14 // number of outer smoothing filter iterations

M 10 128 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.13.2 DIVEMesh: geo.dat

In the 'geo.dat' file all bathymetry coordinates are given in ASCII format (three columns x, y, z) and are then used to interpolate the bed in DIVEMesh. The bed can be visualized with the zero contour of 'topo' in the .vtu files read by ParaView.

13.13.3 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 51 // wave type: wave reconstruction, 1st-order irregular waves from waverecon.dat
B 98 3 // wave generation: Dirichlet (DWG)
B 99 0 // numerical beach off

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

```
F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.75 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 200.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 128 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.05 // print interval in simulated time [s]
P 40 1 // print state files for hot-start
P 42 0.50 // state file interval [s]

T 10 2 // k-ω turbulence model
T 31 0.816 // eddy viscosity limiter factor in phase 1
T 36 2 // free surface turbulence damping on, including wall distance

W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 0.0 0.005 // wave gauge 1 at x, y [m]
P 51 4.0 0.005 // wave gauge 2 at x, y [m]
P 51 8.0 0.005 // wave gauge 3 at x, y [m]
P 51 12.0 0.005 // wave gauge 4 at x, y [m]
P 51 16.0 0.005 // wave gauge 5 at x, y [m]
P 51 20.0 0.005 // wave gauge 6 at x, y [m]

P 61 14.37 0.005 0.74 // point probe 1 at x, y, z [m]
```

13.13.4 REEF3D: waverecon.dat

The 'waverecon.dat' file contains the individual wave components, with the amplitude, angular frequency and phase shift in three columns, in order to generate phase-accurate irregular waves.

13.13.5 What to look at

Compare the six P 51 gauges at x = 0, 4, 8, 12, 16 and 20 m in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF) with the measurements; the gauge at x = 0 m lies on the wave generation boundary. The point probe (P 61) is written to REEF3D_CFD_ProbePoint. In ParaView, REEF3D_CFD_VTU (every 0.05 s) shows the shoaling and breaking waves; the bed is the zero contour of topo.

13.13.6 Results

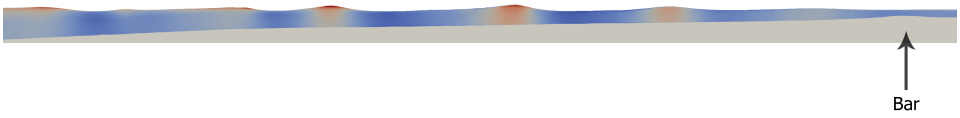


Figure 13.40: Shoaling of irregular waves for the Boers case Elakel [2018]

Figure 13.41: Breaking waves on the bar for the Boers case Elakel [2018]

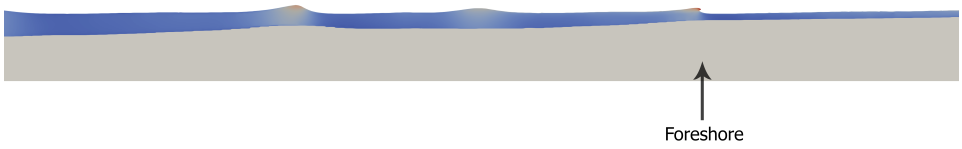


Figure 13.42: Breaking waves on the foreshore for the Boers case Elakel [2018]

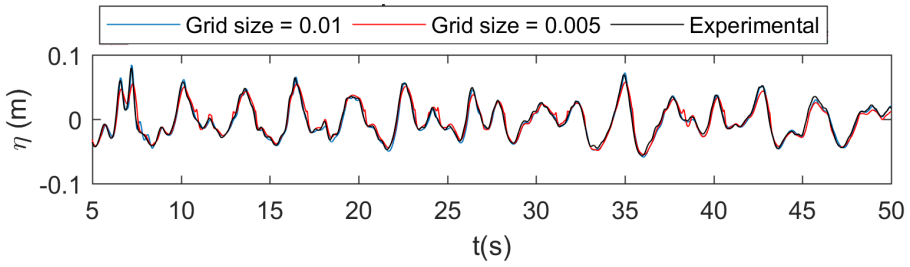


Figure 13.43: Wave gage 1

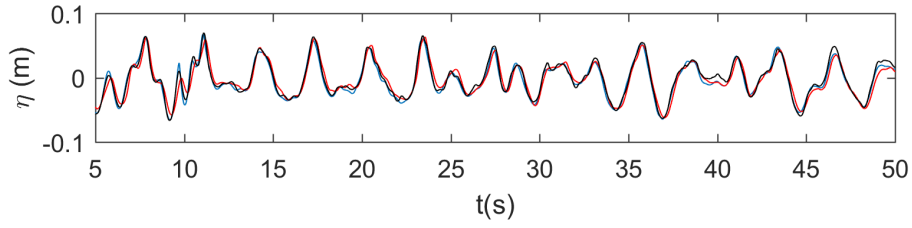


Figure 13.44: Wave gage 2

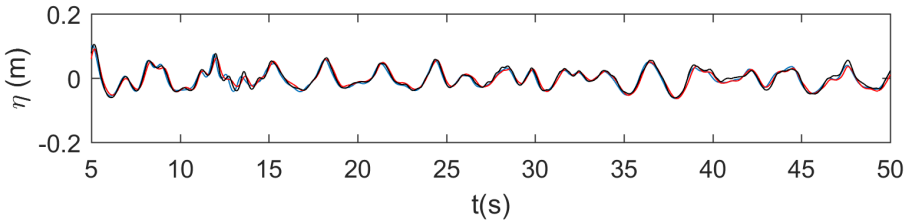


Figure 13.45: Wave gage 3

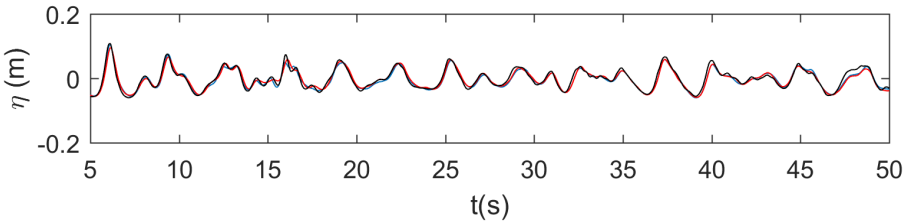


Figure 13.46: Wave gage 4

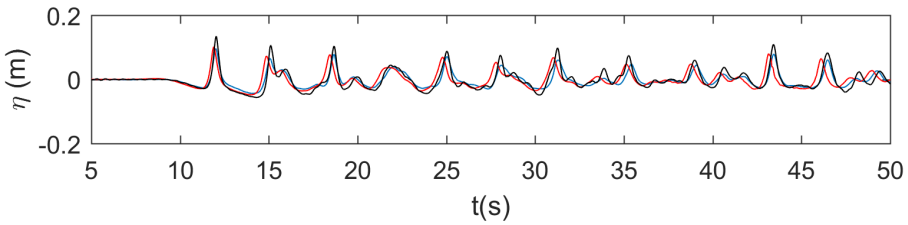


Figure 13.47: Wave gage 5

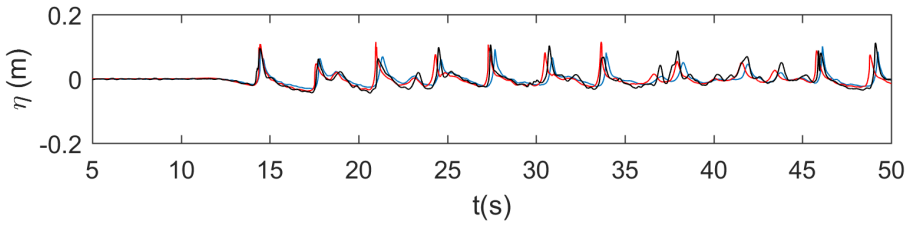


Figure 13.48: Wave gage 6

13.14 Non-Breaking Wave Forces

Repository folder: `Tutorials/REEF3D_CFD/11_14 Non-Breaking Wave Forces`

In many scenarios, wave forces need to be computed. Based on the experiments by Chen et al. [2014], numerical benchmark results were produced, see Bihs et al. [2016a] for more detail. In addition to the previous NWT examples, the use of the force box is shown in this example. The forces on a solid within the force box are calculated and printed out. It is important to give some margin around the solid in order to capture all surface facets in the

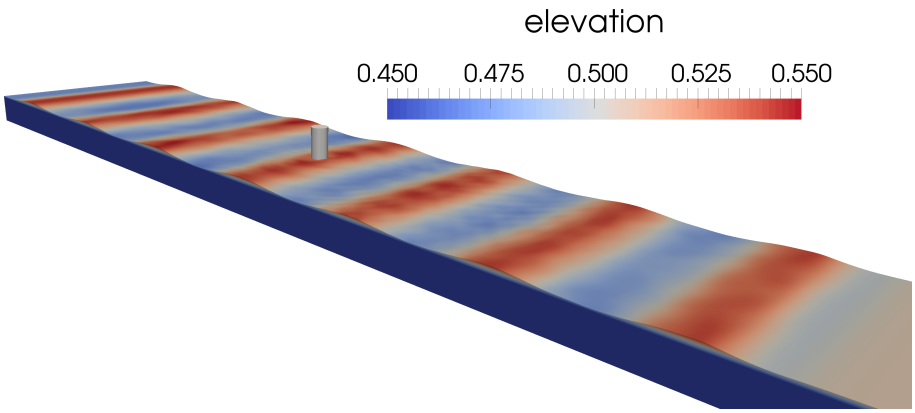


Figure 13.49: Non-breaking wave forces

pressure integration algorithm. More REEF3D related literature on non-breaking wave forces can be found here: Kamath et al. [2015b], Kamath et al. [2016b] and Kamath et al. [2015a].

- Wave: 2nd-order Stokes, H = 0.07 m, T = 1.22 s (B 93), water depth d = 0.505 m
- Domain: 18 m × 3 m × 1 m, cell size dx = 0.025 m (720 × 120 × 40 cells), side walls
- Cylinder at x = 7.5 m, y = 1.5 m, diameter 0.25 m; wave generation zone 2.11 m, numerical beach 4.22 m
- Simulated time 50 s; the folder also contains the measured data 'expt_data_wsf.txt' and 'expt_data_force.txt' (two columns: time, value)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.14.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 7 // side 4 (x_{max}): numerical beach
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.025 // uniform cell size dx [m]

B 10 0.0 18.0 0.0 3.0 0.0 1.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 33 7.50 1.50 0.125 // vertical cylinder at x = 7.5 m, y = 1.5 m, radius 0.125 m

M 20 2 // decomposition method 2 (improved rectangular)

M 10 6 // number of processes (partitions)

13.14.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 4 // wave type: 2nd-order Stokes waves
B 93 0.07 1.22 // wave height H [m], wave period T [s]
B 96 2.11 4.22 // lengths of wave generation zone and numerical beach [m]
B 98 2 // wave generation: relaxation method 2
B 99 2 // numerical beach: relaxation method 2

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 1.0 // length for the initial reinitialization instead of the domain length [m]
F 60 0.505 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 50.0 // maximum simulated time [s]
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 15.0 30.0 0.05 // print window: start [s], end [s], interval [s]
P 40 1 // print state files for hot-start
P 42 1.0 // state file interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 22 -9.81 // gravity in z-direction [m/s^2]

P 55 1.0 // wsfline/line probe interval [s]; no effect here, no P 52 or P 62 is given

P 51 5.00 1.50 // wave gauge 1 at x, y [m]

P 81 7.325 7.675 1.325 1.675 0.0 1.0 // force box around the cylinder

13.14.3 What to look at

The forces from the P 81 force box are written to 'REEF3D_CFD_Force-1.dat' in REEF3D_CFD_Force; plot the inline force F_x over time and compare it with 'expt_data_force.txt'. The gauge at x = 5 m (P 51) in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF) can be compared with 'expt_data_wsf.txt'. REEF3D_CFD_VTU contains the 3D field every 0.05 s between 15 s and 30 s (P 35).

13.14.4 Results

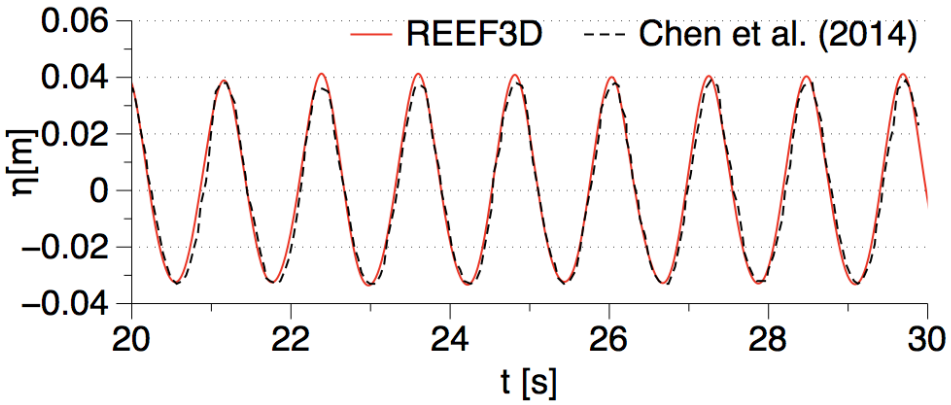


Figure 13.50: Free surface comparison between REEF3D and the experimental data Chen et al. [2014]

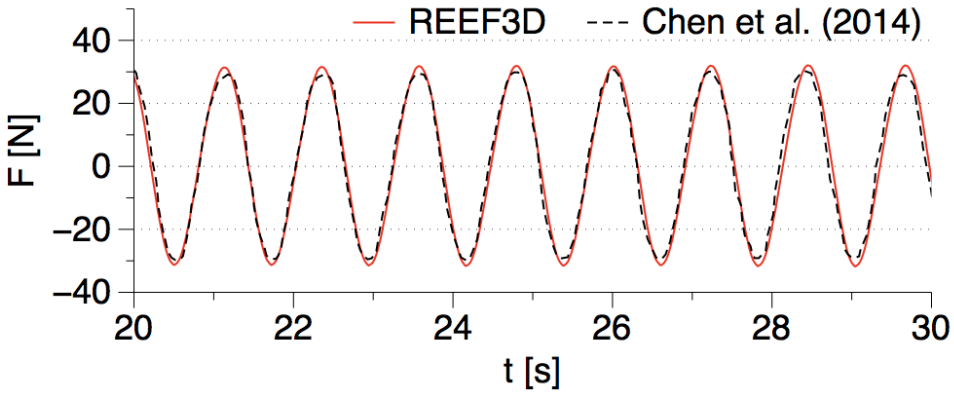


Figure 13.51: Wave Force comparison between REEF3D and the experimental data Chen et al. [2014]

13.15 Breaking Wave Forces

Repository folder: `Tutorials/REEF3D_CFD/11_15 Breaking Wave Forces`

Extending the previous scenario, breaking wave forces are simulated by using the case from Kamath et al. [2016a] based on an experimental study in the GWK in Hannover Irschik et al. [2002]. The waves are shoaling on a 1:10 slope and are breaking directly at the vertical cylinder. The slamming forces are calculated using the force box approach. REEF3D has been extensively validated and applied for breaking wave forces: Bihs et al. [2016b], Alagan Chella et al. [2019c], Alagan Chella et al. [2019b], Alagan Chella et al. [2019a].

- Wave: 5th-order Stokes, H = 1.3 m, T = 4.0 s (B 93), water depth d = 3.8 m, 1.5 m at the cylinder
- Domain: 54 m × 5 m × 7 m, cell size dx = 0.05 m (1080 × 100 × 140 cells), 1:10 slope from x = 21 m to 44 m

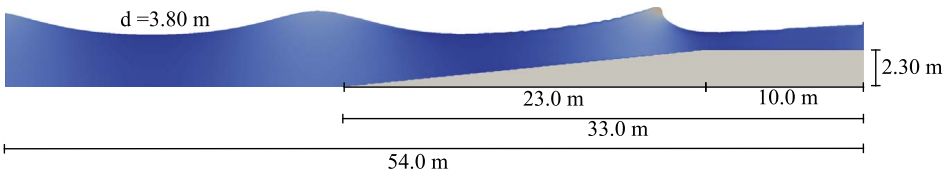


Figure 13.52: Breaking wave forces setup

- Cylinder at x = 44.0 m, y = 2.5 m, diameter 0.7 m; wave generation zone 21 m, active wave absorption at the outlet
- Simulated time 50 s (at most 200000 iterations); the folder also contains the measured data 'expt_data_wsf.txt' and 'expt_data_force.txt'
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.15.1 DIVEMesh: control.txt

C 11 6 // side 1 (x_{min}): wave generation
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 8 // side 4 (x_{max}): numerical beach, type 8 (handled like type 7 in REEF3D)
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // uniform cell size dx [m]

B 10 0.0 54.0 0.0 5.0 0.0 7.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 33 44.0 2.50 0.35 // vertical cylinder at x = 44.0 m, y = 2.5 m, radius 0.35 m

S 61 21.0 44.0 0.0 5.0 0.0 2.30 // 1:10 slope from x = 21 m to 44 m
S 10 44.0 54.0 0.0 5.0 0.0 2.30 // horizontal bed 2.3 m high behind the slope

M 20 2 // decomposition method 2 (improved rectangular)

M 10 6 // number of processes (partitions)

13.15.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.0001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (numerical wave tank)
B 92 5 // wave type: 5th-order Stokes waves (Fenton)
B 93 1.3 4.0 // wave height H [m], wave period T [s]
B 96 21.0 0.0 // lengths of wave generation zone and numerical beach [m]

B 98 2 // wave generation: relaxation method 2
B 99 3 // numerical beach: active wave absorption (AWA)

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 42 7.0 // length for the initial reinitialization instead of the domain length [m]
F 60 3.80 // still water level [m]

I 12 1 // hydrostatic pressure initialization

N 40 3 // momentum: 3rd-order TVD Runge-Kutta
N 41 50.0 // maximum simulated time [s]
N 45 200000 // maximum number of iterations
N 47 0.1 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 35 23.5 25.2 0.05 // print window: start [s], end [s], interval [s]
P 180 1 // print free surface .vtp files
P 182 0.1 // free surface .vtp print interval [s]

T 10 2 // k-ω turbulence model
T 36 2 // free surface turbulence damping on, including wall distance

W 22 -9.81 // gravity in z-direction [m/s^2]

P 51 22.0 2.50 // wave gauge 1 at x, y [m]
P 51 30.0 2.50 // wave gauge 2 at x, y [m]
P 51 35.0 2.50 // wave gauge 3 at x, y [m]
P 51 40.0 2.50 // wave gauge 4 at x, y [m]
P 51 43.65 4.80 // wave gauge 5 at x, y [m]
P 51 43.65 2.50 // wave gauge 6 at x, y [m]
P 51 44.5 2.505 // wave gauge 7 at x, y [m]
P 51 47.0 2.5 // wave gauge 8 at x, y [m]
P 51 50.5 2.5 // wave gauge 9 at x, y [m]
P 51 52.0 2.5 // wave gauge 10 at x, y [m]
P 51 53.0 2.5 // wave gauge 11 at x, y [m]

P 81 43.30 44.70 1.80 3.20 2.30 8.0 // force box around the cylinder

13.15.3 What to look at

Plot F_x from 'REEF3D_CFD_Force-1.dat' in REEF3D_CFD_Force together with 'expt_data_force.txt', and the 11 gauges in 'REEF3D-CFD-WSF-HG.dat' (REEF3D_CFD_WSF) together with 'expt_data_wsf.txt'. For the visualization of the breaker, REEF3D_CFD_FSF contains the free surface as .vtp every 0.1 s (P 180, P 182), and REEF3D_CFD_VTU the 3D field every 0.05 s between 23.5 s and 25.2 s (P 35).

13.15.4 Results

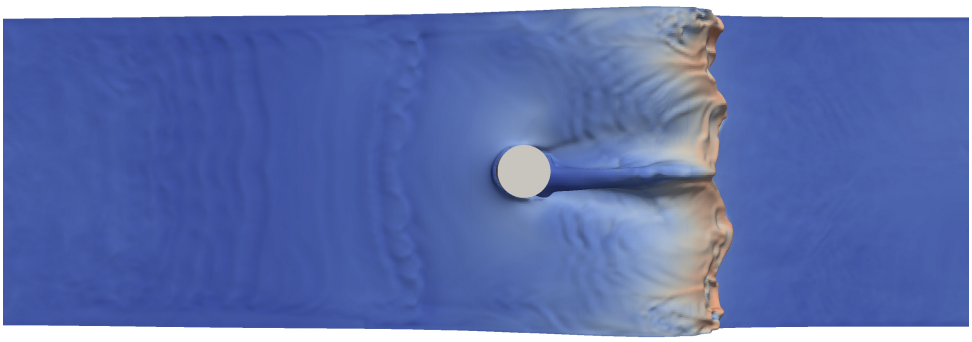


Figure 13.53: Breaking wave impact with REEF3D, plane view

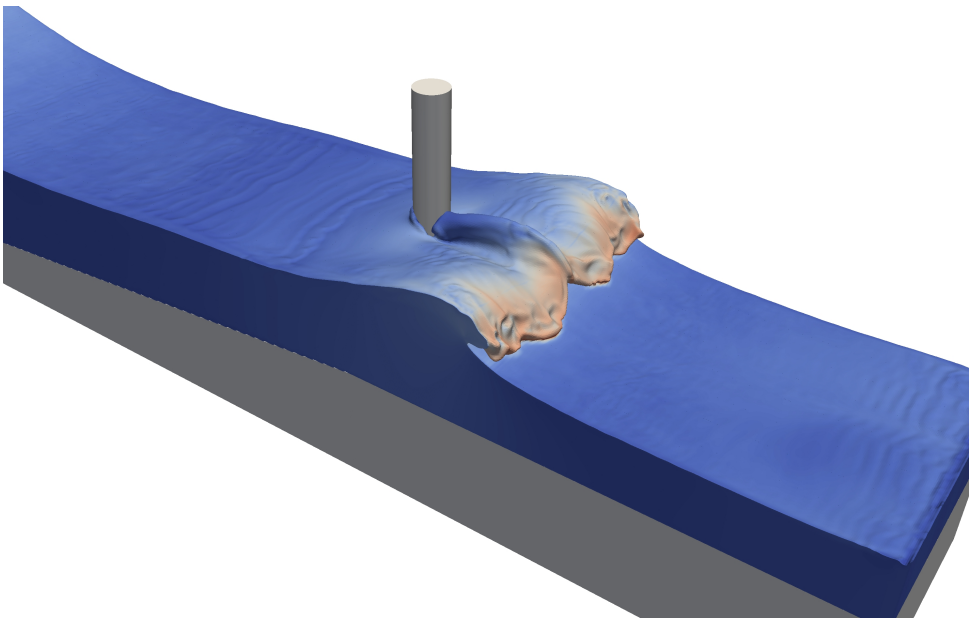


Figure 13.54: Breaking wave impact with REEF3D, side view

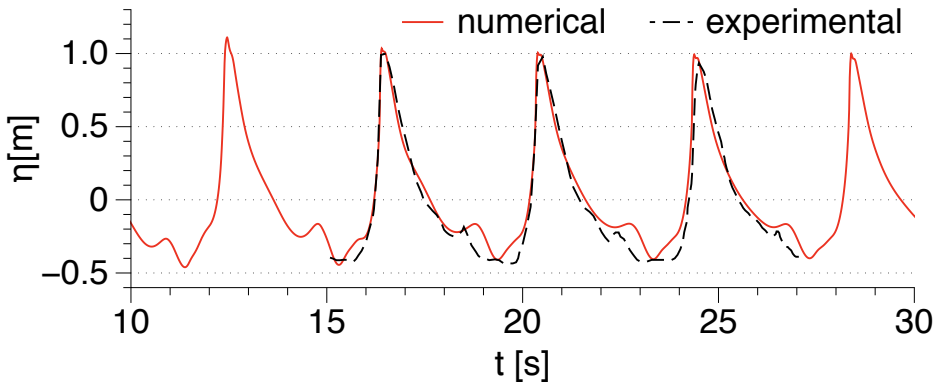


Figure 13.55: Free surface comparison between REEF3D and the experimental data Irschik et al. [2002]

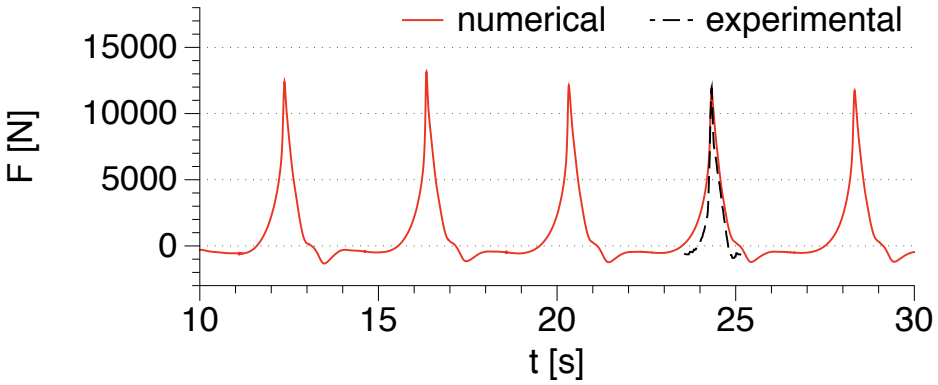


Figure 13.56: Wave Force comparison between REEF3D and the experimental data Irschik et al. [2002]

13.16 Heave Decay of a Sphere with Non-uniform Grid

Repository folder: `Tutorials/REEF3D_CFD/11_16 Heave Decay of Sphere`

Setup for simulations including floating bodies using the continuous direct forcing approach. As an example, the heave decay of a sphere can be simulated using the following input files together with the STL file 'floating.stl' provided in the tutorial folder. The sphere has a radius of 0.5 m and a density of 500 kg/m^3 (X 21), i.e. half the water density, so at rest its centre lies at the still water level z = 2.0 m. X 182 places the centre at z = 2.2 m, and the sphere is released from this position with only the translational degrees of freedom active (X 11). Relaxation zones along the four side walls (B 107) damp the radiated waves. It is recommended to use the cell-based stretching shown in this example to define a box with uniform cell sizes around the floating body.

- Domain: 6 m × 6 m × 4 m, cell-based stretching (B 101–B 103 11) with Δx_{min} = 0.05 m in a 1.5 m focus zone around $x = y$ = 3 m, z = 2 m and Δx_{max} = 0.3 m
- Still water level 2.0 m, initial heave offset 0.2 m, simulated time 10 s
- Extra input file: floating.stl (read by REEF3D)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.16.1 DIVEMesh: control.txt

C 11 21 // side 1 (x_{min}): wall
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 21 // side 4 (x_{max}): wall
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.05 // cell size [m], same as Δx_{min} of the cell-based stretching
B 10 0.0 6.0 0.0 6.0 0.0 4.0 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

B 101 11 // x-direction stretching: cell-size based, uses B 127
B 127 0.05 0.3 3.0 1.5 1.1 // x: Δx_{min}, Δx_{max}, focus point x_f, focus zone length δ_f, cell ratio r
B 102 11 // y-direction stretching: cell-size based, uses B 128
B 128 0.05 0.3 3.0 1.5 1.1 // y: Δy_{min}, Δy_{max}, focus point y_f, focus zone length δ_f, cell ratio r
B 103 11 // z-direction stretching: cell-size based, uses B 129
B 129 0.05 0.3 2.0 1.5 1.1 // z: Δz_{min}, Δz_{max}, focus point z_f, focus zone length δ_f, cell ratio r

M 10 6 // number of processes (partitions)
M 20 2 // decomposition method 2 (improved rectangular)

13.16.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 50 0.00001 // wall roughness k_s [m]
B 90 1 // turn on ioWave (used here for the damping zones only)
B 99 2 // numerical beach: relaxation method 2, zones from B 107
B 107 0.0 0.0 0.0 6.0 0.3 // damping zone along the wall x = 0, 0.3 m wide
B 107 6.0 6.0 0.0 6.0 0.3 // damping zone along the wall x = 6 m
B 107 0.0 6.0 0.0 0.0 0.3 // damping zone along the wall y = 0
B 107 0.0 6.0 6.0 6.0 0.3 // damping zone along the wall y = 6 m

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 35 5 // level set convection: 5th-order WENO (HJ)
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 60 2.0 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 4 // momentum: 3rd-order low-storage Runge-Kutta (used with 6DOF)
N 41 10 // maximum simulated time [s]
N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 30 0.1 // print interval in simulated time [s]

W 1 1000.0 // density of water [kg/m^3]
W 22 -9.81 // gravity in z-direction [m/s^2]

X 10 1 // turn on the 6DOF floating body algorithm
X 11 1 1 1 0 0 0 // degrees of freedom u, v, w, p, q, r: translations free, rotations fixed
X 21 500.0 // density of the floating body [kg/m^3]
X 180 1 // read the floating body geometry from floating.stl
X 182 3.0 3.0 2.2 // translate the STL geometry by $\Delta x, \Delta y, \Delta z$ [m]

13.16.3 What to look at

The motion of the body is written to REEF3D_CFD_6DOF: plot the vertical position over time from the 'REEF3D_6DOF_position' file to obtain the heave decay curve, and use the 'REEF3D_6DOF_forces' file for the forces on the body. The body surface is written as .vtp to REEF3D_CFD_6DOF_VTP, which can be loaded in ParaView together with REEF3D_CFD_VTU (every 0.1 s) to show the radiated waves (contour phi = 0).

13.17 Abutment Scour

Repository folder: `Tutorials/REEF3D_CFD/11_17 Abutment Scour`

Local scour at a vertical-wall abutment in an open channel flume. The abutment is a box of 0.1 m length in flow direction that protrudes 0.2 m from the side wall into the 0.6 m wide channel, standing on a sediment bed with its top at z = 0 (S 57). The discharge is prescribed with ioFlow; the bed evolves with the Engelund-Fredsøe bedload formula, critical shear stress reduction on slopes, bedload direction correction and the sandslide algorithm. Near the inlet the bed is kept at its initial level with a bed relaxation zone (S 73). The case demonstrates the coupling of the CFD flow solver with the sediment transport module.

- Domain: x = 0.5–4.0 m, y = 0–0.6 m, z = -0.3–0.2 m, cell size dx = 0.02 m (175 × 30 × 25 cells)
- Flow depth 0.095 m above the bed (F 60), discharge 0.02679 m^3/s (mean velocity about 0.47 m/s)
- Sediment: d_{50} = 1.9 mm, density 2570 kg/m^3, critical Shields parameter 0.03; sediment transport starts after 100 iterations and the bed is updated every 10 iterations
- The run ends after 7200 s of sediment time (S 19) or 500000 iterations (N 45)
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

Note: this ctrl.txt contains S 17 0. With the default S 15 0 the sediment time step is S 17 times the computed value, i.e. zero, so in the current code the bed does not change and the run is only ended by N 45. Use the default S 17 1.0 to obtain a developing scour hole.

13.17.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
b 2 140 24 30 // lower-case letter: ignored, the grid follows B 1
B 10 0.5 4.0 0.0 0.6 -0.3 0.2 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 10 1.70 1.8 0.4 0.6 -0.3 0.2 // abutment: box attached to the side wall $y = 0.6$ m

B 103 0 // no vertical stretching (B 113 and B 116 are then not used)
B 113 6.0 // stretching factor in z-direction (not used with B 103 0)
B 116 0.312 // focus point of the z-stretching (not used with B 103 0)

M 10 6 // number of processes (partitions)

13.17.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 20 2 // no-slip velocity boundary condition
B 50 0.001 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10

D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 46 0 // no Picard volume correction after reinitialization
F 60 0.095 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 2 // momentum: 2nd-order TVD Runge-Kutta
N 45 500000 // maximum number of iterations
N 47 0.2 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 100 // print interval in iterations
P 23 1 // add the test array to the ParaView output
P 26 1 // add the sediment concentration (ST_conc) to the .vtu files
P 76 1 // add bedload q_b, q_{be} to the .vtu files

```
P 77 1 // add bed slope parameters to the .vtu files
P 78 1 // add sediment parameters (dh, bed change, reduction, ...) to the .vtu files
P 79 1 // add bed shear stress to the ParaView output

T 10 2 // k-ω turbulence model
T 36 1 // free surface turbulence damping on

W 10 0.02679 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

S 10 1 // sediment transport on
S 11 3 // bedload formula: Engelund and Fredsøe
S 13 5.0 // maximum sediment time step [s]
S 14 0.5 // CFL factor for the sediment time step
S 16 4 // bed shear stress from turbulent kinetic energy
S 17 0 // decoupling factor for the sediment time step; 0 gives a zero sediment time step
(see note)
S 19 7200 // maximum sediment (morphological) time [s], ends the run
S 20 0.0019 // sediment d50 [m]
S 21 10.0 // factor on d50 for ks in the bed shear stress
S 22 2570.0 // sediment density [kg/m^3]
S 24 0.75 // porosity of the sediment layer
S 30 0.03 // critical Shields parameter

S 31 2 // Exner formulation, version 2

S 41 1 // sediment start criterion: iterations
S 42 1 // bed update interval criterion: iterations
S 43 100 // flow iterations before sediment transport starts
S 44 10 // flow iterations between bed updates
S 50 3 // bed level fixed at inflow and outflow
S 57 0.0 // initial sediment bed level (top of the sediment) [m]
S 73 0.00 0.75 0.0 0.5 0.0 // bed kept at z = 0 within 0.75 m of the inlet x = 0.5 m

S 80 4 // critical shear stress reduction on slopes: Fredsøe
S 81 30.0 // angle of repose [deg]
S 82 7.5 // deltaphi parameter [deg]
S 85 1 // bedload direction correction (Koch and Flokstra)
S 90 1 // sandslide algorithm: explicit full correction
S 91 5 // number of sandslide iterations
S 92 0.25 // sandslide correction factor
```

13.17.3 What to look at

With sediment transport active, the bed is written to the .vtu files as the variable topo: in ParaView open REEF3D_CFD_VTU (every 100 iterations, P 20) and show the contour topo = 0 colored by elevation to follow the development of the scour hole at the upstream corner of the abutment. P 76–P 79 add the sediment variables (prefix ST_, e.g. bedload, bed change and bed shear stress) and P 26 the sediment concentration to the same files; P 23 adds the test array. The free surface is the contour phi = 0.

13.18 Pier Scour

Repository folder: `Tutorials/REEF3D_CFD/11_18 Pier Scour`

Local scour around a vertical circular pier in an open channel flume. The pier stands in the middle of a 2 m wide channel on a sediment bed with its top at z = 0 (S 57). The discharge is prescribed with ioFlow with a logarithmic inflow profile (B 61 2), and the bed evolves with the Engelund-Fredsøe bedload formula, slope effects and the sandslide algorithm, with a bed update after every flow iteration after the first 100 iterations. The bed is kept at its initial level near the inlet (S 73); the second, lower-case s 73 line for the outlet is deactivated.

- Domain: x = 1.0–5.5 m, y = 0–2.0 m, z = -0.3–0.5 m, cell size dx = 0.02 m (225 × 100 × 40 cells)
- Pier at x = 2.5 m, y = 1.0 m, diameter 0.2 m; flow depth 0.3 m, discharge 0.18 m^3/s (mean velocity 0.3 m/s)
- Sediment: d_{50} = 0.97 mm, density 2570 kg/m^3, critical Shields parameter 0.035
- The run ends after 75600 s of sediment time (S 19); no N 41 or N 45 limit is set
- Run: **./DiveMESH**, then **mpirun -n 6 reef3d**

13.18.1 DIVEMesh: control.txt

C 11 1 // side 1 (x_{min}): inflow
C 12 21 // side 2 (y_{max}): wall
C 13 21 // side 3 (y_{min}): wall
C 14 2 // side 4 (x_{max}): outflow
C 15 21 // side 5 (bottom): wall
C 16 3 // side 6 (top): symmetry plane

B 1 0.02 // uniform cell size dx [m]
B 10 1.0 5.5 0.0 2.0 -0.3 0.5 // rectangular domain $x_s, x_e, y_s, y_e, z_s, z_e$ [m]

S 33 2.5 1.0 0.1 // pier: vertical cylinder at x = 2.5 m, y = 1.0 m, radius 0.1 m

M 10 6 // number of processes (partitions)
M 13 0 // no partitioning in z-direction

13.18.2 REEF3D: ctrl.txt

B 10 1 // wall functions for the velocities on
B 11 1 // wall functions for the turbulence model on
B 50 0.006 // wall roughness k_s [m]
B 60 1 // ioFlow: constant inflow discharge from W 10
B 61 2 // ioFlow inflow profile: logarithmic

```
D 10 4 // convection: 5th-order WENO (conservative flux form)
D 20 2 // diffusion: implicit
D 30 1 // pressure: projection method

F 30 3 // level set: 3rd-order TVD Runge-Kutta
F 40 3 // reinitialization: 3rd-order TVD Runge-Kutta
F 60 0.3 // still water level [m]

I 10 1 // initialize everything (potential flow velocities, pressure, turbulence)

N 40 2 // momentum: 2nd-order TVD Runge-Kutta

N 47 0.3 // CFL number for adaptive time stepping

M 10 6 // number of processes (same as in control.txt)

P 10 1 // print ParaView files (.vtu)
P 20 100 // print interval in iterations
P 23 1 // add the test array to the ParaView output
P 26 1 // add the sediment concentration (ST_conc) to the .vtu files
P 79 1 // add bed shear stress to the ParaView output
P 122 1 // print maximum bed change

T 10 2 // k-ω turbulence model
T 36 1 // free surface turbulence damping on

W 10 0.18 // inflow discharge [m^3/s]
W 22 -9.81 // gravity in z-direction [m/s^2]

S 10 1 // sediment transport on
S 11 3 // bedload formula: Engelund and Fredsøe
S 12 0 // suspended load off
S 13 5.0 // maximum sediment time step [s]
S 14 0.3 // CFL factor for the sediment time step
S 16 4 // bed shear stress from turbulent kinetic energy
S 19 75600 // maximum sediment (morphological) time [s], ends the run
S 20 0.00097 // sediment d_50 [m]
S 21 5.0 // factor on d_50 for k_s in the bed shear stress
S 22 2570.0 // sediment density [kg/m^3]
S 24 0.75 // porosity of the sediment layer
S 30 0.035 // critical Shields parameter

S 41 1 // sediment start criterion: iterations
S 42 1 // bed update interval criterion: iterations
S 43 100 // flow iterations before sediment transport starts
S 44 1 // flow iterations between bed updates
S 50 1 // bed level fixed at the inflow
S 57 0.0 // initial sediment bed level (top of the sediment) [m]
S 73 0.00 0.5 0.0 1.0 0.0 // bed kept at z = 0 within 0.5 m of the inlet x = 1.0 m
s 73 0.00 0.25 0.0 5.5 0.0 // lower-case letter: ignored (would fix the bed near the outlet x
= 5.5 m)
S 80 4 // critical shear stress reduction on slopes: Fredsøe
S 81 35.0 // angle of repose [deg]
S 82 10.0 // deltaphi parameter [deg]
S 90 1 // sandslide algorithm: explicit full correction
S 91 10 // number of sandslide iterations
```

13.18.3 What to look at

The maximum bed change is written to 'REEF3D-CFD-Sediment-Max.dat' in REEF3D_CFD_Sediment (P 122); plot it over time to follow the development of the scour depth. In ParaView, open REEF3D_CFD_VTU (every 100 iterations, P 20) and show the contour topo = 0 for the bed with the scour hole and the deposition downstream of the pier; P 79 adds the bed shear stress (ST_ variables), P 26 the sediment concentration and P 23 the test array.

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